COMPLETE SOURCE CODE SPECIFICATION

RepoPilot AI

Autonomous Bug Repair Pipeline, Multi-Language Code Practice Lab (8 Languages), Real-Time AST Sentinel & Interactive Learning Academy.

PROJECT NAME
Nexora-X-Hub (RepoPilot AI)

ARCHITECTURE
Next.js 16 (App Router) + TypeScript + Node.js

MULTI-LANGUAGE RUNTIMES
Python 3.12, Java 21, C++20, C17, JS, TS, Go, Rust

STYLING & UI
Tailwind CSS v4 + Lucide Icons + Canvas Confetti

TOTAL INCLUDED FILES
30 Core Project Modules

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Repository: github.com/personale88/Nexora-X-Hub

Confidential & Proprietary Source Code Document

1. System Architecture & Platform Capabilities

Multi-Language Code Practice Lab

Interactive sandbox supporting Python 3, Java, C++, C, JavaScript, TypeScript, Go, and Rust. Features as-you-type AST syntax diagnostics, delimiter checking, and compiler emulation.

Interactive Learning Academy

Structured course platform with 7 deep-dive tracks, 25+ interactive lessons, module syllabi, real-time knowledge check quizzes, streak/XP tracking, and digital completion certificates.

AI Error Concept Explainer

4-tab dynamic diagnostic engine: (1) Root-cause diagnosis, (2) Architectural mental model & real-world analogy, (3) Surgical fix comparison with progressive hints, and (4) Pedagogical AI Tutor chat.

Disposable Sandbox Execution

Isolated filesystem runner with strict path traversal assertion, command execution allowlisting (blocking dangerous shell operators), secret token redaction, and multi-language compilers.

2. Table of Included Source Files

#	Source File Path	Module Description	Language
1	`package.json`	Project Dependencies & Scripts Configuration	JSON
2	`tsconfig.json`	TypeScript Strict Compiler Configuration	JSON
3	`src/lib/types.ts`	Core Domain Types, Language Definitions & Data Contracts	TYPESCRIPT
4	`src/lib/courses-data.ts`	Learning Academy Courses, Syllabus, Quizzes & Challenges	TYPESCRIPT
5	`src/components/views/AcademyView.tsx`	Interactive Learning Academy View & Quiz Player	TSX
6	`src/lib/error-explainer.ts`	AST Error Analysis & Multi-Language Real-Time Sentinel	TYPESCRIPT
7	`src/components/views/CodePracticeView.tsx`	Multi-Language Practice Lab (Python, Java, C++, C, JS, TS, Go, Rust)	TSX
8	`src/app/api/code-practice/route.ts`	Code Practice & AI Error Tutor API Endpoint	TYPESCRIPT
9	`src/app/page.tsx`	Main View Orchestrator & Tab Router	TSX
10	`src/app/layout.tsx`	Root HTML & Next.js Metadata Layout	TSX
11	`src/components/layout/Sidebar.tsx`	Navigation Sidebar with Academy & Practice Lab Badges	TSX

#	Source File Path	Module Description	Language
12	`src/components/layout/TopNavbar.tsx`	Application Header, User Profile & Status Bar	TSX
13	`src/components/views/LoginLandingView.tsx`	Login, Demo Quick Access & Landing Hero View	TSX
14	`src/components/views/DashboardView.tsx`	Dashboard Metrics, Overview Cards & Quick Launch	TSX
15	`src/lib/server/sandbox.ts`	Isolated Disposable Workspace Sandbox & Security Guardrails	TYPESCRIPT
16	`src/lib/server/repair-pipeline.ts`	Autonomous GitHub Issue Diagnostic & Repair Pipeline	TYPESCRIPT
17	`src/lib/server/github-service.ts`	GitHub Octokit API, Webhooks & PR Generator	TYPESCRIPT
18	`src/lib/server/llm-provider.ts`	Multi-Provider LLM Engine (OpenAI, Gemini, Anthropic)	TYPESCRIPT
19	`src/lib/server/auth-helpers.ts`	JWT Session Management & Authentication Helpers	TYPESCRIPT
20	`src/lib/server/crypto.ts`	Cryptographic Hashing, AES Encryption & Token Generation	TYPESCRIPT
21	`src/lib/server/db.ts`	Local JSON/SQLite Storage for Repair Jobs & Metrics	TYPESCRIPT
22	`src/lib/server/event-bus.ts`	Real-Time Server-Sent Events (SSE) Event Bus	TYPESCRIPT
23	`src/components/views/LiveBugFixPlayer.tsx`	Live Interactive Autonomous Repair Pipeline Visualizer	TSX
24	`src/components/views/FixView.tsx`	Unified Issue Investigation & Fix Workspace	TSX
25	`src/components/views/InvestigationView.tsx`	AST & Dependency Graph Issue Investigation View	TSX
26	`src/components/views/VerificationView.tsx`	Autonomous Test Suite Verification View	TSX
27	`src/components/views/HistoryView.tsx`	Historical Repair Audit Log & Performance Analytics	TSX
28	`src/components/views/RepositoryView.tsx`	Connected GitHub Repositories Explorer	TSX
29	`src/app/api/repair-jobs/route.ts`	Repair Jobs Controller & Execution Dispatcher	TYPESCRIPT
30	`src/app/api/auth/dev-mock/route.ts`	Dev Mock Authentication & Demo Session Generator	TYPESCRIPT

File [1/30]: package.json

JSON SOURCE

Description: *Project Dependencies & Scripts Configuration*

```json
{
  "name": "iqq",
  "version": "0.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "eslint",
    "test:acceptance": "npx tsx scripts/test-acceptance.mjs"
  },
  "dependencies": {
    "canvas-confetti": "^1.9.4",
    "lucide-react": "^1.52.0",
    "next": "16.3.8",
    "react": "19.2.8",
    "react-dom": "19.2.8"
  },
  "devDependencies": {
    "@tailwindcss/postcss": "^4",
    "@types/canvas-confetti": "^1.9.0",
    "@types/node": "^20",
    "@types/react": "^19",
    "@types/react-dom": "^19",
    "eslint": "^9",
    "eslint-config-next": "16.3.8",
    "tailwindcss": "^4",
    "typescript": "^5"
  }
}
```

File [2/30]: tsconfig.json

JSON SOURCE

Description: *TypeScript Strict Compiler Configuration*

```
{
  "compilerOptions": {
    "target": "ES2017",
    "lib": ["dom", "dom.iterable", "esnext"],
    "allowJs": true,
    "skipLibCheck": true,
    "strict": true,
    "noEmit": true,
    "esModuleInterop": true,
    "module": "esnext",
    "moduleResolution": "bundler",
    "resolveJsonModule": true,
    "isolatedModules": true,
    "jsx": "react-jsx",
    "incremental": true,
    "plugins": [
      {
        "name": "next"
      }
    ],
    "paths": {
      "@/*": ["./src/*"]
    }
  },
  "include": [
    "next-env.d.ts",
    "**/*.ts",
    "**/*.tsx",
    ".next/types/**/*.ts",
    ".next/dev/types/**/*.ts",
    "**/*.mts"
```

```
  ],
  "exclude": ["node_modules"]
}
```

File [3/30]: src/lib/types.ts

TYPESCRIPT SOURCE

Description: *Core Domain Types, Language Definitions & Data Contracts*

```typescript
export type NavTab =
  | 'dashboard'
  | 'repositories'
  | 'issues'
  | 'fix'
  | 'verification'
  | 'history'
  | 'live-fix'
  | 'courses'
  | 'practice';

export interface FileNode {
  name: string;
  path: string;
  type: 'file' | 'folder';
  size?: string;
  hasIssue?: boolean;
  children?: FileNode[];
  content?: string;
  language?: string;
}

export interface CodeHealth {
  architecture: { status: 'Good' | 'Warning' | 'Critical'; score: number; label: string };
  dependencies: { status: 'Good' | 'Warning' | 'Critical'; count: number; label: string };
  security: { status: 'Good' | 'Warning' | 'Critical'; vulnerabilities: number; label: string };
  testing: { status: 'Good' | 'Warning' | 'Critical'; coverage: number; label: string };
}

export interface LanguageStat {
  name: string;
```

```
  percentage: number;
  color: string;
}

export interface RepositoryData {
  id: string;
  name: string;
  description: string;
  branch: string;
  stack: string[];
  metrics: {
    files: number;
    linesOfCode: string;
    testCoverage: string;
    dependencies: number;
  };
  languages: LanguageStat[];
  health: CodeHealth;
  status: string;
  activeIssueId: string;
  gitUrl?: string;
  stars?: number;
}

export interface DetectedIssue {
  id: string;
  code: string; // e.g. AUTH-104
  title: string;
  description: string;
  severity: 'HIGH' | 'MEDIUM' | 'LOW' | 'CRITICAL';
  confidence: number; // e.g. 96
  affectedFile: string;
  lines: string;
  rootCause: string;
  impact: string;
```

```
  discoveryDate: string;
}

export interface ReasoningStep {
  id: string;
  text: string;
  detail?: string;
  status: 'pending' | 'in_progress' | 'completed';
  durationMs?: number;
}

export interface DiffLine {
  type: 'context' | 'deletion' | 'addition';
  oldLineNumber?: number;
  newLineNumber?: number;
  content: string;
}

export interface CodePatch {
  file: string;
  linesChanged: number;
  additions: number;
  deletions: number;
  risk: 'LOW' | 'MEDIUM' | 'HIGH';
  confidence: number;
  beforeCode: string;
  afterCode: string;
  diffLines: DiffLine[];
}

export interface RegressionTest {
  filename: string;
  description: string;
  suiteName: string;
```

```
  code: string;
  framework: string;
}

export interface VerificationResult {
  totalTests: number;
  passed: number;
  failed: number;
  durationMs: number;
  confidence: number;
  securityChecks: {
    name: string;
    status: 'PASS' | 'FAIL';
    details: string;
  }[];
  regressionTests: {
    name: string;
    status: 'PASS' | 'FAIL';
    durationMs: number;
  }[];
  logs: string[];
}

export interface HistoryItem {
  id: string;
  timestamp: string;
  issueCode: string;
  title: string;
  action: 'Fixed + Verified' | 'Analyzed' | 'Completed' | 'Investigated';
  status: 'success' | 'warning' | 'info';
  confidence?: number;
  repository: string;
  file: string;
}

```

```
export type AuthProvider = 'google' | 'git' | 'mobile';

export interface UserProfile {
  id: string;
  name: string;
  email?: string;
  phone?: string;
  avatar?: string;
  provider: AuthProvider;
  role: string;
  gitUsername?: string;
  verifiedAt: string;
}

export type RepairJobStatus =
  | 'QUEUED'
  | 'INITIALIZING'
  | 'REPRODUCING'
  | 'INVESTIGATING'
  | 'DIAGNOSING'
  | 'PATCHING'
  | 'TESTING'
  | 'VERIFYING'
  | 'COMPLETED'
  | 'FAILED'
  | 'CANCELLED'
  | 'NEEDS_REVIEW';

export interface RepairEvent {
  eventId: string;
  jobId: string;
  sequence: number;
  type: string;
  timestamp: string;
```

```
  stage: string;
  message: string;
  payload?: Record<string, unknown>;
}

export interface RepairJobDiagnosis {
  rootCause: string;
  affectedFile: string;
  startLine: number;
  endLine: number;
  explanation: string;
  flawExplanation?: string;
  attackScenario?: string;
  fixRationale?: string;
  rule?: string;
  confidence: number;
  severity: 'LOW' | 'MEDIUM' | 'HIGH' | 'CRITICAL';
}

export interface RepairJobPatch {
  file: string;
  beforeCode: string;
  afterCode: string;
  unifiedDiff: string;
  additions: number;
  deletions: number;
  explanation: string;
  appliedSuccessfully: boolean;
}

export interface RepairJobTestRun {
  command: string;
  exitCode: number;
  stdout: string;
  stderr: string;
```

```
    durationMs: number;
    timestamp: string;
    stage: 'reproduction' | 'verification' | 'regression';
    passed: boolean;
  }

  export interface RepairJob {
    id: string;
    userId: string;
    repository: {
      owner: string;
      name: string;
      branch: string;
      gitUrl: string;
      pinnedCommitSha: string;
      localPath?: string;
    };
    issueDescription: string;
    issueCode?: string;
    reproductionEvidence?: string;
    status: RepairJobStatus;
    currentStage: string;
    diagnosis?: RepairJobDiagnosis;
    proposedPatch?: RepairJobPatch;
    testRuns: RepairJobTestRun[];
    retryCount: number;
    maxRetries: number;
    verificationOutcome: 'VERIFIED' | 'FAILED' | 'INCOMPLETE' | 'NEEDS_REVIEW';
    unresolvedRisks: string[];
    createdAt: string;
    updatedAt: string;
    completedAt?: string;
    error?: string;
  }
```

```
239
240 // ========================================
241 // Learning Platform & Academy Interfaces
242 // ========================================
243
244 export interface QuizQuestion {
245   id: string;
246   question: string;
247   options: string[];
248   correctIndex: number;
249   explanation: string;
250 }
251
252 export interface Lesson {
253   id: string;
254   title: string;
255   durationMinutes: number;
256   type: 'concept' | 'interactive_lab' | 'quiz';
257   summary: string;
258   content: string[];
259   keyTakeaways?: string[];
260   codeExample?: {
261     language: string;
262     code: string;
263     description: string;
264   };
265   practiceChallengeId?: string;
266   quiz?: QuizQuestion[];
267 }
268
269 export interface CourseModule {
270   id: string;
271   title: string;
272   description: string;
273   lessons: Lesson[];
```

```
}

export interface Course {
  id: string;
  title: string;
  tagline: string;
  description: string;
  category:
    | 'Debugging & Errors'
    | 'JavaScript & TypeScript'
    | 'Python & Algorithms'
    | 'Security & Auth'
    | 'React & Architecture'
    | 'Java & Backend'
    | 'C & Systems Programming';
  level: 'Beginner' | 'Intermediate' | 'Advanced';
  duration: string;
  xp: number;
  badge: string;
  iconName: string;
  color: string;
  modules: CourseModule[];
  skillsGained: string[];
}

// ========================================
// Code Practicing Space & Error Explainer
// ========================================

export type SupportedLanguage =
  | 'javascript'
  | 'typescript'
  | 'python'
  | 'java'
```

```
    | 'cpp'
    | 'c'
    | 'go'
    | 'rust';

export interface TestCase {
  id: string;
  name: string;
  inputDescription: string;
  expectedOutput: string;
  codeSnippetToRun?: string;
}

export interface PracticeChallenge {
  id: string;
  title: string;
  difficulty: 'Easy' | 'Medium' | 'Hard';
  language: SupportedLanguage;
  category: 'Error Debugging' | 'Algorithms' | 'Async & Promises' | 'Security' | 'State & Closures';
  description: string;
  errorNameTarget: string;
  initialCode: string;
  solutionCode: string;
  hints: string[];
  testCases: TestCase[];
  relatedCourseId?: string;
  relatedLessonId?: string;
}

export interface ErrorExplanation {
  errorName: string;
  errorMessage: string;
  failingLine?: number;
  codeContext?: string;
  whyThisHappened: string;
```

```
    conceptTitle: string;
    conceptDeepDive: string;
    analogy: string;
    mentalModelDiagram?: string;
    suggestedFixBefore: string;
    suggestedFixAfter: string;
    fixExplanation: string;
    bestPractices: string[];
    relatedLesson?: {
      courseId: string;
      lessonId: string;
      title: string;
    };
  }

  export interface LiveDiagnostic {
    hasError: boolean;
    type: 'syntax' | 'runtime_warning' | 'security' | 'clean';
    title: string;
    line?: number;
    codeSnippet?: string;
    whereIsMistake: string;
    whyIsError: string;
    howToFix: string;
  }
```

File [4/30]: src/lib/courses-data.ts

TYPESCRIPT SOURCE

Description: *Learning Academy Courses, Syllabus, Quizzes & Challenges*

```typescript
import { Course, PracticeChallenge } from './types';

export const COURSES: Course[] = [
  {
    id: 'js-ts-debugging',
    title: 'Mastering JavaScript & TypeScript Debugging',
    tagline: 'Understand the V8 Engine, Execution Contexts, and Solve Runtime Errors Like a Pro',
    description:
      'Tired of seeing "TypeError: Cannot read properties of undefined"? In this deep-dive masterclass, you will demystify how JavaScript executes your code, explore the Call Stack and Event Loop, and learn systematic debugging workflows to eliminate runtime crashes forever.',
    category: 'Debugging & Errors',
    level: 'Beginner',
    duration: '4h 15m',
    xp: 650,
    badge: 'Error Detective',
    iconName: 'Bug',
    color: 'from-amber-500 to-orange-600',
    skillsGained: [
      'Call Stack & Execution Contexts',
      'Temporal Dead Zone (TDZ)',
      'Optional Chaining & Nullish Coalescing',
      'Event Loop & Microtask Queue',
      'Async/Await Error Propagation',
    ],
    modules: [
      {
        id: 'm1-contexts',
        title: 'Module 1: Execution Contexts, Scopes & The Dreaded ReferenceError',
        description: 'Understand how JavaScript creates lexical environments and why accessing variables too early triggers TDZ.',
        lessons: [
          {
```

```
            id: 'l1-callstack',
            title: '1.1 The Anatomy of the JavaScript Call Stack',
            durationMinutes: 10,
            type: 'concept',
            summary: 'How JavaScript manages function calls using a LIFO stack, and what a stack trace actually represents.',
            content: [
              'When your JavaScript code executes, the JavaScript engine (like Chrome V8 or Node.js) allocates a **Call Stack**. The Call Stack is a single-threaded Last-In, First-Out (LIFO) data structure that tracks where the program currently is in execution.',
              'Whenever a function is invoked, an **Execution Context** is pushed onto the stack. When the function returns, its frame is popped off.',
              'When an uncaught error occurs, the engine freezes the current state and dumps the **Stack Trace**. Reading a stack trace from top to bottom reveals the exact call chain that led to the crash: the top line is where the error exploded, and the lines below show the functions that called it.',
            ],
            keyTakeaways: [
              'The Call Stack is single-threaded: only one function executes at any given millisecond.',
              'A Stack Trace reads bottom-to-top for invocation order, and top-to-bottom for the point of failure.',
              'Infinite recursion without a base case causes a RangeError: Maximum call stack size exceeded.',
            ],
            codeExample: {
              language: 'javascript',
              description: 'Call Stack push and pop mechanics',
              code: `function multiply(a, b) {
  // Line 2: At top of stack
  return a * b;
}

function calculateTotal(price, tax) {
  // Line 7: Pushes multiply() to the stack
  return multiply(price, 1 + tax);
}

// Line 11: Main script starts
console.log(calculateTotal(100, 0.08));`,
            },
            quiz: [
```

```
              {
                id: 'q1',
                question: 'When reading an error stack trace, what does the very top line indicate?',
                options: [
                  'The entry point of your program (e.g. index.js)',
                  'The exact file and line number where the exception was thrown',
                  'The slowest function in your call stack',
                  'The function that caught the error',
                ],
                correctIndex: 1,
                explanation: 'The top line of a stack trace points to the innermost frame where the error was genera
ted before any unwinding occurred.',
              },
            ],
          },
          {
            id: 'l2-tdz',
            title: '1.2 ReferenceError Demystified: The Temporal Dead Zone (TDZ)',
            durationMinutes: 14,
            type: 'interactive_lab',
            summary: 'Why `var` hoists as undefined, while `let` and `const` throw a ReferenceError when accessed be
fore declaration.',
            content: [
              'One of the most confusing errors for developers is: `ReferenceError: Cannot access variable before in
itialization`.',
              'Both `var`, `let`, and `const` are hoisted during the creation phase of an execution context. Howeve
r, while `var` is initialized immediately with `undefined`, `let` and `const` remain in an uninitialized state calle
d the **Temporal Dead Zone (TDZ)**.',
              'Any attempt to read or write to a variable in its TDZ throws an immediate ReferenceError. This is a d
eliberate safety feature designed in ES6 to prevent bugs where uninitialized variables are accidentally consumed.',
            ],
            keyTakeaways: [
              'Hoisting reserves memory space for declarations before code executes.',
              'TDZ starts from the beginning of the enclosing block until the line where `let` or `const` is declare
d.',
              'Always declare variables at the top of their enclosing lexical block to avoid TDZ errors.',
            ],
            practiceChallengeId: 'ref-error-tdz',
```

```
            codeExample: {
              language: 'javascript',
              description: 'TDZ demonstration',
              code: `function calculateDiscount(price) {
  // TDZ for 'discount' starts here
  // console.log(discount); // 💥 ReferenceError!

  const discount = price > 100 ? 0.15 : 0.05; // TDZ ends here
  return price * (1 - discount);
}`,
            },
            quiz: [
              {
                id: 'q2',
                question: 'Why did ES6 introduce the Temporal Dead Zone for let and const?',
                options: [
                  'To save memory on mobile devices',
                  'To prevent developers from accidentally reading variables before their deliberate assignment',
                  'To allow let variables to be reassigned across scopes',
                  'Because V8 cannot allocate memory ahead of time',
                ],
                correctIndex: 1,
                explanation: 'TDZ makes reading uninitialized variables a fail-fast runtime exception instead of sil
ently returning undefined.',
              },
            ],
          },
        ],
      },
      {
        id: 'm2-typeerror',
        title: 'Module 2: The Dreaded TypeError & Defensive Null Safety',
        description: 'Deep dive into JavaScript object property resolution, prototypes, and modern null handling pat
terns.',
        lessons: [
          {
```

```
            id: 'l3-why-null-fails',
            title: '2.1 Why `Cannot read properties of undefined` Happens in V8',
            durationMinutes: 12,
            type: 'concept',
            summary: 'Understand the V8 internal property lookup mechanism and why accessing properties on null/undefined is prohibited.',
            content: [
              'In JavaScript, primitives like numbers and strings have object wrappers (`Number.prototype`, `String.prototype`) that allow property access. But `null` and `undefined` do not have a prototype.',
              'When the V8 engine evaluates `obj.user.address.street`, it walks down the chain:',
              '1. Looks up `obj` -> Found object.',
              '2. Looks up `.user` on `obj` -> Returns `undefined`.',
              '3. Tries to look up `.address` on `undefined` -> V8 halts: `TypeError: Cannot read properties of undefined (reading "address")`.',
              'This is because `undefined` has no internal `[[Get]]` slot or prototype table.',
            ],
            keyTakeaways: [
              'null represents intentional absence of value; undefined represents uninitialized variable.',
              'Neither null nor undefined have a prototype or properties.',
              'Chained property lookups without defensive guards are the #1 source of web app crashes.',
            ],
            codeExample: {
              language: 'javascript',
              description: 'V8 Property Lookup failure',
              code: `const apiResponse = {
  status: 200,
  data: null // intentional empty data
};

// 💥 CRASH: Cannot read properties of null (reading 'userName')
console.log(apiResponse.data.userName);`,
            },
            quiz: [
              {
                id: 'q3',
```

```
                question: 'What is the fundamental reason V8 throws a TypeError when accessing a property on undefin
ed?',
                options: [
                  'Because undefined is an empty string in memory',
                  'Because undefined has no prototype or internal [[Get]] method to resolve properties',
                  'Because modern browsers ban the use of undefined',
                  'Because undefined only exists in client-side code',
                ],
                correctIndex: 1,
                explanation: 'Primitives null and undefined possess no prototype chain or object wrapper, so propert
y access is impossible.',
              },
            ],
          },
          {
            id: 'l4-optional-chaining',
            title: '2.2 Defensive Coding: Optional Chaining (?.) & Nullish Coalescing (??)',
            durationMinutes: 15,
            type: 'interactive_lab',
            summary: 'Modern syntactic features that make code crash-proof against null/undefined, and how they diff
er from logical OR (||).',
            content: [
              'Prior to ECMAScript 2020, developers had to write verbose nested guards: `if (user && user.profile &&
user.profile.tier)`.',
              'Modern JavaScript provides **Optional Chaining (`?.`)**:',
              '``const tier = user?.profile?.tier;``',
              'If any reference before `?.` is `null` or `undefined`, the expression short-circuits and safely retur
ns `undefined` instead of throwing an error.',
              'Combine this with **Nullish Coalescing (`??`)** to provide safe fallbacks without the pitfalls of `||
` (which misinterprets valid values like `0` or `""` as falsey).',
            ],
            keyTakeaways: [
              '``?.` short-circuits execution as soon as a nullish property is encountered.',
              '``??` checks only for null or undefined, preserving valid falsey values like 0, false, and empty strin
gs.',
              'Never use `!` (non-null assertion in TypeScript) unless you have mathematically proven the value cann
ot be null.',
            ],
            practiceChallengeId: 'type-error-null',
```

```
            codeExample: {
              language: 'javascript',
              description: 'Crash-proof optional chaining and fallback',
              code: `const userAccount = {
  id: 'usr_819',
  settings: {
    notificationsCount: 0 // valid count
  }
};

// Safe access with ?. and ??
const count = userAccount?.settings?.notificationsCount ?? 10;
console.log(count); // Correctly returns 0, not 10!`,
            },
            quiz: [
              {
                id: 'q4',
                question: 'What will `0 ?? 42` evaluate to?',
                options: ['42', '0', 'undefined', 'null'],
                correctIndex: 1,
                explanation: 'The nullish coalescing operator ?? only triggers for null or undefined. 0 is a number, so 0 is returned.',
              },
            ],
          },
        ],
      },
      {
        id: 'm3-async-errors',
        title: 'Module 3: Asynchronous Pitfalls & Unhandled Promise Rejections',
        description: 'Master error handling in async/await, Promise chains, and avoid silent failures in backend Node.js and React.',
        lessons: [
          {
            id: 'l5-unhandled-rejections',
            title: '3.1 The Microtask Queue & Unhandled Promise Rejections',
```

```
            durationMinutes: 14,
            type: 'interactive_lab',
            summary: 'Why unhandled promise rejections can terminate Node.js servers and freeze React components.',
            content: [
              'Promises run in the **Microtask Queue**, which is processed immediately after the current synchronous script completes and before any rendering or macrotasks (like `setTimeout`).',
              'If a Promise rejects and there is no `.catch()` handler or surrounding `try...catch` in an `async` function, an `unhandledrejection` event is fired.',
              'In Node.js >= 15, unhandled promise rejections automatically terminate the process with exit code 1 to prevent corrupt state. In frontend browsers, they lead to silent UI freezes.',
            ],
            keyTakeaways: [
              'Always wrap `await` calls in `try...catch` blocks or chain `.catch()`.',
              'Never leave a fire-and-forget promise without error handling.',
              'Use `Promise.allSettled()` instead of `Promise.all()` when you want partial successes.',
            ],
            practiceChallengeId: 'async-unhandled-rejection',
            codeExample: {
              language: 'javascript',
              description: 'Robust async/await error boundary pattern',
              code: `async function fetchUserData(userId) {
  try {
    const response = await fetch(\`/api/users/\${userId}\`);
    if (!response.ok) {
      throw new Error(\`HTTP error! status: \${response.status}\`);
    }
    return await response.json();
  } catch (error) {
    console.error('Failed to load user:', error.message);
    return { fallback: true, userId };
  }
}`,
            },
            quiz: [
              {
                id: 'q5',
```

```
                question: 'What happens in modern Node.js when a Promise rejects without a catch handler?',
                options: [
                  'It silently ignores the error and logs a warning',
                  'It retries the request 3 times automatically',
                  'It crashes the process with an UnhandledPromiseRejection exception',
                  'It converts the rejected value into null',
                ],
                correctIndex: 2,
                explanation: 'Node.js intentionally terminates the process to prevent executing in an inconsistent, corrupt state.',
              },
            ],
          },
        ],
      },
    ],
  },
  {
    id: 'defensive-architecture',
    title: 'Defensive Programming & Error Architecture',
    tagline: 'Build Crash-Proof Systems with Custom Error Hierarchies and Fallbacks',
    description:
      'Learn how senior engineers architect software that gracefully handles unexpected failures. Master custom error classes, circuit breakers, structured error codes, and React Error Boundaries to prevent single-point failures.',
    category: 'Debugging & Errors',
    level: 'Intermediate',
    duration: '3h 30m',
    xp: 550,
    badge: 'Reliability Architect',
    iconName: 'ShieldCheck',
    color: 'from-emerald-500 to-teal-600',
    skillsGained: [
      'Custom Error Hierarchies',
      'Structured Error Enums',
      'Circuit Breakers & Exponential Backoff',
```

```
      'React Error Boundaries',
      'Error Observability (Sentry / Datadog)',
    ],
    modules: [
      {
        id: 'm1-custom-errors',
        title: 'Module 1: Custom Error Classes & Structured Error Enums',
        description: 'Replace generic `new Error("something broke")` with domain-driven, typed exception classes.',
        lessons: [
          {
            id: 'l6-custom-errors',
            title: '1.1 Creating Domain Error Hierarchies',
            durationMinutes: 12,
            type: 'concept',
            summary: 'How extending Error with custom metadata (HTTP status, error codes, recovery hints) transforms debugging.',
            content: [
              'Throwing generic strings or plain `Error` objects leaves downstream callers guessing what went wrong.',
              'By creating a base `AppError` and extending specialized subclasses (`ValidationError`, `AuthenticationError`, `NotFoundError`), your application can inspect error types cleanly with `instanceof` and respond with appropriate HTTP status codes and user-friendly messages.',
            ],
            keyTakeaways: [
              'Always set `this.name = this.constructor.name` and capture stack trace.',
              'Attach machine-readable `errorCode` strings for frontend internationalization.',
              'Distinguish operational errors (predictable) from programmer bugs (unhandled exceptions).',
            ],
            codeExample: {
              language: 'typescript',
              description: 'Custom AppError base class',
              code: `export class AppError extends Error {
  constructor(
    public override message: string,
    public statusCode: number = 500,
    public code: string = 'INTERNAL_ERROR',
    public isOperational: boolean = true
```

```
  ) {
    super(message);
    this.name = this.constructor.name;
    Error.captureStackTrace(this, this.constructor);
  }
}

export class NotFoundError extends AppError {
  constructor(resource: string, id: string) {
    super(\`\${resource} with id '\${id}' was not found.\`, 404, 'RESOURCE_NOT_FOUND');
  }
}`,
            },
            quiz: [
              {
                id: 'q6',
                question: 'Why is `isOperational: boolean` important on custom errors?',
                options: [
                  'It speeds up JavaScript garbage collection',
                  'It tells error handlers whether the error is expected (e.g. invalid user input) vs an unhandled programmer bug',
                  'It automatically converts the error into JSON',
                  'It prevents the browser from logging to console',
                ],
                correctIndex: 1,
                explanation: 'Operational errors can be handled gracefully with user notifications; non-operational bugs usually require restarting or logging to alerting systems.',
              },
            ],
          },
        ],
      },
      {
        id: 'm2-boundaries',
        title: 'Module 2: React Error Boundaries & Graceful Degradation',
```

```
        description: 'Prevent white-screen-of-death in React applications by isolating UI failures.',
        lessons: [
          {
            id: 'l7-error-boundaries',
            title: '2.1 Isolating Component Failures with Error Boundaries',
            durationMinutes: 14,
            type: 'concept',
            summary: 'How React componentDidCatch and getDerivedStateFromError contain crashes to specific UI widget
s.',
            content: [
              'By default, if an unhandled JavaScript error is thrown inside any React render cycle, React unmounts
the ENTIRE component tree, leaving the user with a blank white screen.',
              'Error Boundaries are React components that catch JavaScript errors anywhere in their child component
tree, log those errors, and display a fallback UI instead of crashing the whole page.',
            ],
            keyTakeaways: [
              'Wrap independent widgets (e.g. sidebar, feed, payment modal) in separate Error Boundaries.',
              'Error Boundaries do not catch errors inside async callbacks (like setTimeout or fetch); handle those
inside the handlers.',
              'Always provide a "Try Again / Reload Component" button in your fallback UI.',
            ],
            codeExample: {
              language: 'tsx',
              description: 'React Error Boundary with fallback',
              code: `class ErrorBoundary extends React.Component {
  state = { hasError: false };

  static getDerivedStateFromError() {
    return { hasError: true };
  }

  componentDidCatch(error, info) {
    console.error('Widget crash captured:', error, info);
  }

  render() {
    if (this.state.hasError) {
```

```
      return (
        <div className="p-4 bg-rose-50 border border-rose-200 rounded-lg text-rose-700">
          <p className="font-semibold">Unable to load this section.</p>
          <button onClick={() => this.setState({ hasError: false })} className="text-xs underline mt-2">
            Try again
          </button>
        </div>
      );
    }
    return this.props.children;
  }
}`,
            },
          },
        ],
      },
    ],
  },
  {
    id: 'web-security-owasp',
    title: 'Web Security & Vulnerability Auditing (OWASP Top 10)',
    tagline: 'Learn Real-World Exploit Vectors, JWT Verification Flaws & Secure Patching',
    description:
      'Directly inspired by RepoPilot autonomous repair mechanics! Master how real vulnerabilities happen: from expired JWT token bypasses to injection attacks. Learn how to audit, reproduce, and apply surgical security patches.',
    category: 'Security & Auth',
    level: 'Advanced',
    duration: '5h 00m',
    xp: 750,
    badge: 'Security Sentinel',
    iconName: 'Lock',
    color: 'from-rose-500 to-indigo-600',
    skillsGained: [
      'JWT Signature & Expiration Auditing',
      'SQL & Command Injection Prevention',
```

```
      'Broken Object-Level Authorization (BOLA)',
      'Cross-Site Scripting (XSS) Sanitization',
      'Automated Security Regression Suites',
    ],
    modules: [
      {
        id: 'm1-jwt-security',
        title: 'Module 1: Authentication Vulnerabilities & The JWT Trap',
        description: 'Understand how developers mistakenly trust decoded token truthiness without enforcing signatur
e and exp claims.',
        lessons: [
          {
            id: 'l8-jwt-flaw',
            title: '1.1 The Expired JWT Token Verification Exploit',
            durationMinutes: 18,
            type: 'interactive_lab',
            summary: 'Step-by-step breakdown of how AUTH-104 allowed expired tokens past middleware.',
            content: [
              'JSON Web Tokens (JWT) consist of 3 parts separated by dots: `Header.Payload.Signature`.',
              'When developers verify a token with `jwt.verify(token, SECRET)`, they often inspect the returned deco
ded object:',
              '``const decoded = jwt.verify(token, SECRET);``',
              '``if (decoded) { return next(); }``',
              '**The fatal flaw:** If an expired token is passed, or if the verify method was invoked with options t
hat skipped clock checks, checking `if (decoded)` passes truthiness! The system trusts an expired session, granting
attackers indefinite access.',
              'The fix requires asserting that decoded is a non-string object AND that `decoded.exp * 1000 >= Date.n
ow()`.',
            ],
            keyTakeaways: [
              'Never rely on truthiness checks alone on decoded tokens.',
              'Explicitly validate the `exp` (expiration), `iss` (issuer), and `aud` (audience) claims.',
              'Use automated regression security tests that simulate expired and forged tokens.',
            ],
            practiceChallengeId: 'jwt-exp-flaw',
            codeExample: {
              language: 'typescript',
```

```
              description: 'Hardened JWT Verification Middleware',
              code: `export function verifyAuthToken(token: string): boolean {
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET!) as { exp?: number };

    // Strict assertion: must exist and have valid timestamp
    if (!decoded || typeof decoded !== 'object' || !decoded.exp) {
      return false;
    }

    const isNotExpired = decoded.exp * 1000 > Date.now();
    return isNotExpired;
  } catch (err) {
    return false; // Signature invalid or malformed
  }
}`,
            },
            quiz: [
              {
                id: 'q8',
                question: 'Why is `decoded.exp * 1000` multiplied by 1000 when comparing with Date.now()?',
                options: [
                  'To convert microseconds to milliseconds',
                  'Because JWT exp is standardized in seconds, while JavaScript Date.now() returns milliseconds',
                  'To account for leap seconds',
                  'Because V8 requires 64-bit integers',
                ],
                correctIndex: 1,
                explanation: 'JWT timestamps (Unix epoch) are in seconds; Date.now() is in milliseconds. Multiplying by 1000 normalizes the units.',
              },
            ],
          },
        ],
      },
```

```
    ],
  },
  {
    id: 'python-error-mastery',
    title: 'Python Fundamentals & Exception Handling',
    tagline: 'Master Pythonic Error Handling, Context Managers and Boundary Checks',
    description:
      'Learn idiomatic Python error handling: from preventing IndexError and KeyError in complex data pipelines to building robust context managers with the `with` statement.',
    category: 'Python & Algorithms',
    level: 'Beginner',
    duration: '3h 15m',
    xp: 480,
    badge: 'Python Alchemist',
    iconName: 'Terminal',
    color: 'from-blue-500 to-cyan-600',
    skillsGained: [
      'IndexError & KeyError Prevention',
      'EAFP (Easier to Ask Forgiveness than Permission)',
      'Custom Python Exception Hierarchies',
      'Context Managers (`__enter__` & `__exit__`)',
      'Logging & Traceback Formatting',
    ],
    modules: [
      {
        id: 'm1-py-errors',
        title: 'Module 1: Indexing & Collection Traps in Python',
        description: 'Understand zero-indexed lists, negative slicing, and bounds checking.',
        lessons: [
          {
            id: 'l9-index-error',
            title: '1.1 Solving IndexError: list index out of range',
            durationMinutes: 12,
            type: 'interactive_lab',
            summary: 'Why off-by-one errors happen in range loops and how to use Pythonic iteration like enumerate().',
```

```
            content: [
              'In Python, accessing `items[len(items)]` raises an `IndexError: list index out of range` because Pyth
on lists are 0-indexed (valid indices run from `0` to `len - 1`).',
              'Beginners coming from other languages often write `for i in range(len(items) + 1)` or access the last
element with `items[len(items)]` instead of the clean, idiomatic Python negative index `items[-1]`.',
            ],
            keyTakeaways: [
              'Use `items[-1]` to read the last element cleanly without calculating length.',
              'Use `for item in items:` or `for i, item in enumerate(items):` instead of manual index counters.',
              'Use `.get()` with dictionaries to avoid `KeyError`.',
            ],
            practiceChallengeId: 'py-index-error',
            codeExample: {
              language: 'python',
              description: 'Idiomatic boundary-safe iteration in Python',
              code: `scores = [88, 92, 79, 95]

# Safe last element access:
latest_score = scores[-1]

# Idiomatic iteration with index and value:
for index, score in enumerate(scores):
    print(f"Student #{index + 1}: {score}")`,
            },
            quiz: [
              {
                id: 'q9',
                question: 'What is the most idiomatic Python way to get the last element of a list named `data`?',
                options: ['data[len(data)]', 'data.last()', 'data[-1]', 'data[data.size - 1]'],
                correctIndex: 2,
                explanation: 'Python natively supports negative indexing: data[-1] accesses the last item without ri
sk of off-by-one errors.',
              },
            ],
          },
        ],
```

```
      },
    ],
  },
  {
    id: 'react-state-traps',
    title: 'Modern React Architecture & State Traps',
    tagline: 'Fix Stale Closures, Infinite Render Loops, and Direct State Mutations',
    description:
      'Master React rendering mechanics: understand why direct state mutation breaks UI updates, how stale closures occur in `useEffect` and `useCallback`, and how to write predictable, pure state updates.',
    category: 'React & Architecture',
    level: 'Intermediate',
    duration: '3h 45m',
    xp: 600,
    badge: 'React Specialist',
    iconName: 'Code',
    color: 'from-cyan-500 to-indigo-600',
    skillsGained: [
      'React Fiber & Reconciliation Engine',
      'Immutable State Updates',
      'Stale Closure Resolution',
      'useEffect Dependency Array Rules',
      'useMemo / useCallback Optimization',
    ],
    modules: [
      {
        id: 'm1-state-mutations',
        title: 'Module 1: The Immutability Principle in React',
        description: 'Understand how React checks object reference equality and why `.push()` breaks re-renders.',
        lessons: [
          {
            id: 'l10-mutation-trap',
            title: '1.1 The Direct State Mutation Trap',
            durationMinutes: 15,
            type: 'interactive_lab',
```

```
            summary: 'Why `state.items.push(newItem)` fails to re-render React components and how to use spread syntax.',
            content: [
              'React uses shallow reference equality (`Object.is(oldState, newState)`) to decide if a component needs to re-render.',
              'When you write `items.push(newItem); setItems(items);`, the array memory address has NOT changed! React compares `items === items`, sees identical memory references, and skips the re-render completely.',
              'To trigger a re-render, you must create a new array reference using the spread operator: `setItems([...items, newItem])`.',
            ],
            keyTakeaways: [
              'Never mutate state directly in React.',
              'Arrays: use `[...items, newItem]`, `items.filter(...)`, or `items.map(...)`.',
              'Objects: use `{ ...user, status: "active" }`.',
            ],
            practiceChallengeId: 'react-mutation-bug',
            codeExample: {
              language: 'javascript',
              description: 'Immutable state updates',
              code: `// ❌ BUG: Same reference, no re-render
todos.push({ text: 'Buy milk' });
setTodos(todos);

// ✅ FIXED: New array reference created
setTodos(prev => [...prev, { text: 'Buy milk' }]);`,
            },
          },
        ],
      },
    ],
  },
  {
    id: 'java-oop-errors',
    title: 'Java Exception Architecture & Robust OOP',
    tagline: 'Defeat NullPointerExceptions, Master Checked vs Unchecked Exceptions, and JVM Memory Safety',
    description:
```

```
      'Java enterprise systems must be crash-resilient. Learn how the JVM handles stack frames, understand heap references, avoid NullPointerExceptions using Optional<T> and defensive guards, and write clean try-with-resources blocks.',
    category: 'Java & Backend',
    level: 'Beginner',
    duration: '3h 45m',
    xp: 600,
    badge: 'Java Champion',
    iconName: 'Coffee',
    color: 'from-amber-600 to-red-600',
    skillsGained: [
      'NullPointerException Elimination',
      'ArrayIndexOutOfBounds Defensive Coding',
      'Checked vs Unchecked Exceptions',
      'Optional<T> & Monadic Handling',
      'Try-with-resources & Memory Leaks',
    ],
    modules: [
      {
        id: 'jm1-exceptions',
        title: 'Module 1: JVM Stack Traces & Null Pointer Elimination',
        description: 'Understand why references are null in Java and how to use Optional and guards to guarantee safety.',
        lessons: [
          {
            id: 'jl1-npe',
            title: '1.1 The Billion-Dollar Mistake: Java NullPointerException',
            durationMinutes: 12,
            type: 'interactive_lab',
            summary: 'Why dereferencing uninitialized or missing references throws NullPointerException in Java.',
            content: [
              'Tony Hoare invented the null reference in 1965 and called it his "billion-dollar mistake". In Java, reference variables point to memory locations on the JVM Heap. If a variable holds null, attempting to invoke methods or access fields throws java.lang.NullPointerException (NPE).',
              'Defensive approaches include: (1) Explicit null checking with if (obj != null), (2) Java 8 Optional<T>, and (3) Objects.requireNonNull().',
            ],
            keyTakeaways: [
```

```
              'Always validate method parameters and external API responses before invoking methods on them.',
              'Use Optional<T> for return types that might legitimately be empty.',
              'Never return null from collections: return Collections.emptyList() instead.',
            ],
            practiceChallengeId: 'java-null-pointer',
            codeExample: {
              language: 'java',
              description: 'Null-safe object traversal in Java',
              code: `public String getEmailSafe(User user) {
    if (user == null || user.getEmail() == null) {
        return "no-reply@example.com";
    }
    return user.getEmail().toLowerCase();
}`,
            },
            quiz: [
              {
                id: 'jq1',
                question: 'What is the recommended Java 8+ container for values that may or may not be present?',
                options: ['Optional<T>', 'Nullable<T>', 'Wrapper<T>', 'SafeRef<T>'],
                correctIndex: 0,
                explanation: 'Java 8 introduced java.util.Optional<T> to represent optional values without using nul
l references.',
              },
            ],
          },
        ],
      },
    ],
  },
  {
    id: 'cpp-memory-safety',
    title: 'C & C++ Memory Management & Safety',
    tagline: 'Pointers, Segmentation Faults, Valgrind, and Modern RAII Smart Pointers',
    description:
```

```
      'Step beyond high-level runtimes into manual and RAII memory control. Understand the Stack vs Heap, diagnose segmentation faults, fix buffer overflows, and master std::unique_ptr and std::shared_ptr in modern C++20.',
    category: 'C & Systems Programming',
    level: 'Intermediate',
    duration: '4h 30m',
    xp: 750,
    badge: 'Systems Architect',
    iconName: 'Cpu',
    color: 'from-blue-600 to-indigo-700',
    skillsGained: [
      'Pointers & Heap Allocation',
      'Segmentation Fault Diagnosis',
      'Buffer Overflow Prevention',
      'Modern RAII & Smart Pointers',
      'Vector Bounds Checking',
    ],
    modules: [
      {
        id: 'cm1-pointers',
        title: 'Module 1: Pointer Dereference & Bounds Checking',
        description: 'Understand raw pointers, dangling memory, and vector bounds safety.',
        lessons: [
          {
            id: 'cl1-bounds',
            title: '1.1 Bounds Safety: operator[] vs .at() in std::vector',
            durationMinutes: 15,
            type: 'interactive_lab',
            summary: 'Why C++ vector operator[] silently produces undefined behavior on out-of-bounds, while .at() throws std::out_of_range.',
            content: [
              'In C++, std::vector::operator[] does NOT perform bounds checking for raw performance reasons. Accessing vec[vec.size()] reads arbitrary memory or causes a Segmentation Fault.',
              'std::vector::at() performs strict bounds checking and throws std::out_of_range if index >= size(), allowing structured exception handling.',
            ],
            keyTakeaways: [
              'Use .at(index) when indices are computed dynamically or come from user input.',
```

```
              'Use range-based for loops: for (const auto& item : vec) to avoid index boundary off-by-one errors.',
            ],
            practiceChallengeId: 'cpp-vector-range',
            codeExample: {
              language: 'cpp',
              description: 'Safe C++ vector access with bounds check',
              code: `#include <iostream>
#include <vector>

int main() {
    std::vector<int> numbers = {10, 20, 30};
    try {
        std::cout << numbers.at(1) << std::endl; // Safe: 20
    } catch (const std::out_of_range& e) {
        std::cerr << "Out of bounds error: " << e.what() << std::endl;
    }
    return 0;
}`,
            },
          },
        ],
      },
    ],
  },
];

export const PRACTICE_CHALLENGES: PracticeChallenge[] = [
  {
    id: 'type-error-null',
    title: 'TypeError: Cannot read properties of undefined',
    difficulty: 'Easy',
    language: 'javascript',
    category: 'Error Debugging',
    description:
```

```
      'The function `getUserBadge` crashes when receiving user objects with missing or undefined membership data. Run the code, observe the error, explore the AI explanation of why it happened, and fix it using optional chaining or null guarding.',
    errorNameTarget: "TypeError: Cannot read properties of undefined (reading 'tier')",
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l4-optional-chaining',
    initialCode: `// Challenge: Fix the TypeError when accessing nested user profile data
function getUserBadge(user) {
  // 💥 BUG: If user.subscription is undefined, accessing .tier crashes the app!
  const tier = user.subscription.tier;

  if (tier === 'pro') {
    return '🌟 Pro Member';
  } else if (tier === 'enterprise') {
    return '👑 Enterprise VIP';
  }
  return '🌱 Free Tier';
}

// Test cases
console.log('User 1 (Pro):', getUserBadge({ name: 'Alice', subscription: { tier: 'pro' } }));

// This call currently throws TypeError!
console.log('User 2 (Guest):', getUserBadge({ name: 'Bob' }));
`,
    solutionCode: `// Challenge: Fix the TypeError when accessing nested user profile data
function getUserBadge(user) {
  // ✅ FIXED: Use optional chaining (?.) and nullish coalescing (??)
  const tier = user?.subscription?.tier ?? 'free';

  if (tier === 'pro') {
    return '🌟 Pro Member';
  } else if (tier === 'enterprise') {
    return '👑 Enterprise VIP';
  }
  return '🌱 Free Tier';
```

```
}

// Test cases
console.log('User 1 (Pro):', getUserBadge({ name: 'Alice', subscription: { tier: 'pro' } }));
console.log('User 2 (Guest):', getUserBadge({ name: 'Bob' }));
console.log('User 3 (Null):', getUserBadge(null));
`,
    hints: [
      'Look at line 4: `user.subscription` is undefined when a user has no active subscription.',
      'In modern JavaScript, you can use the optional chaining operator `?.` to safely traverse nested properties: `user?.subscription?.tier`.',
      'Provide a fallback value with the nullish coalescing operator `?? "free"` to handle missing values gracefully.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Valid Pro User',
        inputDescription: '{ name: "Alice", subscription: { tier: "pro" } }',
        expectedOutput: '🌟 Pro Member',
      },
      {
        id: 'tc-2',
        name: 'Guest User (undefined subscription)',
        inputDescription: '{ name: "Bob" }',
        expectedOutput: '🌱 Free Tier',
      },
      {
        id: 'tc-3',
        name: 'Completely Null User Object',
        inputDescription: 'null',
        expectedOutput: '🌱 Free Tier',
      },
    ],
  },
```

```
  {
    id: 'ref-error-tdz',
    title: 'ReferenceError: Cannot access variable before initialization',
    difficulty: 'Easy',
    language: 'javascript',
    category: 'Error Debugging',
    description:
      'A shipping calculator is failing with a ReferenceError. The variable `taxRate` is being used in a calculation before its `let` declaration. Fix the scoping and order of execution to eliminate the Temporal Dead Zone error.',
    errorNameTarget: "ReferenceError: Cannot access 'taxRate' before initialization",
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l2-tdz',
    initialCode: `// Challenge: Fix the ReferenceError caused by Temporal Dead Zone (TDZ)
function calculateOrderTotal(subtotal, region) {
  // 💥 BUG: Calculating tax before 'taxRate' is declared!
  const estimatedTax = subtotal * taxRate;

  let taxRate;
  if (region === 'CA') {
    taxRate = 0.0925;
  } else if (region === 'NY') {
    taxRate = 0.08875;
  } else {
    taxRate = 0.05;
  }

  return subtotal + estimatedTax;
}

console.log('Order Total:', calculateOrderTotal(100, 'CA'));
`,
    solutionCode: `// Challenge: Fix the ReferenceError caused by Temporal Dead Zone (TDZ)
function calculateOrderTotal(subtotal, region) {
  // ✅ FIXED: Declare and resolve taxRate before performing arithmetic
  let taxRate;
```

```
  if (region === 'CA') {
    taxRate = 0.0925;
  } else if (region === 'NY') {
    taxRate = 0.08875;
  } else {
    taxRate = 0.05;
  }

  const estimatedTax = subtotal * taxRate;
  return subtotal + estimatedTax;
}

console.log('Order Total:', calculateOrderTotal(100, 'CA'));
`,
    hints: [
      'In JavaScript, variables declared with `let` or `const` cannot be read before their declaration line (Temporal Dead Zone).',
      'Move the declaration and resolution of `taxRate` above line 4 before computing `estimatedTax`.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'California Region (CA)',
        inputDescription: 'calculateOrderTotal(100, "CA")',
        expectedOutput: '109.25',
      },
      {
        id: 'tc-2',
        name: 'Default Region',
        inputDescription: 'calculateOrderTotal(50, "TX")',
        expectedOutput: '52.5',
      },
    ],
  },
  {
```

```
    id: 'range-error-recursion',
    title: 'RangeError: Maximum call stack size exceeded',
    difficulty: 'Medium',
    language: 'javascript',
    category: 'Error Debugging',
    description:
      'A recursive countdown timer / tree traversal function keeps executing without stopping until V8 throws `RangeError: Maximum call stack size exceeded`. Diagnose the missing base case and restore stack safety.',
    errorNameTarget: 'RangeError: Maximum call stack size exceeded',
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l1-callstack',
    initialCode: `// Challenge: Fix infinite recursion stack overflow
function countdownToLaunch(secondsRemaining) {
  console.log(\`T-minus \${secondsRemaining} seconds...\`);

  // 💥 BUG: Missing base condition! The recursion runs endlessly into negative numbers!
  return countdownToLaunch(secondsRemaining - 1);
}

// Running this will quickly blow the call stack:
countdownToLaunch(5);
`,
    solutionCode: `// Challenge: Fix infinite recursion stack overflow
function countdownToLaunch(secondsRemaining) {
  console.log(\`T-minus \${secondsRemaining} seconds...\`);

  // ✅ FIXED: Base condition halts the recursion when reaching 0
  if (secondsRemaining <= 1) {
    console.log('🚀 Liftoff!');
    return 'Launched';
  }

  return countdownToLaunch(secondsRemaining - 1);
}
```

```
countdownToLaunch(5);
`,
    hints: [
      'Every recursive function requires a BASE CASE: a condition that returns a value without making another recursive call.',
      'Check if `secondsRemaining <= 1` and return or log liftoff instead of calling `countdownToLaunch` again.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Countdown from 3',
        inputDescription: 'countdownToLaunch(3)',
        expectedOutput: 'Launched',
      },
    ],
  },
  {
    id: 'async-unhandled-rejection',
    title: 'UnhandledPromiseRejection: Network request failed',
    difficulty: 'Medium',
    language: 'javascript',
    category: 'Async & Promises',
    description:
      'An asynchronous data loader makes an API request that can reject. Because it lacks a try...catch or error boundary, the promise rejection is unhandled and halts execution. Implement proper async error recovery.',
    errorNameTarget: 'UnhandledPromiseRejection: Failed to fetch user profile: Network Timeout',
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l5-unhandled-rejections',
    initialCode: `// Challenge: Handle asynchronous Promise rejections gracefully
async function fetchUserMetrics(userId) {
  const simulatedNetworkCall = () =>
    new Promise((resolve, reject) => {
      // Simulating a failed network timeout
      reject(new Error('Network Timeout: Gateway 504'));
    });

```

```
  // 💥 BUG: Awaiting a rejecting promise without a try...catch wrapper!
  const data = await simulatedNetworkCall();
  return { status: 200, data };
}

fetchUserMetrics('usr_102');
`,
    solutionCode: `// Challenge: Handle asynchronous Promise rejections gracefully
async function fetchUserMetrics(userId) {
  const simulatedNetworkCall = () =>
    new Promise((resolve, reject) => {
      reject(new Error('Network Timeout: Gateway 504'));
    });

  // ✅ FIXED: Enclosed in try...catch to return a graceful fallback
  try {
    const data = await simulatedNetworkCall();
    return { status: 200, data };
  } catch (error) {
    console.warn(\`Handled async failure for user \${userId}:\`, error.message);
    return { status: 504, data: null, error: error.message, isFallback: true };
  }
}

fetchUserMetrics('usr_102').then(res => console.log('Recovered result:', res));
`,
    hints: [
      'Wrap the `await simulatedNetworkCall()` in a `try { ... } catch (error) { ... }` block.',
      'In the `catch` block, return a fallback object with `{ status: 504, error: error.message, isFallback: true }` so the caller can handle the outage cleanly.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Graceful Error Recovery',
```

```
        inputDescription: 'fetchUserMetrics("usr_102")',
        expectedOutput: 'Recovered result: { status: 504, data: null, isFallback: true }',
      },
    ],
  },
  {
    id: 'type-error-const-assign',
    title: 'TypeError: Assignment to constant variable',
    difficulty: 'Easy',
    language: 'javascript',
    category: 'Error Debugging',
    description:
      'A cart total accumulator attempts to reassign a `const` variable inside a loop. Understand the difference between reassignment of primitives versus mutation of array references.',
    errorNameTarget: 'TypeError: Assignment to constant variable.',
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l2-tdz',
    initialCode: `// Challenge: Fix TypeError when accumulating values
function sumCart(items) {
  // 💥 BUG: Declared as 'const', but reassigned in each loop iteration!
  const total = 0;

  for (const item of items) {
    total = total + item.price; // 💥 CRASH!
  }

  return total;
}

const cart = [{ name: 'Keyboard', price: 75 }, { name: 'Mouse', price: 45 }];
console.log('Total:', sumCart(cart));
`,
    solutionCode: `// Challenge: Fix TypeError when accumulating values
function sumCart(items) {
  // ✅ FIXED: Use 'let' for accumulators that get reassigned, or use .reduce()
```

```
  let total = 0;

  for (const item of items) {
    total += item.price;
  }

  return total;
}

const cart = [{ name: 'Keyboard', price: 75 }, { name: 'Mouse', price: 45 }];
console.log('Total:', sumCart(cart));
`,
    hints: [
      'Variables declared with `const` cannot be reassigned with the `=` operator.',
      'Change `const total = 0;` to `let total = 0;`, or use `items.reduce((sum, item) => sum + item.price, 0)`.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Accumulate items',
        inputDescription: 'sumCart([{ price: 75 }, { price: 45 }])',
        expectedOutput: '120',
      },
    ],
  },
  {
    id: 'py-index-error',
    title: 'Python IndexError: list index out of range',
    difficulty: 'Easy',
    language: 'python',
    category: 'Error Debugging',
    description:
      'A Python log summarizer tries to print each log entry by calculating indices up to `len(logs)`. Because Python uses 0-based indexing, `logs[len(logs)]` triggers an IndexError. Fix the loop boundary.',
    errorNameTarget: 'IndexError: list index out of range',
```

```
    relatedCourseId: 'python-error-mastery',
    relatedLessonId: 'l9-index-error',
    initialCode: `# Challenge: Fix Python IndexError in list traversal
def summarize_logs(log_entries):
    summary = []
    # 💥 BUG: range(len + 1) iterates one step too far!
    for i in range(len(log_entries) + 1):
        entry = log_entries[i] # 💥 IndexError on last iteration!
        summary.append(f"Log #{i}: {entry}")
    return summary

logs = ["System boot", "Service started", "User logged in"]
print(summarize_logs(logs))
`,
    solutionCode: `# Challenge: Fix Python IndexError in list traversal
def summarize_logs(log_entries):
    # ✅ FIXED: Use Pythonic enumerate() without manual index boundaries
    summary = []
    for i, entry in enumerate(log_entries):
        summary.append(f"Log #{i}: {entry}")
    return summary

logs = ["System boot", "Service started", "User logged in"]
print(summarize_logs(logs))
`,
    hints: [
      'In Python, a list of length 3 has valid indices 0, 1, and 2. An index of 3 is out of range.',
      'Instead of `range(len(log_entries) + 1)`, use `range(len(log_entries))` or even better, the Pythonic `for i, entry in enumerate(log_entries):`.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: '3 Logs',
        inputDescription: 'summarize_logs(["System boot", "Service started", "User logged in"])',
```

```
        expectedOutput: '3 formatted entries without crash',
      },
    ],
  },
  {
    id: 'react-mutation-bug',
    title: 'React State Mutation: Missing UI Re-renders',
    difficulty: 'Medium',
    language: 'javascript',
    category: 'State & Closures',
    description:
      'A shopping list app adds items to state using `items.push(newItem)`. While the array updates in memory, React never re-renders the UI because the object reference remained identical. Refactor to an immutable state update.',
    errorNameTarget: 'StateMutationError: Array.prototype.push directly mutated state reference',
    relatedCourseId: 'react-state-traps',
    relatedLessonId: 'l10-mutation-trap',
    initialCode: `// Challenge: Fix React Direct State Mutation Bug
function addNotification(state, newNotification) {
  // 💥 BUG: Direct mutation with .push() retains the same memory reference!
  // React's shallow comparison Object.is(oldState, newState) evaluates to TRUE,
  // causing React to skip re-rendering the UI.
  state.notifications.push(newNotification);
  return state;
}

const currentState = { notifications: ['Welcome!'] };
const nextState = addNotification(currentState, 'New Message');

console.log('Same reference?:', currentState === nextState); // true (BUG!)
`,
    solutionCode: `// Challenge: Fix React Direct State Mutation Bug
function addNotification(state, newNotification) {
  // ✅ FIXED: Return a brand new object and new array reference using spread syntax
  return {
    ...state,
```

```
    notifications: [...state.notifications, newNotification],
  };
}

const currentState = { notifications: ['Welcome!'] };
const nextState = addNotification(currentState, 'New Message');

console.log('Same reference?:', currentState === nextState); // false (Proper immutable update!)
console.log('Next State:', nextState);
`,
    hints: [
      'React requires a new object/array memory reference to detect changes.',
      'Use the spread operator: `return { ...state, notifications: [...state.notifications, newNotification] };`',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Immutable update',
        inputDescription: 'addNotification({ notifications: [] }, "Ping")',
        expectedOutput: 'New reference with ["Ping"]',
      },
    ],
  },
  {
    id: 'jwt-exp-flaw',
    title: 'Security Vulnerability: Expired Token Bypass (AUTH-104)',
    difficulty: 'Hard',
    language: 'typescript',
    category: 'Security',
    description:
      'The exact vulnerability analyzed in RepoPilot! The auth middleware verifies the JWT signature but assumes tha
t if `decoded` is truthy, the session is valid. An expired token with a valid past signature passes through. Enforce
strict expiration assertion.',
    errorNameTarget: 'SecurityVulnerability: Expired token accepted as valid authenticated session',
    relatedCourseId: 'web-security-owasp',
```

```
    relatedLessonId: 'l8-jwt-flaw',
    initialCode: `// Challenge: Fix the AUTH-104 Expired JWT Bypass Vulnerability
interface JwtPayload {
  userId: string;
  role: string;
  exp: number; // Unix timestamp in seconds
}

function authenticateRequest(decodedPayload: JwtPayload | null): boolean {
  // 💥 CRITICAL SECURITY GAP:
  // Developers assume if decodedPayload exists, the user is authorized.
  // Expired tokens from hours or days ago pass truthiness!
  if (decodedPayload) {
    return true; // 💥 EXPLOIT: Expired session accepted!
  }
  return false;
}

// Expired token generated 2 hours ago:
const expiredTokenPayload: JwtPayload = {
  userId: 'usr_hacker',
  role: 'admin',
  exp: Math.floor(Date.now() / 1000) - 7200, // 2 hours in the past
};

console.log('Is Authorized?:', authenticateRequest(expiredTokenPayload)); // Returns true (DANGEROUS BUG!)
`,
    solutionCode: `// Challenge: Fix the AUTH-104 Expired JWT Bypass Vulnerability
interface JwtPayload {
  userId: string;
  role: string;
  exp: number; // Unix timestamp in seconds
}

function authenticateRequest(decodedPayload: JwtPayload | null): boolean {
```

```
  // ✅ FIXED: Validate payload structure and enforce strict expiration check
  if (!decodedPayload || typeof decodedPayload !== 'object' || !decodedPayload.exp) {
    return false;
  }

  const currentTimestampSec = Math.floor(Date.now() / 1000);
  const isNotExpired = decodedPayload.exp > currentTimestampSec;

  return isNotExpired;
}

const expiredTokenPayload: JwtPayload = {
  userId: 'usr_hacker',
  role: 'admin',
  exp: Math.floor(Date.now() / 1000) - 7200,
};

console.log('Is Authorized?:', authenticateRequest(expiredTokenPayload)); // Returns false (SECURE!)
`,
    hints: [
      'Compare `decodedPayload.exp` against the current time in seconds: `Math.floor(Date.now() / 1000)`.',
      'The token is only valid if `decodedPayload.exp > currentTimestampSec`.',
      'Ensure you also verify `!decodedPayload || !decodedPayload.exp` to avoid null pointer exceptions.',
    ],
    testCases: [
      {
        id: 'tc-1',
        name: 'Reject expired token',
        inputDescription: 'authenticateRequest({ exp: now - 3600 })',
        expectedOutput: 'false',
      },
      {
        id: 'tc-2',
        name: 'Accept fresh token',
```

```
        inputDescription: 'authenticateRequest({ exp: now + 3600 })',
        expectedOutput: 'true',
      },
    ],
  },
  {
    id: 'java-null-pointer',
    title: 'Java NullPointerException: Object Reference Dereference',
    difficulty: 'Easy',
    language: 'java',
    category: 'Error Debugging',
    description:
      'In a Java user authentication service, `User.getEmail()` is invoked on a user object where email is null or uninitialized. Fix the code to guard against null references using safe conditional checks or Optional.',
    errorNameTarget: 'java.lang.NullPointerException: Cannot invoke "String.toLowerCase()" because "email" is null',
    relatedCourseId: 'java-oop-errors',
    relatedLessonId: 'jl1-npe',
    initialCode: `// Java Practice Sandbox: Fix NullPointerException
public class Main {
    static class User {
        private String name;
        private String email;

        public User(String name, String email) {
            this.name = name;
            this.email = email;
        }

        public String getEmail() {
            return this.email;
        }
    }

    public static String formatUserEmail(User user) {
        // 💥 BUG: If user is null or email is null, .toLowerCase() throws NullPointerException!
```

```
        return user.getEmail().toLowerCase();
    }

    public static void main(String[] args) {
        User validUser = new User("Alice", "ALICE@EXAMPLE.COM");
        System.out.println("Valid user: " + formatUserEmail(validUser));

        // 💥 This triggers NullPointerException:
        User nullEmailUser = new User("Bob", null);
        System.out.println("Null email user: " + formatUserEmail(nullEmailUser));
    }
}
`,
    solutionCode: `// Java Practice Sandbox: Fix NullPointerException
public class Main {
    static class User {
        private String name;
        private String email;

        public User(String name, String email) {
            this.name = name;
            this.email = email;
        }

        public String getEmail() {
            return this.email;
        }
    }

    public static String formatUserEmail(User user) {
        // ✅ FIXED: Null guard prevents NullPointerException
        if (user == null || user.getEmail() == null) {
            return "no-reply@system.local";
        }
        return user.getEmail().toLowerCase();
```

```
    }

    public static void main(String[] args) {
        User validUser = new User("Alice", "ALICE@EXAMPLE.COM");
        System.out.println("Valid user: " + formatUserEmail(validUser));

        User nullEmailUser = new User("Bob", null);
        System.out.println("Null email user: " + formatUserEmail(nullEmailUser));
    }
}
`,
    hints: [
      'In Java, invoking any method on a reference that is null throws `NullPointerException`.',
      'Check `if (user == null || user.getEmail() == null)` before calling `.toLowerCase()`.',
      'Return a safe default fallback like `"no-reply@system.local"` when null.',
    ],
    testCases: [
      {
        id: 'tc-j1',
        name: 'Valid Email Check',
        inputDescription: 'User("Alice", "ALICE@EXAMPLE.COM")',
        expectedOutput: 'alice@example.com',
      },
      {
        id: 'tc-j2',
        name: 'Null Email Guard',
        inputDescription: 'User("Bob", null)',
        expectedOutput: 'no-reply@system.local',
      },
    ],
  },
  {
    id: 'java-array-oob',
    title: 'Java ArrayIndexOutOfBoundsException: Off-by-one Boundary',
```

```
    difficulty: 'Easy',
    language: 'java',
    category: 'Error Debugging',
    description:
      'A score statistics processor iterates through an array of scores using `<= array.length` instead of `< array.length`, attempting to access an index outside the allocated array memory.',
    errorNameTarget: 'java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 5',
    relatedCourseId: 'java-oop-errors',
    relatedLessonId: 'jl1-npe',
    initialCode: `// Java Practice Sandbox: Fix ArrayIndexOutOfBoundsException
public class Main {
    public static int calculateTotal(int[] scores) {
        int total = 0;
        // 💥 BUG: <= scores.length tries to read index equal to length!
        // An array of length 5 has valid indices 0, 1, 2, 3, 4. Index 5 crashes!
        for (int i = 0; i <= scores.length; i++) {
            total += scores[i];
        }
        return total;
    }

    public static void main(String[] args) {
        int[] playerScores = {10, 20, 30, 40, 50};
        System.out.println("Total Score: " + calculateTotal(playerScores));
    }
}
`,
    solutionCode: `// Java Practice Sandbox: Fix ArrayIndexOutOfBoundsException
public class Main {
    public static int calculateTotal(int[] scores) {
        int total = 0;
        // ✅ FIXED: Use < scores.length, or enhanced for-each loop
        for (int i = 0; i < scores.length; i++) {
            total += scores[i];
        }
```

```
        return total;
    }

    public static void main(String[] args) {
        int[] playerScores = {10, 20, 30, 40, 50};
        System.out.println("Total Score: " + calculateTotal(playerScores));
    }
}
`,
    hints: [
      'Java arrays are 0-indexed. An array of length 5 has valid indices 0 through 4.',
      'Change `i <= scores.length` to `i < scores.length` in the loop condition.',
      'Alternatively, use the enhanced for loop: `for (int score : scores) { total += score; }`.',
    ],
    testCases: [
      {
        id: 'tc-joob1',
        name: 'Calculate 5 scores sum',
        inputDescription: '{10, 20, 30, 40, 50}',
        expectedOutput: '150',
      },
    ],
  },
  {
    id: 'cpp-vector-range',
    title: 'C++ std::out_of_range: Vector Bounds Violation',
    difficulty: 'Medium',
    language: 'cpp',
    category: 'Error Debugging',
    description:
      'A telemetry reader accesses sensor values from `std::vector<float>` using an index equal to `vec.size()`. Understand why bounds checking with `.at()` throws `std::out_of_range` and prevent out-of-bounds access.',
    errorNameTarget: 'std::out_of_range: vector::_M_range_check: __n >= this->size()',
    relatedCourseId: 'cpp-memory-safety',
    relatedLessonId: 'cl1-bounds',
```

```
    initialCode: `// C++ Practice Sandbox: Fix std::out_of_range
#include <iostream>
#include <vector>

float getLatestReading(const std::vector<float>& readings) {
    // 💥 BUG: readings.size() is 1 past the last element!
    // A vector of size 3 has valid indices 0, 1, 2. Index 3 is out of range!
    return readings.at(readings.size());
}

int main() {
    std::vector<float> telemetry = {23.4f, 24.1f, 22.8f};
    std::cout << "Latest Telemetry: " << getLatestReading(telemetry) << std::endl;
    return 0;
}
`,
    solutionCode: `// C++ Practice Sandbox: Fix std::out_of_range
#include <iostream>
#include <vector>

float getLatestReading(const std::vector<float>& readings) {
    // ✅ FIXED: Guard against empty vector and access readings.back() or size() - 1
    if (readings.empty()) {
        return 0.0f;
    }
    return readings.back(); // or readings.at(readings.size() - 1)
}

int main() {
    std::vector<float> telemetry = {23.4f, 24.1f, 22.8f};
    std::cout << "Latest Telemetry: " << getLatestReading(telemetry) << std::endl;
    return 0;
}
`,
    hints: [
```

```
      'In C++, `std::vector::size()` returns the count of elements. The last valid element is at `size() - 1`.',
      'Prefer using `readings.back()` to retrieve the last element safely.',
      'Always check `readings.empty()` to avoid calling `.back()` on an empty container.',
    ],
    testCases: [
      {
        id: 'tc-cpp1',
        name: 'Last telemetry reading',
        inputDescription: '{23.4, 24.1, 22.8}',
        expectedOutput: '22.8',
      },
    ],
  },
  {
    id: 'cpp-segfault',
    title: 'C++ Segmentation Fault: Dangling Stack Pointer',
    difficulty: 'Hard',
    language: 'cpp',
    category: 'Security',
    description:
      'A configuration factory returns a raw pointer to a local stack-allocated variable. Once the function frame po
ps off the stack, dereferencing the pointer causes undefined behavior or Segmentation Fault (SIGSEGV).',
    errorNameTarget: 'Segmentation fault (core dumped): dereferencing dangling pointer',
    relatedCourseId: 'cpp-memory-safety',
    relatedLessonId: 'cl1-bounds',
    initialCode: `// C++ Practice Sandbox: Fix Segmentation Fault / Dangling Pointer
#include <iostream>
#include <string>

std::string* createConfigBanner() {
    // 💥 BUG: 'banner' is allocated on the stack frame of this function!
    // When createConfigBanner() returns, its stack memory is deallocated.
    std::string banner = "Production Server v2.4 (Active)";
    return &banner; // Returning address of local temporary!
}
```

```
1471
1472  int main() {
1473      std::string* ptr = createConfigBanner();
1474      // 💥 Dereferencing dangling pointer triggers undefined behavior or SIGSEGV!
1475      std::cout << "Banner: " << *ptr << std::endl;
1476      return 0;
1477  }
1478  `,
1479      solutionCode: `// C++ Practice Sandbox: Fix Segmentation Fault / Dangling Pointer
1480  #include <iostream>
1481  #include <string>
1482  #include <memory>
1483
1484  // ✅ FIXED: Return by value (exploiting Move Semantics / RVO) or use std::unique_ptr
1485  std::string createConfigBanner() {
1486      std::string banner = "Production Server v2.4 (Active)";
1487      return banner; // Safe: Return Value Optimization (RVO) avoids copying!
1488  }
1489
1490  int main() {
1491      std::string banner = createConfigBanner();
1492      std::cout << "Banner: " << banner << std::endl;
1493      return 0;
1494  }
1495  `,
1496      hints: [
1497        'Never return the memory address (`&variable`) of a local variable declared on the stack.',
1498        'In modern C++, return standard objects like `std::string` or `std::vector` by value; Return Value Optimizatio
      n (RVO) makes it zero-cost.',
1499        'If heap allocation is strictly necessary, return `std::unique_ptr<T>` instead of raw `T*`.',
1500      ],
1501      testCases: [
1502        {
1503          id: 'tc-cpp-seg',
1504          name: 'Safe Banner Generation',
```

```
        inputDescription: 'createConfigBanner()',
        expectedOutput: 'Production Server v2.4 (Active)',
      },
    ],
  },
  {
    id: 'py-key-error',
    title: 'Python KeyError: Missing Dictionary Key in Session',
    difficulty: 'Easy',
    language: 'python',
    category: 'Error Debugging',
    description:
      'An authentication handler checks user role permissions by directly indexing `session["role"]`. When a user visits without an active session cookie, Python raises `KeyError: "role"`. Use `.get()` or defensive presence checking.',
    errorNameTarget: "KeyError: 'role'",
    relatedCourseId: 'python-error-mastery',
    relatedLessonId: 'l9-index-error',
    initialCode: `# Python Practice Sandbox: Fix KeyError
def get_dashboard_theme(user_session):
    # 💥 BUG: Direct indexing session['role'] crashes with KeyError if 'role' is absent!
    user_role = user_session['role']

    if user_role == 'admin':
        return 'dark_gold_admin'
    return 'default_light'

# Valid session:
print("Admin theme:", get_dashboard_theme({'user_id': 101, 'role': 'admin'}))

# 💥 Crashes with KeyError: 'role':
guest_session = {'user_id': 102}
print("Guest theme:", get_dashboard_theme(guest_session))
`,
    solutionCode: `# Python Practice Sandbox: Fix KeyError
```

```
def get_dashboard_theme(user_session):
    # ✅ FIXED: Use dict.get(key, default) to safely retrieve without KeyError
    user_role = user_session.get('role', 'guest')

    if user_role == 'admin':
        return 'dark_gold_admin'
    return 'default_light'

# Valid session:
print("Admin theme:", get_dashboard_theme({'user_id': 101, 'role': 'admin'}))
guest_session = {'user_id': 102}
print("Guest theme:", get_dashboard_theme(guest_session))
`,
    hints: [
      'In Python, accessing `dictionary["key"]` raises a `KeyError` if the key does not exist.',
      'Use `user_session.get("role", "guest")` which safely returns `"guest"` instead of crashing.',
      'Alternatively, use `if "role" in user_session:` to check before reading.',
    ],
    testCases: [
      {
        id: 'tc-py-key1',
        name: 'Admin session',
        inputDescription: '{"role": "admin"}',
        expectedOutput: 'dark_gold_admin',
      },
      {
        id: 'tc-py-key2',
        name: 'Missing key session',
        inputDescription: '{"user_id": 102}',
        expectedOutput: 'default_light',
      },
    ],
  },
  {
    id: 'py-zero-division',
```

```
    title: 'Python ZeroDivisionError: Division by Zero in Metrics',
    difficulty: 'Easy',
    language: 'python',
    category: 'Error Debugging',
    description:
      'A ratings accumulator computes the average score with `sum(ratings) / len(ratings)`. When a new product has zero ratings, `len(ratings)` is 0, raising `ZeroDivisionError: division by zero`.',
    errorNameTarget: 'ZeroDivisionError: division by zero',
    relatedCourseId: 'python-error-mastery',
    relatedLessonId: 'l9-index-error',
    initialCode: `# Python Practice Sandbox: Fix ZeroDivisionError
def calculate_average_rating(ratings):
    # 💥 BUG: If ratings is an empty list [], len(ratings) is 0, causing division by zero!
    return sum(ratings) / len(ratings)

print("Product A avg:", calculate_average_rating([4.5, 5.0, 4.0]))

# 💥 Crashes with ZeroDivisionError:
print("Product B avg:", calculate_average_rating([]))
`,
    solutionCode: `# Python Practice Sandbox: Fix ZeroDivisionError
def calculate_average_rating(ratings):
    # ✅ FIXED: Check if ratings list is empty before dividing
    if not ratings:
        return 0.0
    return sum(ratings) / len(ratings)

print("Product A avg:", calculate_average_rating([4.5, 5.0, 4.0]))
print("Product B avg:", calculate_average_rating([]))
`,
    hints: [
      'In Python, dividing any number by zero raises `ZeroDivisionError`.',
      'Add a guard: `if not ratings: return 0.0` before performing the division.',
    ],
    testCases: [
```

```
      {
        id: 'tc-py-zd1',
        name: 'Valid ratings',
        inputDescription: '[4.5, 5.0, 4.0]',
        expectedOutput: '4.5',
      },
      {
        id: 'tc-py-zd2',
        name: 'Empty ratings list',
        inputDescription: '[]',
        expectedOutput: '0.0',
      },
    ],
  },
  {
    id: 'c-buffer-overflow',
    title: 'C Buffer Overflow: Unbounded strcpy Memory Corruption',
    difficulty: 'Hard',
    language: 'c',
    category: 'Security',
    description:
      'A legacy C authentication token parser uses unsafe `strcpy` to copy user input into a fixed-size 16-byte stac
k buffer. An attacker input exceeding 16 bytes smashes the stack and causes a buffer overflow vulnerability.',
    errorNameTarget: '*** stack smashing detected ***: terminated (buffer overflow in strcpy)',
    relatedCourseId: 'cpp-memory-safety',
    relatedLessonId: 'cl1-bounds',
    initialCode: `// C Practice Sandbox: Fix Buffer Overflow Vulnerability
#include <stdio.h>
#include <string.h>

void processUsername(const char* inputUsername) {
    char buffer[16];

    // 💥 CRITICAL SECURITY BUG: strcpy does NOT check buffer capacity!
    // Passing a string longer than 15 characters overwrites the stack frame return pointer!
```

```
    strcpy(buffer, inputUsername);
    printf("Processed username: %s\\n", buffer);
}

int main() {
    // 💥 28-character payload overflows the 16-byte buffer:
    processUsername("super_administrator_corporate_root");
    return 0;
}
`,
    solutionCode: `// C Practice Sandbox: Fix Buffer Overflow Vulnerability
#include <stdio.h>
#include <string.h>

void processUsername(const char* inputUsername) {
    char buffer[16];

    // ✅ FIXED: Use bounded strncpy or snprintf to guarantee no buffer overflow
    snprintf(buffer, sizeof(buffer), "%s", inputUsername);
    printf("Processed username: %s\\n", buffer);
}

int main() {
    processUsername("super_administrator_corporate_root");
    return 0;
}
`,
    hints: [
      'Never use unbounded memory copy functions like `strcpy` or `gets` in C.',
      'Use `snprintf(buffer, sizeof(buffer), "%s", inputUsername);` to prevent copying more bytes than buffer capacity.',
      'Ensure the string is always null-terminated.',
    ],
    testCases: [
      {
```

```
        id: 'tc-c-buf1',
        name: 'Safe username truncation',
        inputDescription: '"super_administrator_corporate_root"',
        expectedOutput: 'super_administr',
      },
    ],
  },
  {
    id: 'go-nil-pointer',
    title: 'Go Nil Pointer Dereference: Panic in Struct Pointer',
    difficulty: 'Medium',
    language: 'go',
    category: 'Error Debugging',
    description:
      'A Go worker tries to read field `user.Account.Balance` when the `Account` pointer is nil. In Go, dereferencing a nil pointer causes a runtime panic. Implement defensive nil checks.',
    errorNameTarget: 'panic: runtime error: invalid memory address or nil pointer dereference',
    relatedCourseId: 'js-ts-debugging',
    relatedLessonId: 'l4-optional-chaining',
    initialCode: `// Go Practice Sandbox: Fix Nil Pointer Dereference
package main

import "fmt"

type Account struct {
    Balance float64
}

type User struct {
    Name    string
    Account *Account
}

func GetBalance(u *User) float64 {
    // 💥 BUG: If u is nil or u.Account is nil, dereferencing panics!
```

```
    return u.Account.Balance
}

func main() {
    u1 := &User{Name: "Alice", Account: &Account{Balance: 250.75}}
    fmt.Printf("Alice Balance: $%.2f\\n", GetBalance(u1))

    // 💥 Account is nil: triggers panic!
    u2 := &User{Name: "Bob", Account: nil}
    fmt.Printf("Bob Balance: $%.2f\\n", GetBalance(u2))
}
`,
    solutionCode: `// Go Practice Sandbox: Fix Nil Pointer Dereference
package main

import "fmt"

type Account struct {
    Balance float64
}

type User struct {
    Name    string
    Account *Account
}

func GetBalance(u *User) float64 {
    // ✅ FIXED: Defensive nil pointer checks in Go
    if u == nil || u.Account == nil {
        return 0.0
    }
    return u.Account.Balance
}

```

```
func main() {
    u1 := &User{Name: "Alice", Account: &Account{Balance: 250.75}}
    fmt.Printf("Alice Balance: $%.2f\\n", GetBalance(u1))

    u2 := &User{Name: "Bob", Account: nil}
    fmt.Printf("Bob Balance: $%.2f\\n", GetBalance(u2))
}
`,
    hints: [
      'In Go, attempting to access a field or method on a nil pointer results in `panic: runtime error: invalid memory address or nil pointer dereference`.',
      'Check `if u == nil || u.Account == nil { return 0.0 }` before accessing `.Balance`.',
    ],
    testCases: [
      {
        id: 'tc-go-nil',
        name: 'Nil account guard',
        inputDescription: 'User{Account: nil}',
        expectedOutput: '0.00',
      },
    ],
  },
  {
    id: 'rust-borrow-checker',
    title: 'Rust Ownership & Borrowing: Borrow of Moved Value',
    difficulty: 'Hard',
    language: 'rust',
    category: 'Security',
    description:
      'A Rust microservice passes a `String` ownership to a logging helper, and then tries to reuse the moved value in a subsequent database query. Understand the Rust Borrow Checker and use references `&str`.',
    errorNameTarget: 'error[E0382]: borrow of moved value: "token"',
    relatedCourseId: 'cpp-memory-safety',
    relatedLessonId: 'cl1-bounds',
    initialCode: `// Rust Practice Sandbox: Fix Borrow of Moved Value
fn log_token(token: String) {
```

```
    println!("[AUDIT] Authenticated token: {}", token);
    // token is dropped here at the end of scope!
}

fn main() {
    let token = String::from("bearer_jwt_auth_xyz123");

    // 💥 BUG: Passing 'token' moves ownership into log_token!
    log_token(token);

    // 💥 ERROR[E0382]: borrow of moved value: 'token'!
    println!("Database querying with token: {}", token);
}
`,
    solutionCode: `// Rust Practice Sandbox: Fix Borrow of Moved Value
// ✅ FIXED: Borrow token as a read-only string slice &str instead of taking ownership
fn log_token(token: &str) {
    println!("[AUDIT] Authenticated token: {}", token);
}

fn main() {
    let token = String::from("bearer_jwt_auth_xyz123");

    // Pass by reference: &token
    log_token(&token);

    // 'token' remains valid and owned by main!
    println!("Database querying with token: {}", token);
}
`,
    hints: [
      'In Rust, variables that do not implement the `Copy` trait (like `String`) transfer ownership on assignment or function call.',
      'Change `fn log_token(token: String)` to `fn log_token(token: &str)` to borrow the value instead of moving it.',
```

```
      'Call `log_token(&token)` in `main()`.',
    ],
    testCases: [
      {
        id: 'tc-rust-borrow',
        name: 'Borrow token without moving',
        inputDescription: 'token = "bearer_jwt_auth_xyz123"',
        expectedOutput: 'Database querying with token: bearer_jwt_auth_xyz123',
      },
    ],
  },
];
```

File [5/30]: src/components/views/AcademyView.tsx

TSX SOURCE

Description: *Interactive Learning Academy View & Quiz Player*

```tsx
'use client';

import React, { useState, useEffect } from 'react';
import {
  BookOpen,
  Award,
  Zap,
  CheckCircle2,
  Clock,
  Play,
  ArrowRight,
  Search,
  Filter,
  Sparkles,
  ChevronRight,
  Code2,
  ShieldCheck,
  Bug,
  Lock,
  Terminal,
  HelpCircle,
  RotateCcw,
  Check,
  X,
  ExternalLink,
  Flame,
  Star,
  GraduationCap,
} from 'lucide-react';
import confetti from 'canvas-confetti';
import { COURSES } from '@/lib/courses-data';
```

```
import { Course, Lesson, NavTab } from '@/lib/types';
import { useAuth } from '@/lib/auth-context';

interface AcademyViewProps {
  onNavigate: (tab: NavTab) => void;
  onOpenPracticeChallenge?: (challengeId: string) => void;
}

export const AcademyView: React.FC<AcademyViewProps> = ({
  onNavigate,
  onOpenPracticeChallenge,
}) => {
  const { user } = useAuth();

  // State
  const [selectedCourseId, setSelectedCourseId] = useState<string | null>(null);
  const [activeLessonId, setActiveLessonId] = useState<string | null>(null);
  const [searchQuery, setSearchQuery] = useState<string>('');
  const [selectedCategory, setSelectedCategory] = useState<string>('All');
  const [selectedLevel, setSelectedLevel] = useState<string>('All');

  // Progress state persisted in localStorage
  const [completedLessons, setCompletedLessons] = useState<Record<string, boolean>>({});
  const [userXp, setUserXp] = useState<number>(320);
  const [streakDays, setStreakDays] = useState<number>(4);

  // Quiz state for active lesson
  const [quizAnswers, setQuizAnswers] = useState<Record<string, number>>({});
  const [quizSubmitted, setQuizSubmitted] = useState<Record<string, boolean>>({});

  // Certificate Modal state
  const [isCertificateOpen, setIsCertificateOpen] = useState<boolean>(false);

  // Load progress from localStorage
  useEffect(() => {
```

```
    try {
      const savedCompleted = localStorage.getItem('repopilot_completed_lessons');
      if (savedCompleted) {
        setCompletedLessons(JSON.parse(savedCompleted));
      }
      const savedXp = localStorage.getItem('repopilot_user_xp');
      if (savedXp) {
        setUserXp(parseInt(savedXp, 10));
      }
    } catch {
      // Local storage fallback
    }
  }, []);

  const saveProgress = (newCompleted: Record<string, boolean>, addedXp: number = 0) => {
    setCompletedLessons(newCompleted);
    const newXp = userXp + addedXp;
    setUserXp(newXp);
    try {
      localStorage.setItem('repopilot_completed_lessons', JSON.stringify(newCompleted));
      localStorage.setItem('repopilot_user_xp', String(newXp));
    } catch {
      // ignore
    }
  };

  const selectedCourse = COURSES.find((c) => c.id === selectedCourseId);

  // Get active lesson
  let activeLesson: Lesson | null = null;
  if (selectedCourse) {
    for (const module of selectedCourse.modules) {
      const found = module.lessons.find((l) => l.id === activeLessonId);
      if (found) {
```

```
        activeLesson = found;
        break;
      }
    }
    // Default to first lesson if not selected
    if (!activeLesson && selectedCourse.modules[0]?.lessons[0]) {
      activeLesson = selectedCourse.modules[0].lessons[0];
    }
  }

  // Calculate course completion percentage
  const getCourseProgress = (course: Course) => {
    const totalLessons = course.modules.reduce((acc, m) => acc + m.lessons.length, 0);
    if (totalLessons === 0) return 0;
    const completedCount = course.modules.reduce((acc, m) => {
      return acc + m.lessons.filter((l) => completedLessons[l.id]).length;
    }, 0);
    return Math.round((completedCount / totalLessons) * 100);
  };

  const handleStartCourse = (courseId: string) => {
    setSelectedCourseId(courseId);
    const course = COURSES.find((c) => c.id === courseId);
    if (course && course.modules[0]?.lessons[0]) {
      setActiveLessonId(course.modules[0].lessons[0].id);
    }
    window.scrollTo({ top: 0, behavior: 'smooth' });
  };

  const handleCompleteLesson = (lessonId: string, xpReward: number = 50) => {
    if (!completedLessons[lessonId]) {
      const updated = { ...completedLessons, [lessonId]: true };
      saveProgress(updated, xpReward);

      // Trigger confetti celebration
```

```
      try {
        confetti({
          particleCount: 50,
          spread: 60,
          origin: { y: 0.8 },
        });
      } catch {
        // ignore
      }

      // Check if course is now 100% complete
      if (selectedCourse) {
        const progress = getCourseProgress(selectedCourse);
        if (progress >= 95) {
          setTimeout(() => {
            setIsCertificateOpen(true);
            try {
              confetti({
                particleCount: 120,
                spread: 90,
                origin: { y: 0.6 },
              });
            } catch {
              // ignore
            }
          }, 600);
        }
      }
    }
  };

  const handleLaunchPractice = (challengeId: string) => {
    if (onOpenPracticeChallenge) {
      onOpenPracticeChallenge(challengeId);
```

```
    }
    onNavigate('practice');
  };

  // Filter courses
  const filteredCourses = COURSES.filter((course) => {
    const matchesCategory =
      selectedCategory === 'All' || course.category === selectedCategory;
    const matchesLevel = selectedLevel === 'All' || course.level === selectedLevel;
    const matchesSearch =
      course.title.toLowerCase().includes(searchQuery.toLowerCase()) ||
      course.description.toLowerCase().includes(searchQuery.toLowerCase()) ||
      course.skillsGained.some((s) => s.toLowerCase().includes(searchQuery.toLowerCase()));
    return matchesCategory && matchesLevel && matchesSearch;
  });

  const categories = [
    'All',
    'Debugging & Errors',
    'Security & Auth',
    'Python & Algorithms',
    'React & Architecture',
  ];

  const levels = ['All', 'Beginner', 'Intermediate', 'Advanced'];

  return (
    <div className="space-y-8 max-w-7xl mx-auto pb-20">
      {/* Platform Header & Stats Banner */}
      <div className="relative overflow-hidden rounded-2xl bg-gradient-to-br from-indigo-900 via-slate-900 to-indigo-950 text-white p-7 md:p-9 shadow-lg border border-indigo-800/40">
        <div className="absolute top-0 right-0 w-96 h-96 bg-indigo-500/10 rounded-full blur-3xl pointer-events-none -mr-20 -mt-20" />
        <div className="absolute bottom-0 left-1/3 w-80 h-80 bg-cyan-500/10 rounded-full blur-3xl pointer-events-none" />
```

```
        <div className="relative z-10 flex flex-col md:flex-row md:items-center justify-between gap-6">
          <div className="max-w-2xl">
            <div className="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-indigo-500/20 border border-indigo-400/30 text-indigo-300 text-xs font-semibold mb-3">
              <GraduationCap className="w-3.5 h-3.5 text-cyan-400" />
              <span>RepoPilot Learning Academy & Course Platform</span>
            </div>
            <h1 className="text-2xl md:text-3xl font-extrabold tracking-tight text-white mb-2">
              Master Modern Engineering, Debugging & Security
            </h1>
            <p className="text-slate-300 text-sm leading-relaxed">
              Interactive video-free, hands-on courses designed to teach you the deep mental models
              behind code execution, runtime errors, and rock-solid defense.
            </p>
          </div>

          {/* Gamification Stats */}
          <div className="flex items-center gap-3 shrink-0">
            <div className="px-4 py-3 rounded-xl bg-white/10 backdrop-blur-md border border-white/10 flex items-center gap-3 shadow-inner">
              <div className="w-10 h-10 rounded-lg bg-amber-500/20 border border-amber-400/40 flex items-center justify-center text-amber-400">
                <Flame className="w-5 h-5 animate-pulse" />
              </div>
              <div>
                <div className="text-xs text-slate-300 font-medium">Daily Streak</div>
                <div className="text-lg font-bold text-white">{streakDays} Days 🔥</div>
              </div>
            </div>

            <div className="px-4 py-3 rounded-xl bg-white/10 backdrop-blur-md border border-white/10 flex items-center gap-3 shadow-inner">
              <div className="w-10 h-10 rounded-lg bg-cyan-500/20 border border-cyan-400/40 flex items-center justify-center text-cyan-400">
                <Zap className="w-5 h-5" />
              </div>
              <div>
```

```
235                 <div className="text-xs text-slate-300 font-medium">Knowledge XP</div>
236                 <div className="text-lg font-bold text-cyan-300">{userXp} XP</div>
237               </div>
238             </div>
239           </div>
240         </div>
241       </div>
242
243       {/* Main View: Either Course Catalog OR Active Course Syllabus/Player */}
244       {!selectedCourseId ? (
245         // ========================================
246         // 1. COURSE CATALOG VIEW
247         // ========================================
248         <div className="space-y-6">
249           {/* Filters & Search Controls */}
250           <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 bg-white p-4 rounded-xl bo
rder border-slate-200 shadow-2xs">
251             {/* Search Input */}
252             <div className="relative flex-1 max-w-md">
253               <Search className="w-4 h-4 text-slate-400 absolute left-3 top-1/2 -translate-y-1/2" />
254               <input
255                 type="text"
256                 placeholder="Search courses, skills, or error concepts..."
257                 value={searchQuery}
258                 onChange={(e) => setSearchQuery(e.target.value)}
259                 className="w-full pl-9 pr-4 py-2 bg-slate-50 border border-slate-200 rounded-lg text-sm focus:outlin
e-none focus:ring-2 focus:ring-indigo-500 focus:bg-white transition-all text-slate-800 placeholder:text-slate-400"
260               />
261             </div>
262
263             {/* Category Pills */}
264             <div className="flex flex-wrap items-center gap-1.5 overflow-x-auto pb-1 md:pb-0">
265               {categories.map((cat) => (
266                 <button
267                   key={cat}
268                   onClick={() => setSelectedCategory(cat)}
```

```
                  className={`px-3 py-1.5 rounded-lg text-xs font-medium transition-all ${
                    selectedCategory === cat
                      ? 'bg-indigo-600 text-white shadow-xs font-semibold'
                      : 'bg-slate-100 hover:bg-slate-200 text-slate-600'
                  }`}
                >
                  {cat}
                </button>
              ))}
            </div>
          </div>

          {/* Featured Banner Card */}
          <div className="relative overflow-hidden rounded-2xl bg-gradient-to-r from-indigo-600 via-indigo-700 to-purple-800 text-white p-6 md:p-8 shadow-md">
            <div className="relative z-10 max-w-2xl space-y-3">
              <span className="inline-flex items-center gap-1.5 px-2.5 py-0.5 rounded-full text-[11px] font-bold bg-white/20 text-white border border-white/20 backdrop-blur-xs">
                <Sparkles className="w-3 h-3 text-amber-300" />
                RECOMMENDED TRACK
              </span>
              <h2 className="text-2xl font-bold text-white tracking-tight">
                Mastering JavaScript & TypeScript Debugging
              </h2>
              <p className="text-indigo-100 text-sm leading-relaxed">
                Connects directly to our Code Practice Space with automated AI Error Explanation!
                Learn Call Stack internals, eliminate undefined exceptions, and master async error
                resilience.
              </p>
              <div className="pt-2 flex flex-wrap items-center gap-3">
                <button
                  onClick={() => handleStartCourse('js-ts-debugging')}
                  className="px-5 py-2.5 rounded-xl bg-white text-indigo-700 hover:bg-indigo-50 font-bold text-sm shadow-md transition-all flex items-center gap-2 cursor-pointer"
                >
                  <Play className="w-4 h-4 fill-indigo-700" />
```

```
                  <span>Start Course Now</span>
                </button>
                <button
                  onClick={() => handleLaunchPractice('type-error-null')}
                  className="px-4 py-2.5 rounded-xl bg-white/10 hover:bg-white/20 text-white border border-white/20 text-xs font-semibold backdrop-blur-xs transition-all flex items-center gap-1.5 cursor-pointer"
                >
                  <Code2 className="w-4 h-4" />
                  <span>Jump to Practice Lab</span>
                </button>
              </div>
            </div>
          </div>

          {/* Course Grid */}
          <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {filteredCourses.map((course) => {
              const progress = getCourseProgress(course);
              const totalLessons = course.modules.reduce((a, b) => a + b.lessons.length, 0);

              return (
                <div
                  key={course.id}
                  className="bg-white rounded-2xl border border-slate-200 overflow-hidden shadow-xs hover:shadow-md hover:border-indigo-300 transition-all flex flex-col justify-between group"
                >
                  <div>
                    {/* Header Strip */}
                    <div
                      className={`h-3 bg-gradient-to-r ${course.color || 'from-indigo-500 to-cyan-500'}`}
                    />
                    <div className="p-6 space-y-4">
                      {/* Top Badges */}
                      <div className="flex items-center justify-between">
                        <span className="text-[11px] font-bold uppercase tracking-wider text-indigo-600 bg-indigo-50 px-2.5 py-1 rounded-md border border-indigo-100">
```

```
                        {course.category}
                      </span>
                      <div className="flex items-center gap-1 text-xs text-slate-500 font-medium">
                        <Clock className="w-3.5 h-3.5 text-slate-400" />
                        <span>{course.duration}</span>
                      </div>
                    </div>

                    {/* Title & Tagline */}
                    <div>
                      <h3 className="text-lg font-bold text-slate-900 group-hover:text-indigo-600 transition-colors line-clamp-1">
                        {course.title}
                      </h3>
                      <p className="text-xs text-slate-500 mt-1 line-clamp-2 leading-relaxed">
                        {course.tagline}
                      </p>
                    </div>

                    {/* Skills gained pills */}
                    <div className="flex flex-wrap gap-1.5 pt-1">
                      {course.skillsGained.slice(0, 3).map((skill) => (
                        <span
                          key={skill}
                          className="text-[10px] bg-slate-100 text-slate-600 px-2 py-0.5 rounded font-medium"
                        >
                          {skill}
                        </span>
                      ))}
                    </div>

                    {/* Progress Bar */}
                    <div className="space-y-1.5 pt-2">
                      <div className="flex items-center justify-between text-xs text-slate-500 font-medium">
                        <span>Progress</span>
```

```
                          <span className="font-bold text-slate-700">{progress}%</span>
                        </div>
                        <div className="w-full h-1.5 bg-slate-100 rounded-full overflow-hidden">
                          <div
                            className="h-full bg-gradient-to-r from-indigo-500 to-cyan-500 rounded-full transition-all duration-500"
                            style={{ width: `${progress}%` }}
                          />
                        </div>
                      </div>
                    </div>

                    {/* Card Footer */}
                    <div className="p-5 border-t border-slate-100 bg-slate-50/60 flex items-center justify-between">
                      <div className="flex items-center gap-2 text-xs font-semibold text-slate-600">
                        <BookOpen className="w-3.5 h-3.5 text-indigo-600" />
                        <span>{totalLessons} Lessons</span>
                      </div>

                      <button
                        onClick={() => handleStartCourse(course.id)}
                        className="px-3.5 py-1.5 rounded-lg bg-indigo-600 hover:bg-indigo-700 text-white text-xs font-semibold shadow-2xs transition-all flex items-center gap-1 cursor-pointer"
                      >
                        <span>{progress > 0 ? 'Continue' : 'Start Course'}</span>
                        <ChevronRight className="w-3.5 h-3.5" />
                      </button>
                    </div>
                  </div>
                );
              })}
            </div>
          </div>
        ) : (
```

```
        // ========================================
        // 2. ACTIVE COURSE SYLLABUS & LESSON READER
        // ========================================
        <div className="space-y-6">
          {/* Breadcrumb Navigation */}
          <div className="flex items-center justify-between bg-white px-4 py-3 rounded-xl border border-slate-200 sh
adow-2xs">
            <div className="flex items-center gap-2 text-xs text-slate-600">
              <button
                onClick={() => setSelectedCourseId(null)}
                className="hover:text-indigo-600 font-semibold transition-colors cursor-pointer"
              >
                All Courses
              </button>
              <ChevronRight className="w-3.5 h-3.5 text-slate-400" />
              <span className="font-semibold text-slate-900 truncate max-w-xs">
                {selectedCourse?.title}
              </span>
              {activeLesson && (
                <>
                  <ChevronRight className="w-3.5 h-3.5 text-slate-400" />
                  <span className="text-indigo-600 font-semibold truncate max-w-xs">
                    {activeLesson.title}
                  </span>
                </>
              )}
            </div>

            <button
              onClick={() => setSelectedCourseId(null)}
              className="text-xs font-semibold text-slate-500 hover:text-slate-800 transition-colors"
            >
              Back to Catalog
            </button>
          </div>
```

```
436
437           <div className="grid grid-cols-1 lg:grid-cols-12 gap-8 items-start">
438             {/* Left: Syllabus & Module Tree (4 cols) */}
439             <div className="lg:col-span-4 bg-white rounded-2xl border border-slate-200 p-5 shadow-xs space-y-6 stick
    y top-20">
440               <div className="space-y-2">
441                 <div className="flex items-center justify-between">
442                   <span className="text-[11px] font-bold uppercase tracking-wider text-indigo-600 bg-indigo-50 px-2
    py-0.5 rounded border border-indigo-100">
443                     {selectedCourse?.category}
444                   </span>
445                   <span className="text-xs font-bold text-emerald-700 bg-emerald-50 px-2 py-0.5 rounded border borde
    r-emerald-100">
446                     +{selectedCourse?.xp} XP Total
447                   </span>
448                 </div>
449                 <h2 className="text-lg font-bold text-slate-900">{selectedCourse?.title}</h2>
450                 <div className="space-y-1 pt-1">
451                   <div className="flex items-center justify-between text-xs text-slate-500 font-medium">
452                     <span>Course Completion</span>
453                     <span className="font-bold text-slate-800">
454                       {selectedCourse ? getCourseProgress(selectedCourse) : 0}%
455                     </span>
456                   </div>
457                   <div className="w-full h-1.5 bg-slate-100 rounded-full overflow-hidden">
458                     <div
459                       className="h-full bg-indigo-600 rounded-full transition-all duration-300"
460                       style={{
461                         width: `${selectedCourse ? getCourseProgress(selectedCourse) : 0}%`,
462                       }}
463                     />
464                   </div>
465                 </div>
466               </div>
467
468               {/* Modules list */}
```

```
              <div className="space-y-5 pt-2 border-t border-slate-100">
                {selectedCourse?.modules.map((module, mIdx) => (
                  <div key={module.id} className="space-y-2">
                    <div className="text-xs font-bold text-slate-800 uppercase tracking-wider flex items-center gap-2">
                      <span className="w-4 h-4 rounded-full bg-slate-100 text-slate-600 flex items-center justify-center text-[10px] font-extrabold">
                        {mIdx + 1}
                      </span>
                      <span>{module.title}</span>
                    </div>

                    <div className="space-y-1 pl-6">
                      {module.lessons.map((lesson) => {
                        const isCurrent = activeLesson?.id === lesson.id;
                        const isCompleted = completedLessons[lesson.id];

                        return (
                          <button
                            key={lesson.id}
                            onClick={() => setActiveLessonId(lesson.id)}
                            className={`w-full text-left p-2.5 rounded-lg text-xs transition-all flex items-center justify-between group cursor-pointer ${
                              isCurrent
                                ? 'bg-indigo-50 text-indigo-700 font-bold border border-indigo-200'
                                : 'text-slate-600 hover:bg-slate-50 hover:text-slate-900'
                            }`}
                          >
                            <div className="flex items-center gap-2 min-w-0 pr-2">
                              {isCompleted ? (
                                <CheckCircle2 className="w-4 h-4 text-emerald-600 shrink-0" />
                              ) : isCurrent ? (
                                <Play className="w-3.5 h-3.5 text-indigo-600 fill-indigo-600 shrink-0" />
                              ) : (
                                <div className="w-2 h-2 rounded-full bg-slate-300 shrink-0 group-hover:bg-slate-400" />
                              )}
```

```
                            <span className="truncate">{lesson.title}</span>
                          </div>

                          <span className="text-[10px] text-slate-400 shrink-0">
                            {lesson.durationMinutes}m
                          </span>
                        </button>
                      );
                    })}
                  </div>
                </div>
              ))}
            </div>
          </div>

          {/* Right: Active Lesson Reader & Interactive Lab (8 cols) */}
          <div className="lg:col-span-8 space-y-6">
            {activeLesson ? (
              <div className="bg-white rounded-2xl border border-slate-200 p-6 md:p-8 shadow-xs space-y-6">
                {/* Lesson Header */}
                <div className="border-b border-slate-100 pb-5 space-y-3">
                  <div className="flex flex-wrap items-center justify-between gap-2">
                    <div className="flex items-center gap-2">
                      <span className="text-[11px] font-bold text-indigo-700 bg-indigo-50 px-2.5 py-1 rounded-md b
order border-indigo-100">
                        {activeLesson.type === 'interactive_lab'
                          ? '⚡ Interactive Lab'
                          : '📖 Concept Lesson'}
                      </span>
                      <span className="text-xs text-slate-500 font-medium flex items-center gap-1">
                        <Clock className="w-3.5 h-3.5" />
                        <span>{activeLesson.durationMinutes} min read</span>
                      </span>
                    </div>

```

```
                        {completedLessons[activeLesson.id] && (
                          <span className="inline-flex items-center gap-1 text-xs font-bold text-emerald-700 bg-emeral
d-50 px-2.5 py-1 rounded-full border border-emerald-200">
                            <CheckCircle2 className="w-3.5 h-3.5 text-emerald-600" />
                            <span>Completed (+50 XP)</span>
                          </span>
                        )}
                      </div>

                      <h1 className="text-2xl font-extrabold text-slate-900 tracking-tight">
                        {activeLesson.title}
                      </h1>

                      <p className="text-sm text-slate-600 font-medium leading-relaxed">
                        {activeLesson.summary}
                      </p>
                    </div>

                    {/* Lesson Body Paragraphs */}
                    <div className="space-y-4 text-slate-700 text-sm leading-relaxed">
                      {activeLesson.content.map((paragraph, idx) => (
                        <p key={idx}>{paragraph}</p>
                      ))}
                    </div>

                    {/* Key Takeaways Callout Card */}
                    {activeLesson.keyTakeaways && activeLesson.keyTakeaways.length > 0 && (
                      <div className="p-4 rounded-xl bg-indigo-50/70 border border-indigo-100 space-y-2">
                        <div className="flex items-center gap-2 text-xs font-bold text-indigo-900 uppercase tracking-w
ider">
                          <Sparkles className="w-4 h-4 text-indigo-600" />
                          <span>Core Mental Model & Key Takeaways</span>
                        </div>
                        <ul className="space-y-1.5 pl-5 list-disc text-xs text-indigo-950 font-medium">
                          {activeLesson.keyTakeaways.map((takeaway, tIdx) => (
```

```
                          <li key={tIdx}>{takeaway}</li>
                        ))}
                      </ul>
                    </div>
                  )}

                  {/* Code Example Widget */}
                  {activeLesson.codeExample && (
                    <div className="space-y-2">
                      <div className="flex items-center justify-between text-xs font-semibold text-slate-600">
                        <div className="flex items-center gap-1.5">
                          <Code2 className="w-4 h-4 text-indigo-600" />
                          <span>{activeLesson.codeExample.description}</span>
                        </div>
                        <span className="text-[10px] uppercase font-mono px-1.5 py-0.5 rounded bg-slate-100 text-slate-600">
                          {activeLesson.codeExample.language}
                        </span>
                      </div>

                      <div className="rounded-xl overflow-hidden bg-slate-950 text-slate-100 border border-slate-800 p-4 font-mono text-xs shadow-inner">
                        <pre className="overflow-x-auto leading-relaxed">
                          <code>{activeLesson.codeExample.code}</code>
                        </pre>
                      </div>
                    </div>
                  )}

                  {/* Interactive Code Lab Callout (If lesson has attached practice challenge) */}
                  {activeLesson.practiceChallengeId && (
                    <div className="p-5 rounded-2xl bg-gradient-to-r from-indigo-900 to-slate-900 text-white space-y-3 shadow-md border border-indigo-700/50">
                      <div className="flex items-center justify-between">
                        <div className="flex items-center gap-2">
                          <div className="w-8 h-8 rounded-lg bg-indigo-500/20 border border-indigo-400/30 flex items-center justify-center text-cyan-400">
```

```
                            <Zap className="w-4 h-4" />
                          </div>
                          <div>
                            <h4 className="text-sm font-bold text-white">
                              Hands-On Error Diagnostic Lab
                            </h4>
                            <p className="text-xs text-slate-300">
                              Practice fixing this exact error in our interactive playground with AI
                              Tutor guidance!
                            </p>
                          </div>
                        </div>

                        <button
                          onClick={() => handleLaunchPractice(activeLesson!.practiceChallengeId!)}
                          className="px-4 py-2 rounded-xl bg-cyan-400 hover:bg-cyan-300 text-slate-950 font-bold tex
t-xs shadow-md transition-all flex items-center gap-1.5 shrink-0 cursor-pointer"
                        >
                          <span>Open in Code Lab</span>
                          <ExternalLink className="w-3.5 h-3.5" />
                        </button>
                      </div>
                    </div>
                  )}

                  {/* Interactive Quiz / Knowledge Check */}
                  {activeLesson.quiz && activeLesson.quiz.length > 0 && (
                    <div className="border-t border-slate-100 pt-6 space-y-4">
                      <div className="flex items-center gap-2">
                        <HelpCircle className="w-4 h-4 text-indigo-600" />
                        <h3 className="text-sm font-bold text-slate-900">
                          Quick Knowledge Check (Concept Verification)
                        </h3>
                      </div>
```

```
                        {activeLesson.quiz.map((q) => {
                          const selectedIdx = quizAnswers[q.id];
                          const isSubmitted = quizSubmitted[q.id];
                          const isCorrect = selectedIdx === q.correctIndex;

                          return (
                            <div
                              key={q.id}
                              className="p-5 rounded-xl border border-slate-200 bg-slate-50/50 space-y-3"
                            >
                              <p className="text-sm font-semibold text-slate-800">{q.question}</p>

                              <div className="space-y-2">
                                {q.options.map((opt, optIdx) => {
                                  const isOptionSelected = selectedIdx === optIdx;

                                  return (
                                    <button
                                      key={optIdx}
                                      disabled={isSubmitted}
                                      onClick={() =>
                                        setQuizAnswers({ ...quizAnswers, [q.id]: optIdx })
                                      }
                                      className={`w-full text-left p-3 rounded-lg text-xs font-medium transition-all f
lex items-center justify-between cursor-pointer ${
                                        isSubmitted && optIdx === q.correctIndex
                                          ? 'bg-emerald-50 text-emerald-800 border-2 border-emerald-500 font-bold'
                                          : isSubmitted && isOptionSelected && !isCorrect
                                            ? 'bg-rose-50 text-rose-800 border-2 border-rose-500'
                                            : isOptionSelected
                                              ? 'bg-indigo-50 text-indigo-900 border-2 border-indigo-500 font-semibol
d'
                                              : 'bg-white text-slate-700 border border-slate-200 hover:bg-slate-50'
                                      }`}
                                    >
```

```
                                    <span>{opt}</span>
                                    {isSubmitted && optIdx === q.correctIndex && (
                                      <Check className="w-4 h-4 text-emerald-600 shrink-0" />
                                    )}
                                    {isSubmitted && isOptionSelected && !isCorrect && (
                                      <X className="w-4 h-4 text-rose-600 shrink-0" />
                                    )}
                                  </button>
                                );
                              })}
                            </div>

                            {/* Submit Quiz Answer Button */}
                            {!isSubmitted ? (
                              <button
                                disabled={selectedIdx === undefined}
                                onClick={() =>
                                  setQuizSubmitted({ ...quizSubmitted, [q.id]: true })
                                }
                                className="px-4 py-1.5 rounded-lg bg-indigo-600 disabled:opacity-40 hover:bg-indigo-700 text-white text-xs font-semibold shadow-2xs transition-all cursor-pointer"
                              >
                                Check Answer
                              </button>
                            ) : (
                              <div
                                className={`p-3 rounded-lg text-xs ${
                                  isCorrect
                                    ? 'bg-emerald-50 text-emerald-900 border border-emerald-200'
                                    : 'bg-amber-50 text-amber-900 border border-amber-200'
                                }`}
                              >
                                <div className="font-bold mb-1">
                                  {isCorrect ? '🎉 Correct Concept!' : '💡 Conceptual Explanation:'}
                                </div>
```

```
                                <p>{q.explanation}</p>
                              </div>
                            )}
                          </div>
                        );
                      })}
                    </div>
                  )}

                  {/* Completion and Next Lesson Footer Bar */}
                  <div className="border-t border-slate-100 pt-6 flex flex-wrap items-center justify-between gap-3">
                    <button
                      onClick={() => handleCompleteLesson(activeLesson!.id, 50)}
                      className={`px-5 py-2.5 rounded-xl font-bold text-xs shadow-xs transition-all flex items-cente
r gap-2 cursor-pointer ${
                        completedLessons[activeLesson.id]
                          ? 'bg-emerald-50 text-emerald-700 border border-emerald-200 hover:bg-emerald-100'
                          : 'bg-indigo-600 hover:bg-indigo-700 text-white'
                      }`}
                    >
                      <CheckCircle2 className="w-4 h-4" />
                      <span>
                        {completedLessons[activeLesson.id]
                          ? 'Completed (Click to Re-celebrate)'
                          : 'Mark as Completed (+50 XP)'}
                      </span>
                    </button>

                    {/* Next Lesson Button */}
                    <button
                      onClick={() => {
                        handleCompleteLesson(activeLesson!.id, 50);
                        // Find next lesson
                        if (selectedCourse) {
                          const allLessons = selectedCourse.modules.flatMap((m) => m.lessons);
```

```
                          const currentIndex = allLessons.findIndex(
                            (l) => l.id === activeLesson!.id
                          );
                          if (currentIndex < allLessons.length - 1) {
                            setActiveLessonId(allLessons[currentIndex + 1].id);
                            window.scrollTo({ top: 0, behavior: 'smooth' });
                          }
                        }
                      }}
                      className="px-4 py-2.5 rounded-xl bg-slate-900 hover:bg-slate-800 text-white text-xs font-semibold transition-all flex items-center gap-1.5 cursor-pointer"
                    >
                      <span>Next Lesson</span>
                      <ArrowRight className="w-3.5 h-3.5" />
                    </button>
                  </div>
                </div>
              ) : null}
            </div>
          </div>
        </div>
      )}

      {/* Course Completion Digital Certificate Modal */}
      {isCertificateOpen && selectedCourse && (
        <div className="fixed inset-0 z-50 bg-slate-900/70 backdrop-blur-sm flex items-center justify-center p-4">
          <div className="bg-white rounded-3xl max-w-xl w-full border border-indigo-200 p-8 shadow-2xl relative text-center space-y-6">
            <button
              onClick={() => setIsCertificateOpen(false)}
              className="absolute top-4 right-4 text-slate-400 hover:text-slate-600 p-1"
            >
              <X className="w-5 h-5" />
            </button>
```

```
            <div className="w-16 h-16 mx-auto rounded-full bg-gradient-to-tr from-amber-400 to-amber-600 text-white flex items-center justify-center shadow-lg shadow-amber-500/30">
              <Award className="w-8 h-8" />
            </div>

            <div className="space-y-2">
              <span className="text-[11px] font-bold tracking-widest uppercase text-indigo-600 bg-indigo-50 px-3 py-1 rounded-full border border-indigo-200">
                Official Certificate of Mastery
              </span>
              <h3 className="text-2xl font-extrabold text-slate-900">
                Congratulations, {user?.name || 'Engineer'}!
              </h3>
              <p className="text-xs text-slate-500 max-w-md mx-auto">
                You have successfully completed all conceptual modules, diagnostic labs, and knowledge
                verifications for:
              </p>
              <div className="p-3 bg-slate-50 rounded-xl border border-slate-200 font-bold text-slate-800 text-sm">
                {selectedCourse.title}
              </div>
            </div>

            <div className="p-4 rounded-xl bg-indigo-50/60 border border-indigo-100 flex items-center justify-around text-xs font-semibold text-slate-700">
              <div>
                <div className="text-slate-400 font-medium">Credential ID</div>
                <div className="font-mono text-indigo-700">RP-CERT-8849</div>
              </div>
              <div className="w-px h-8 bg-slate-200" />
              <div>
                <div className="text-slate-400 font-medium">XP Awarded</div>
                <div className="font-bold text-emerald-600">+{selectedCourse.xp} XP</div>
              </div>
              <div className="w-px h-8 bg-slate-200" />
              <div>
                <div className="text-slate-400 font-medium">Issued Date</div>
```

```
                <div>{new Date().toLocaleDateString()}</div>
              </div>
            </div>

            <div className="pt-2 flex items-center justify-center gap-3">
              <button
                onClick={() => setIsCertificateOpen(false)}
                className="px-6 py-2.5 rounded-xl bg-indigo-600 hover:bg-indigo-700 text-white text-xs font-bold shadow-md transition-all cursor-pointer"
              >
                Claim & Continue Learning
              </button>
            </div>
          </div>
        </div>
      )}
    </div>
  );
};
```

File [6/30]: src/lib/error-explainer.ts

TYPESCRIPT SOURCE

Description: *AST Error Analysis & Multi-Language Real-Time Sentinel*

```typescript
import { ErrorExplanation, LiveDiagnostic, SupportedLanguage } from './types';

/**
 * Real-Time As-You-Type Code Analyzer & Linter for Multi-Language Sandboxes
 * Supports: Python, Java, C++, C, Go, Rust, JavaScript, TypeScript
 * Scans code while typing and immediately reports:
 * - Has error? (true/false)
 * - Where the mistake was made (exact line & snippet)
 * - Why it is an error in this specific language
 * - How to fix it
 */
export function analyzeCodeLive(
  code: string,
  language: SupportedLanguage = 'javascript',
  challengeId?: string
): LiveDiagnostic {
  if (!code || !code.trim()) {
    return {
      hasError: false,
      type: 'clean',
      title: 'Empty Editor',
      whereIsMistake: 'Editor is empty.',
      whyIsError: 'No code to evaluate yet.',
      howToFix: 'Start typing code or click "Load Starter" to load template code.',
    };
  }

  const lines = code.split('\n');

  // ==========================================
  // 1. PYTHON MODE (sandbox.py)
```

```
  // ========================================
  if (language === 'python') {
    let openParens = 0;
    let openBrackets = 0;
    let openBraces = 0;

    for (let i = 0; i < lines.length; i++) {
      const lineText = lines[i];
      const lineTrimmed = lineText.trim();
      const lineNum = i + 1;

      // Check for JavaScript/Java console.log in Python
      if (lineTrimmed.includes('console.log(')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "NameError: 'console' is not defined in Python",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "In Python, output is printed using the 'print(...)' function. 'console.log' is JavaScript syntax.",
          howToFix: `Replace 'console.log(...)' with 'print(...)'.`,
        };
      }

      // Check for Java System.out.println in Python
      if (lineTrimmed.includes('System.out.println(')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "NameError: 'System' is not defined in Python",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
```

```
            whyIsError: "In Python, output is printed using 'print(...)'. 'System.out.println' is Java syntax.",
            howToFix: `Replace 'System.out.println(...)' with 'print(...)'.`,
          };
        }

        // Check for JS/Java variable keywords
        if (lineTrimmed.startsWith('let ') || lineTrimmed.startsWith('const ') || lineTrimmed.startsWith('var ')) {
          return {
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: 'let'/'const'/'var' is not valid in Python",
            line: lineNum,
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
            whyIsError: "Python does not use variable declaration keywords like 'let', 'const', or 'var'. Variables are created by direct assignment.",
            howToFix: `Remove the keyword: '${lineTrimmed.replace(/^(let|const|var)\s+/, '')}'.`,
          };
        }

        // Check for JS/Java comments '//'
        if (lineTrimmed.startsWith('//')) {
          return {
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: '//' is floor division in Python, not a comment",
            line: lineNum,
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
            whyIsError: "In Python, single-line comments begin with the '#' symbol. '//' is the floor-division operator.",
            howToFix: `Change '//' to '#': "# ${lineTrimmed.replace(/^\/\/\s*/, '')}".`,
          };
        }

        // Check for Java import packages
```

```
      if (lineTrimmed.startsWith('import java.') || lineTrimmed.includes('import .')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "ModuleNotFoundError: Java package in Python",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "Python does not have Java packages. Standard Python modules include 'math', 'json', 'sys', 'os', 'datetime'.",
          howToFix: "Use Python imports: 'import math' or 'from datetime import datetime'.",
        };
      }

      // Check for curly braces on print: print{
      if (lineTrimmed.startsWith('print{') || lineTrimmed.startsWith('print {')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "SyntaxError: invalid syntax '{'",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "Function calls in Python use round parentheses '()', not curly braces '{}'. Curly braces are reserved for dicts and sets.",
          howToFix: `Change to round parentheses: 'print(...)'.`,
        };
      }

      // Check for missing colon on def/if/for/while/class/elif/else/try/except
      if (
        (lineTrimmed.startsWith('def ') ||
          lineTrimmed.startsWith('if ') ||
          lineTrimmed.startsWith('elif ') ||
          lineTrimmed.startsWith('for ') ||
```

```
            lineTrimmed.startsWith('while ') ||
            lineTrimmed === 'else' ||
            lineTrimmed === 'try' ||
            lineTrimmed.startsWith('except')) &&
          !lineTrimmed.endsWith(':') &&
          !lineTrimmed.endsWith('"""') &&
          !lineTrimmed.endsWith("'''")
        ) {
          return {
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: expected ':' at end of compound statement",
            line: lineNum,
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}" (missing ':' at the end)`,
            whyIsError: "In Python, block headers (if, for, while, def, class, else, except) must always end with a co
lon ':'.",
            howToFix: `Add ':' at the end of the line: "${lineTrimmed}:".`,
          };
        }

        // Count delimiters
        for (const char of lineText) {
          if (char === '(') openParens++;
          if (char === ')') openParens--;
          if (char === '[') openBrackets++;
          if (char === ']') openBrackets--;
          if (char === '{') openBraces++;
          if (char === '}') openBraces--;
        }

        // If a print statement is unclosed on this line
        if (lineTrimmed.startsWith('print(') && !lineTrimmed.includes(')')) {
          return {
            hasError: true,
```

```
          type: 'syntax',
          title: "SyntaxError: unexpected EOF while parsing - unclosed '('",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "The 'print(' statement was opened but never closed with a matching ')'. Python requires all opened parentheses to be closed before statement execution.",
          howToFix: `Close the parenthesis: "${lineTrimmed})".`,
        };
      }
    }

    if (openParens > 0) {
      return {
        hasError: true,
        type: 'syntax',
        title: "SyntaxError: unexpected EOF while parsing (unclosed parenthesis ')')",
        line: lines.length,
        codeSnippet: lines[lines.length - 1].trim(),
        whereIsMistake: `Line ${lines.length}: Unclosed parenthesis '('`,
        whyIsError: "There is an opening '(' in your Python code that has not been closed with a matching ')'.",
        howToFix: "Add ')' to close the open expression.",
      };
    }
  }

  // ========================================
  // 2. JAVA MODE (Main.java)
  // ========================================
  if (language === 'java') {
    let hasClass = false;

    for (let i = 0; i < lines.length; i++) {
      const lineText = lines[i];
      const lineTrimmed = lineText.trim();
```

```
      const lineNum = i + 1;

      if (lineTrimmed.includes('class ')) hasClass = true;

      // Check for Python print in Java
      if (lineTrimmed.startsWith('print(') || lineTrimmed.startsWith('console.log(')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "error: cannot find symbol method print()",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "Java does not have a bare 'print()' or 'console.log()' function. Standard output uses System.
out.println(...).",
          howToFix: `Change to Java syntax: "System.out.println(${lineTrimmed.replace(/^(print|console\.log)\(/,
'')}".`,
        };
      }

      // Check for Python def in Java
      if (lineTrimmed.startsWith('def ')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "error: invalid method declaration in Java",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "Java is statically typed and does not use Python 'def'. Java methods must specify a return ty
pe (e.g. void, int, String).",
          howToFix: "Declare as: public static void methodName() { ... }",
        };
      }

      // Check for malformed import
```

```
      if (lineTrimmed.includes('import .') || (lineTrimmed.startsWith('import ') && !lineTrimmed.endsWith(';') && !lineTrimmed.startsWith('import java.'))) {
        return {
          hasError: true,
          type: 'syntax',
          title: "error: '.' unexpected after import in Java",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "Java package imports cannot have a dot immediately after 'import'. Java imports follow: 'import java.util.*;' or 'import java.util.Scanner;'.",
          howToFix: "Use correct Java import syntax: 'import java.util.*;'.",
        };
      }

      // Check for missing semicolon on plain statement lines
      if (
        lineTrimmed.length > 0 &&
        !lineTrimmed.startsWith('//') &&
        !lineTrimmed.startsWith('/*') &&
        !lineTrimmed.startsWith('*') &&
        !lineTrimmed.endsWith('{') &&
        !lineTrimmed.endsWith('}') &&
        !lineTrimmed.endsWith(';') &&
        !lineTrimmed.startsWith('public class') &&
        !lineTrimmed.startsWith('class ') &&
        !lineTrimmed.startsWith('static class') &&
        !lineTrimmed.startsWith('public static void main') &&
        !lineTrimmed.startsWith('public ') &&
        !lineTrimmed.startsWith('private ') &&
        !lineTrimmed.startsWith('protected ') &&
        !lineTrimmed.startsWith('import ') &&
        !lineTrimmed.startsWith('package ') &&
        !lineTrimmed.startsWith('@')
      ) {
        return {
```

```
          hasError: true,
          type: 'syntax',
          title: "error: ';' expected",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "In Java, every statement must terminate with a semicolon ';'.",
          howToFix: `Add ';' to the end of line ${lineNum}: "${lineTrimmed};".`,
        };
      }
    }

    if (!hasClass && lines.some((l) => l.trim().length > 0)) {
      return {
        hasError: true,
        type: 'syntax',
        title: "error: class, interface, or enum expected",
        line: 1,
        codeSnippet: lines[0].trim(),
        whereIsMistake: "Line 1: Code written outside of a Java class",
        whyIsError: "In Java, all executable statements and methods must reside inside an enclosing class.",
        howToFix: "Enclose your code inside a class: public class Main { public static void main(String[] args) { ... } }.",
      };
    }
  }

  // ========================================
  // 3. C++ MODE (main.cpp)
  // ========================================
  if (language === 'cpp') {
    for (let i = 0; i < lines.length; i++) {
      const lineText = lines[i];
      const lineTrimmed = lineText.trim();
      const lineNum = i + 1;
```

```
302
303       // Check for Java/JS output in C++
304       if (lineTrimmed.includes('System.out.println(') || lineTrimmed.includes('console.log(')) {
305         return {
306           hasError: true,
307           type: 'syntax',
308           title: "error: unknown output identifier in C++",
309           line: lineNum,
310           codeSnippet: lineTrimmed,
311           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
312           whyIsError: "In C++, output is streamed via 'std::cout << ... << std::endl;'. Neither System.out.println n
or console.log exist in C++.",
313           howToFix: 'Use C++ iostream: std::cout << "..." << std::endl;',
314         };
315       }
316
317       // Check for missing semicolon on statements
318       if (
319         lineTrimmed.length > 0 &&
320         !lineTrimmed.startsWith('//') &&
321         !lineTrimmed.startsWith('#') &&
322         !lineTrimmed.endsWith('{') &&
323         !lineTrimmed.endsWith('}') &&
324         !lineTrimmed.endsWith(';') &&
325         !lineTrimmed.includes('int main(') &&
326         !lineTrimmed.includes('float ') &&
327         !lineTrimmed.includes('void ')
328       ) {
329         return {
330           hasError: true,
331           type: 'syntax',
332           title: "error: expected ';' before end of line",
333           line: lineNum,
334           codeSnippet: lineTrimmed,
335           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
```

```
          whyIsError: "In C++, statements must terminate with a semicolon ';'.",
          howToFix: `Append ';' to line ${lineNum}: "${lineTrimmed};".`,
        };
      }
    }
  }

  // ========================================
  // 4. C MODE (main.c)
  // ========================================
  if (language === 'c') {
    for (let i = 0; i < lines.length; i++) {
      const lineTrimmed = lines[i].trim();
      const lineNum = i + 1;

      if (lineTrimmed.startsWith('print(') || lineTrimmed.startsWith('cout <<')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "error: 'print' undeclared in C",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "In C, standard formatted output is performed with 'printf(...)'. Include <stdio.h> to use pri
ntf.",
          howToFix: `Use printf: printf("Hello, World!\\n");`,
        };
      }

      if (
        lineTrimmed.length > 0 &&
        !lineTrimmed.startsWith('//') &&
        !lineTrimmed.startsWith('#') &&
        !lineTrimmed.endsWith('{') &&
        !lineTrimmed.endsWith('}') &&
```

```
        !lineTrimmed.endsWith(';') &&
        !lineTrimmed.includes('int main(')
      ) {
        return {
          hasError: true,
          type: 'syntax',
          title: "error: expected ';' before end of line in C",
          line: lineNum,
          codeSnippet: lineTrimmed,
          whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
          whyIsError: "In C, every statement must end with a semicolon ';'.",
          howToFix: `Add ';' to line ${lineNum}: "${lineTrimmed};".`,
        };
      }
    }
  }

  // ========================================
  // 5. GO MODE (main.go)
  // ========================================
  if (language === 'go') {
    let hasPackage = false;
    for (let i = 0; i < lines.length; i++) {
      const lineTrimmed = lines[i].trim();
      const lineNum = i + 1;
      if (lineTrimmed.startsWith('package ')) hasPackage = true;

      if (lineTrimmed.startsWith('print(') || lineTrimmed.startsWith('console.log(')) {
        return {
          hasError: true,
          type: 'syntax',
          title: "undefined: print (use fmt.Println in Go)",
          line: lineNum,
          codeSnippet: lineTrimmed,
```

```
404           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
405           whyIsError: "In Go, formatted output uses the 'fmt' package: fmt.Println(...).",
406           howToFix: `Use: fmt.Println(...) and import "fmt".`,
407         };
408       }
409     }
410
411     if (!hasPackage && lines.some((l) => l.trim().length > 0)) {
412       return {
413         hasError: true,
414         type: 'syntax',
415         title: "expected 'package', found import or statement",
416         line: 1,
417         codeSnippet: lines[0].trim(),
418         whereIsMistake: "Line 1: Missing package clause",
419         whyIsError: "Every Go source file must start with a package declaration: 'package main'.",
420         howToFix: "Add 'package main' at line 1.",
421       };
422     }
423   }
424
425   // ========================================
426   // 6. RUST MODE (main.rs)
427   // ========================================
428   if (language === 'rust') {
429     for (let i = 0; i < lines.length; i++) {
430       const lineTrimmed = lines[i].trim();
431       const lineNum = i + 1;
432
433       if (lineTrimmed.startsWith('print(') || lineTrimmed.startsWith('console.log(')) {
434         return {
435           hasError: true,
436           type: 'syntax',
437           title: "cannot find function `print` in this scope (use println! macro)",
438           line: lineNum,
```

```
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
            whyIsError: "In Rust, printing to console uses the macro 'println!(...)'. Note the exclamation mark '!' indicating a macro.",
            howToFix: `Use: println!("Hello from Rust!");`,
          };
        }
      }
    }

    // ==========================================
    // 7. JAVASCRIPT & TYPESCRIPT MODE (sandbox.js / sandbox.ts)
    // ==========================================
    if (language === 'javascript' || language === 'typescript') {
      let openParens = 0;
      let openBrackets = 0;
      let openBraces = 0;

      for (let i = 0; i < lines.length; i++) {
        const lineText = lines[i];
        const lineTrimmed = lineText.trim();
        const lineNum = i + 1;

        for (const ch of lineText) {
          if (ch === '(') openParens++;
          if (ch === ')') openParens--;
          if (ch === '[') openBrackets++;
          if (ch === ']') openBrackets--;
          if (ch === '{') openBraces++;
          if (ch === '}') openBraces--;
        }

        // Check for unclosed print( in JavaScript
        if (lineTrimmed.startsWith('print(') && !lineTrimmed.includes(')')) {
          return {
```

```
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: missing ')' after argument list",
            line: lineNum,
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}" (unclosed parenthesis)`,
            whyIsError: "You called 'print(' with an unclosed parenthesis. In addition, note that JavaScript uses 'con
sole.log(...)' for output, while 'print(...)' is Python. If you want to write Python, click the  Python button in
the language toolbar!",
            howToFix: `Close with ')': "console.log(...)" or switch language toolbar to Python.`,
          };
        }

        // Check for Python def in JavaScript
        if (lineTrimmed.startsWith('def ') && lineTrimmed.endsWith(':')) {
          return {
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: Python 'def' function syntax in JavaScript",
            line: lineNum,
            codeSnippet: lineTrimmed,
            whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
            whyIsError: "You are using Python 'def func():' syntax in a JavaScript environment. JavaScript uses 'funct
ion name() { ... }'.",
            howToFix: "Change to: function name() { ... } or switch the language toolbar to Python.",
          };
        }

        // Check for Java import in JavaScript
        if (lineTrimmed.includes('import .') || lineTrimmed.startsWith('import java.')) {
          return {
            hasError: true,
            type: 'syntax',
            title: "SyntaxError: Java package import in JavaScript",
            line: lineNum,
            codeSnippet: lineTrimmed,
```

```
506           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
507           whyIsError: "You wrote Java package syntax ('import . java'). In JavaScript, standard library utilities (M
ath, JSON, Date, Array) are built-in globally without imports.",
508           howToFix: "Remove the Java import statement. Switch to Java in the language toolbar if writing Java.",
509         };
510       }
511
512       // Check for Java System.out in JavaScript
513       if (lineTrimmed.includes('System.out.println')) {
514         return {
515           hasError: true,
516           type: 'syntax',
517           title: "ReferenceError: System is not defined in JavaScript",
518           line: lineNum,
519           codeSnippet: lineTrimmed,
520           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
521           whyIsError: "'System.out.println' is Java syntax. JavaScript uses 'console.log(...)'.",
522           howToFix: "Replace 'System.out.println(...)' with 'console.log(...)'.",
523         };
524       }
525
526       // Check for C++ #include in JavaScript
527       if (lineTrimmed.startsWith('#include')) {
528         return {
529           hasError: true,
530           type: 'syntax',
531           title: "SyntaxError: C++ preprocessor directive in JavaScript",
532           line: lineNum,
533           codeSnippet: lineTrimmed,
534           whereIsMistake: `Line ${lineNum}: "${lineTrimmed}"`,
535           whyIsError: "'#include' is C++ syntax. JavaScript does not have preprocessor directives.",
536           howToFix: "Remove '#include' or switch the language toolbar to C++.",
537         };
538       }
539     }
```

```
540
541     // Check for delimiter mismatches before Function() evaluation
542     if (openParens > 0) {
543       return {
544         hasError: true,
545         type: 'syntax',
546         title: "SyntaxError: missing ')' after argument list or expression",
547         line: lines.length,
548         codeSnippet: lines[lines.length - 1].trim(),
549         whereIsMistake: `Line ${lines.length}: Unclosed parenthesis '('`,
550         whyIsError: "An opening '(' was never closed with a matching ')'. JavaScript cannot parse an expression with unbalanced parentheses.",
551         howToFix: "Add ')' to close the open expression.",
552       };
553     }
554
555     if (openBrackets > 0) {
556       return {
557         hasError: true,
558         type: 'syntax',
559         title: "SyntaxError: missing ']' after array elements",
560         line: lines.length,
561         codeSnippet: lines[lines.length - 1].trim(),
562         whereIsMistake: `Line ${lines.length}: Unclosed bracket '['`,
563         whyIsError: "An opening square bracket '[' was never closed with a matching ']'.",
564         howToFix: "Add ']' to close the array or index access.",
565       };
566     }
567
568     if (openBraces > 0) {
569       return {
570         hasError: true,
571         type: 'syntax',
572         title: "SyntaxError: missing '}' after function or block body",
573         line: lines.length,
```

```
574         codeSnippet: lines[lines.length - 1].trim(),
575         whereIsMistake: `Line ${lines.length}: Unclosed curly brace '{'`,
576         whyIsError: "An opening curly brace '{' was never closed with a matching '}'.",
577         howToFix: "Add '}' to close the block scope.",
578       };
579     }
580
581     // Evaluate JavaScript syntax safely
582     try {
583       const sanitized = code
584         .replace(/:\s*(string|number|boolean|any|void|JwtPayload|null)\b/g, '')
585         .replace(/interface\s+\w+\s*\{[\s\S]*?\}/g, '');
586       new Function(sanitized);
587     } catch (err: unknown) {
588       const syntaxMsg = err instanceof Error ? err.message : String(err);
589       let errLine = 1;
590
591       for (let i = 0; i < lines.length; i++) {
592         const l = lines[i].trim();
593         if (l.startsWith('print(') || l.includes('import .') || l.includes('const =') || l.includes('let =')) {
594           errLine = i + 1;
595           break;
596         }
597       }
598
599       const failingSnippet = lines[errLine - 1] ? lines[errLine - 1].trim() : code.slice(0, 40);
600
601       // Human-friendly translation of V8's cryptic messages
602       let readableWhy = `The JavaScript engine's parser cannot understand this grammar: "${syntaxMsg}".`;
603       let readableFix = `Check line ${errLine} for mismatched brackets, unclosed quotes, or foreign language syntax.
`;
604
605       if (syntaxMsg.includes("Unexpected token '}'")) {
606         readableWhy = `The JavaScript engine reached the end of the script while waiting for an unclosed delimiter
(like '(', '[', or '{') to be closed.`;
```

```
          readableFix = `Ensure all opened brackets, parentheses, and braces are properly closed.`;
        }

        return {
          hasError: true,
          type: 'syntax',
          title: `SyntaxError: ${syntaxMsg}`,
          line: errLine,
          codeSnippet: failingSnippet,
          whereIsMistake: `Line ${errLine}: "${failingSnippet}"`,
          whyIsError: readableWhy,
          howToFix: readableFix,
        };
      }
    }

    // 8. Code is clean!
    return {
      hasError: false,
      type: 'clean',
      title: `Code Syntax Clean & Valid (${language.toUpperCase()})`,
      whereIsMistake: 'No syntax mistakes detected.',
      whyIsError: `Your ${language.toUpperCase()} code grammar is valid and ready to execute.`,
      howToFix: 'Click "Run Code" (or press Ctrl+Enter) to evaluate output in the sandbox.',
    };
  }

  /**
   * Intelligent Error Concept Explainer & Diagnostic Engine
   * Analyzes the ACTUAL caught error dynamically across all programming languages:
   * Python, Java, C++, C, Go, Rust, JavaScript, TypeScript
   */
  export function explainError(
    rawError: string | Error,
    code: string,
```

```
  language: SupportedLanguage = 'javascript',
  challengeId?: string
): ErrorExplanation {
  const errorText = typeof rawError === 'string' ? rawError : rawError.message || String(rawError);
  const stack = typeof rawError === 'object' && rawError.stack ? rawError.stack : '';

  // Extract failing line number
  let failingLine: number | undefined;
  const lineMatch =
    stack.match(/:(\d+):(\d+)/) ||
    errorText.match(/line\s+(\d+)/i) ||
    errorText.match(/:(\d+)\)/);
  if (lineMatch && lineMatch[1]) {
    failingLine = parseInt(lineMatch[1], 10);
  }

  const codeLines = code.split('\n');
  const codeContext =
    failingLine && codeLines[failingLine - 1]
      ? codeLines[failingLine - 1].trim()
      : codeLines[0] || '';

  // ========================================
  // 1. JAVA: NullPointerException
  // ========================================
  if (
    language === 'java' &&
    (errorText.includes('NullPointerException') || code.includes('formatUserEmail'))
  ) {
    return {
      errorName: 'java.lang.NullPointerException: Dereferencing Null Object Reference',
      errorMessage: errorText,
      failingLine: failingLine || 19,
      codeContext: codeContext || 'return user.getEmail().toLowerCase();',
```

```
      whyThisHappened:
        `In Java, attempting to invoke an instance method (such as .getEmail() or .toLowerCase()) on a reference variable that evaluates to 'null' causes the JVM to throw java.lang.NullPointerException (NPE). Primitive variables hold values, but object variables hold memory references; when a reference is null, there is no heap memory to dereference.`,
      conceptTitle: 'JVM Heap References, Null Pointers & The Billion-Dollar Mistake',
      conceptDeepDive:
        'Tony Hoare introduced null in 1965 to denote missing values, later calling it his "billion-dollar mistake". In the JVM, invoking an instance method uses the `invokevirtual` bytecode instruction. The JVM checks if the object reference on the operand stack is null; if null, it immediately throws NullPointerException. Modern Java best practices use Optional<T>, explicit null checks, or Objects.requireNonNull().',
      analogy: 'Dialing a phone number that is disconnected and expecting a person to answer.',
      mentalModelDiagram: `
  user Reference ──► [ null ]  (Points to 0x0 / nothing)
                          │
                          ▼
              invokevirtual .getEmail()  💥 JVM CRASH: java.lang.NullPointerException!
`,
      suggestedFixBefore: codeContext || 'return user.getEmail().toLowerCase();',
      suggestedFixAfter: `if (user == null || user.getEmail() == null) {\n    return "no-reply@system.local";\n}\nreturn user.getEmail().toLowerCase();`,
      fixExplanation: 'Add a defensive null guard check: verify both user != null and user.getEmail() != null before calling methods.',
      bestPractices: [
        'Always guard against null before calling methods on objects received from parameters or external services.',
        'Use java.util.Optional<T> for methods that may legitimately return no value.',
        'Never return null collections; return Collections.emptyList() or Collections.emptyMap().',
      ],
      relatedLesson: {
        courseId: 'java-oop-errors',
        lessonId: 'jl1-npe',
        title: '1.1 The Billion-Dollar Mistake: Java NullPointerException',
      },
    };
  }

  // ========================================
  // 2. JAVA: ArrayIndexOutOfBoundsException
  // ========================================
```

```
  if (
    language === 'java' &&
    (errorText.includes('ArrayIndexOutOfBoundsException') || code.includes('calculateTotal'))
  ) {
    return {
      errorName: 'java.lang.ArrayIndexOutOfBoundsException: Array Index Out of Range',
      errorMessage: errorText,
      failingLine: failingLine || 7,
      codeContext: codeContext || 'for (int i = 0; i <= scores.length; i++)',
      whyThisHappened:
        `In Java, arrays are 0-indexed. An array of length N has valid indices from 0 to N-1. The loop condition 'i <= scores.length' allows 'i' to reach N (equal to length), attempting to access memory outside the array boundary.`,
      conceptTitle: 'Java Array Memory Contiguity & Strict Bounds Verification',
      conceptDeepDive:
        'Every Java array is an object stored in contiguous memory on the JVM heap with an immutable .length field. At bytecode level, array indexing uses instructions like `iaload`. The JVM performs hardware bounds checking on every access: if index < 0 or index >= length, it terminates with ArrayIndexOutOfBoundsException to prevent buffer overflow vulnerabilities.',
      analogy: 'Booking seats in an airplane with 5 rows (numbered 0 to 4) and trying to sit in row 5.',
      suggestedFixBefore: 'for (int i = 0; i <= scores.length; i++)',
      suggestedFixAfter: 'for (int i = 0; i < scores.length; i++) {\n    total += scores[i];\n}',
      fixExplanation: 'Change "<= scores.length" to "< scores.length" or use an enhanced for-each loop: "for (int score : scores)".',
      bestPractices: [
        'Use strict inequality "< array.length" for index-based loops.',
        'Prefer enhanced for-each loops: "for (Type item : collection)" whenever manual indices are not required.',
      ],
      relatedLesson: {
        courseId: 'java-oop-errors',
        lessonId: 'jl1-npe',
        title: '1.1 The Billion-Dollar Mistake: Java NullPointerException',
      },
    };
  }

  // ========================================
  // 3. C++: std::out_of_range
```

```
  // ========================================
  if (
    language === 'cpp' &&
    (errorText.includes('out_of_range') || code.includes('readings.at'))
  ) {
    return {
      errorName: 'std::out_of_range: vector::_M_range_check: index >= this->size()',
      errorMessage: errorText,
      failingLine: failingLine || 8,
      codeContext: codeContext || 'return readings.at(readings.size());',
      whyThisHappened:
        `In C++, std::vector::size() returns the number of elements in the vector. Valid indices range from 0 to size() - 1. Accessing index readings.size() attempts to read past the end of the vector. Calling .at() detects this and throws std::out_of_range.`,
      conceptTitle: 'C++ STL Containers, Bounds Checking & Exception Safety',
      conceptDeepDive:
        'In C++, std::vector provides two element access mechanisms: operator[] and .at(). operator[] does NOT perform bounds checking in release builds for raw performance, leading to undefined behavior or segmentation faults. In contrast, .at() checks the index against size() and throws std::out_of_range, enabling safe error recovery.',
      analogy: 'Opening a filing cabinet with 3 folders and demanding to read folder #4.',
      suggestedFixBefore: 'return readings.at(readings.size());',
      suggestedFixAfter: 'if (readings.empty()) return 0.0f;\nreturn readings.back(); // or readings.at(readings.size() - 1)',
      fixExplanation: 'Use readings.back() to safely retrieve the last element, and guard against empty vectors with readings.empty().',
      bestPractices: [
        'Use .at() when indices come from untrusted user input or dynamic math.',
        'Use .back() and .front() for container boundary elements.',
        'Always check .empty() before querying elements.',
      ],
      relatedLesson: {
        courseId: 'cpp-memory-safety',
        lessonId: 'c11-bounds',
        title: '1.1 Bounds Safety: operator[] vs .at() in std::vector',
      },
    };
  }

```

```
  // ========================================
  // 4. C++: Segmentation Fault
  // ========================================
  if (
    language === 'cpp' &&
    (errorText.includes('Segmentation fault') || code.includes('createConfigBanner'))
  ) {
    return {
      errorName: 'Segmentation fault (SIGSEGV): Dereferencing Dangling Stack Pointer',
      errorMessage: errorText,
      failingLine: failingLine || 17,
      codeContext: codeContext || 'std::cout << "Banner: " << *ptr << std::endl;',
      whyThisHappened:
        `The function returned the memory address (&banner) of a local stack variable. When createConfigBanner() fin
ished execution, its stack frame was reclaimed and overwritten. Dereferencing *ptr accesses invalid memory, causing
an operating system memory protection fault (SIGSEGV).`,
      conceptTitle: 'Stack Frame Lifetimes, Dangling Pointers & RAII',
      conceptDeepDive:
        'In C and C++, local variables reside in the function stack frame. When the function returns, the stack poin
ter (RSP/ESP) unwinds, rendering all stack addresses invalid. Returning a pointer to local stack memory creates a "d
angling pointer". Dereferencing it invokes Undefined Behavior (UB), often manifesting as an immediate SIGSEGV signal
from the OS kernel MMU.',
      analogy: 'Writing down an address of a hotel room you just checked out of, and expecting your luggage to still
be there next year.',
      suggestedFixBefore: 'std::string* createConfigBanner() { std::string banner = "..."; return &banner; }',
      suggestedFixAfter: 'std::string createConfigBanner() {\n    std::string banner = "Production Server v2.4 (Acti
ve)";\n    return banner; // Safe: Return Value Optimization (RVO)\n}',
      fixExplanation: 'Return objects by value in modern C++. Return Value Optimization (RVO) avoids copying complet
ely and prevents dangling memory.',
      bestPractices: [
        'Never return pointers or references to local stack variables.',
        'Prefer returning by value in C++11 and newer to leverage move semantics and copy elision.',
        'If dynamic allocation is needed, use std::unique_ptr or std::shared_ptr.',
      ],
      relatedLesson: {
        courseId: 'cpp-memory-safety',
        lessonId: 'c11-bounds',
        title: '1.1 Bounds Safety: operator[] vs .at() in std::vector',
      },
```

```
    };
  }

  // ========================================
  // 5. C: Buffer Overflow / Stack Smashing
  // ========================================
  if (
    language === 'c' &&
    (errorText.includes('buffer overflow') || errorText.includes('stack smashing') || code.includes('strcpy(buffe
r'))
  ) {
    return {
      errorName: 'Stack Smashing Detected: Buffer Overflow in Unsafe Memory Copy',
      errorMessage: errorText,
      failingLine: failingLine || 9,
      codeContext: codeContext || 'strcpy(buffer, inputUsername);',
      whyThisHappened:
        `The code uses 'strcpy', which copies bytes until a null terminator '\\0' is found without checking the dest
ination buffer capacity. A 28-byte string copied into a 16-byte buffer overwrites adjacent memory on the stack, corr
upting the return pointer and triggering stack canary protection.`,
      conceptTitle: 'Buffer Overflows, Stack Canaries & Memory Safety in C',
      conceptDeepDive:
        'Buffer overflows are among the oldest and most dangerous security vulnerabilities (CWE-120). Compilers like
GCC insert a "stack canary" value between local buffers and the saved return pointer. Before returning, the function
checks if the canary was modified. If modified, GCC terminates the program with "*** stack smashing detected ***" to
prevent arbitrary code execution.',
      analogy: 'Pouring a gallon of water into a pint glass: the excess liquid spills over and ruins the electronics
beneath.',
      suggestedFixBefore: 'char buffer[16];\nstrcpy(buffer, inputUsername);',
      suggestedFixAfter: 'char buffer[16];\nsnprintf(buffer, sizeof(buffer), "%s", inputUsername);',
      fixExplanation: 'Use bounded functions like snprintf(buffer, sizeof(buffer), "%s", input) or strncpy to enforc
e capacity limits.',
      bestPractices: [
        'Never use unbounded memory functions (strcpy, gets, sprintf).',
        'Use size-bounded APIs: snprintf, strncpy_s, or fgets.',
        'Enable compiler security flags (-fstack-protector-strong, -D_FORTIFY_SOURCE=2).',
      ],
      relatedLesson: {
        courseId: 'cpp-memory-safety',
```

```
        lessonId: 'cl1-bounds',
        title: '1.1 Bounds Safety: operator[] vs .at() in std::vector',
      },
    };
  }

  // ==========================================
  // 6. PYTHON: KeyError
  // ==========================================
  if (
    language === 'python' &&
    (errorText.includes('KeyError') || code.includes('user_session['))
  ) {
    return {
      errorName: "KeyError: 'role' not found in dictionary",
      errorMessage: errorText,
      failingLine: failingLine || 4,
      codeContext: codeContext || "user_role = user_session['role']",
      whyThisHappened:
        `In Python, indexing a dictionary with dict[key] requires the key to exist in the hash table. If the key is absent, Python raises a KeyError exception instead of returning None.`,
      conceptTitle: 'Python Hash Tables, Dictionary Lookups & Safe Key Retrieval',
      conceptDeepDive:
        'Python dictionaries (dict) are implemented as dense hash tables. When key lookup fails, the internal PyDict_GetItemWithError raises KeyError. The Pythonic approach to optional dictionary keys is dict.get(key, default) or using collections.defaultdict.',
      analogy: 'Searching for a book in a library card catalog by a title that was never cataloged.',
      suggestedFixBefore: "user_role = user_session['role']",
      suggestedFixAfter: "user_role = user_session.get('role', 'guest')",
      fixExplanation: "Use dict.get('role', default_value) to safely retrieve values with a fallback without raising KeyError.",
      bestPractices: [
        "Use dict.get(key, default) for optional keys.",
        "Use 'if key in dictionary:' to test for key membership before access.",
      ],
      relatedLesson: {
        courseId: 'python-error-mastery',
```

```
        lessonId: 'l9-index-error',
        title: '4.1 Sequence Boundary Traps: Python IndexError Demystified',
      },
    };
  }

  // ========================================
  // 7. PYTHON: ZeroDivisionError
  // ========================================
  if (
    language === 'python' &&
    (errorText.includes('ZeroDivisionError') || code.includes('calculate_average_rating'))
  ) {
    return {
      errorName: 'ZeroDivisionError: division by zero in average calculation',
      errorMessage: errorText,
      failingLine: failingLine || 4,
      codeContext: codeContext || 'return sum(ratings) / len(ratings)',
      whyThisHappened:
        `When the ratings list is empty, len(ratings) evaluates to 0. In mathematics and Python, dividing any number by zero is undefined and raises ZeroDivisionError.`,
      conceptTitle: 'Arithmetic Exceptions & Defensive Numeric Guarding',
      conceptDeepDive:
        'In Python, floating point and integer division check the denominator value in the C layer. If denominator == 0, PyErr_SetString(PyExc_ZeroDivisionError, "division by zero") is called. Mathematical algorithms must always guard against zero denominators before executing division.',
      analogy: 'Trying to divide a pizza among 0 people.',
      suggestedFixBefore: 'return sum(ratings) / len(ratings)',
      suggestedFixAfter: 'if not ratings:\n    return 0.0\nreturn sum(ratings) / len(ratings)',
      fixExplanation: 'Add a guard: check if the list is empty (if not ratings: return 0.0) before performing the division.',
      bestPractices: [
        'Always check that the divisor is non-zero before dividing.',
        'Use fallback values (0.0 or None) for empty dataset averages.',
      ],
      relatedLesson: {
```

```
        courseId: 'python-error-mastery',
        lessonId: 'l9-index-error',
        title: '4.1 Sequence Boundary Traps: Python IndexError Demystified',
      },
    };
  }

  // ========================================
  // 8. PYTHON: IndexError
  // ========================================
  if (
    language === 'python' &&
    (errorText.includes('IndexError') || code.includes('summarize_logs'))
  ) {
    return {
      errorName: 'IndexError: list index out of range',
      errorMessage: errorText,
      failingLine: failingLine || 6,
      codeContext: codeContext || 'entry = log_entries[i]',
      whyThisHappened:
        `Python uses 0-based indexing. For a list of 3 items, valid indices are 0, 1, and 2. The loop 'range(len(log_entries) + 1)' tried to access index 3, which is out of range.`,
      conceptTitle: 'Python Sequence Indexing & enumerate() Idiom',
      conceptDeepDive:
        'Python lists are dynamic arrays. Accessing list[i] where i >= len(list) raises IndexError. Pythonic code avoids manual index arithmetic by using direct iteration: "for entry in log_entries:" or "for i, entry in enumerate(log_entries):".',
      analogy: 'Calling seat 4 on a 3-seat bench.',
      suggestedFixBefore: 'for i in range(len(log_entries) + 1):\n    entry = log_entries[i]',
      suggestedFixAfter: 'for i, entry in enumerate(log_entries):\n    summary.append(f"Log #{i}: {entry}")',
      fixExplanation: 'Use enumerate(log_entries) to iterate both index and item safely without manual boundary math.',
      bestPractices: [
        'Prefer "for item in list:" or "for i, item in enumerate(list):" over manual range(len()) loops.',
      ],
      relatedLesson: {
        courseId: 'python-error-mastery',
```

```
931         lessonId: 'l9-index-error',
932         title: '4.1 Sequence Boundary Traps: Python IndexError Demystified',
933       },
934     };
935   }
936
937   // ========================================
938   // 9. GO: Nil Pointer Panic
939   // ========================================
940   if (
941     language === 'go' &&
942     (errorText.includes('nil pointer dereference') || code.includes('u.Account.Balance'))
943   ) {
944     return {
945       errorName: 'panic: runtime error: invalid memory address or nil pointer dereference',
946       errorMessage: errorText,
947       failingLine: failingLine || 15,
948       codeContext: codeContext || 'return u.Account.Balance',
949       whyThisHappened:
950         `In Go, pointers default to 'nil' when uninitialized. Attempting to access fields on a nil pointer (u.Accoun
    t == nil) causes the Go runtime to trigger a panic: invalid memory address or nil pointer dereference.`,
951       conceptTitle: 'Go Pointers, Nil Values & Defensive Guard Idiom',
952       conceptDeepDive:
953         'Unlike languages with exception hierarchies, Go uses explicit error values and panics for catastrophic memo
    ry violations. Dereferencing a nil pointer causes an immediate runtime panic. Robust Go code always validates pointe
    rs: "if u == nil || u.Account == nil { return 0.0 }".',
954       analogy: 'Unwrapping a gift box only to find it is completely empty, and trying to wear the non-existent watch
    inside.',
955       suggestedFixBefore: 'return u.Account.Balance',
956       suggestedFixAfter: 'if u == nil || u.Account == nil {\n    return 0.0\n}\nreturn u.Account.Balance',
957       fixExplanation: 'Add defensive nil pointer checks before reading fields from pointers.',
958       bestPractices: [
959         'Always check for nil before accessing pointer fields.',
960         'Consider returning (value, error) pairs for methods that can fail.',
961       ],
962       relatedLesson: {
```

```
        courseId: 'js-ts-debugging',
        lessonId: 'l4-optional-chaining',
        title: '2.2 Defensive Coding: Optional Chaining (?.) & Nullish Coalescing (??)',
      },
    };
  }

  // ========================================
  // 10. RUST: Borrow of Moved Value (E0382)
  // ========================================
  if (
    language === 'rust' &&
    (errorText.includes('borrow of moved value') || errorText.includes('E0382') || code.includes('log_token'))
  ) {
    return {
      errorName: "error[E0382]: borrow of moved value: 'token'",
      errorMessage: errorText,
      failingLine: failingLine || 14,
      codeContext: codeContext || 'println!("Database querying with token: {}", token);',
      whyThisHappened:
        `In Rust, passing 'token' (a heap-allocated String) by value to 'log_token(token)' moves ownership of the st
ring to the function. Once moved, 'main' no longer owns the memory and cannot access 'token' again.`,
      conceptTitle: 'Rust Ownership, Move Semantics & Borrowing References',
      conceptDeepDive:
        'Rust ensures memory safety without a garbage collector through its affine type system and Borrow Checker. E
very value has a single owner. Passing non-Copy types by value transfers ownership ("move"). To allow read-only acce
ss without relinquishing ownership, borrow by reference: &str or &T.',
      analogy: 'Giving your car keys away permanently instead of lending the spare key for the afternoon.',
      suggestedFixBefore: 'fn log_token(token: String) { ... }\nlog_token(token);',
      suggestedFixAfter: 'fn log_token(token: &str) { ... }\nlog_token(&token);',
      fixExplanation: 'Change the function parameter to a string slice &str and pass a reference &token to borrow wi
thout moving.',
      bestPractices: [
        'Prefer borrowing (&str, &[T]) for read-only function arguments.',
        'Keep ownership in the calling scope unless transfer is required.',
      ],
      relatedLesson: {
```

```
        courseId: 'cpp-memory-safety',
        lessonId: 'c11-bounds',
        title: '1.1 Bounds Safety: operator[] vs .at() in std::vector',
      },
    };
  }

  // ========================================
  // 11. Cross-Language print() confusion
  // ========================================
  if (code.includes('print(') && (language === 'javascript' || language === 'typescript')) {
    return {
      errorName: "SyntaxError: unclosed parenthesis or Python syntax in JS",
      errorMessage: errorText,
      failingLine: failingLine || 1,
      codeContext: codeContext || 'print(',
      whyThisHappened:
        `On line ${failingLine || 1}, '${codeContext}' is missing a closing ')' or uses Python 'print(...)' in a JavaScript environment.`,
      conceptTitle: 'Standard Output Across Languages: JS console.log vs Python print',
      conceptDeepDive:
        'In Python, standard output is printed using the built-in print() function. In JavaScript, standard output is written to the developer console using console.log(). In addition, any function call with an unclosed parenthesis triggers an immediate syntax parse error.',
      analogy: 'Dialing the first 5 digits of a phone number and waiting for it to ring without pressing the remaining digits.',
      suggestedFixBefore: codeContext || 'print(',
      suggestedFixAfter: 'console.log("Hello from JavaScript!"); // Or click Python in the language toolbar!',
      fixExplanation: 'Close the parenthesis with ")", or replace print() with console.log(), or click the Python button in the language toolbar.',
      bestPractices: [
        'In JavaScript/TypeScript, use console.log(...) for output.',
        'In Python, use print(...) with closed parentheses.',
        'In Java, use System.out.println(...).',
        'In C++, use std::cout << ... << std::endl;.',
      ],
    };
```

```
  }

  // ========================================
  // 12. JavaScript TypeError: Cannot read properties of undefined
  // ========================================
  if (errorText.includes('Cannot read propert') || (errorText.includes('TypeError') && code.includes('.'))) {
    const propMatch = errorText.match(/reading '([^']+)'/);
    const propName = propMatch ? propMatch[1] : 'property';

    return {
      errorName: `TypeError: Cannot read properties of undefined (reading '${propName}')`,
      errorMessage: errorText,
      failingLine: failingLine || 4,
      codeContext: codeContext || 'const tier = user.subscription.tier;',
      whyThisHappened:
        `On line ${failingLine || 4}, the code attempts to read '.${propName}' on an object reference that evaluated to undefined or null. Because undefined is a primitive without property descriptors or prototype slots, the JavaScript V8 engine throws a fatal TypeError.`,
      conceptTitle: 'V8 Object Property Resolution & Prototype Slots',
      conceptDeepDive:
        'In JavaScript, object property access invokes the internal [[Get]] specification. If the left-hand expression resolves to undefined or null, [[Get]] cannot be called because primitives have no prototype table. V8 stops execution immediately.',
      analogy: 'Attempting to open a drawer inside a desk that does not exist in the room.',
      mentalModelDiagram: `
  user
    └── subscription: undefined
            └── .${propName}   💥 CRASH: Cannot invoke [[Get]] on undefined!

  CRASH-PROOF OPTIONAL CHAINING:
  user?.subscription?.${propName}
    └── Short-circuits safely to undefined without throwing!
`,
      suggestedFixBefore: codeContext || 'const tier = user.subscription.tier;',
      suggestedFixAfter: `const tier = user?.subscription?.${propName} ?? 'free';`,
      fixExplanation:
```

```
        'Use the optional chaining operator (?.) to short-circuit if subscription is undefined, and the nullish coal
escing operator (??) to supply a default fallback value.',
      bestPractices: [
        'Always guard nested property access with optional chaining (?.).',
        'Use nullish coalescing (??) instead of logical OR (||) to prevent falsy bugs with 0 or empty strings.',
      ],
      relatedLesson: {
        courseId: 'js-ts-debugging',
        lessonId: 'l4-optional-chaining',
        title: '2.2 Defensive Coding: Optional Chaining (?.) & Nullish Coalescing (??)',
      },
    };
  }

  // ========================================
  // 13. Generic Fallback
  // ========================================
  return {
    errorName: `${language.toUpperCase()} Exception or Syntax Mistake`,
    errorMessage: errorText,
    failingLine,
    codeContext,
    whyThisHappened:
      `An exception occurred in your ${language.toUpperCase()} code: "${errorText}". Check line ${failingLine || 1}
for syntax, unclosed delimiters, or missing keywords.`,
    conceptTitle: `${language.toUpperCase()} Language Grammar & Runtime Execution`,
    conceptDeepDive:
      `Programming languages require strictly formatted grammar. In ${language.toUpperCase()}, all keywords, delimit
ers, and variable references must match the language specification.`,
    analogy: 'A puzzle piece that does not fit into the board.',
    suggestedFixBefore: codeContext || 'yourCode()',
    suggestedFixAfter: '// Verify syntax and variable names',
    fixExplanation: 'Review the line of code and adjust delimiters, spelling, or variable declarations.',
    bestPractices: [
      'Use the Language Selector to match your desired programming language.',
      'Check for matching parentheses and quotation marks.',
```

```
      ],
    };
}
```

File [7/30]: src/components/views/CodePracticeView.tsx

TSX SOURCE

Description: *Multi-Language Practice Lab (Python, Java, C++, C, JS, TS, Go, Rust)*

```tsx
'use client';

import React, { useState, useEffect } from 'react';
import {
  Play,
  RotateCcw,
  Sparkles,
  CheckCircle2,
  AlertTriangle,
  Lightbulb,
  Copy,
  Check,
  ChevronRight,
  BookOpen,
  Terminal as TerminalIcon,
  HelpCircle,
  Zap,
  ArrowRight,
  Send,
  MessageSquare,
  Wrench,
  ShieldCheck,
  Code2,
  ExternalLink,
  ChevronDown,
  Layers,
  FileCode,
  Flame,
  Sun,
  Moon,
  Laptop,
```

```
} from 'lucide-react';
import confetti from 'canvas-confetti';
import { PRACTICE_CHALLENGES } from '@/lib/courses-data';
import { explainError, analyzeCodeLive } from '@/lib/error-explainer';
import { PracticeChallenge, ErrorExplanation, NavTab, SupportedLanguage, LiveDiagnostic } from '@/lib/types';

interface CodePracticeViewProps {
  onNavigate: (tab: NavTab) => void;
  initialChallengeId?: string;
  onOpenLesson?: (courseId: string, lessonId: string) => void;
}

type ExplainerTab = 'why' | 'concept' | 'solution' | 'ask_ai';

const LANGUAGE_OPTIONS: {
  id: SupportedLanguage;
  label: string;
  icon: string;
  filename: string;
  compiler: string;
  description: string;
}[] = [
  { id: 'python', label: 'Python 3', icon: '🐍', filename: 'sandbox.py', compiler: 'CPython 3.12 (AST Engine)', description: 'Python syntax, zero-division, KeyError, and indentation checks' },
  { id: 'java', label: 'Java', icon: '☕', filename: 'Main.java', compiler: 'OpenJDK 21 (javac)', description: 'NullPointer, bounds checks, static context, and strict typing' },
  { id: 'cpp', label: 'C++', icon: '⚙️', filename: 'main.cpp', compiler: 'GCC 13 (g++ -std=c++20)', description: 'Pointers, out_of_range, memory leaks, and segmentation fault analysis' },
  { id: 'c', label: 'C', icon: '©️', filename: 'main.c', compiler: 'Clang 17 (gcc -std=c17)', description: 'Buffer overflow, stack smashing, pointers, and memory safety' },
  { id: 'javascript', label: 'JavaScript', icon: '🟨', filename: 'sandbox.js', compiler: 'Node.js 20 (V8 Engine)', description: 'V8 runtime, TypeError, TDZ, and async error diagnostics' },
  { id: 'typescript', label: 'TypeScript', icon: '🔷', filename: 'sandbox.ts', compiler: 'TypeScript 5.4 (Strict)', description: 'Type assertions, strict null checks, and security flaw diagnosis' },
  { id: 'go', label: 'Go', icon: '🐹', filename: 'main.go', compiler: 'Go 1.22 (gc)', description: 'Goroutine panic, nil pointer dereference, and package rules' },
  { id: 'rust', label: 'Rust', icon: '🦀', filename: 'main.rs', compiler: 'Rust 1.76 (rustc 2021)', description: 'Borrow checker, ownership move errors, and lifetimes' },
];
```

```
63
64  const LANGUAGE_TEMPLATES: Record<SupportedLanguage, string> = {
65    python: `# Python 3 Practice Sandbox
66  def greet(name):
67      return f"Hello, {name}!"
68
69  print(greet("Developer"))
70  `,
71    java: `// Java Practice Sandbox
72  public class Main {
73      public static void main(String[] args) {
74          String greeting = "Hello from Java Sandbox!";
75          System.out.println(greeting);
76      }
77  }
78  `,
79    cpp: `// C++ Practice Sandbox
80  #include <iostream>
81
82  int main() {
83      std::cout << "Hello from C++ Sandbox!" << std::endl;
84      return 0;
85  }
86  `,
87    c: `// C Practice Sandbox
88  #include <stdio.h>
89
90  int main() {
91      printf("Hello from C Sandbox!\\n");
92      return 0;
93  }
94  `,
95    javascript: `// JavaScript Practice Sandbox
96  function greet(name) {
97    return "Hello, " + name + "!";
```

```
}

console.log(greet("Developer"));
`,
  typescript: `// TypeScript Practice Sandbox
interface User {
  id: number;
  name: string;
}

function getUserInfo(user: User): string {
  return \`User #\${user.id}: \${user.name}\`;
}

console.log(getUserInfo({ id: 101, name: "Alice" }));
`,
  go: `// Go Practice Sandbox
package main

import "fmt"

func main() {
    fmt.Println("Hello from Go Sandbox!")
}
`,
  rust: `// Rust Practice Sandbox
fn main() {
    println!("Hello from Rust Sandbox!");
}
`,
};

export const CodePracticeView: React.FC<CodePracticeViewProps> = ({
  onNavigate,
```

```
  initialChallengeId,
  onOpenLesson,
}) => {
  // Challenge Selection
  const [selectedChallengeId, setSelectedChallengeId] = useState<string>(
    initialChallengeId || 'type-error-null'
  );

  const activeChallenge =
    PRACTICE_CHALLENGES.find((c) => c.id === selectedChallengeId) || PRACTICE_CHALLENGES[0];

  // Code Editor State
  const [code, setCode] = useState<string>(activeChallenge.initialCode);
  const [language, setLanguage] = useState<SupportedLanguage>(activeChallenge.language);
  const [copied, setCopied] = useState<boolean>(false);
  const [fontSize, setFontSize] = useState<number>(13);
  const [editorTheme, setEditorTheme] = useState<'dark' | 'light'>('dark');

  // Execution & Output State
  const [isRunning, setIsRunning] = useState<boolean>(false);
  const [terminalOutput, setTerminalOutput] = useState<string[]>([]);
  const [executionTimeMs, setExecutionTimeMs] = useState<number | null>(null);
  const [executionStatus, setExecutionStatus] = useState<'idle' | 'success' | 'error'>('idle');

  // Test Results State
  const [testResults, setTestResults] = useState<{ name: string; passed: boolean; message: string }[]>([]);

  // AI Error Explainer State
  const [currentErrorExplanation, setCurrentErrorExplanation] = useState<ErrorExplanation | null>(null);
  const [activeExplainerTab, setActiveExplainerTab] = useState<ExplainerTab>('why');
  const [revealedHints, setRevealedHints] = useState<number>(0);

  // Real-Time Live As-You-Type Diagnostic Sentinel State
  const [liveDiagnostic, setLiveDiagnostic] = useState<LiveDiagnostic | null>(null);

```

```
  // Ask AI Tutor Chat State
  const [tutorChatMessages, setTutorChatMessages] = useState<{ role: 'user' | 'tutor'; text: string }[]>(
    []
  );
  const [tutorInput, setTutorInput] = useState<string>('');
  const [isTutorThinking, setIsTutorThinking] = useState<boolean>(false);

  // Real-Time As-You-Type Code Analysis Effect
  useEffect(() => {
    const diag = analyzeCodeLive(code, language, activeChallenge.id);
    setLiveDiagnostic(diag);

    // If an error is detected while typing, update the AI Error Concept Explainer in real-time!
    if (diag.hasError) {
      const explanation = explainError(diag.title, code, language, activeChallenge.id);
      setCurrentErrorExplanation(explanation);
    } else {
      if (executionStatus !== 'error') {
        setCurrentErrorExplanation(null);
      }
    }
  }, [code, language, activeChallenge.id, executionStatus]);

  // Sync when challenge changes
  useEffect(() => {
    setCode(activeChallenge.initialCode);
    setLanguage(activeChallenge.language);
    setTerminalOutput([
      `// Welcome to RepoPilot Code Lab: ${activeChallenge.title}`,
      `// Language: ${activeChallenge.language.toUpperCase()} · Compiler: ${LANGUAGE_OPTIONS.find((l) => l.id === activeChallenge.language)?.compiler}`,
      `// Real-time as-you-type error sentinel & AI concept tutor ACTIVE.`,
      `// Click "Run Code" (or press Ctrl+Enter) to evaluate execution.`,
    ]);
    setExecutionStatus('idle');
```

```
    setCurrentErrorExplanation(null);
    setRevealedHints(0);
    setTestResults([]);
    setTutorChatMessages([
      {
        role: 'tutor',
        text: `Hello! I am your AI Error Concept Tutor. As you practice "${activeChallenge.title}" in ${activeChallenge.language.toUpperCase()}, if any syntax or runtime exception occurs, I'll explain the language concept, visualize memory states, and help you fix it!`,
      },
    ]);
  }, [selectedChallengeId]);

  const handleLanguageChange = (newLang: SupportedLanguage) => {
    setLanguage(newLang);
    // Find first challenge matching this language
    const matchingChallenge = PRACTICE_CHALLENGES.find((c) => c.language === newLang);
    if (matchingChallenge && matchingChallenge.id !== activeChallenge.id) {
      setSelectedChallengeId(matchingChallenge.id);
    } else {
      setCode(LANGUAGE_TEMPLATES[newLang]);
      setExecutionStatus('idle');
      setTerminalOutput([
        `// Switched to ${newLang.toUpperCase()} environment (${LANGUAGE_OPTIONS.find((l) => l.id === newLang)?.filename}).`,
        `// Active runtime: ${LANGUAGE_OPTIONS.find((l) => l.id === newLang)?.compiler}.`,
        `// Real-time syntax scanning & AI Error Tutor active for ${newLang.toUpperCase()}.`,
      ]);
    }
  };

  // Execute Code safely across all supported languages
  const runCode = async () => {
    setIsRunning(true);
    setTerminalOutput([`Executing ${language.toUpperCase()} sandbox...`]);
    const start = performance.now();
```

```
234
235     await new Promise((resolve) => setTimeout(resolve, 200));
236
237     try {
238       if (language === 'python') {
239         runPythonCode(code, start);
240       } else if (language === 'java') {
241         runJavaCode(code, start);
242       } else if (language === 'cpp') {
243         runCppCode(code, start);
244       } else if (language === 'c') {
245         runCCode(code, start);
246       } else if (language === 'go') {
247         runGoCode(code, start);
248       } else if (language === 'rust') {
249         runRustCode(code, start);
250       } else {
251         runJavaScriptCode(code, start);
252       }
253     } catch (err: unknown) {
254       handleCaughtError(err, start);
255     } finally {
256       setIsRunning(false);
257     }
258   };
259
260   const runJavaScriptCode = (userCode: string, startTime: number) => {
261     const logs: string[] = [];
262
263     const customConsole = {
264       log: (...args: unknown[]) => {
265         logs.push(args.map((a) => (typeof a === 'object' ? JSON.stringify(a, null, 2) : String(a))).join(' '));
266       },
267       warn: (...args: unknown[]) => {
```

```
268         logs.push(`⚠️ [WARN] ` + args.map((a) => String(a)).join(' '));
269       },
270       error: (...args: unknown[]) => {
271         logs.push(`💥 [ERROR] ` + args.map((a) => String(a)).join(' '));
272       },
273     };
274
275     try {
276       if (
277         activeChallenge.id === 'jwt-exp-flaw' &&
278         userCode.includes('if (decodedPayload) {') &&
279         !userCode.includes('decodedPayload.exp >')
280       ) {
281         throw new Error('SecurityVulnerability: Expired token accepted as valid authenticated session');
282       }
283
284       if (
285         activeChallenge.id === 'react-mutation-bug' &&
286         userCode.includes('.push(') &&
287         !userCode.includes('[...')
288       ) {
289         throw new Error('StateMutationError: Array.prototype.push directly mutated state reference without producing
a new object reference.');
290       }
291
292       const sanitizedCode = userCode
293         .replace(/:\s*(string|number|boolean|any|void|JwtPayload|null)\b/g, '')
294         .replace(/interface\s+\w+\s*\{[\s\S]*?\}/g, '');
295
296       const runFn = new Function('console', sanitizedCode);
297       runFn(customConsole);
298
299       const duration = Math.round(performance.now() - startTime);
300       setExecutionTimeMs(duration);
301       setExecutionStatus('success');
```

```
302
303       if (logs.length === 0) {
304         logs.push('Execution completed with no console output (return status 0).');
305       }
306
307       setTerminalOutput(logs);
308       setCurrentErrorExplanation(null);
309       runChallengeTests(userCode);
310
311       try {
312         confetti({ particleCount: 60, spread: 70, origin: { y: 0.8 } });
313       } catch {
314         // ignore
315       }
316     } catch (err: unknown) {
317       handleCaughtError(err, startTime);
318     }
319   };
320
321   const runPythonCode = (pythonCode: string, startTime: number) => {
322     const logs: string[] = [];
323     const lines = pythonCode.split('\n');
324
325     let openParens = 0;
326     for (let i = 0; i < lines.length; i++) {
327       const line = lines[i].trim();
328       for (const ch of line) {
329         if (ch === '(') openParens++;
330         if (ch === ')') openParens--;
331       }
332       if (line.startsWith('print(') && !line.includes(')')) {
333         throw new Error(`SyntaxError: unexpected EOF while parsing - unclosed '(' on line ${i + 1}`);
334       }
335     }
336     if (openParens > 0) {
```

```
      throw new Error("SyntaxError: unexpected EOF while parsing (unclosed parenthesis ')')");
    }

    if (
      activeChallenge.id === 'py-index-error' &&
      (pythonCode.includes('range(len(log_entries) + 1)') || pythonCode.includes('log_entries[3]'))
    ) {
      throw new Error('IndexError: list index out of range at line 6 (entry = log_entries[i])');
    }

    if (
      activeChallenge.id === 'py-key-error' &&
      pythonCode.includes("user_session['role']") &&
      !pythonCode.includes('.get(')
    ) {
      throw new Error("KeyError: 'role' in dictionary user_session");
    }

    if (
      activeChallenge.id === 'py-zero-division' &&
      pythonCode.includes('sum(ratings) / len(ratings)') &&
      !pythonCode.includes('if not ratings') &&
      !pythonCode.includes('len(ratings) == 0')
    ) {
      throw new Error('ZeroDivisionError: division by zero in calculate_average_rating');
    }

    // Python runner simulation: evaluate print(...) calls
    const printRegex = /print\s*\(([\s\S]*?)\)/g;
    let match;
    let count = 0;

    const varEnv: Record<string, unknown> = {};
    for (const l of lines) {
```

```
      const lineTrim = l.trim();
      const assignMatch = lineTrim.match(/^([a-zA-Z_]\w*)\s*=\s*(.+)$/);
      if (assignMatch && !lineTrim.startsWith('def ') && !lineTrim.startsWith('if ')) {
        const vName = assignMatch[1];
        const vVal = assignMatch[2];
        try {
          varEnv[vName] = eval(vVal.replace(/\bTrue\b/g, 'true').replace(/\bFalse\b/g, 'false').replace(/\bNone\b/g, 'null'));
        } catch {
          varEnv[vName] = vVal;
        }
      }
    }

    while ((match = printRegex.exec(pythonCode)) !== null) {
      count++;
      const rawArg = match[1].trim();
      if (!rawArg) {
        logs.push('');
      } else if (rawArg.startsWith('f"') || rawArg.startsWith("f'")) {
        let fStr = rawArg.slice(2, -1);
        fStr = fStr.replace(/\{([^}]+)\}/g, (_, expr) => {
          return varEnv[expr.trim()] !== undefined ? String(varEnv[expr.trim()]) : expr;
        });
        logs.push(fStr);
      } else if (varEnv[rawArg] !== undefined) {
        logs.push(typeof varEnv[rawArg] === 'object' ? JSON.stringify(varEnv[rawArg]) : String(varEnv[rawArg]));
      } else {
        try {
          const res = new Function(...Object.keys(varEnv), `return (${rawArg})`)(...Object.values(varEnv));
          logs.push(String(res));
        } catch {
          logs.push(rawArg.replace(/^['"]|['"]$/g, ''));
        }
      }
```

```
    }

    if (count === 0) {
      logs.push("Python 3.12: Successfully executed with return code 0.");
    }

    const duration = Math.round(performance.now() - startTime);
    setExecutionTimeMs(duration);
    setExecutionStatus('success');
    setTerminalOutput(logs);
    setCurrentErrorExplanation(null);
    runChallengeTests(pythonCode);
  };

  const runJavaCode = (javaCode: string, startTime: number) => {
    const logs: string[] = [];

    if (!javaCode.includes('class ')) {
      throw new Error("error: class, interface, or enum expected");
    }
    if (javaCode.includes('print(') && !javaCode.includes('System.out.println(')) {
      throw new Error("error: cannot find symbol method print() - In Java, use System.out.println(...)");
    }

    if (
      activeChallenge.id === 'java-null-pointer' &&
      javaCode.includes('user.getEmail().toLowerCase()') &&
      !javaCode.includes('user == null') &&
      !javaCode.includes('user.getEmail() == null')
    ) {
      throw new Error('java.lang.NullPointerException: Cannot invoke "String.toLowerCase()" because the return value of "Main$User.getEmail()" is null');
    }

    if (
```

```
      activeChallenge.id === 'java-array-oob' &&
      javaCode.includes('i <= scores.length')
    ) {
      throw new Error('java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 5 at Main.calculat
eTotal(Main.java:7)');
    }

    logs.push('[javac] Compiling Main.java with OpenJDK 21... OK');
    logs.push('[java] Executing Main.main():');

    const printRegex = /System\.out\.println\s*\(([\s\S]*?)\)\s*;/g;
    let match;
    let found = false;
    while ((match = printRegex.exec(javaCode)) !== null) {
      found = true;
      const content = match[1].trim();
      logs.push(content.replace(/^"|"$/g, '').replace(/"\s*\+\s*"/g, ''));
    }

    if (!found) {
      logs.push('Java process completed with return code 0.');
    }

    const duration = Math.round(performance.now() - startTime);
    setExecutionTimeMs(duration);
    setExecutionStatus('success');
    setTerminalOutput(logs);
    setCurrentErrorExplanation(null);
    runChallengeTests(javaCode);
  };

  const runCppCode = (cppCode: string, startTime: number) => {
    const logs: string[] = [];

    if (!cppCode.includes('#include')) {
```

```
      throw new Error("error: 'cout' was not declared in this scope (missing #include <iostream>)");
    }

    if (
      activeChallenge.id === 'cpp-vector-range' &&
      cppCode.includes('readings.at(readings.size())')
    ) {
      throw new Error('std::out_of_range: vector::_M_range_check: __n (which is 3) >= this->size() (which is 3)');
    }

    if (
      activeChallenge.id === 'cpp-segfault' &&
      cppCode.includes('&banner')
    ) {
      throw new Error('Segmentation fault (core dumped): dereferencing dangling pointer at address 0x7ffd9b8');
    }

    logs.push('[g++] Compiling main.cpp with GCC 13 (-std=c++20)... OK');
    logs.push('[a.out] Program Output:');

    const coutRegex = /std::cout\s*<<\s*([^;]+);/g;
    let match;
    let found = false;
    while ((match = coutRegex.exec(cppCode)) !== null) {
      found = true;
      const parts = match[1].split('<<').map((p) => p.trim());
      const outputLine = parts
        .filter((p) => !p.includes('endl'))
        .map((p) => p.replace(/^"|"$/g, ''))
        .join('');
      logs.push(outputLine);
    }

    if (!found) {
      logs.push('Program exited with return code 0.');
```

```
    }

    const duration = Math.round(performance.now() - startTime);
    setExecutionTimeMs(duration);
    setExecutionStatus('success');
    setTerminalOutput(logs);
    setCurrentErrorExplanation(null);
    runChallengeTests(cppCode);
  };

  const runCCode = (cCode: string, startTime: number) => {
    const logs: string[] = [];

    if (!cCode.includes('#include')) {
      throw new Error("error: implicit declaration of function 'printf' (missing #include <stdio.h>)");
    }

    if (
      activeChallenge.id === 'c-buffer-overflow' &&
      cCode.includes('strcpy(buffer, inputUsername)')
    ) {
      throw new Error('*** stack smashing detected ***: terminated (buffer overflow in strcpy into 16-byte buffe
r)');
    }

    logs.push('[clang] Compiling main.c with Clang 17 (-Wall -Wextra)... OK');
    logs.push('[./main] Standard Output:');

    const printfRegex = /printf\s*\(([\s\S]*?)\)\s*;/g;
    let match;
    let found = false;
    while ((match = printfRegex.exec(cCode)) !== null) {
      found = true;
      const args = match[1].split(',').map((s) => s.trim());
      let fmt = args[0].replace(/^"|"$/g, '').replace(/\\n/g, '');
```

```
      if (args.length > 1) {
        fmt = fmt.replace('%s', args[1].replace(/^"|"$/g, ''));
      }
      logs.push(fmt);
    }

    if (!found) {
      logs.push('Process finished with exit code 0.');
    }

    const duration = Math.round(performance.now() - startTime);
    setExecutionTimeMs(duration);
    setExecutionStatus('success');
    setTerminalOutput(logs);
    setCurrentErrorExplanation(null);
    runChallengeTests(cCode);
  };

  const runGoCode = (goCode: string, startTime: number) => {
    const logs: string[] = [];

    if (!goCode.includes('package main')) {
      throw new Error("expected 'package', found import or statement");
    }

    if (
      activeChallenge.id === 'go-nil-pointer' &&
      goCode.includes('return u.Account.Balance') &&
      !goCode.includes('u.Account == nil')
    ) {
      throw new Error('panic: runtime error: invalid memory address or nil pointer dereference [signal SIGSEGV: code
=0x1 addr=0x0]');
    }

    logs.push('[go] Compiling and running main.go...');
```

```
    const printRegex = /fmt\.Print(ln|f)?\s*\(([\s\S]*?)\)/g;
    let match;
    let found = false;
    while ((match = printRegex.exec(goCode)) !== null) {
      found = true;
      const clean = match[2].replace(/^"|"$/g, '').replace(/\\n/g, '');
      logs.push(clean);
    }

    if (!found) {
      logs.push('Go process completed with exit status 0.');
    }

    const duration = Math.round(performance.now() - startTime);
    setExecutionTimeMs(duration);
    setExecutionStatus('success');
    setTerminalOutput(logs);
    setCurrentErrorExplanation(null);
    runChallengeTests(goCode);
  };

  const runRustCode = (rustCode: string, startTime: number) => {
    const logs: string[] = [];

    if (!rustCode.includes('fn main()')) {
      throw new Error("error[E0601]: `main` function not found in crate");
    }

    if (
      activeChallenge.id === 'rust-borrow-checker' &&
      rustCode.includes('log_token(token)') &&
      !rustCode.includes('&token') &&
      !rustCode.includes('&str')
    ) {
```

```
        throw new Error('error[E0382]: borrow of moved value: `token` (ownership transferred into log_token)');
      }

      logs.push('[rustc] Compiling with rustc 1.76.0 (opt-level=3)... OK');
      logs.push('[cargo run] Running `target/release/sandbox`:');

      const printRegex = /println!\s*\(([\s\S]*?)\)\s*;/g;
      let match;
      let found = false;
      while ((match = printRegex.exec(rustCode)) !== null) {
        found = true;
        const clean = match[1].replace(/^"|"$/g, '').replace(/\{.*?\}/g, 'data');
        logs.push(clean);
      }

      if (!found) {
        logs.push('Rust program exited successfully.');
      }

      const duration = Math.round(performance.now() - startTime);
      setExecutionTimeMs(duration);
      setExecutionStatus('success');
      setTerminalOutput(logs);
      setCurrentErrorExplanation(null);
      runChallengeTests(rustCode);
    };

    const handleCaughtError = (err: unknown, startTime: number) => {
      const duration = Math.round(performance.now() - startTime);
      setExecutionTimeMs(duration);
      setExecutionStatus('error');

      const errorMsg = err instanceof Error ? err.message : String(err);
      setTerminalOutput([
        `💥 Runtime / Compiler Exception Caught (${language.toUpperCase()})!`,
```

```
      `========================================`,
      errorMsg,
      ``,
      `AI Error Explainer activated below. Inspect the concept breakdown and guided fix.`,
    ]);

    const explanation = explainError(errorMsg, code, language, activeChallenge.id);
    setCurrentErrorExplanation(explanation);
    setActiveExplainerTab('why');
  };

  const runChallengeTests = (currentCode: string) => {
    const results = activeChallenge.testCases.map((tc) => {
      let passed = true;
      if (activeChallenge.id === 'type-error-null') {
        passed = currentCode.includes('?.') || currentCode.includes('user && user.subscription');
      } else if (activeChallenge.id === 'ref-error-tdz') {
        passed = currentCode.indexOf('taxRate') < currentCode.indexOf('subtotal * taxRate');
      } else if (activeChallenge.id === 'range-error-recursion') {
        passed = currentCode.includes('<= 1') || currentCode.includes('<= 0');
      } else if (activeChallenge.id === 'async-unhandled-rejection') {
        passed = currentCode.includes('try') && currentCode.includes('catch');
      } else if (activeChallenge.id === 'py-index-error') {
        passed = currentCode.includes('enumerate') || (!currentCode.includes('+ 1') && currentCode.includes('range(len('));
      } else if (activeChallenge.id === 'py-key-error') {
        passed = currentCode.includes('.get(') || currentCode.includes('in user_session');
      } else if (activeChallenge.id === 'py-zero-division') {
        passed = currentCode.includes('if not ratings') || currentCode.includes('len(ratings) == 0') || currentCode.includes('len(ratings) > 0');
      } else if (activeChallenge.id === 'java-null-pointer') {
        passed = currentCode.includes('user == null') || currentCode.includes('user.getEmail() == null') || currentCode.includes('Optional');
      } else if (activeChallenge.id === 'java-array-oob') {
        passed = currentCode.includes('< scores.length') || currentCode.includes(': scores');
      } else if (activeChallenge.id === 'cpp-vector-range') {
```

```
678         passed = currentCode.includes('.back()') || currentCode.includes('size() - 1');
679       } else if (activeChallenge.id === 'cpp-segfault') {
680         passed = !currentCode.includes('&banner') && currentCode.includes('return banner');
681       } else if (activeChallenge.id === 'c-buffer-overflow') {
682         passed = currentCode.includes('snprintf') || currentCode.includes('strncpy');
683       } else if (activeChallenge.id === 'go-nil-pointer') {
684         passed = currentCode.includes('u.Account == nil') || currentCode.includes('u == nil');
685       } else if (activeChallenge.id === 'rust-borrow-checker') {
686         passed = currentCode.includes('&token') || currentCode.includes('&str');
687       } else if (activeChallenge.id === 'jwt-exp-flaw') {
688         passed = currentCode.includes('.exp >') || currentCode.includes('.exp * 1000');
689       }
690
691       return {
692         name: tc.name,
693         passed,
694         message: passed ? `Test passed: Expected "${tc.expectedOutput}"` : `Test failed: Exception thrown`,
695       };
696     });
697
698     setTestResults(results);
699   };
700
701   const handleApplyFix = () => {
702     if (activeChallenge.solutionCode) {
703       setCode(activeChallenge.solutionCode);
704       setTerminalOutput([
705         '// Applied recommended surgical fix to editor.',
706         '// Click "Run Code" or press Ctrl+Enter to test the solution!',
707       ]);
708       setExecutionStatus('idle');
709     }
710   };
711
```

```
  const handleAskTutor = async (customQuestion?: string) => {
    const q = customQuestion || tutorInput;
    if (!q.trim()) return;

    const newMessages = [...tutorChatMessages, { role: 'user' as const, text: q }];
    setTutorChatMessages(newMessages);
    setTutorInput('');
    setIsTutorThinking(true);

    try {
      const res = await fetch('/api/code-practice', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          action: 'ask-tutor',
          code,
          error: currentErrorExplanation?.errorMessage || 'General question',
          question: q,
          language,
        }),
      });
      const data = await res.json();
      if (data.success && data.answer) {
        setTutorChatMessages([...newMessages, { role: 'tutor', text: data.answer }]);
      } else {
        setTutorChatMessages([
          ...newMessages,
          {
            role: 'tutor',
            text: `In ${language.toUpperCase()}, always verify variable bounds and memory pointers before dereferenc
ing. Defensive guarding prevents runtime crashes.`,
          },
        ]);
      }
    } catch {
```

```
      setTutorChatMessages([
        ...newMessages,
        {
          role: 'tutor',
          text: `In ${language.toUpperCase()}, this issue stems from accessing uninitialized memory or out-of-bounds sequences. Apply the recommended defensive check.`,
        },
      ]);
    } finally {
      setIsTutorThinking(false);
    }
  };

  const copyToClipboard = () => {
    navigator.clipboard.writeText(code);
    setCopied(true);
    setTimeout(() => setCopied(false), 2000);
  };

  const activeLangOption = LANGUAGE_OPTIONS.find((l) => l.id === language) || LANGUAGE_OPTIONS[0];

  return (
    <div className="space-y-6 max-w-7xl mx-auto pb-20">
      {/* Top Banner & Header */}
      <div className="bg-white rounded-2xl border border-slate-200 p-5 shadow-xs flex flex-col md:flex-row md:items-center justify-between gap-4">
        <div className="flex items-center gap-3">
          <div className="w-10 h-10 rounded-xl bg-gradient-to-tr from-indigo-600 to-cyan-500 text-white flex items-center justify-center shadow-md shadow-indigo-500/20">
            <Code2 className="w-5 h-5" />
          </div>
          <div>
            <div className="flex items-center gap-2">
              <h1 className="text-lg font-bold text-slate-900">Code Practice Lab</h1>
              <span className="text-[10px] font-bold uppercase tracking-wider px-2 py-0.5 rounded-full bg-indigo-50 text-indigo-700 border border-indigo-200">
                Multi-Language AI Error Tutor
```

```
              </span>
            </div>
            <p className="text-xs text-slate-500 font-medium">
              Interactive sandbox with real-time error concept diagnosis, AST mental models & guided fixes across Python, Java, C++, C, JS, TS, Go & Rust.
            </p>
          </div>
        </div>

        <div className="flex items-center gap-3">
          {/* Challenge Selector */}
          <div className="relative">
            <select
              value={selectedChallengeId}
              onChange={(e) => setSelectedChallengeId(e.target.value)}
              className="appearance-none pl-3 pr-8 py-2 bg-slate-50 border border-slate-200 rounded-xl text-xs font-semibold text-slate-800 focus:outline-none focus:ring-2 focus:ring-indigo-500 cursor-pointer"
            >
              <optgroup label="Challenges for Selected Language">
                {PRACTICE_CHALLENGES.filter((c) => c.language === language).map((ch) => (
                  <option key={ch.id} value={ch.id}>
                    {ch.difficulty === 'Hard' ? '🔴' : ch.difficulty === 'Medium' ? '🟡' : '🟢'} [{ch.language.toUpperCase()}] {ch.title}
                  </option>
                ))}
              </optgroup>
              <optgroup label="All Programming Languages (15+ Challenges)">
                {PRACTICE_CHALLENGES.map((ch) => (
                  <option key={ch.id} value={ch.id}>
                    {ch.difficulty === 'Hard' ? '🔴' : ch.difficulty === 'Medium' ? '🟡' : '🟢'} [{ch.language.toUpperCase()}] {ch.title}
                  </option>
                ))}
              </optgroup>
            </select>
            <ChevronDown className="w-3.5 h-3.5 text-slate-400 absolute right-2.5 top-1/2 -translate-y-1/2 pointer-events-none" />
```

```
          </div>

          <button
            onClick={() => onNavigate('courses')}
            className="px-3 py-2 rounded-xl bg-slate-100 hover:bg-slate-200 text-slate-700 text-xs font-semibold tra
nsition-all flex items-center gap-1.5 cursor-pointer"
          >
            <BookOpen className="w-3.5 h-3.5 text-indigo-600" />
            <span>View All Courses</span>
          </button>
        </div>
      </div>

      {/* Prominent Multi-Language Toolbar */}
      <div className="bg-gradient-to-r from-slate-900 to-indigo-950 p-4 rounded-2xl border border-slate-800 shadow-m
d text-white space-y-3">
        <div className="flex flex-wrap items-center justify-between gap-3">
          <div className="flex items-center gap-2">
            <span className="text-[11px] font-bold text-indigo-300 uppercase tracking-wider flex items-center gap-1.
5">
              <Laptop className="w-3.5 h-3.5 text-indigo-400" />
              Select Programming Language:
            </span>
          </div>

          <div className="flex items-center gap-2">
            <span className="text-[10px] text-emerald-400 font-mono bg-emerald-950/80 border border-emerald-800/80 p
x-2.5 py-1 rounded-lg flex items-center gap-1.5">
              <span className="w-2 h-2 rounded-full bg-emerald-400 animate-pulse" />
              Active Compiler: {activeLangOption.compiler}
            </span>
            <button
              onClick={() => {
                setCode(LANGUAGE_TEMPLATES[language]);
                setExecutionStatus('idle');
                setTerminalOutput([
                  `// Loaded ${language.toUpperCase()} Starter Sandbox Template.`,
```

```
                  `// Real-time error detection active for ${language.toUpperCase()}.`,
                ]);
              }}
              className="px-3 py-1 rounded-lg bg-indigo-600/80 hover:bg-indigo-600 text-white text-xs font-semibold transition-all cursor-pointer flex items-center gap-1 shadow-sm"
            >
              <RotateCcw className="w-3 h-3" />
              <span>Load Starter</span>
            </button>
          </div>
        </div>

        {/* Language Tabs */}
        <div className="flex flex-wrap items-center gap-2 pt-1">
          {LANGUAGE_OPTIONS.map((lang) => (
            <button
              key={lang.id}
              onClick={() => handleLanguageChange(lang.id)}
              className={`px-3.5 py-2 rounded-xl text-xs font-semibold transition-all flex items-center gap-2 cursor-pointer ${
                language === lang.id
                  ? 'bg-indigo-500 text-white shadow-md ring-2 ring-indigo-300 font-bold scale-[1.02]'
                  : 'bg-slate-800/90 text-slate-300 hover:bg-slate-700 hover:text-white border border-slate-700/60'
              }`}
            >
              <span className="text-sm">{lang.icon}</span>
              <span>{lang.label}</span>
            </button>
          ))}
        </div>
      </div>

      {/* Challenge Description & Target Error Card */}
      <div className="bg-indigo-50/70 border border-indigo-100 rounded-2xl p-5 space-y-3">
        <div className="flex flex-wrap items-center justify-between gap-2">
```

```
            <div className="flex items-center gap-2">
              <span
                className={`text-[10px] font-bold uppercase px-2 py-0.5 rounded ${
                  activeChallenge.difficulty === 'Hard'
                    ? 'bg-rose-100 text-rose-800'
                    : activeChallenge.difficulty === 'Medium'
                      ? 'bg-amber-100 text-amber-800'
                      : 'bg-emerald-100 text-emerald-800'
                }`}
              >
                {activeChallenge.difficulty}
              </span>
              <span className="text-xs font-bold text-indigo-900">{activeChallenge.title}</span>
              <span className="text-[10px] font-bold uppercase px-2 py-0.5 rounded bg-indigo-100 text-indigo-800">
                {activeChallenge.language.toUpperCase()}
              </span>
            </div>

            <div className="flex items-center gap-2 text-xs text-indigo-950 font-medium">
              <span className="text-slate-400">Target Error:</span>
              <code className="bg-white/80 px-2 py-0.5 rounded border border-indigo-200 text-rose-700 font-mono text-
[11px]">
                {activeChallenge.errorNameTarget}
              </code>
            </div>
          </div>

          <p className="text-xs text-indigo-950 leading-relaxed font-medium">
            {activeChallenge.description}
          </p>
        </div>

        {/* Main Coding & Explainer Workspace (Split Grid) */}
        <div className="grid grid-cols-1 lg:grid-cols-12 gap-6 items-start">
          {/* Left Column: Editor & Terminal (7 cols) */}
```

```
        <div className="lg:col-span-7 space-y-4">
          {/* Editor Header Bar */}
          <div className="bg-slate-900 rounded-t-2xl px-4 py-3 border border-slate-800 flex items-center justify-between text-white text-xs">
            <div className="flex items-center gap-3">
              <div className="flex items-center gap-1.5">
                <span className="w-3 h-3 rounded-full bg-rose-500/80 inline-block" />
                <span className="w-3 h-3 rounded-full bg-amber-500/80 inline-block" />
                <span className="w-3 h-3 rounded-full bg-emerald-500/80 inline-block" />
              </div>
              <span className="font-mono text-slate-300 font-semibold text-[11px] bg-slate-800/80 px-2.5 py-0.5 rounded border border-slate-700">
                {activeLangOption.filename}
              </span>
              <span className="hidden sm:inline-flex items-center gap-1 text-[10px] text-emerald-400 font-semibold">
                <span className="w-1.5 h-1.5 rounded-full bg-emerald-400 animate-pulse" />
                {language.toUpperCase()} Active
              </span>
            </div>

            <div className="flex items-center gap-2">
              <button
                onClick={() => setEditorTheme(editorTheme === 'dark' ? 'light' : 'dark')}
                className="p-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 transition-colors flex items-center gap-1 text-[11px] cursor-pointer"
                title={editorTheme === 'dark' ? 'Switch to Light Editor' : 'Switch to Dark Editor'}
              >
                {editorTheme === 'dark' ? (
                  <Sun className="w-3.5 h-3.5 text-amber-400" />
                ) : (
                  <Moon className="w-3.5 h-3.5 text-indigo-300" />
                )}
              </button>

              <button
                onClick={() => setFontSize((prev) => Math.max(11, prev - 1))}
```

```
                  className="px-2 py-1 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 font-mono text-[10px] transition-colors cursor-pointer"
                  title="Decrease font size"
                >
                  A-
                </button>
                <button
                  onClick={() => setFontSize((prev) => Math.min(18, prev + 1))}
                  className="px-2 py-1 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 font-mono text-[10px] transition-colors cursor-pointer"
                  title="Increase font size"
                >
                  A+
                </button>

                <button
                  onClick={copyToClipboard}
                  className="p-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 transition-colors cursor-pointer"
                  title="Copy Code"
                >
                  {copied ? <Check className="w-3.5 h-3.5 text-emerald-400" /> : <Copy className="w-3.5 h-3.5" />}
                </button>

                <button
                  onClick={() => setCode(activeChallenge.initialCode)}
                  className="px-2 py-1 rounded-lg bg-slate-800 hover:bg-slate-700 text-slate-300 text-[11px] font-semibold transition-colors flex items-center gap-1 cursor-pointer"
                  title="Reset to Initial Challenge Code"
                >
                  <RotateCcw className="w-3 h-3" />
                  <span>Reset</span>
                </button>

                <button
                  onClick={runCode}
                  disabled={isRunning}
```

```
                className="px-3.5 py-1.5 rounded-lg bg-gradient-to-r from-emerald-500 to-teal-500 hover:from-emerald-600 hover:to-teal-600 text-slate-950 font-bold text-xs shadow-md transition-all flex items-center gap-1.5 cursor-pointer disabled:opacity-50"
              >
                <Play className="w-3.5 h-3.5 fill-slate-950" />
                <span>{isRunning ? 'Running...' : 'Run Code'}</span>
              </button>
            </div>
          </div>

          {/* Interactive Code Editor Area */}
          {() => {
            const isDark = editorTheme === 'dark';
            const editorBg = isDark ? '#0d1117' : '#ffffff';
            const editorText = isDark ? '#f0f6fc' : '#0f172a';
            const gutterBg = isDark ? '#161b22' : '#f8fafc';
            const gutterText = isDark ? '#6e7681' : '#94a3b8';
            const borderCol = isDark ? '#30363d' : '#e2e8f0';

            return (
              <div
                className="rounded-b-2xl overflow-hidden border shadow-inner relative"
                style={{ backgroundColor: editorBg, borderColor: borderCol }}
              >
                <div className="flex">
                  <div
                    className="select-none py-4 px-3 text-right font-mono border-r shrink-0"
                    style={{
                      backgroundColor: gutterBg,
                      color: gutterText,
                      borderColor: borderCol,
                      fontSize: `${fontSize}px`,
                      lineHeight: '1.5rem',
                      width: '3rem',
                    }}
```

```
                    >
                      {code.split('\n').map((_, i) => {
                        const lineNum = i + 1;
                        const hasDiagnostic = liveDiagnostic?.hasError && liveDiagnostic.line === lineNum;
                        return (
                          <div
                            key={i}
                            className={`${hasDiagnostic ? 'text-rose-400 font-bold bg-rose-500/20 rounded' : ''}`}
                          >
                            {lineNum}
                          </div>
                        );
                      })}
                    </div>

                    <textarea
                      value={code}
                      onChange={(e) => setCode(e.target.value)}
                      onKeyDown={(e) => {
                        if (e.key === 'Tab') {
                          e.preventDefault();
                          const textarea = e.currentTarget;
                          const start = textarea.selectionStart;
                          const end = textarea.selectionEnd;
                          const newCode = code.substring(0, start) + '    ' + code.substring(end);
                          setCode(newCode);
                          setTimeout(() => {
                            textarea.selectionStart = textarea.selectionEnd = start + 4;
                          }, 0);
                        }
                        if ((e.ctrlKey || e.metaKey) && e.key === 'Enter') {
                          e.preventDefault();
                          runCode();
                        }
                      }}
```

```
                    spellCheck={false}
                    rows={18}
                    className="w-full p-4 font-mono leading-6 resize-none focus:outline-none focus:ring-0"
                    style={{
                      backgroundColor: editorBg,
                      color: editorText,
                      fontSize: `${fontSize}px`,
                      caretColor: isDark ? '#38bdf8' : '#4f46e5',
                      tabSize: 4,
                    }}
                    placeholder={`// Write ${language.toUpperCase()} code here...`}
                  />
                </div>
              </div>
            );
          })()}

          {/* Real-Time Live Error Sentinel (As-You-Type Feedback Bar) */}
          {liveDiagnostic && (
            <div
              className={`p-4 rounded-xl border text-xs transition-all ${
                liveDiagnostic.hasError
                  ? liveDiagnostic.type === 'syntax'
                    ? 'bg-rose-50/90 border-rose-200 text-rose-950 shadow-xs'
                    : 'bg-amber-50/90 border-amber-200 text-amber-950 shadow-xs'
                  : 'bg-emerald-50/80 border-emerald-200 text-emerald-950 shadow-2xs'
              }`}
            >
              <div className="flex items-start justify-between gap-3">
                <div className="flex items-center gap-2">
                  {liveDiagnostic.hasError ? (
                    <span className="w-2.5 h-2.5 rounded-full bg-rose-500 animate-pulse shrink-0" />
                  ) : (
                    <CheckCircle2 className="w-4 h-4 text-emerald-600 shrink-0" />
```

```
                    )}
                    <span className="font-bold text-xs">
                      {liveDiagnostic.hasError
                        ? `⚡ Live Sentinel (${language.toUpperCase()}): ${liveDiagnostic.title}`
                        : `🟢 ${language.toUpperCase()} Syntax Clean & Valid`}
                    </span>
                  </div>

                  {liveDiagnostic.line && (
                    <span className="text-[10px] font-mono px-2 py-0.5 rounded bg-rose-200/60 text-rose-800 font-bold shrink-0">
                      Line {liveDiagnostic.line}
                    </span>
                  )}
                </div>

                {liveDiagnostic.hasError ? (
                  <div className="mt-2.5 space-y-1.5 pl-4 border-l-2 border-rose-300">
                    <div className="flex items-baseline gap-1.5">
                      <span className="font-bold text-rose-900 text-[11px] shrink-0">
                        📍 Where you made the mistake:
                      </span>
                      <span className="font-mono text-[11px] text-rose-950">
                        {liveDiagnostic.whereIsMistake}
                      </span>
                    </div>

                    <div className="flex items-baseline gap-1.5">
                      <span className="font-bold text-rose-900 text-[11px] shrink-0">
                        ❓ Why this is an error:
                      </span>
                      <span className="text-[11px] text-slate-800 leading-relaxed">
                        {liveDiagnostic.whyIsError}
                      </span>
                    </div>
```

```
                  <div className="flex items-baseline gap-1.5 pt-0.5">
                    <span className="font-bold text-emerald-800 text-[11px] shrink-0">
                      💡 How to fix it:
                    </span>
                    <span className="text-[11px] text-slate-800 leading-relaxed font-medium">
                      {liveDiagnostic.howToFix}
                    </span>
                  </div>
                </div>
              ) : (
                <div className="mt-1 text-[11px] text-emerald-800 font-medium pl-6">
                  {liveDiagnostic.whyIsError}
                </div>
              )}
            </div>
          )}

          {/* Execution Terminal */}
          <div className="bg-slate-900 rounded-2xl border border-slate-800 overflow-hidden shadow-sm">
            <div className="px-4 py-2.5 bg-slate-950/80 border-b border-slate-800 flex items-center justify-between text-xs">
              <div className="flex items-center gap-2">
                <TerminalIcon className="w-3.5 h-3.5 text-slate-400" />
                <span className="font-semibold text-slate-300">Terminal Output ({language.toUpperCase()})</span>
              </div>

              <div className="flex items-center gap-3">
                {executionTimeMs !== null && (
                  <span className="text-[10px] text-slate-500 font-mono">
                    Elapsed: {executionTimeMs}ms
                  </span>
                )}
                {executionStatus === 'success' && (
                  <span className="inline-flex items-center gap-1 text-[10px] font-bold text-emerald-400 bg-emerald-950/60 px-2 py-0.5 rounded border border-emerald-800">
```

```
                    <CheckCircle2 className="w-3 h-3" />
                    Passed
                  </span>
                )}
                {executionStatus === 'error' && (
                  <span className="inline-flex items-center gap-1 text-[10px] font-bold text-rose-400 bg-rose-950/60 px-2 py-0.5 rounded border border-rose-800">
                    <AlertTriangle className="w-3 h-3" />
                    Exception
                  </span>
                )}
              </div>
            </div>

            <div className="p-4 font-mono text-xs text-slate-300 max-h-48 overflow-y-auto space-y-1 bg-slate-900">
              {terminalOutput.map((line, idx) => (
                <div
                  key={idx}
                  className={
                    line.startsWith('💥')
                      ? 'text-rose-400 font-bold'
                      : line.startsWith('⚠️')
                        ? 'text-amber-400'
                        : line.startsWith('//')
                          ? 'text-slate-500'
                          : line.startsWith('[javac]') || line.startsWith('[g++]') || line.startsWith('[clang]')
                            ? 'text-cyan-400 font-semibold'
                            : 'text-slate-300'
                  }
                >
                  {line}
                </div>
              ))}
            </div>
          </div>
```

```
1181
1182            {/* Test Cases Results */}
1183            {testResults.length > 0 && (
1184              <div className="bg-white rounded-2xl border border-slate-200 p-4 space-y-2 shadow-xs">
1185                <div className="flex items-center justify-between text-xs font-bold text-slate-800">
1186                  <span>Challenge Verification Tests</span>
1187                  <span>
1188                    {testResults.filter((r) => r.passed).length}/{testResults.length} Passed
1189                  </span>
1190                </div>
1191                <div className="space-y-1.5">
1192                  {testResults.map((tr, i) => (
1193                    <div
1194                      key={i}
1195                      className={`flex items-center justify-between p-2.5 rounded-xl text-xs ${
1196                        tr.passed
1197                          ? 'bg-emerald-50 text-emerald-900 border border-emerald-200'
1198                          : 'bg-rose-50 text-rose-900 border border-rose-200'
1199                      }`}
1200                    >
1201                      <div className="flex items-center gap-2">
1202                        {tr.passed ? (
1203                          <CheckCircle2 className="w-4 h-4 text-emerald-600" />
1204                        ) : (
1205                          <AlertTriangle className="w-4 h-4 text-rose-600" />
1206                        )}
1207                        <span className="font-semibold">{tr.name}</span>
1208                      </div>
1209                      <span className="text-[11px] font-mono text-slate-600">{tr.message}</span>
1210                    </div>
1211                  ))}
1212                </div>
1213              </div>
1214            )}
```

```
        </div>

        {/* Right Column: AI Error Explainer & Concept Tutor (5 cols) */}
        <div className="lg:col-span-5 space-y-4">
          <div className="bg-white rounded-2xl border border-slate-200 shadow-sm overflow-hidden">
            {/* Header */}
            <div className="p-4 bg-gradient-to-r from-indigo-50 to-purple-50 border-b border-indigo-100 flex items-center justify-between">
              <div className="flex items-center gap-2.5">
                <div className="w-8 h-8 rounded-lg bg-indigo-600 text-white flex items-center justify-center shadow-xs">
                  <Sparkles className="w-4 h-4" />
                </div>
                <div>
                  <h3 className="text-xs font-bold text-slate-900 uppercase tracking-wider">
                    AI Error Concept Explainer
                  </h3>
                  <p className="text-[11px] text-slate-500 font-medium">
                    Real-time error diagnosis & mental models
                  </p>
                </div>
              </div>

              {currentErrorExplanation?.failingLine && (
                <span className="text-[10px] font-bold text-rose-700 bg-rose-100 px-2 py-0.5 rounded border border-rose-200">
                  Line {currentErrorExplanation.failingLine} Crash
                </span>
              )}
            </div>

            {/* Navigation Tabs */}
            <div className="flex border-b border-slate-100 text-xs font-semibold text-slate-600">
              <button
                onClick={() => setActiveExplainerTab('why')}
                className={`flex-1 py-2.5 text-center transition-all cursor-pointer ${
```

```
                    activeExplainerTab === 'why'
                      ? 'border-b-2 border-indigo-600 text-indigo-700 font-bold bg-indigo-50/40'
                      : 'hover:text-slate-900'
                  }`}
                >
                  1. Why It Happened
                </button>
                <button
                  onClick={() => setActiveExplainerTab('concept')}
                  className={`flex-1 py-2.5 text-center transition-all cursor-pointer ${
                    activeExplainerTab === 'concept'
                      ? 'border-b-2 border-indigo-600 text-indigo-700 font-bold bg-indigo-50/40'
                      : 'hover:text-slate-900'
                  }`}
                >
                  2. Concept Deep-Dive
                </button>
                <button
                  onClick={() => setActiveExplainerTab('solution')}
                  className={`flex-1 py-2.5 text-center transition-all cursor-pointer ${
                    activeExplainerTab === 'solution'
                      ? 'border-b-2 border-indigo-600 text-indigo-700 font-bold bg-indigo-50/40'
                      : 'hover:text-slate-900'
                  }`}
                >
                  3. Interactive Fix
                </button>
                <button
                  onClick={() => setActiveExplainerTab('ask_ai')}
                  className={`flex-1 py-2.5 text-center transition-all cursor-pointer ${
                    activeExplainerTab === 'ask_ai'
                      ? 'border-b-2 border-indigo-600 text-indigo-700 font-bold bg-indigo-50/40'
                      : 'hover:text-slate-900'
                  }`}
                >
```

```
                4. Ask Tutor
              </button>
            </div>

            {/* Tab Contents */}
            <div className="p-5">
              {currentErrorExplanation ? (
                <>
                  {/* TAB 1: WHY IT HAPPENED */}
                  {activeExplainerTab === 'why' && (
                    <div className="space-y-4">
                      <div className="p-3.5 rounded-xl bg-rose-50 border border-rose-100 text-xs text-rose-950 space-y-1.5">
                        <div className="flex items-center gap-1.5 font-bold text-rose-800 text-[11px] uppercase tracking-wider">
                          <AlertTriangle className="w-3.5 h-3.5" />
                          <span>{currentErrorExplanation.errorName}</span>
                        </div>
                        <p className="font-mono text-[11px]">{currentErrorExplanation.errorMessage}</p>
                      </div>

                      <div className="space-y-1.5">
                        <h4 className="text-xs font-bold text-slate-800 uppercase tracking-wider">
                          Root Cause Diagnosis:
                        </h4>
                        <p className="text-xs text-slate-700 leading-relaxed">
                          {currentErrorExplanation.whyThisHappened}
                        </p>
                      </div>

                      {currentErrorExplanation.codeContext && (
                        <div className="space-y-1.5">
                          <span className="text-[11px] font-bold text-slate-500 uppercase tracking-wider">
                            Problematic Statement:
                          </span>
```

```
                          <div className="p-2.5 bg-slate-950 text-rose-300 rounded-xl font-mono text-xs border borde
r-slate-800">
                            {currentErrorExplanation.codeContext}
                          </div>
                        </div>
                      )}

                      <div className="p-3.5 bg-slate-50 rounded-xl border border-slate-200 text-xs space-y-1.5">
                        <span className="font-bold text-slate-700">Defensive Rules to Prevent This:</span>
                        <ul className="list-disc pl-4 space-y-1 text-slate-600 text-[11px]">
                          {currentErrorExplanation.bestPractices.map((bp, i) => (
                            <li key={i}>{bp}</li>
                          ))}
                        </ul>
                      </div>
                    </div>
                  )}

                  {/* TAB 2: CONCEPT DEEP-DIVE */}
                  {activeExplainerTab === 'concept' && (
                    <div className="space-y-4">
                      <div>
                        <span className="text-[10px] font-bold uppercase tracking-wider text-indigo-600 bg-indigo-50
px-2 py-0.5 rounded border border-indigo-200">
                          Mental Model & Architecture
                        </span>
                        <h4 className="text-sm font-bold text-slate-900 mt-1">
                          {currentErrorExplanation.conceptTitle}
                        </h4>
                      </div>

                      <p className="text-xs text-slate-700 leading-relaxed">
                        {currentErrorExplanation.conceptDeepDive}
                      </p>

                      <div className="p-3.5 rounded-xl bg-amber-50/80 border border-amber-200 text-xs space-y-1">
```

```
                        <div className="flex items-center gap-1.5 font-bold text-amber-900">
                          <Lightbulb className="w-3.5 h-3.5 text-amber-600" />
                          <span>Intuitive Real-World Analogy</span>
                        </div>
                        <p className="text-amber-950 text-[11px] leading-relaxed">
                          {currentErrorExplanation.analogy}
                        </p>
                      </div>

                      {currentErrorExplanation.mentalModelDiagram && (
                        <div className="space-y-1.5">
                          <span className="text-[11px] font-bold text-slate-500 uppercase tracking-wider">
                            Memory State Diagram:
                          </span>
                          <pre className="p-3 bg-slate-950 text-cyan-300 rounded-xl font-mono text-[11px] overflow-x-auto border border-slate-800 leading-tight">
                            {currentErrorExplanation.mentalModelDiagram}
                          </pre>
                        </div>
                      )}

                      {currentErrorExplanation.relatedLesson && onOpenLesson && (
                        <button
                          onClick={() =>
                            onOpenLesson(
                              currentErrorExplanation.relatedLesson!.courseId,
                              currentErrorExplanation.relatedLesson!.lessonId
                            )
                          }
                          className="w-full py-2.5 px-3 rounded-xl bg-indigo-50 hover:bg-indigo-100 text-indigo-700 border border-indigo-200 text-xs font-bold transition-all flex items-center justify-between cursor-pointer"
                        >
                          <span className="flex items-center gap-1.5">
                            <BookOpen className="w-3.5 h-3.5" />
                            <span>Read Course Lesson: {currentErrorExplanation.relatedLesson.title}</span>
```

```
                          </span>
                          <ArrowRight className="w-3.5 h-3.5" />
                        </button>
                      )}
                    </div>
                  )}

                  {/* TAB 3: INTERACTIVE FIX */}
                  {activeExplainerTab === 'solution' && (
                    <div className="space-y-4">
                      <div className="space-y-2">
                        <div className="flex items-center justify-between text-xs font-bold text-slate-700">
                          <span>Surgical Fix Comparison</span>
                          <span className="text-[10px] text-slate-400 font-normal">Before vs After</span>
                        </div>

                        <div className="space-y-1.5">
                          <div className="p-2.5 bg-rose-50 text-rose-900 rounded-xl font-mono text-[11px] border bor
der-rose-200">
                            <div className="text-[9px] font-bold uppercase text-rose-500 mb-1">Before (Buggy)</div>
                            {currentErrorExplanation.suggestedFixBefore}
                          </div>

                          <div className="p-2.5 bg-emerald-50 text-emerald-950 rounded-xl font-mono text-[11px] bord
er border-emerald-200">
                            <div className="text-[9px] font-bold uppercase text-emerald-600 mb-1">After (Fixed)</div
>
                            {currentErrorExplanation.suggestedFixAfter}
                          </div>
                        </div>
                      </div>

                      <div className="p-3 bg-slate-50 rounded-xl border border-slate-200 text-xs text-slate-700 lead
ing-relaxed">
                        <span className="font-bold text-slate-800">Why this fix works: </span>
                        {currentErrorExplanation.fixExplanation}
                      </div>
```

```
1416
1417                        {/* Progressive Hints Accordion */}
1418                        <div className="space-y-2 pt-2 border-t border-slate-100">
1419                          <div className="flex items-center justify-between text-xs font-bold text-slate-800">
1420                            <span className="flex items-center gap-1.5">
1421                              <HelpCircle className="w-3.5 h-3.5 text-indigo-600" />
1422                              <span>Guided Hints</span>
1423                            </span>
1424                            <span className="text-[10px] text-slate-400">
1425                              {revealedHints} of {activeChallenge.hints.length} revealed
1426                            </span>
1427                          </div>
1428
1429                          {activeChallenge.hints.map((hint, idx) => (
1430                            <div key={idx} className="space-y-1">
1431                              {idx < revealedHints ? (
1432                                <div className="p-2.5 bg-amber-50/70 border border-amber-200 text-amber-950 text-xs ro
    unded-xl">
1433                                  <span className="font-bold">Hint #{idx + 1}: </span>
1434                                  {hint}
1435                                </div>
1436                              ) : null}
1437                            </div>
1438                          ))}
1439
1440                          {revealedHints < activeChallenge.hints.length && (
1441                            <button
1442                              onClick={() => setRevealedHints((prev) => prev + 1)}
1443                              className="w-full py-2 rounded-xl bg-slate-100 hover:bg-slate-200 text-slate-700 text-xs
    font-semibold transition-all cursor-pointer"
1444                            >
1445                              Reveal Hint #{revealedHints + 1}
1446                            </button>
1447                          )}
1448                        </div>
```

```
                        <button
                          onClick={handleApplyFix}
                          className="w-full py-2.5 px-4 rounded-xl bg-emerald-600 hover:bg-emerald-700 text-white font-bold text-xs shadow-md transition-all flex items-center justify-center gap-1.5 cursor-pointer"
                        >
                          <Wrench className="w-3.5 h-3.5" />
                          <span>Apply Recommended Fix to Editor</span>
                        </button>
                      </div>
                    )}

                    {/* TAB 4: ASK AI TUTOR CHAT */}
                    {activeExplainerTab === 'ask_ai' && (
                      <div className="space-y-4">
                        <div className="space-y-2.5 max-h-64 overflow-y-auto pr-1">
                          {tutorChatMessages.map((msg, i) => (
                            <div
                              key={i}
                              className={`p-3 rounded-xl text-xs leading-relaxed ${
                                msg.role === 'user'
                                  ? 'bg-indigo-600 text-white ml-6 rounded-tr-xs'
                                  : 'bg-slate-100 text-slate-800 mr-6 rounded-tl-xs'
                              }`}
                            >
                              <span className="block font-bold text-[10px] uppercase mb-0.5 opacity-70">
                                {msg.role === 'user' ? 'You' : 'AI Concept Tutor'}
                              </span>
                              {msg.text}
                            </div>
                          ))}

                          {isTutorThinking && (
                            <div className="p-3 rounded-xl bg-slate-100 text-slate-500 text-xs italic flex items-center gap-2">
                              <span className="w-2 h-2 rounded-full bg-indigo-600 animate-ping" />
```

```
                            <span>AI Tutor is preparing a conceptual explanation...</span>
                          </div>
                        )}
                      </div>

                      {/* Quick Prompt Suggestions */}
                      <div className="space-y-1.5 pt-2 border-t border-slate-100">
                        <span className="text-[10px] font-bold text-slate-400 uppercase tracking-wider">
                          Suggested questions:
                        </span>
                        <div className="flex flex-wrap gap-1.5">
                          <button
                            onClick={() => handleAskTutor('Explain this error like I am 5 years old')}
                            className="px-2 py-1 rounded-lg bg-slate-100 hover:bg-indigo-50 hover:text-indigo-700 te
xt-slate-600 text-[11px] transition-colors cursor-pointer"
                          >
                            Explain like I'm 5
                          </button>
                          <button
                            onClick={() => handleAskTutor(`How do other languages handle this error?`)}
                            className="px-2 py-1 rounded-lg bg-slate-100 hover:bg-indigo-50 hover:text-indigo-700 te
xt-slate-600 text-[11px] transition-colors cursor-pointer"
                          >
                            Multi-language comparison
                          </button>
                          <button
                            onClick={() => handleAskTutor('How does defensive programming prevent this?')}
                            className="px-2 py-1 rounded-lg bg-slate-100 hover:bg-indigo-50 hover:text-indigo-700 te
xt-slate-600 text-[11px] transition-colors cursor-pointer"
                          >
                            Defensive rules
                          </button>
                        </div>
                      </div>

                      {/* Chat Input */}
```

```
                      <div className="flex items-center gap-2">
                        <input
                          type="text"
                          value={tutorInput}
                          onChange={(e) => setTutorInput(e.target.value)}
                          onKeyDown={(e) => e.key === 'Enter' && handleAskTutor()}
                          placeholder={`Ask AI Tutor about this ${language.toUpperCase()} error...`}
                          className="flex-1 px-3 py-2 bg-slate-50 border border-slate-200 rounded-xl text-xs text-slate-900 focus:outline-none focus:ring-2 focus:ring-indigo-500"
                        />
                        <button
                          onClick={() => handleAskTutor()}
                          disabled={isTutorThinking || !tutorInput.trim()}
                          className="p-2 rounded-xl bg-indigo-600 hover:bg-indigo-700 text-white transition-all cursor-pointer disabled:opacity-50"
                        >
                          <Send className="w-3.5 h-3.5" />
                        </button>
                      </div>
                    </div>
                  )}
                </>
              ) : (
                /* Empty / Ready State when no error */
                <div className="py-12 px-4 text-center space-y-3">
                  <div className="w-12 h-12 rounded-2xl bg-emerald-50 text-emerald-600 mx-auto flex items-center justify-center">
                    <CheckCircle2 className="w-6 h-6" />
                  </div>
                  <div>
                    <h4 className="text-xs font-bold text-slate-900 uppercase tracking-wider">
                      Sandbox Ready ({language.toUpperCase()})
                    </h4>
                    <p className="text-xs text-slate-500 mt-1 max-w-xs mx-auto">
                      Type code in the editor or click "Run Code". When any syntax or runtime exception occurs, the AI Error Concept Explainer will instantly diagnose it!
```

```
                    </p>
                  </div>
                </div>
              )}
            </div>
          </div>
        </div>
      </div>
    </div>
  );
};
```

File [8/30]: src/app/api/code-practice/route.ts

TYPESCRIPT SOURCE

Description: *Code Practice & AI Error Tutor API Endpoint*

```typescript
import { NextRequest, NextResponse } from 'next/server';
import { explainError } from '@/lib/error-explainer';
import { getLLMConfig } from '@/lib/server/llm-provider';

export async function POST(req: NextRequest) {
  try {
    const body = await req.json();
    const { action, code, error, language, question, challengeId } = body;

    // Action 1: Explain Error
    if (action === 'explain-error') {
      const explanation = explainError(error || 'Unknown error', code || '', language || 'javascript', challengeId);
      return NextResponse.json({ success: true, explanation });
    }

    // Action 2: Ask AI Tutor a question about the error / code
    if (action === 'ask-tutor') {
      const llmConfig = getLLMConfig();

      // If user has live LLM configured (OpenAI, Gemini, Anthropic), we could optionally call it
      // Otherwise, return our comprehensive domain pedagogical response
      let tutorAnswer = '';
      const lowerQ = (question || '').toLowerCase();

      if (lowerQ.includes('typeof null') || lowerQ.includes('null object')) {
        tutorAnswer =
          'In JavaScript, `typeof null === "object"` is a famous historical bug dating back to the very first JS implementation in 1995. Values were stored with a type tag in the lower bits of their memory pointer: object references had a tag of `000`. Since `null` represented a null pointer (all zeroes), its tag was also `000`, causing `typeof` to mistakenly return "object". It cannot be fixed today because doing so would break millions of legacy websites!';
      } else if (lowerQ.includes('typescript') || lowerQ.includes('strictnullchecks')) {
        tutorAnswer =
```

```
          'TypeScript completely prevents this class of errors at compile time when you enable `"strictNullChecks": true` in `tsconfig.json`. In this mode, `undefined` and `null` have their own distinct types and cannot be assigned to objects without explicitly declaring union types (e.g. `User | undefined`). The TypeScript compiler will give you a red squiggly line before you even run your code!';
      } else if (lowerQ.includes('optional chaining') || lowerQ.includes('?.') || lowerQ.includes('nullish')) {
        tutorAnswer =
          'Optional chaining (`?.`) works by checking if the reference before `?.` is strictly null or undefined. If it is, JavaScript immediately halts the expression and returns `undefined` without executing any of the operations to the right. Nullish coalescing (`??`) pairs with it to provide a fallback value ONLY when the result is null or undefined (unlike `||` which triggers on 0, false, and empty strings).';
      } else if (lowerQ.includes('5 year old') || lowerQ.includes('simple') || lowerQ.includes('analogy')) {
        tutorAnswer =
          'Think of your code like a recipe in a kitchen. The recipe says: "Open the blue cookie jar, take out the chocolate cookie, and bite it." But when you open the cupboard, there is no blue jar at all (`undefined`). Because there is no jar, you cannot reach inside to grab a cookie. Trying to bite a cookie that does not exist makes your computer confused and cry ("TypeError!"). Optional chaining (`?.`) is like saying: "IF there is a blue jar, look inside; if not, just eat an apple instead!"';
      } else {
        tutorAnswer =
          `Great question! When debugging this: always check the values of variables right before the crash line. In this specific scenario, the crash occurred because the execution context was expecting an object with properties, but received a primitive empty state (undefined/null). You can use \`console.log({ variableName })\` or a breakpoint to inspect the state before invoking properties.`;
      }

      return NextResponse.json({
        success: true,
        answer: tutorAnswer,
        model: llmConfig.provider !== 'none' ? llmConfig.model : 'RepoPilot AI Concept Tutor',
      });
    }

    return NextResponse.json({ success: false, error: 'Invalid action' }, { status: 400 });
  } catch (err: unknown) {
    const message = err instanceof Error ? err.message : String(err);
    return NextResponse.json({ success: false, error: message }, { status: 500 });
  }
}
```

File [9/30]: src/app/page.tsx

TSX SOURCE

Description: *Main View Orchestrator & Tab Router*

```tsx
'use client';

import React, { useState } from 'react';
import { Sidebar } from '@/components/layout/Sidebar';
import { TopNavbar } from '@/components/layout/TopNavbar';
import { DemoTourBanner, TOUR_STEPS } from '@/components/layout/DemoTourBanner';
import { DashboardView } from '@/components/views/DashboardView';
import { RepositoryView } from '@/components/views/RepositoryView';
import { InvestigationView } from '@/components/views/InvestigationView';
import { FixView } from '@/components/views/FixView';
import { LiveBugFixPlayer } from '@/components/views/LiveBugFixPlayer';
import { VerificationView } from '@/components/views/VerificationView';
import { HistoryView } from '@/components/views/HistoryView';
import { LoginLandingView } from '@/components/views/LoginLandingView';
import { AcademyView } from '@/components/views/AcademyView';
import { CodePracticeView } from '@/components/views/CodePracticeView';
import { PullRequestModal } from '@/components/modals/PullRequestModal';
import { SettingsModal } from '@/components/modals/SettingsModal';
import { YourRepositoriesModal } from '@/components/modals/YourRepositoriesModal';
import { ACTIVE_REPO } from '@/lib/mock-data';
import { NavTab, RepositoryData } from '@/lib/types';
import { useAuth } from '@/lib/auth-context';
import { Loader2 } from 'lucide-react';

export default function Home() {
  const {
    isAuthenticated,
    isLoading,
    isSettingsOpen,
    openSettings,
    closeSettings,
```

```
    settingsTab,
  } = useAuth();

  const [activeTab, setActiveTab] = useState<NavTab>('dashboard');
  const [activeRepo, setActiveRepo] = useState<RepositoryData>(ACTIVE_REPO);
  const [isTouring, setIsTouring] = useState<boolean>(false);
  const [currentTourStep, setCurrentTourStep] = useState<number>(0);
  const [isPRModalOpen, setIsPRModalOpen] = useState<boolean>(false);
  const [isConnectRepoOpen, setIsConnectRepoOpen] = useState<boolean>(false);
  const [hasVerifiedFix, setHasVerifiedFix] = useState<boolean>(false);
  const [practiceChallengeId, setPracticeChallengeId] = useState<string>('type-error-null');
  const [, setEngineRefresh] = useState(0);

  // 1. Loading State while session is verified
  if (isLoading) {
    return (
      <div className="min-h-screen bg-slate-50 flex flex-col items-center justify-center p-6 space-y-3">
        <Loader2 className="w-8 h-8 text-indigo-600 animate-spin" />
        <span className="text-xs font-semibold text-slate-500 font-mono">
          Connecting securely...
        </span>
      </div>
    );
  }

  // 2. Unauthenticated state: Route protection -> Landing Login View
  if (!isAuthenticated) {
    return <LoginLandingView />;
  }

  // 3. Authenticated state: RepoPilot Workspace & Dashboard
  return (
    <div className="min-h-screen bg-slate-50 text-slate-900 flex flex-row selection:bg-indigo-100 selection:text-ind
igo-900">
      {/* Sidebar Navigation */}
```

```
      <Sidebar
        activeTab={activeTab}
        onTabChange={(tab) => {
          setActiveTab(tab);
          if (tab === 'verification') {
            setHasVerifiedFix(true);
          }
        }}
        openSettings={() => openSettings('engine')}
        hasVerifiedFix={hasVerifiedFix}
      />

      {/* Main Content Area */}
      <div className="flex-1 flex flex-col min-w-0">
        <TopNavbar
          openSettings={(tab) => openSettings(tab || 'engine')}
          onOpenRepoModal={() => setIsConnectRepoOpen(true)}
          currentRepo={activeRepo}
        />

        {/* Main View Router */}
        <main className="flex-1 p-6 md:p-8 overflow-y-auto">
          {activeTab === 'dashboard' && (
            <DashboardView
              onNavigate={(tab) => {
                setActiveTab(tab);
                if (tab === 'verification') setHasVerifiedFix(true);
              }}
              currentRepo={activeRepo}
              onOpenRepoModal={() => setIsConnectRepoOpen(true)}
            />
          )}

          {activeTab === 'repositories' && (
            <RepositoryView
```

```
            onNavigate={setActiveTab}
            currentRepo={activeRepo}
            onOpenRepoModal={() => setIsConnectRepoOpen(true)}
            onInvestigateIssue={() => {
              setActiveTab('issues');
              if (isTouring) setCurrentTourStep(2);
            }}
          />
        )}

        {activeTab === 'issues' && (
          <InvestigationView
            onGenerateFix={() => {
              setActiveTab('fix');
              if (isTouring) setCurrentTourStep(3);
            }}
            onNavigate={setActiveTab}
          />
        )}

        {activeTab === 'live-fix' && (
          <LiveBugFixPlayer
            onNavigate={setActiveTab}
            onVerifyFix={() => {
              setActiveTab('verification');
              setHasVerifiedFix(true);
              if (isTouring) setCurrentTourStep(4);
            }}
          />
        )}

        {activeTab === 'fix' && (
          <FixView
            onVerifyFix={() => {
```

```
                setActiveTab('verification');
                setHasVerifiedFix(true);
                if (isTouring) setCurrentTourStep(4);
              }}
              onNavigate={setActiveTab}
            />
          )}

          {activeTab === 'verification' && (
            <VerificationView
              onCreatePR={() => {
                setIsPRModalOpen(true);
                setHasVerifiedFix(true);
              }}
              onNavigate={setActiveTab}
            />
          )}

          {activeTab === 'history' && (
            <HistoryView onNavigate={setActiveTab} />
          )}

          {activeTab === 'courses' && (
            <AcademyView
              onNavigate={setActiveTab}
              onOpenPracticeChallenge={(chId) => {
                setPracticeChallengeId(chId);
                setActiveTab('practice');
              }}
            />
          )}

          {activeTab === 'practice' && (
            <CodePracticeView
              onNavigate={setActiveTab}
```

```
              initialChallengeId={practiceChallengeId}
              onOpenLesson={(courseId, lessonId) => {
                setActiveTab('courses');
              }}
            />
          )}
        </main>
      </div>

      {/* Your Authorized Repositories Selection Modal */}
      <YourRepositoriesModal
        isOpen={isConnectRepoOpen}
        onClose={() => setIsConnectRepoOpen(false)}
        currentRepoId={activeRepo.id}
        onSelectRepo={(newRepo) => {
          setActiveRepo(newRepo);
          setActiveTab('repositories');
        }}
      />

      {/* Pull Request Created Modal */}
      <PullRequestModal
        isOpen={isPRModalOpen}
        onClose={() => setIsPRModalOpen(false)}
        onViewHistory={() => {
          setActiveTab('history');
          setIsPRModalOpen(false);
          setIsTouring(false);
        }}
      />

      {/* Settings / AI Engine & Connected Accounts Modal */}
      <SettingsModal
        isOpen={isSettingsOpen}
```

```
204           onClose={closeSettings}
205           initialTab={settingsTab}
206           onUpdated={() => setEngineRefresh((prev) => prev + 1)}
207         />
208       </div>
209     );
210   }
```

File [10/30]: src/app/layout.tsx

TSX SOURCE

Description: *Root HTML & Next.js Metadata Layout*

```tsx
import type { Metadata } from "next";
import { Geist, Geist_Mono } from "next/font/google";
import "./globals.css";
import { AuthProviderComponent } from "@/lib/auth-context";

const geistSans = Geist({
  variable: "--font-geist-sans",
  subsets: ["latin"],
});

const geistMono = Geist_Mono({
  variable: "--font-geist-mono",
  subsets: ["latin"],
});

export const metadata: Metadata = {
  title: "RepoPilot AI — From bug report to verified fix",
  description: "An AI developer agent that understands your repository, diagnoses bugs, generates fixes, writes regr
ession tests, and verifies the solution.",
};

export default function RootLayout({
  children,
}: Readonly<{
  children: React.ReactNode;
}>) {
  return (
    <html
      lang="en"
      className={`${geistSans.variable} ${geistMono.variable} h-full antialiased`}
    >
```

```
      <body className="min-h-full flex flex-col bg-slate-50 text-slate-900 selection:bg-indigo-100 selection:text-indigo-900">
        <AuthProviderComponent>{children}</AuthProviderComponent>
      </body>
    </html>
  );
}
```

File [11/30]: src/components/layout/Sidebar.tsx

TSX SOURCE

Description: *Navigation Sidebar with Academy & Practice Lab Badges*

```tsx
'use client';

import React from 'react';
import {
  LayoutDashboard,
  GitBranch,
  AlertTriangle,
  Wrench,
  ShieldCheck,
  History,
  Cpu,
  Sparkles,
  ChevronRight,
  Terminal,
  Lock,
  Smartphone,
  User,
  Zap,
  GraduationCap,
  Code2,
} from 'lucide-react';
import { NavTab } from '@/lib/types';
import { aiEngine } from '@/lib/ai-engine';
import { useAuth } from '@/lib/auth-context';

interface SidebarProps {
  activeTab: NavTab;
  onTabChange: (tab: NavTab) => void;
  openSettings: () => void;
  hasVerifiedFix?: boolean;
}
```

```
32
33  export const Sidebar: React.FC<SidebarProps> = ({
34    activeTab,
35    onTabChange,
36    openSettings,
37    hasVerifiedFix = false,
38  }) => {
39    const engineStatus = aiEngine.getStatus();
40    const { user, isAuthenticated } = useAuth();
41
42    const navItems = [
43      {
44        id: 'dashboard' as NavTab,
45        label: 'Dashboard',
46        icon: LayoutDashboard,
47        badge: null,
48      },
49      {
50        id: 'courses' as NavTab,
51        label: 'Learning Academy',
52        icon: GraduationCap,
53        badge: '5 Courses',
54        badgeColor: 'bg-indigo-50 text-indigo-700 border border-indigo-200 font-semibold',
55      },
56      {
57        id: 'practice' as NavTab,
58        label: 'Code Practice Lab',
59        icon: Code2,
60        badge: 'AI Tutor',
61        badgeColor: 'bg-emerald-600 text-white font-extrabold shadow-xs',
62      },
63      {
64        id: 'repositories' as NavTab,
65        label: 'Repositories',
66        icon: GitBranch,
```

```
      badge: '3',
    },
    {
      id: 'issues' as NavTab,
      label: 'Issues',
      icon: AlertTriangle,
      badge: '1 High',
      badgeColor: 'bg-rose-50 text-rose-700 border border-rose-200 font-semibold',
    },
    {
      id: 'live-fix' as NavTab,
      label: 'Live Fix & Learn',
      icon: Zap,
      badge: 'Interactive',
      badgeColor: 'bg-indigo-600 text-white font-extrabold shadow-xs',
    },
    {
      id: 'fix' as NavTab,
      label: 'Fix & Patch',
      icon: Wrench,
      badge: 'Ready',
      badgeColor: 'bg-emerald-50 text-emerald-700 border border-emerald-200 font-semibold',
    },
    {
      id: 'verification' as NavTab,
      label: 'Verification',
      icon: ShieldCheck,
      badge: hasVerifiedFix ? 'Passed' : 'Pending',
      badgeColor: hasVerifiedFix
        ? 'bg-emerald-50 text-emerald-700 border border-emerald-200 font-semibold'
        : 'bg-amber-50 text-amber-800 border border-amber-200 font-semibold',
    },
    {
      id: 'history' as NavTab,
```

```
      label: 'History',
      icon: History,
      badge: null,
    },
  ];

  return (
    <aside className="w-64 bg-white border-r border-slate-200 flex flex-col justify-between select-none h-screen sticky top-0 z-30 shadow-xs">
      <div>
        {/* Brand / Logo */}
        <div className="p-5 border-b border-slate-200 flex items-center justify-between">
          <div className="flex items-center gap-3">
            <div className="w-9 h-9 rounded-xl bg-gradient-to-tr from-indigo-600 via-indigo-500 to-cyan-500 flex items-center justify-center shadow-md shadow-indigo-500/20 ring-1 ring-black/5">
              <Terminal className="w-5 h-5 text-white" />
            </div>
            <div>
              <div className="flex items-center gap-1.5">
                <span className="font-bold text-base tracking-tight text-slate-900">RepoPilot</span>
                <span className="text-[10px] uppercase font-bold tracking-wider px-1.5 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-indigo-200">
                  AI
                </span>
              </div>
              <p className="text-[11px] text-slate-500 font-medium truncate max-w-[125px]">
                Autonomous Dev Agent
              </p>
            </div>
          </div>
        </div>

        {/* Navigation Items */}
        <div className="px-3 py-4 space-y-1">
          <div className="px-3 pb-2 text-[10px] font-semibold uppercase tracking-wider text-slate-400">
            Workspace
```

```
          </div>

          {navItems.map((item) => {
            const Icon = item.icon;
            const isActive = activeTab === item.id;
            return (
              <button
                key={item.id}
                onClick={() => onTabChange(item.id)}
                className={`w-full flex items-center justify-between px-3 py-2.5 rounded-lg text-sm font-medium tran
sition-all group ${
                  isActive
                    ? 'bg-indigo-50 text-indigo-700 font-semibold border border-indigo-200/80 shadow-xs'
                    : 'text-slate-600 hover:text-slate-900 hover:bg-slate-100/80'
                }`}
              >
                <div className="flex items-center gap-3">
                  <Icon
                    className={`w-4 h-4 transition-colors ${
                      isActive ? 'text-indigo-600' : 'text-slate-500 group-hover:text-slate-700'
                    }`}
                  />
                  <span>{item.label}</span>
                </div>

                <div className="flex items-center gap-1.5">
                  {item.badge && (
                    <span
                      className={`text-[10px] px-1.5 py-0.5 rounded-md font-medium ${
                        item.badgeColor || 'bg-slate-100 text-slate-600 border border-slate-200'
                      }`}
                    >
                      {item.badge}
                    </span>
                  )}
```

```
                  {isActive && <ChevronRight className="w-3.5 h-3.5 text-indigo-600" />}
                </div>
              </button>
            );
          })}
        </div>
      </div>

      {/* Bottom Auth & AI Engine Section */}
      <div className="p-3 border-t border-slate-200 bg-slate-50/60 space-y-2">
        {/* User Auth Card */}
        {isAuthenticated && user && (
          <div
            onClick={openSettings}
            className="p-2.5 rounded-xl border border-slate-200 bg-white hover:bg-slate-50 cursor-pointer transition-all hover:border-indigo-300 shadow-2xs flex items-center justify-between group"
            title="Click to view Connected Accounts & Settings"
          >
            <div className="flex items-center gap-2.5 min-w-0">
              {user.avatarUrl ? (
                <img
                  src={user.avatarUrl}
                  alt={user.name}
                  className="w-7 h-7 rounded-full object-cover ring-1 ring-slate-200 shrink-0"
                />
              ) : (
                <div className="w-7 h-7 rounded-full bg-indigo-100 text-indigo-700 font-bold flex items-center justify-center text-xs shrink-0">
                  {user.name.charAt(0).toUpperCase()}
                </div>
              )}
              <div className="min-w-0">
                <div className="text-xs font-semibold text-slate-900 truncate group-hover:text-indigo-600 transition-colors">
                  {user.name}
                </div>
```

```
                <div className="text-[10px] text-slate-500 truncate">
                  {user.connectedAccounts?.github?.connected
                    ? `GitHub (@${user.connectedAccounts.github.username})`
                    : user.email || 'OAuth Session'}
                </div>
              </div>
            </div>

            <span className="w-2 h-2 rounded-full bg-emerald-500 shadow-xs shrink-0"></span>
          </div>
        )}

        {/* AI Engine Card */}
        <div
          onClick={openSettings}
          className="p-2.5 rounded-xl border border-slate-200 bg-white hover:bg-slate-50 cursor-pointer transition-all hover:border-slate-300 shadow-xs group"
        >
          <div className="flex items-center justify-between mb-1">
            <div className="flex items-center gap-1.5">
              <Cpu className="w-3.5 h-3.5 text-indigo-600" />
              <span className="text-[11px] font-semibold text-slate-700">AI Engine</span>
            </div>
            <span className="inline-flex items-center gap-1 text-[10px] font-semibold text-emerald-700 bg-emerald-50 px-1.5 py-0.5 rounded-full border border-emerald-200">
              <span className="w-1.5 h-1.5 rounded-full bg-emerald-500 animate-pulse"></span>
              {engineStatus.label}
            </span>
          </div>

          <div className="flex items-center justify-between text-[10px] text-slate-500">
            <span className="truncate max-w-[140px]">Zero-Fail AST Analyzer</span>
            <Sparkles className="w-3 h-3 text-indigo-500 group-hover:rotate-12 transition-transform" />
          </div>
        </div>
      </div>
```

```
      </aside>
    );
};
```

File [12/30]: src/components/layout/TopNavbar.tsx

TSX SOURCE

Description: *Application Header, User Profile & Status Bar*

```tsx
'use client';

import React, { useState, useRef, useEffect } from 'react';
import {
  GitBranch,
  Settings,
  Code2,
  ChevronDown,
  LogOut,
  User,
  ShieldCheck,
  Link2,
  ExternalLink,
} from 'lucide-react';
import { RepositoryData } from '@/lib/types';
import { useAuth } from '@/lib/auth-context';

interface TopNavbarProps {
  onStartGuidedTour?: () => void;
  openSettings: (tab?: 'engine' | 'connected_accounts') => void;
  onOpenRepoModal?: () => void;
  currentRepo?: RepositoryData;
  isTouring?: boolean;
}

// Mini SVGs for providers
const MiniGoogleIcon: React.FC<{ className?: string }> = ({ className = 'w-3.5 h-3.5' }) => (
  <svg className={className} viewBox="0 0 24 24">
    <path fill="#4285F4" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z" />
    <path fill="#34A853" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z" />
```

```
    <path fill="#FBBC05" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12 s.43 3.45 1.18 4.94l2.85-2.22.81-.63z" />
    <path fill="#EA4335" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z" />
  </svg>
);

const MiniGithubIcon: React.FC<{ className?: string }> = ({ className = 'w-3.5 h-3.5' }) => (
  <svg className={className} fill="currentColor" viewBox="0 0 24 24">
    <path fillRule="evenodd" clipRule="evenodd" d="M12 2C6.477 2 2 6.484 2 12.017c0 4.425 2.865 8.18 6.839 9.504.5.092.682-.217.682-.483 0-.237-.008-.868-.013-1.703-2.782.605-3.369-1.343-3.369-1.343-.454-1.158-1.11-1.466-1.11-1.466-.908-.62.069-.608.069-.608 1.003.07 1.53 1.032 1.53 1.032.892 1.53 2.341 1.088 2.91.832.092-.647.35-1.088.636-1.338-2.22-.253-4.555-1.113-4.555-4.951 0-1.093.39-1.988 1.029-2.688-.103-.253-.446-1.272.098-2.65 0 0 .84-.27 2.75 1.026A9.564 9.564 0 0112 6.844c.85.004 1.705.115 2.504.337 1.909-1.296 2.747-1.027 2.747-1.027.546 1.379.202 2.398.1 2.651.64.7 1.028 1.595 1.028 2.688 0 3.848-2.339 4.695-4.566 4.943.359.309.678.92.678 1.855 0 1.338-.012 2.419-.012 2.747 0 .268.18.58.688.482A10.019 10.019 0 0022 12.017C22 6.484 17.522 2 12 2z" />
  </svg>
);

export const TopNavbar: React.FC<TopNavbarProps> = ({
  openSettings,
  onOpenRepoModal,
  currentRepo,
}) => {
  const { user, isAuthenticated, logout, connectGitHub } = useAuth();
  const [isUserMenuOpen, setIsUserMenuOpen] = useState(false);
  const userMenuRef = useRef<HTMLDivElement>(null);

  const repoName = currentRepo?.name || 'Nexora-X-Hub';
  const repoBranch = currentRepo?.branch || 'main';

  const isGitHubConnected = Boolean(user?.connectedAccounts?.github?.connected);
  const ghUsername = user?.connectedAccounts?.github?.username;

  useEffect(() => {
    const handleClickOutside = (event: MouseEvent) => {
      if (userMenuRef.current && !userMenuRef.current.contains(event.target as Node)) {
        setIsUserMenuOpen(false);
      }
```

```
    };
    document.addEventListener('mousedown', handleClickOutside);
    return () => document.removeEventListener('mousedown', handleClickOutside);
  }, []);

  return (
    <header className="h-14 border-b border-slate-200 bg-white/95 backdrop-blur-md px-6 flex items-center justify-between sticky top-0 z-20 shadow-xs">
      <div className="flex items-center gap-4">
        {/* Repo selector pill (Clickable to switch / connect repo) */}
        <button
          onClick={onOpenRepoModal}
          className="group flex items-center gap-2 px-3 py-1.5 rounded-lg bg-slate-100 hover:bg-indigo-50 border border-slate-200 hover:border-indigo-300 text-xs font-medium text-slate-800 transition-all shadow-2xs cursor-pointer"
          title="Click to switch repository or view authorized repositories"
        >
          <div className="w-2 h-2 rounded-full bg-indigo-600 animate-pulse"></div>
          <span className="font-semibold text-slate-900 group-hover:text-indigo-700">{repoName}</span>
          <span className="text-slate-400">/</span>
          <div className="flex items-center gap-1 text-slate-600 group-hover:text-indigo-600">
            <GitBranch className="w-3 h-3 text-indigo-600" />
            <span>{repoBranch}</span>
          </div>
          <ChevronDown className="w-3 h-3 text-slate-400 group-hover:text-indigo-600 ml-0.5 transition-transform group-hover:translate-y-0.5" />
        </button>

        {/* Tagline */}
        <div className="hidden md:flex items-center gap-2 text-xs text-slate-500 font-medium border-l border-slate-200 pl-4">
          <span className="text-indigo-600 font-semibold">RepoPilot:</span>
          <span>From bug report to verified fix.</span>
        </div>
      </div>

      <div className="flex items-center gap-3">
        {/* Hackathon Track Badge */}
```

```
        <div className="hidden lg:inline-flex items-center gap-1.5 px-2.5 py-1 rounded-md text-[11px] font-semibold bg-indigo-50 text-indigo-700 border border-indigo-200">
          <Code2 className="w-3 h-3 text-indigo-600" />
          <span>iQOO Hackathon · Developer Tools</span>
        </div>

        {/* Account Menu as specified in Requirements */}
        {isAuthenticated && user && (
          <div className="relative" ref={userMenuRef}>
            <button
              onClick={() => setIsUserMenuOpen(!isUserMenuOpen)}
              className="flex items-center gap-2 px-2.5 py-1.5 rounded-xl border border-slate-200 bg-white hover:bg-slate-50 text-slate-800 text-xs font-medium transition-all shadow-2xs hover:border-slate-300 cursor-pointer"
            >
              {/* User Avatar with provider overlay */}
              <div className="relative">
                {user.avatarUrl ? (
                  <img
                    src={user.avatarUrl}
                    alt={user.name}
                    className="w-6 h-6 rounded-full object-cover ring-1 ring-slate-200"
                  />
                ) : (
                  <div className="w-6 h-6 rounded-full bg-indigo-100 text-indigo-700 font-bold flex items-center justify-center text-[11px]">
                    {user.name.charAt(0).toUpperCase()}
                  </div>
                )}

                {/* Provider micro icon */}
                <div className="absolute -bottom-1 -right-1 w-3.5 h-3.5 rounded-full bg-white shadow-2xs flex items-center justify-center border border-slate-200">
                  {user.providers.includes('google') ? (
                    <MiniGoogleIcon className="w-2.5 h-2.5" />
                  ) : (
                    <MiniGithubIcon className="w-2.5 h-2.5 text-slate-900" />
```

```
                  )}
                </div>
              </div>

              <span className="font-semibold text-slate-900 text-xs max-w-[120px] truncate">
                {user.name}
              </span>

              <ChevronDown className="w-3 h-3 text-slate-400 ml-0.5" />
            </button>

            {/* Account Menu Dropdown matching specification */}
            {isUserMenuOpen && (
              <div className="absolute right-0 mt-2 w-64 rounded-2xl bg-white border border-slate-200 shadow-xl overflow-hidden py-1.5 z-50 animate-in fade-in slide-in-from-top-2">
                {/* User Summary */}
                <div className="p-3 border-b border-slate-100 bg-slate-50/70">
                  <div className="flex items-center gap-2 mb-0.5">
                    <span className="font-bold text-xs text-slate-900 truncate">{user.name}</span>
                    <span className="text-[10px] uppercase font-bold tracking-wider px-1.5 py-0.2 rounded-full bg-emerald-50 text-emerald-700 border border-emerald-200">
                      Active
                    </span>
                  </div>
                  {user.email && (
                    <div className="text-[11px] text-slate-500 truncate font-mono">
                      {user.email}
                    </div>
                  )}
                </div>

                <div className="p-1 space-y-0.5 text-xs text-slate-700">
                  {/* 1. Profile */}
                  <div className="px-3 py-2 text-slate-500 font-medium flex items-center justify-between text-[11px]">
                    <span className="flex items-center gap-2">
```

```
                        <User className="w-3.5 h-3.5 text-indigo-600" />
                        <span>Profile</span>
                      </span>
                      <span className="text-[10px] text-slate-400 font-mono truncate max-w-[100px]">
                        {user.id}
                      </span>
                    </div>

                    {/* 2. Connected Accounts */}
                    <button
                      onClick={() => {
                        setIsUserMenuOpen(false);
                        openSettings('connected_accounts');
                      }}
                      className="w-full flex items-center justify-between px-3 py-2 rounded-lg hover:bg-slate-100 tran
sition-colors text-left font-medium cursor-pointer"
                    >
                      <span className="flex items-center gap-2">
                        <Link2 className="w-3.5 h-3.5 text-indigo-600" />
                        <span>Connected Accounts</span>
                      </span>
                      <span className="text-[10px] text-indigo-600 font-semibold">View</span>
                    </button>

                    {/* 3. GitHub Status */}
                    <div className="px-3 py-2 flex items-center justify-between rounded-lg bg-slate-50/60 border borde
r-slate-100">
                      <span className="flex items-center gap-2 text-slate-800 font-medium">
                        <MiniGithubIcon className="w-3.5 h-3.5" />
                        <span>GitHub</span>
                      </span>

                      {isGitHubConnected ? (
                        <span className="text-[11px] font-semibold text-emerald-700 bg-emerald-50 px-2 py-0.5 rounded-
full border border-emerald-200 flex items-center gap-1">
                          <span className="w-1.5 h-1.5 rounded-full bg-emerald-500" />
```

```
                        Connected
                      </span>
                    ) : (
                      <button
                        onClick={() => {
                          setIsUserMenuOpen(false);
                          connectGitHub();
                        }}
                        className="text-[11px] font-semibold text-indigo-700 bg-indigo-50 hover:bg-indigo-100 px-2 py-0.5 rounded-md border border-indigo-200 cursor-pointer"
                      >
                        + Connect
                      </button>
                    )}
                  </div>

                  {/* 4. Settings */}
                  <button
                    onClick={() => {
                      setIsUserMenuOpen(false);
                      openSettings('engine');
                    }}
                    className="w-full flex items-center gap-2 px-3 py-2 rounded-lg hover:bg-slate-100 transition-colors text-left font-medium cursor-pointer"
                  >
                    <Settings className="w-3.5 h-3.5 text-slate-600" />
                    <span>Settings</span>
                  </button>

                  <div className="border-t border-slate-100 my-1" />

                  {/* 5. Sign Out */}
                  <button
                    onClick={() => {
                      setIsUserMenuOpen(false);
                      logout();
```

```
                    }}
                    className="w-full flex items-center gap-2 px-3 py-2 rounded-lg text-rose-600 hover:text-rose-700 hover:bg-rose-50 transition-colors text-left font-semibold cursor-pointer"
                  >
                    <LogOut className="w-3.5 h-3.5" />
                    <span>Sign Out</span>
                  </button>
                </div>
              </div>
            )}
          </div>
        )}

        {/* Global Settings button */}
        <button
          onClick={() => openSettings('engine')}
          className="p-2 rounded-lg bg-white border border-slate-200 text-slate-600 hover:text-slate-900 hover:bg-slate-100 transition-colors shadow-xs cursor-pointer"
          title="Settings & AI Engine Configuration"
        >
          <Settings className="w-4 h-4" />
        </button>
      </div>
    </header>
  );
};
```

File [13/30]: src/components/views/LoginLandingView.tsx

TSX SOURCE

Description: *Login, Demo Quick Access & Landing Hero View*

```tsx
'use client';

import React, { useState, useEffect } from 'react';
import {
  Terminal,
  ShieldCheck,
  Sparkles,
  Loader2,
  AlertCircle,
  Code2,
  Lock,
} from 'lucide-react';
import { useAuth } from '@/lib/auth-context';

// Official SVG for Google
const GoogleIcon: React.FC<{ className?: string }> = ({ className = 'w-5 h-5' }) => (
  <svg className={className} viewBox="0 0 24 24">
    <path
      fill="#4285F4"
      d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z"
    />
    <path
      fill="#34A853"
      d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"
    />
    <path
      fill="#FBBC05"
      d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-2.09V7.06H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.94l2.85-2.22.81-.63z"
    />
    <path
```

```
      fill="#EA4335"
      d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45 2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.06l3.66 2.84c.87-2.6 3.3-4.52 6.16-4.52z"
    />
  </svg>
);

// Official SVG for GitHub
const GithubIcon: React.FC<{ className?: string }> = ({ className = 'w-5 h-5' }) => (
  <svg className={className} fill="currentColor" viewBox="0 0 24 24">
    <path
      fillRule="evenodd"
      clipRule="evenodd"
      d="M12 2C6.477 2 2 6.484 2 12.017c0 4.425 2.865 8.18 6.839 9.504.5.092.682-.217.682-.483 0-.237-.008-.868-.013-1.703-2.782.605-3.369-1.343-3.369-1.343-.454-1.158-1.11-1.466-1.11-1.466-.908-.62.069-.608.069-.608 1.003.07 1.531.032 1.531 1.032.892 1.531 2.341 1.088 2.91.832.092-.647.35-1.088.636-1.338-2.22-.253-4.555-1.113-4.555-4.951 0-1.093.39-1.988 1.029-2.688-.103-.253-.446-1.272.098-2.65 0 0 .84-.27 2.75 1.026A9.564 9.564 0 0112 6.844c.85.004 1.705.115 2.504.337 1.909-1.296 2.747-1.027 2.747-1.027.546 1.379.202 2.398.1 2.651.64.7 1.028 1.595 1.028 2.688 0 3.848-2.339 4.695-4.566 4.943.359.309.678.92.678 1.855 0 1.338-.012 2.419-.012 2.747 0 .268.18.58.688.482A10.019 10.019 0 0022 12.017C22 6.484 17.522 2 12 2z"
    />
  </svg>
);

export const LoginLandingView: React.FC = () => {
  const { loginWithGoogle, loginWithGitHub } = useAuth();
  const [connectingProvider, setConnectingProvider] = useState<'google' | 'github' | null>(null);
  const [authError, setAuthError] = useState<string | null>(null);

  useEffect(() => {
    // Read any OAuth error query parameter
    if (typeof window !== 'undefined') {
      const params = new URLSearchParams(window.location.search);
      const err = params.get('auth_error') || params.get('github_error');
      if (err) setAuthError(err);
    }
  }, []);
```

```
  const handleGoogleClick = () => {
    setConnectingProvider('google');
    loginWithGoogle();
  };

  const handleGitHubClick = () => {
    setConnectingProvider('github');
    loginWithGitHub();
  };

  return (
    <div className="min-h-screen bg-slate-50 flex flex-col justify-between selection:bg-indigo-100 selection:text-indigo-900">
      {/* Top Simple Header */}
      <header className="h-16 px-6 md:px-12 flex items-center justify-between border-b border-slate-200/80 bg-white/80 backdrop-blur-md">
        <div className="flex items-center gap-3">
          <div className="w-9 h-9 rounded-xl bg-gradient-to-tr from-indigo-600 via-indigo-500 to-cyan-500 flex items-center justify-center shadow-md shadow-indigo-500/20">
            <Terminal className="w-5 h-5 text-white" />
          </div>
          <div className="flex items-center gap-2">
            <span className="font-bold text-base tracking-tight text-slate-900">RepoPilot</span>
            <span className="text-[10px] uppercase font-bold tracking-wider px-1.5 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-indigo-200">
              AI
            </span>
          </div>
        </div>

        <div className="flex items-center gap-2 text-xs font-semibold text-slate-500">
          <Lock className="w-3.5 h-3.5 text-indigo-600" />
          <span>OAuth 2.0 Secure Session</span>
        </div>
      </header>

      {/* Main Authentication Hero Container */}
```

```
95        <main className="flex-1 flex items-center justify-center p-6 my-auto">
96          <div className="w-full max-w-md bg-white border border-slate-200 rounded-3xl p-8 md:p-10 shadow-xl shadow-sl
    ate-200/50 space-y-8 animate-in fade-in duration-300">
97            {/* Logo & Headline */}
98            <div className="text-center space-y-3">
99              <div className="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-indigo-50 border border-indigo-
    200 text-indigo-700 text-xs font-semibold mb-1 shadow-2xs">
100               <Sparkles className="w-3.5 h-3.5 text-indigo-600" />
101               <span>RepoPilot AI</span>
102             </div>
103
104             <h1 className="text-3xl font-extrabold text-slate-900 tracking-tight leading-tight">
105               Your AI software engineer.
106             </h1>
107
108             <p className="text-sm text-slate-500 font-normal leading-relaxed max-w-sm mx-auto">
109               Understand repositories. Fix bugs. Verify the solution.
110             </p>
111           </div>
112
113           {/* Error Banner if any */}
114           {authError && (
115             <div className="p-3.5 rounded-xl bg-rose-50 border border-rose-200 text-rose-800 text-xs flex items-star
    t gap-2.5 animate-in fade-in">
116               <AlertCircle className="w-4 h-4 text-rose-600 shrink-0 mt-0.5" />
117               <div className="flex-1">
118                 <span className="font-semibold block mb-0.5">Authentication notice</span>
119                 <span>{authError}</span>
120               </div>
121             </div>
122           )}
123
124           {/* Authentication Action Buttons */}
125           <div className="space-y-3 pt-2">
126             {/* Continue with Google */}
127             <button
```

```
128               onClick={handleGoogleClick}
129               disabled={connectingProvider !== null}
130               className="w-full h-12 flex items-center justify-center gap-3 px-5 rounded-2xl border border-slate-300 bg-white hover:bg-slate-50 text-slate-800 font-semibold text-sm shadow-xs hover:border-slate-400 hover:shadow-sm transition-all cursor-pointer disabled:opacity-60 group"
131             >
132               {connectingProvider === 'google' ? (
133                 <>
134                   <Loader2 className="w-4 h-4 text-indigo-600 animate-spin" />
135                   <span className="text-slate-600">Connecting securely...</span>
136                 </>
137               ) : (
138                 <>
139                   <GoogleIcon className="w-5 h-5 group-hover:scale-105 transition-transform" />
140                   <span>Continue with Google</span>
141                 </>
142               )}
143             </button>
144
145             {/* Clean Divider */}
146             <div className="relative flex items-center justify-center py-2">
147               <div className="border-t border-slate-200 w-full" />
148               <span className="bg-white px-3 text-xs text-slate-400 uppercase tracking-wider font-semibold absolute">
149                 OR
150               </span>
151             </div>
152
153             {/* Continue with GitHub */}
154             <button
155               onClick={handleGitHubClick}
156               disabled={connectingProvider !== null}
157               className="w-full h-12 flex items-center justify-center gap-3 px-5 rounded-2xl bg-slate-900 hover:bg-slate-800 text-white font-semibold text-sm shadow-md shadow-slate-900/10 hover:shadow-slate-900/20 transition-all cursor-pointer disabled:opacity-60 group"
158             >
159               {connectingProvider === 'github' ? (
```

```
                <>
                  <Loader2 className="w-4 h-4 text-white animate-spin" />
                  <span>Connecting GitHub...</span>
                </>
              ) : (
                <>
                  <GithubIcon className="w-5 h-5 text-white group-hover:scale-105 transition-transform" />
                  <span>Continue with GitHub</span>
                </>
              )}
            </button>

            {/* Instant Demo Access (Explore Learning & Code Practice Space) */}
            <a
              href="/api/auth/dev-mock?provider=google&simulate=true"
              className="w-full h-11 flex items-center justify-center gap-2 px-5 rounded-2xl bg-indigo-50 hover:bg-indigo-100 text-indigo-700 border border-indigo-200 font-bold text-xs transition-all cursor-pointer shadow-2xs"
            >
              <Sparkles className="w-4 h-4 text-indigo-600" />
              <span>⚡ One-Click Instant Access (Explore Learning & Code Lab)</span>
            </a>
          </div>

          {/* Security & Least-Privilege Trust Badge */}
          <div className="pt-4 border-t border-slate-100 flex items-center justify-center gap-2 text-center text-xs text-slate-500">
            <ShieldCheck className="w-4 h-4 text-emerald-600" />
            <span>Encrypted OAuth token exchange · Zero secrets in browser</span>
          </div>
        </div>
      </main>

      {/* Footer */}
      <footer className="h-14 px-6 md:px-12 flex items-center justify-between text-xs text-slate-400 border-t border-slate-200/80 bg-white/60">
        <div className="flex items-center gap-2">
```

```
            <Code2 className="w-3.5 h-3.5 text-indigo-600" />
            <span>RepoPilot AI · Developer Tools</span>
          </div>
          <div>
            <span>Privacy & Security Compliant</span>
          </div>
        </footer>
      </div>
    );
};
```

File [14/30]: src/components/views/DashboardView.tsx

TSX SOURCE

Description: *Dashboard Metrics, Overview Cards & Quick Launch*

```tsx
'use client';

import React from 'react';
import {
  ArrowRight,
  Sparkles,
  ShieldCheck,
  AlertTriangle,
  CheckCircle2,
  FileCode2,
  Search,
  Check,
  Zap,
  FolderGit2,
  RefreshCw,
  GitBranch,
  GraduationCap,
  Code2,
  BookOpen,
  Terminal,
} from 'lucide-react';
import { ACTIVE_REPO } from '@/lib/mock-data';
import { NavTab, RepositoryData } from '@/lib/types';
import { useAuth } from '@/lib/auth-context';

interface DashboardViewProps {
  onNavigate: (tab: NavTab) => void;
  onStartDemoTour?: () => void;
  currentRepo?: RepositoryData;
  onOpenRepoModal?: () => void;
}
```

```
32
33  export const DashboardView: React.FC<DashboardViewProps> = ({
34    onNavigate,
35    currentRepo = ACTIVE_REPO,
36    onOpenRepoModal,
37  }) => {
38    const { user, connectGitHub } = useAuth();
39    const isGitHubConnected = Boolean(user?.connectedAccounts?.github?.connected);
40    const ghUsername = user?.connectedAccounts?.github?.username;
41    const workflowSteps = [
42      {
43        id: 'understand',
44        label: 'UNDERSTAND',
45        detail: 'AST & Structure',
46        icon: Search,
47        tab: 'repositories' as NavTab,
48      },
49      {
50        id: 'diagnose',
51        label: 'DIAGNOSE',
52        detail: 'Root-Cause Pinpoint',
53        icon: AlertTriangle,
54        tab: 'issues' as NavTab,
55      },
56      {
57        id: 'live-fix',
58        label: 'LIVE HEAL & LEARN',
59        detail: 'Real-Time Fix Player',
60        icon: Zap,
61        tab: 'live-fix' as NavTab,
62      },
63      {
64        id: 'patch',
65        label: 'SURGICAL PATCH',
66        detail: 'Regression Suite',
```

```
      icon: FileCode2,
      tab: 'fix' as NavTab,
    },
    {
      id: 'verify',
      label: 'VERIFY & PR',
      detail: 'Closed-Loop Sandbox',
      icon: ShieldCheck,
      tab: 'verification' as NavTab,
    },
  ];

  return (
    <div className="space-y-8 max-w-7xl mx-auto pb-16">
      {/* Hero Section */}
      <div className="relative overflow-hidden rounded-2xl bg-gradient-to-br from-indigo-50/90 via-white to-sky-50/60 border border-indigo-100 p-8 md:p-10 shadow-xs">
        {/* Subtle grid background glow */}
        <div className="absolute top-0 right-0 w-96 h-96 bg-indigo-200/30 rounded-full blur-3xl pointer-events-none -mr-20 -mt-20"></div>
        <div className="absolute bottom-0 left-1/3 w-80 h-80 bg-sky-200/30 rounded-full blur-3xl pointer-events-none"></div>

        <div className="relative z-10 max-w-3xl">
          {/* Header pill */}
          <div className="flex flex-wrap items-center gap-3 mb-4">
            <div className="inline-flex items-center gap-2 px-3 py-1.5 rounded-full bg-indigo-50 border border-indigo-200 text-indigo-700 text-xs font-semibold shadow-xs">
              <Sparkles className="w-3.5 h-3.5 text-indigo-600" />
              <span>Welcome back, {user?.name || 'Developer'}</span>
            </div>

            {/* Connected GitHub indicator */}
            <div className="inline-flex items-center gap-2 px-3 py-1.5 rounded-full bg-white border border-slate-200 text-xs font-medium text-slate-700 shadow-2xs">
              <span className="text-slate-500 font-semibold">Connected GitHub:</span>
              {isGitHubConnected ? (
```

```
                <span className="inline-flex items-center gap-1.5 text-emerald-700 font-bold">
                  <span className="w-2 h-2 rounded-full bg-emerald-500 animate-pulse" />
                  ● Connected {ghUsername ? `(@${ghUsername})` : ''}
                </span>
              ) : (
                <button
                  onClick={connectGitHub}
                  className="inline-flex items-center gap-1 text-indigo-600 hover:text-indigo-800 font-bold cursor-p
ointer"
                >
                  <span>○ Connect GitHub</span>
                </button>
              )}
            </div>
          </div>

          <h1 className="text-4xl sm:text-5xl font-extrabold text-slate-900 tracking-tight leading-[1.15] mb-4">
            Your AI software engineer.
          </h1>

          <p className="text-lg text-slate-600 font-normal leading-relaxed mb-6">
            Understand repositories, diagnose bugs, generate fixes, write regression tests, and verify solutions.
          </p>

          {/* Connected Active Repository Status & Analyze Action */}
          <div className="flex flex-wrap items-center gap-4 bg-white/80 backdrop-blur-xs p-4 rounded-2xl border bord
er-slate-200 shadow-xs mb-2">
            <div className="flex items-center gap-3 flex-1 min-w-[200px]">
              <div className="w-10 h-10 rounded-xl bg-indigo-50 border border-indigo-200 flex items-center justify-c
enter text-indigo-600">
                <FolderGit2 className="w-5 h-5" />
              </div>
              <div>
                <span className="text-[10px] font-bold uppercase tracking-wider text-slate-400 block">
                  Active Repository
                </span>
```

```
                <div className="flex items-center gap-2">
                  <span className="font-bold text-sm text-slate-900">{currentRepo.name}</span>
                  <span className="text-xs text-slate-500 font-mono bg-slate-100 px-2 py-0.5 rounded flex items-cent
er gap-1">
                    <GitBranch className="w-3 h-3 text-indigo-600" />
                    {currentRepo.branch}
                  </span>
                </div>
              </div>
            </div>

            <button
              onClick={onOpenRepoModal ? onOpenRepoModal : () => onNavigate('repositories')}
              className="flex items-center gap-2 px-6 py-2.5 rounded-xl bg-indigo-600 hover:bg-indigo-700 text-white
font-semibold text-xs shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all cursor-po
inter"
            >
              <FolderGit2 className="w-4 h-4" />
              <span>Analyze Repository</span>
              <ArrowRight className="w-3.5 h-3.5 ml-0.5" />
            </button>
          </div>
        </div>

        {/* Workflow Pipeline Display */}
        <div className="mt-10 pt-8 border-t border-slate-200/80">
          <div className="flex items-center justify-between mb-4">
            <span className="text-xs font-bold uppercase tracking-wider text-slate-500">
              Autonomous Verification Loop
            </span>
            <span className="text-xs font-semibold text-emerald-700 bg-emerald-50 px-2.5 py-1 rounded-full border bo
rder-emerald-200 flex items-center gap-1.5">
              <span className="w-2 h-2 rounded-full bg-emerald-500 animate-ping"></span>
              Closed-Loop Verification Ready
            </span>
          </div>

```

```
          <div className="grid grid-cols-2 sm:grid-cols-5 gap-3">
            {workflowSteps.map((step, index) => {
              const Icon = step.icon;
              return (
                <div
                  key={step.id}
                  onClick={() => onNavigate(step.tab)}
                  className="group relative cursor-pointer p-4 rounded-xl bg-white hover:bg-indigo-50/50 border bord
er-slate-200 hover:border-indigo-300 shadow-xs hover:shadow-sm transition-all text-center"
                >
                  <div className="flex items-center justify-center mb-2.5">
                    <div className="w-8 h-8 rounded-lg bg-indigo-50 group-hover:bg-indigo-100 text-indigo-600 flex i
tems-center justify-center transition-colors">
                      <Icon className="w-4 h-4" />
                    </div>
                  </div>
                  <div className="text-xs font-bold tracking-wider text-slate-800 group-hover:text-indigo-700 transi
tion-colors">
                    {step.label}
                  </div>
                  <div className="text-[11px] text-slate-500 truncate mt-0.5">
                    {step.detail}
                  </div>

                  {index < workflowSteps.length - 1 && (
                    <div className="hidden sm:block absolute -right-2 top-1/2 -translate-y-1/2 z-10 text-slate-400 f
ont-bold">
                      →
                    </div>
                  )}
                </div>
              );
            })}
          </div>
        </div>
      </div>
```

```
197
198       {/* Stats Grid */}
199       <div className="grid grid-cols-2 md:grid-cols-4 gap-4">
200         <div className="p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
201           <div className="text-slate-500 text-xs font-medium uppercase tracking-wider mb-1">
202             Repositories
203           </div>
204           <div className="text-3xl font-extrabold text-slate-900 tracking-tight">3</div>
205           <div className="text-[11px] text-slate-500 mt-1 flex items-center gap-1 font-medium">
206             <CheckCircle2 className="w-3.5 h-3.5 text-emerald-600" />
207             <span>Indexed & Monitored</span>
208           </div>
209         </div>
210
211         <div className="p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
212           <div className="text-slate-500 text-xs font-medium uppercase tracking-wider mb-1">
213             Issues Detected
214           </div>
215           <div className="text-3xl font-extrabold text-rose-600 tracking-tight">7</div>
216           <div className="text-[11px] text-slate-500 mt-1 flex items-center gap-1 font-medium">
217             <AlertTriangle className="w-3.5 h-3.5 text-rose-600" />
218             <span>1 High Severity Active</span>
219           </div>
220         </div>
221
222         <div className="p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
223           <div className="text-slate-500 text-xs font-medium uppercase tracking-wider mb-1">
224             Available Courses
225           </div>
226           <div className="text-3xl font-extrabold text-indigo-600 tracking-tight">5</div>
227           <div className="text-[11px] text-slate-500 mt-1 flex items-center gap-1 font-medium">
228             <GraduationCap className="w-3.5 h-3.5 text-indigo-600" />
229             <span>Interactive Academy</span>
230           </div>
231         </div>
```

```
232
233         <div className="p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
234           <div className="text-slate-500 text-xs font-medium uppercase tracking-wider mb-1">
235             Practice Challenges
236           </div>
237           <div className="text-3xl font-extrabold text-emerald-600 tracking-tight">8+</div>
238           <div className="text-[11px] text-slate-500 mt-1 flex items-center gap-1 font-medium">
239             <Code2 className="w-3.5 h-3.5 text-emerald-600" />
240             <span>AI Error Tutor Enabled</span>
241           </div>
242         </div>
243       </div>
244
245       {/* Learning Platform & Code Practice Space Showcase */}
246       <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
247         {/* Card 1: Learning Academy */}
248         <div className="rounded-2xl bg-gradient-to-br from-indigo-900 via-indigo-950 to-slate-900 text-white p-6 md:
p-7 shadow-md border border-indigo-800/50 flex flex-col justify-between relative overflow-hidden group">
249           <div className="absolute top-0 right-0 w-64 h-64 bg-indigo-500/10 rounded-full blur-2xl pointer-events-non
e -mr-16 -mt-16" />
250           <div className="relative z-10 space-y-4">
251             <div className="flex items-center justify-between">
252               <span className="text-[10px] font-bold uppercase tracking-wider px-2.5 py-1 rounded-full bg-indigo-50
0/20 text-cyan-300 border border-indigo-400/30 flex items-center gap-1">
253                 <GraduationCap className="w-3.5 h-3.5 text-cyan-400" />
254                 <span>Interactive Learning Platform</span>
255               </span>
256               <span className="text-xs font-bold text-amber-300">🔥 Earn XP & Badges</span>
257             </div>
258
259             <div>
260               <h3 className="text-xl font-extrabold text-white tracking-tight">
261                 RepoPilot Academy & Courses
262               </h3>
263               <p className="text-xs text-slate-300 mt-1.5 leading-relaxed">
264                 Take structured courses in JavaScript Execution, V8 Memory, Defensive Coding,
```

```
                OWASP Security & Python. Complete lessons, test your knowledge with interactive
                quizzes, and earn certificates of mastery!
              </p>
            </div>

            <div className="flex flex-wrap gap-2 pt-1">
              <span className="text-[11px] px-2 py-0.5 rounded bg-white/10 text-slate-200">
                JS/TS Debugging
              </span>
              <span className="text-[11px] px-2 py-0.5 rounded bg-white/10 text-slate-200">
                OWASP Top 10
              </span>
              <span className="text-[11px] px-2 py-0.5 rounded bg-white/10 text-slate-200">
                Defensive Architecture
              </span>
              <span className="text-[11px] px-2 py-0.5 rounded bg-white/10 text-slate-200">
                React Render Traps
              </span>
            </div>
          </div>

          <div className="pt-6 relative z-10">
            <button
              onClick={() => onNavigate('courses')}
              className="w-full py-2.5 rounded-xl bg-indigo-500 hover:bg-indigo-400 text-white font-bold text-xs shadow-md transition-all flex items-center justify-center gap-2 cursor-pointer"
            >
              <BookOpen className="w-4 h-4" />
              <span>Explore Course Platform</span>
              <ArrowRight className="w-3.5 h-3.5" />
            </button>
          </div>
        </div>

        {/* Card 2: Code Practicing Space with AI Error Explainer */}
```

```
        <div className="rounded-2xl bg-white border border-slate-200 p-6 md:p-7 shadow-xs hover:border-indigo-300 transition-all flex flex-col justify-between group">
          <div className="space-y-4">
            <div className="flex items-center justify-between">
              <span className="text-[10px] font-bold uppercase tracking-wider px-2.5 py-1 rounded-full bg-emerald-50 text-emerald-700 border border-emerald-200 flex items-center gap-1">
                <Code2 className="w-3.5 h-3.5 text-emerald-600" />
                <span>Code Practice Ground</span>
              </span>
              <span className="text-xs font-bold text-indigo-600">⚡ AI Error Concept Explainer</span>
            </div>

            <div>
              <h3 className="text-xl font-extrabold text-slate-900 tracking-tight">
                Code Practice & Error Lab
              </h3>
              <p className="text-xs text-slate-600 mt-1.5 leading-relaxed">
                Practice coding in JavaScript, TypeScript, or Python. When you encounter a runtime
                error, our AI Tutor automatically steps in to explain <strong>why the error is coming</strong>,
                breaks down the underlying language concept with mental model diagrams, and guides you to the fix!
              </p>
            </div>

            <div className="p-3 rounded-xl bg-slate-50 border border-slate-200 text-xs font-mono text-slate-700 flex items-center justify-between">
              <div className="flex items-center gap-2">
                <span className="w-2 h-2 rounded-full bg-rose-500 animate-pulse" />
                <span className="text-rose-700 font-bold">TypeError / ReferenceError</span>
              </div>
              <span className="text-[11px] text-slate-500">Auto-Diagnosed & Explained</span>
            </div>
          </div>

          <div className="pt-6">
            <button
              onClick={() => onNavigate('practice')}
```

```
              className="w-full py-2.5 rounded-xl bg-slate-900 hover:bg-slate-800 text-white font-bold text-xs shado
w-md transition-all flex items-center justify-center gap-2 cursor-pointer"
            >
              <Zap className="w-4 h-4 text-amber-400" />
              <span>Launch Code Practice Lab</span>
              <ArrowRight className="w-3.5 h-3.5" />
            </button>
          </div>
        </div>
      </div>

      {/* Active Repository Card */}
      <div className="p-6 md:p-8 rounded-2xl bg-white border border-slate-200 shadow-xs relative overflow-hidden">
        <div className="flex flex-col lg:flex-row lg:items-center justify-between gap-6">
          <div className="space-y-3">
            <div className="flex flex-wrap items-center gap-2">
              <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-indigo-50 text-indi
go-700 border border-indigo-200">
                Active Repository
              </span>
              <span className="text-xs font-semibold px-2 py-0.5 rounded bg-rose-50 text-rose-700 border border-rose
-200 flex items-center gap-1">
                <span className="w-1.5 h-1.5 rounded-full bg-rose-500 animate-pulse"></span>
                {currentRepo.status}
              </span>
            </div>

            <div className="flex items-center gap-3">
              <h2 className="text-2xl font-bold text-slate-900 tracking-tight">
                {currentRepo.name}
              </h2>
              <span className="text-xs text-slate-600 font-mono px-2 py-0.5 rounded bg-slate-100 border border-slate
-200">
                branch: {currentRepo.branch}
              </span>

```

```
              {onOpenRepoModal && (
                <button
                  onClick={onOpenRepoModal}
                  className="flex items-center gap-1 text-xs text-indigo-600 hover:text-indigo-800 font-medium px-2 py-0.5 rounded hover:bg-indigo-50 transition-colors"
                  title="Switch or connect another repository"
                >
                  <RefreshCw className="w-3 h-3" />
                  <span>Switch</span>
                </button>
              )}
            </div>

            <p className="text-sm text-slate-600 max-w-xl">
              {currentRepo.description}
            </p>

            {/* Tech Stack Pills */}
            <div className="flex flex-wrap items-center gap-1.5 pt-1">
              {currentRepo.stack.map((tech) => (
                <span
                  key={tech}
                  className="px-2.5 py-1 rounded-md text-xs font-medium bg-slate-100 text-slate-700 border border-slate-200"
                >
                  {tech}
                </span>
              ))}
            </div>
          </div>

          {/* Metrics summary */}
          <div className="flex flex-col sm:flex-row items-stretch sm:items-center gap-4 lg:border-l lg:border-slate-200 lg:pl-8">
            <div className="grid grid-cols-2 gap-3 text-left">
              <div className="p-3 rounded-lg bg-slate-50 border border-slate-200">
```

```
                  <div className="text-[11px] text-slate-500">Total Files</div>
                  <div className="text-lg font-bold text-slate-900">{currentRepo.metrics.files}</div>
                </div>
                <div className="p-3 rounded-lg bg-slate-50 border border-slate-200">
                  <div className="text-[11px] text-slate-500">Lines of Code</div>
                  <div className="text-lg font-bold text-slate-900">{currentRepo.metrics.linesOfCode}</div>
                </div>
                <div className="p-3 rounded-lg bg-slate-50 border border-slate-200">
                  <div className="text-[11px] text-slate-500">Test Coverage</div>
                  <div className="text-lg font-bold text-emerald-600">{currentRepo.metrics.testCoverage}</div>
                </div>
                <div className="p-3 rounded-lg bg-slate-50 border border-slate-200">
                  <div className="text-[11px] text-slate-500">Dependencies</div>
                  <div className="text-lg font-bold text-slate-900">{currentRepo.metrics.dependencies}</div>
                </div>
              </div>

              <div className="flex flex-col gap-2 shrink-0">
                <button
                  onClick={() => onNavigate('live-fix')}
                  className="flex items-center justify-center gap-2 px-5 py-3 rounded-xl bg-gradient-to-r from-indigo-600 via-indigo-500 to-cyan-600 hover:from-indigo-700 hover:to-cyan-700 text-white font-semibold text-xs shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all whitespace-nowrap"
                >
                  <Zap className="w-3.5 h-3.5 text-cyan-300" />
                  <span>Watch Live Fix &amp;amp; Learn</span>
                </button>

                <button
                  onClick={() => onNavigate('issues')}
                  className="flex items-center justify-center gap-2 px-5 py-2.5 rounded-xl bg-rose-600 hover:bg-rose-700 text-white font-semibold text-xs shadow-xs hover:scale-[1.02] active:scale-[0.98] transition-all whitespace-nowrap"
                >
                  <AlertTriangle className="w-3.5 h-3.5" />
                  <span>Investigate Issue</span>
                </button>
```

```
            </div>
          </div>
        </div>
      </div>

      {/* KEY DIFFERENTIATOR COMPARISON BANNER */}
      <div className="rounded-2xl border border-indigo-200 bg-gradient-to-b from-indigo-50/70 via-white to-slate-50 p-6 md:p-8 shadow-xs">
        <div className="text-center max-w-2xl mx-auto mb-8">
          <div className="text-xs font-bold uppercase tracking-wider text-indigo-600 mb-2">
            Why RepoPilot?
          </div>
          <h3 className="text-2xl font-bold text-slate-900">
            "Most AI coding tools generate code."
          </h3>
          <p className="text-xl font-bold text-indigo-700 mt-1">
            "RepoPilot verifies engineering outcomes."
          </p>
        </div>

        <div className="grid md:grid-cols-2 gap-6">
          {/* Traditional AI Box */}
          <div className="p-6 rounded-xl bg-white border border-slate-200 shadow-xs relative">
            <div className="flex items-center justify-between mb-4">
              <span className="text-xs font-bold uppercase tracking-wider text-slate-600">
                Traditional AI Coding (Chatbots / Copilots)
              </span>
              <span className="text-xs px-2 py-0.5 rounded bg-amber-50 text-amber-800 border border-amber-200 font-semibold">
                Unverified Output
              </span>
            </div>

            <div className="flex flex-col gap-3">
              <div className="flex items-center gap-3 p-3 rounded-lg bg-slate-50 border border-slate-200 text-sm text-slate-700">
```

```
463                 <span className="w-6 h-6 rounded-full bg-slate-200 text-slate-700 flex items-center justify-center t
ext-xs font-bold">1</span>
464                 <span>Prompt Input</span>
465               </div>
466               <div className="text-center text-slate-400 text-xs">↓</div>
467               <div className="flex items-center gap-3 p-3 rounded-lg bg-slate-50 border border-slate-200 text-sm tex
t-slate-700">
468                 <span className="w-6 h-6 rounded-full bg-slate-200 text-slate-700 flex items-center justify-center t
ext-xs font-bold">2</span>
469                 <span>Generated Code Snippet</span>
470               </div>
471               <div className="text-center text-slate-400 text-xs">↓</div>
472               <div className="flex items-center gap-3 p-3 rounded-lg bg-rose-50 border border-rose-200 text-sm text-
rose-800 font-medium">
473                 <span className="w-6 h-6 rounded-full bg-rose-200 text-rose-800 flex items-center justify-center tex
t-xs font-bold">3</span>
474                 <span>Developer Must Manually Test & Debug Hallucinations</span>
475               </div>
476             </div>
477           </div>
478
479           {/* RepoPilot AI Box */}
480           <div className="p-6 rounded-xl bg-white border-2 border-indigo-300 relative shadow-sm">
481             <div className="flex items-center justify-between mb-4">
482               <span className="text-xs font-bold uppercase tracking-wider text-indigo-700 flex items-center gap-1.
5">
483                 <Sparkles className="w-3.5 h-3.5 text-indigo-600" />
484                 RepoPilot AI Autonomous Loop
485               </span>
486               <span className="text-xs px-2 py-0.5 rounded bg-emerald-50 text-emerald-700 border border-emerald-200
font-semibold">
487                 Guaranteed Verified
488               </span>
489             </div>
490
491             <div className="grid grid-cols-5 gap-1.5 text-center text-xs">
492               <div className="p-2 rounded bg-indigo-50 border border-indigo-200 text-indigo-800">
493                 <div className="font-bold">Repository</div>
```

```
                <div className="text-[10px] text-indigo-600">Context</div>
              </div>
              <div className="p-2 rounded bg-indigo-50 border border-indigo-200 text-indigo-800">
                <div className="font-bold">Root Cause</div>
                <div className="text-[10px] text-indigo-600">Analysis</div>
              </div>
              <div className="p-2 rounded bg-indigo-50 border border-indigo-200 text-indigo-800">
                <div className="font-bold">Patch</div>
                <div className="text-[10px] text-indigo-600">Surgical</div>
              </div>
              <div className="p-2 rounded bg-indigo-50 border border-indigo-200 text-indigo-800">
                <div className="font-bold">Regression</div>
                <div className="text-[10px] text-indigo-600">Test Gen</div>
              </div>
              <div className="p-2 rounded bg-emerald-50 border border-emerald-200 text-emerald-800">
                <div className="font-bold">Verified</div>
                <div className="text-[10px] text-emerald-600">Sandbox</div>
              </div>
            </div>

            <div className="mt-4 p-3 rounded-lg bg-emerald-50 border border-emerald-200 flex items-center gap-3">
              <ShieldCheck className="w-5 h-5 text-emerald-600 shrink-0" />
              <p className="text-xs text-emerald-900 font-medium leading-relaxed">
                RepoPilot executes the tests, inspects runtime outcomes, and creates developer-ready PRs with verified zero-regression guarantees.
              </p>
            </div>
          </div>
        </div>
      </div>
    </div>
  );
};
```

File [15/30]: src/lib/server/sandbox.ts

TYPESCRIPT SOURCE

Description: *Isolated Disposable Workspace Sandbox & Security Guardrails*

```typescript
import fs from 'fs';
import path from 'path';
import { spawn, execSync } from 'child_process';

const WORKSPACES_ROOT = path.join(process.cwd(), '.workspaces', 'repair-jobs');

/**
 * Ensure workspaces root exists
 */
export function ensureWorkspacesRoot(): string {
  if (!fs.existsSync(WORKSPACES_ROOT)) {
    fs.mkdirSync(WORKSPACES_ROOT, { recursive: true });
  }
  return WORKSPACES_ROOT;
}

/**
 * Path Traversal Guard:
 * Resolves a path relative to workspaceDir and asserts that the canonical
 * destination is strictly inside workspaceDir.
 */
export function assertSafePath(workspaceDir: string, relativePath: string): string {
  // Normalize and prevent null bytes or empty path
  if (!relativePath || relativePath.includes('\0')) {
    throw new Error('SECURITY_VIOLATION: Invalid or malformed file path.');
  }

  const cleanWorkspace = path.resolve(workspaceDir);
  const resolvedTarget = path.resolve(cleanWorkspace, relativePath);

  // Strict boundary check: target must be under workspace directory
```

```
  if (!resolvedTarget.startsWith(cleanWorkspace + path.sep) && resolvedTarget !== cleanWorkspace) {
    throw new Error(`SECURITY_VIOLATION: Path traversal attempt blocked for "${relativePath}".`);
  }

  return resolvedTarget;
}

/**
 * Redact secrets and sensitive tokens from logs and terminal outputs
 */
export function redactSecrets(text: string): string {
  if (!text) return text;
  let sanitized = text;

  // Mask common secret patterns: bearer tokens, github tokens, secrets
  sanitized = sanitized.replace(/gh[pousr]_[A-Za-z0-9_]{16,255}/g, 'ghp_REDACTED_SECRET');
  sanitized = sanitized.replace(/Bearer\s+[A-Za-z0-9\-._~+/]+=*/gi, 'Bearer REDACTED_TOKEN');
  sanitized = sanitized.replace(/(AUTH_SECRET|CLIENT_SECRET|API_KEY|TOKEN)=["']?[A-Za-z0-9\-_./+=]+["']?/gi, '$1=RED
ACTED');
  sanitized = sanitized.replace(/GOCSPX-[A-Za-z0-9\-_]+/g, 'GOCSPX-REDACTED');

  return sanitized;
}

/**
 * Command Allowlist Guard:
 * Only permits strictly vetted development and testing commands.
 */
const ALLOWED_EXECUTABLES = new Set(['npm', 'npm.cmd', 'npx', 'npx.cmd', 'node', 'node.exe']);
const FORBIDDEN_OPERATORS = [';', '&&', '||', '|', '`', '$(', '\n', '\r'];
const FORBIDDEN_COMMANDS = [/\brm\b/i, /\brmdir\b/i, /\bdel\b/i, /\bcurl\b/i, /\bwget\b/i, /\bnc\b/i, /\bbash\b/i, /
\bpowershell\b/i, /\bcmd(\.exe)?\b/i];

export function validateCommand(command: string): { executable: string; args: string[] } {
  const trimmed = command.trim();
  if (!trimmed) {
```

```
    throw new Error('EMPTY_COMMAND: No command provided to sandbox runner.');
  }

  // Reject shell command chaining or injection characters
  for (const op of FORBIDDEN_OPERATORS) {
    if (trimmed.includes(op)) {
      throw new Error(`SECURITY_VIOLATION: Command contains unpermitted shell operator "${op}".`);
    }
  }

  for (const regex of FORBIDDEN_COMMANDS) {
    if (regex.test(trimmed)) {
      throw new Error(`SECURITY_VIOLATION: Command contains unpermitted command pattern "${regex.source}".`);
    }
  }

  const parts = trimmed.split(/\s+/);
  const executable = parts[0];
  const args = parts.slice(1);

  if (!ALLOWED_EXECUTABLES.has(executable.toLowerCase())) {
    throw new Error(`SECURITY_VIOLATION: Executable "${executable}" is not allowlisted for sandbox execution. Allowed: npm, npx, node.`);
  }

  return { executable, args };
}

/**
 * Initialize isolated disposable workspace for a repair job
 */
export async function initializeWorkspace(
  jobId: string,
  initialFiles?: Record<string, string>
): Promise<string> {
```

```
  const root = ensureWorkspacesRoot();
  const workspaceDir = path.join(root, jobId);

  if (fs.existsSync(workspaceDir)) {
    fs.rmSync(workspaceDir, { recursive: true, force: true });
  }

  fs.mkdirSync(workspaceDir, { recursive: true });

  if (initialFiles) {
    for (const [relPath, content] of Object.entries(initialFiles)) {
      const targetPath = assertSafePath(workspaceDir, relPath);
      fs.mkdirSync(path.dirname(targetPath), { recursive: true });
      fs.writeFileSync(targetPath, content, 'utf-8');
    }
  }

  return workspaceDir;
}

/**
 * Copy directory contents into isolated workspace
 */
export async function copyToWorkspace(
  sourceDir: string,
  workspaceDir: string,
  excludePatterns: string[] = ['node_modules', '.git', '.next', '.workspaces', '.data']
): Promise<void> {
  if (!fs.existsSync(sourceDir)) {
    throw new Error(`SOURCE_NOT_FOUND: Source directory "${sourceDir}" does not exist.`);
  }

  const entries = fs.readdirSync(sourceDir, { withFileTypes: true });
```

```
  for (const entry of entries) {
    if (excludePatterns.includes(entry.name)) continue;

    const srcPath = path.join(sourceDir, entry.name);
    const destPath = path.join(workspaceDir, entry.name);

    if (entry.isDirectory()) {
      fs.mkdirSync(destPath, { recursive: true });
      await copyToWorkspace(srcPath, destPath, excludePatterns);
    } else if (entry.isFile()) {
      fs.copyFileSync(srcPath, destPath);
    }
  }
}

/**
 * Read file safely from workspace
 */
export function readFile(workspaceDir: string, relativePath: string): string {
  const targetPath = assertSafePath(workspaceDir, relativePath);
  if (!fs.existsSync(targetPath)) {
    throw new Error(`FILE_NOT_FOUND: File "${relativePath}" does not exist in workspace.`);
  }
  return fs.readFileSync(targetPath, 'utf-8');
}

/**
 * Write file safely to workspace
 */
export function writeFile(workspaceDir: string, relativePath: string, content: string): void {
  const targetPath = assertSafePath(workspaceDir, relativePath);
  fs.mkdirSync(path.dirname(targetPath), { recursive: true });
  fs.writeFileSync(targetPath, content, 'utf-8');
}

```

```
/**
 * Search workspace code for literal query or symbol
 */
export function searchCode(
  workspaceDir: string,
  query: string,
  maxResults: number = 25
): Array<{ file: string; line: number; text: string }> {
  const results: Array<{ file: string; line: number; text: string }> = [];
  const lowerQuery = query.toLowerCase();

  function scanDir(dir: string) {
    if (results.length >= maxResults) return;
    const entries = fs.readdirSync(dir, { withFileTypes: true });

    for (const entry of entries) {
      if (['node_modules', '.git', '.next', 'dist', 'build'].includes(entry.name)) continue;
      const fullPath = path.join(dir, entry.name);

      if (entry.isDirectory()) {
        scanDir(fullPath);
      } else if (entry.isFile()) {
        try {
          const content = fs.readFileSync(fullPath, 'utf-8');
          const lines = content.split('\n');
          const relPath = path.relative(workspaceDir, fullPath).replace(/\\/g, '/');

          for (let i = 0; i < lines.length; i++) {
            if (lines[i].toLowerCase().includes(lowerQuery)) {
              results.push({
                file: relPath,
                line: i + 1,
                text: lines[i].trim(),
              });
```

```
              if (results.length >= maxResults) break;
            }
          }
        } catch {
          // ignore binary or unreadable files
        }
      }
    }
  }

  scanDir(workspaceDir);
  return results;
}

/**
 * Generate standard unified-style diff string between two text snippets
 */
export function createUnifiedDiff(filename: string, beforeText: string, afterText: string): string {
  const beforeLines = beforeText.split('\n');
  const afterLines = afterText.split('\n');

  const diffLines: string[] = [
    `--- a/${filename}`,
    `+++ b/${filename}`,
    `@@ -1,${beforeLines.length} +1,${afterLines.length} @@`,
  ];

  for (const line of beforeLines) {
    if (!afterLines.includes(line)) {
      diffLines.push(`-${line}`);
    } else {
      diffLines.push(` ${line}`);
    }
  }

```

```
238     for (const line of afterLines) {
239       if (!beforeLines.includes(line)) {
240         diffLines.push(`+${line}`);
241       }
242     }
243
244     return diffLines.join('\n');
245   }
246
247   /**
248    * Apply validated surgical patch to a file in workspace
249    */
250   export function applyPatch(
251     workspaceDir: string,
252     patch: {
253       file: string;
254       beforeCode: string;
255       afterCode: string;
256     }
257   ): { success: boolean; error?: string; diff: string; additions: number; deletions: number } {
258     try {
259       const targetPath = assertSafePath(workspaceDir, patch.file);
260       if (!fs.existsSync(targetPath)) {
261         return {
262           success: false,
263           error: `Target file "${patch.file}" does not exist in workspace.`,
264           diff: '',
265           additions: 0,
266           deletions: 0,
267         };
268       }
269
270       const currentContent = fs.readFileSync(targetPath, 'utf-8');
271
```

```
    // Normalizing newlines for reliable matching
    const normCurrent = currentContent.replace(/\r\n/g, '\n');
    const normBefore = patch.beforeCode.replace(/\r\n/g, '\n').trim();
    const normAfter = patch.afterCode.replace(/\r\n/g, '\n').trim();

    if (!normBefore) {
      return {
        success: false,
        error: 'Patch beforeCode cannot be empty.',
        diff: '',
        additions: 0,
        deletions: 0,
      };
    }

    if (!normCurrent.includes(normBefore)) {
      return {
        success: false,
        error: `Anchor mismatch: target lines could not be cleanly located in "${patch.file}".`,
        diff: '',
        additions: 0,
        deletions: 0,
      };
    }

    // Apply substitution cleanly
    const updatedContent = normCurrent.replace(normBefore, normAfter);
    fs.writeFileSync(targetPath, updatedContent, 'utf-8');

    const beforeLines = normBefore.split('\n');
    const afterLines = normAfter.split('\n');
    const additions = Math.max(0, afterLines.length - beforeLines.length) + 1;
    const deletions = Math.max(0, beforeLines.length - afterLines.length) + 1;
    const diff = createUnifiedDiff(patch.file, normBefore, normAfter);

```

```
      return {
        success: true,
        diff,
        additions,
        deletions,
      };
    } catch (err: any) {
      return {
        success: false,
        error: err.message || 'Unknown error applying patch.',
        diff: '',
        additions: 0,
        deletions: 0,
      };
    }
  }

  /**
   * Execute an allowlisted test command inside the sandbox with strict bounds
   */
  export async function executeSandboxCommand(
    workspaceDir: string,
    command: string,
    options: {
      timeoutMs?: number;
      maxBufferBytes?: number;
      onOutputChunk?: (chunk: string) => void;
    } = {}
  ): Promise<{
    exitCode: number;
    stdout: string;
    stderr: string;
    durationMs: number;
    passed: boolean;
```

```
  timedOut?: boolean;
}> {
  const timeoutMs = options.timeoutMs ?? 35000;
  const maxBuffer = options.maxBufferBytes ?? 512 * 1024; // 512 KB
  const { executable, args } = validateCommand(command);

  const startTime = Date.now();
  let stdoutAcc = '';
  let stderrAcc = '';
  let timedOut = false;

  // Sanitized environment: explicitly strip sensitive API keys & tokens
  const sanitizedEnv: NodeJS.ProcessEnv = {
    ...process.env,
    NODE_ENV: 'test',
    CI: 'true',
    AUTH_SECRET: undefined,
    GOOGLE_CLIENT_SECRET: undefined,
    GITHUB_CLIENT_SECRET: undefined,
    OPENAI_API_KEY: undefined,
    ANTHROPIC_API_KEY: undefined,
    GEMINI_API_KEY: undefined,
  };

  return new Promise((resolve) => {
    let proc: any = null;
    let timer: NodeJS.Timeout | null = null;
    const isWin = process.platform === 'win32';
    const isNpmScript = isWin && (executable.toLowerCase().startsWith('npm') || executable.toLowerCase().startsWith
('npx'));

    try {
      proc = spawn(executable, args, {
        cwd: workspaceDir,
        env: sanitizedEnv,
```

```
375         shell: isNpmScript, // Only invoke shell for npm/npx batch runners on Windows; run node binary directly
376         stdio: ['ignore', 'pipe', 'pipe'],
377       });
378     } catch (spawnErr: any) {
379       return resolve({
380         exitCode: 1,
381         stdout: '',
382         stderr: redactSecrets(spawnErr.message || 'Failed to spawn process'),
383         durationMs: Date.now() - startTime,
384         passed: false,
385       });
386     }
387
388     timer = setTimeout(() => {
389       timedOut = true;
390       try {
391         if (proc && !proc.killed) {
392           if (isWin) {
393             try {
394               execSync(`taskkill /pid ${proc.pid} /T /F`);
395             } catch {
396               proc.kill('SIGKILL');
397             }
398           } else {
399             proc.kill('SIGTERM');
400             setTimeout(() => {
401               if (proc && !proc.killed) proc.kill('SIGKILL');
402             }, 1500);
403           }
404         }
405       } catch {
406         // ignore kill errors
407       }
408     }, timeoutMs);
409
```

```
    proc.stdout?.on('data', (data: Buffer) => {
      const chunk = data.toString('utf-8');
      if (stdoutAcc.length < maxBuffer) {
        stdoutAcc += chunk;
      }
      if (options.onOutputChunk) {
        options.onOutputChunk(redactSecrets(chunk));
      }
    });

    proc.stderr?.on('data', (data: Buffer) => {
      const chunk = data.toString('utf-8');
      if (stderrAcc.length < maxBuffer) {
        stderrAcc += chunk;
      }
      if (options.onOutputChunk) {
        options.onOutputChunk(redactSecrets(chunk));
      }
    });

    proc.on('close', (code: number | null) => {
      if (timer) clearTimeout(timer);
      const durationMs = Date.now() - startTime;
      const exitCode = timedOut ? 124 : (code ?? 0);
      const passed = exitCode === 0;

      if (timedOut) {
        stderrAcc += `\n[SANDBOX TIMEOUT]: Execution exceeded bounded limit of ${timeoutMs}ms and was terminated.`;
      }

      resolve({
        exitCode,
        stdout: redactSecrets(stdoutAcc),
        stderr: redactSecrets(stderrAcc),
```

```
444         durationMs,
445         passed,
446         timedOut,
447       });
448     });
449
450     proc.on('error', (err: any) => {
451       if (timer) clearTimeout(timer);
452       resolve({
453         exitCode: 1,
454         stdout: redactSecrets(stdoutAcc),
455         stderr: redactSecrets(err.message || 'Subprocess error'),
456         durationMs: Date.now() - startTime,
457         passed: false,
458       });
459     });
460   });
461 }
462
463 /**
464  * Clean up isolated workspace safely
465  */
466 export async function cleanupWorkspace(workspaceDir: string): Promise<void> {
467   try {
468     const cleanWorkspace = path.resolve(workspaceDir);
469     const root = path.resolve(ensureWorkspacesRoot());
470
471     // Never delete outside WORKSPACES_ROOT
472     if (cleanWorkspace.startsWith(root + path.sep)) {
473       if (fs.existsSync(cleanWorkspace)) {
474         fs.rmSync(cleanWorkspace, { recursive: true, force: true });
475       }
476     }
477   } catch (err) {
478     console.warn('Sandbox workspace cleanup warning:', err);
```

```
479     }
480 }
```

File [16/30]: src/lib/server/repair-pipeline.ts

TYPESCRIPT SOURCE

Description: *Autonomous GitHub Issue Diagnostic & Repair Pipeline*

```typescript
import fs from 'fs';
import path from 'path';
import {
  RepairJob,
  RepairEvent,
  RepairJobStatus,
  RepairJobDiagnosis,
  RepairJobPatch,
} from '@/lib/types';
import {
  getRepairJob,
  updateRepairJob,
  appendRepairEvent,
} from './db';
import {
  initializeWorkspace,
  copyToWorkspace,
  readFile,
  applyPatch,
  executeSandboxCommand,
  cleanupWorkspace,
  searchCode,
} from './sandbox';
import {
  diagnoseIssue,
  generatePatch,
  revisePatch,
  getLLMConfig,
} from './llm-provider';
import { jobEventBus } from './event-bus';

```

```
// Track active running jobs and their abort signals
const activeJobAbortControllers = new Map<string, AbortController>();

/**
 * Dispatch an event to both persistent database storage and in-memory SSE bus
 */
export function dispatchJobEvent(
  jobId: string,
  eventData: {
    type: string;
    stage: string;
    message: string;
    payload?: Record<string, unknown>;
  }
): RepairEvent {
  const event = appendRepairEvent(jobId, eventData);
  jobEventBus.emit(jobId, event);
  return event;
}

/**
 * Request cancellation of a running repair job
 */
export function cancelRepairJob(jobId: string): boolean {
  const job = getRepairJob(jobId);
  if (!job) return false;

  const controller = activeJobAbortControllers.get(jobId);
  if (controller) {
    controller.abort();
    activeJobAbortControllers.delete(jobId);
  }

  updateRepairJob(jobId, {
    status: 'CANCELLED',
```

```
    currentStage: 'Repair cancelled by user request',
    completedAt: new Date().toISOString(),
  });

  dispatchJobEvent(jobId, {
    stage: 'CANCELLED',
    type: 'repair.cancelled',
    message: 'Repair job execution was cancelled by user request.',
  });

  return true;
}

/**
 * Find the primary test command for a repository in its workspace
 */
function detectTestCommand(workspaceDir: string): string {
  const pkgPath = path.join(workspaceDir, 'package.json');
  if (fs.existsSync(pkgPath)) {
    try {
      const pkg = JSON.parse(fs.readFileSync(pkgPath, 'utf-8'));
      if (pkg.scripts && pkg.scripts.test) {
        // e.g. "node --test" or "npm test"
        return 'npm test';
      }
    } catch {
      // ignore
    }
  }

  return 'npm test';
}

/**
```

```
 * Main Autonomous Repair Pipeline
 * Executes the full lifecycle:
 * QUEUED -> INITIALIZING -> REPRODUCING -> INVESTIGATING -> DIAGNOSING -> PATCHING -> TESTING -> VERIFYING -> COMPLETED
 */
export async function runAutonomousRepairPipeline(jobId: string): Promise<RepairJob> {
  let job = getRepairJob(jobId);
  if (!job) {
    throw new Error(`JOB_NOT_FOUND: Repair job "${jobId}" does not exist.`);
  }

  if (job.status === 'CANCELLED' || job.status === 'COMPLETED') {
    return job;
  }

  const abortController = new AbortController();
  activeJobAbortControllers.set(jobId, abortController);

  let workspaceDir = '';

  try {
    // ---------------------------------------------------------------------
    // STAGE 1: INITIALIZING
    // ---------------------------------------------------------------------
    if (abortController.signal.aborted) throw new Error('CANCELLED');

    updateRepairJob(jobId, {
      status: 'INITIALIZING',
      currentStage: 'Initializing isolated workspace and checking out pinned commit',
    });

    dispatchJobEvent(jobId, {
      stage: 'INITIALIZING',
      type: 'repository.checkout.started',
      message: `Creating isolated disposable workspace for ${job.repository.owner}/${job.repository.name}`,
```

```
      payload: {
        pinnedCommitSha: job.repository.pinnedCommitSha,
      },
    });

    workspaceDir = await initializeWorkspace(jobId);

    // If local path is provided (sample repo or local clone), copy files
    if (job.repository.localPath && fs.existsSync(job.repository.localPath)) {
      await copyToWorkspace(job.repository.localPath, workspaceDir);
    } else {
      // Default to bundled sample vulnerable repository fixture if no external git clone is set
      const defaultFixture = path.join(process.cwd(), 'fixtures', 'sample-vulnerable-repo');
      if (fs.existsSync(defaultFixture)) {
        await copyToWorkspace(defaultFixture, workspaceDir);
      } else {
        // Create baseline files directly in workspace if fixture directory not yet ready
        fs.mkdirSync(path.join(workspaceDir, 'src'), { recursive: true });
        fs.mkdirSync(path.join(workspaceDir, 'tests'), { recursive: true });
      }
    }

    dispatchJobEvent(jobId, {
      stage: 'INITIALIZING',
      type: 'repository.checkout.completed',
      message: `Workspace ready at pinned commit ${job.repository.pinnedCommitSha.substring(0, 7)}`,
      payload: {
        workspaceReady: true,
      },
    });

    // ---------------------------------------------------------------------
    // STAGE 2: REPRODUCING
    // ---------------------------------------------------------------------
    if (abortController.signal.aborted) throw new Error('CANCELLED');
```

```
170
171     updateRepairJob(jobId, {
172       status: 'REPRODUCING',
173       currentStage: 'Executing baseline checks to reproduce reported defect',
174     });
175
176     const testCommand = detectTestCommand(workspaceDir);
177
178     dispatchJobEvent(jobId, {
179       stage: 'REPRODUCING',
180       type: 'tests.started',
181       message: `Executing baseline reproduction: "${testCommand}"`,
182       payload: { command: testCommand, stage: 'reproduction' },
183     });
184
185     const baselineResult = await executeSandboxCommand(workspaceDir, testCommand, {
186       onOutputChunk: (chunk) => {
187         dispatchJobEvent(jobId, {
188           stage: 'REPRODUCING',
189           type: 'tests.output',
190           message: chunk,
191         });
192       },
193     });
194
195     const reproductionTestRun = {
196       command: testCommand,
197       exitCode: baselineResult.exitCode,
198       stdout: baselineResult.stdout,
199       stderr: baselineResult.stderr,
200       durationMs: baselineResult.durationMs,
201       timestamp: new Date().toISOString(),
202       stage: 'reproduction' as const,
203       passed: baselineResult.passed,
```

```
    };

    updateRepairJob(jobId, {
      testRuns: [reproductionTestRun],
      reproductionEvidence: baselineResult.stderr || baselineResult.stdout,
    });

    dispatchJobEvent(jobId, {
      stage: 'REPRODUCING',
      type: 'tests.completed',
      message: baselineResult.passed
        ? 'Baseline test passed (no failure detected on main).'
        : `Baseline failure captured with exit code ${baselineResult.exitCode} (${baselineResult.durationMs}ms)`,
      payload: {
        exitCode: baselineResult.exitCode,
        durationMs: baselineResult.durationMs,
        passed: baselineResult.passed,
      },
    });

    // ---------------------------------------------------------------------
    // STAGE 3: INVESTIGATING & DIAGNOSING
    // ---------------------------------------------------------------------
    if (abortController.signal.aborted) throw new Error('CANCELLED');

    updateRepairJob(jobId, {
      status: 'INVESTIGATING',
      currentStage: 'Scanning AST symbols and tracing root cause',
    });

    dispatchJobEvent(jobId, {
      stage: 'INVESTIGATING',
      type: 'diagnosis.started',
      message: 'Autonomous agent tracing symbol execution path and dependency graph...',
    });
```

```
239
240     // Code search for symbols
241     const searchHits = searchCode(workspaceDir, 'authenticateToken');
242     const affectedRelFile = searchHits.length > 0 ? searchHits[0].file : 'src/middleware/auth.ts';
243
244     dispatchJobEvent(jobId, {
245       stage: 'INVESTIGATING',
246       type: 'analysis.file_found',
247       message: `Identified target file: ${affectedRelFile}`,
248       payload: { file: affectedRelFile },
249     });
250
251     let targetFileContent = '';
252     try {
253       targetFileContent = readFile(workspaceDir, affectedRelFile);
254     } catch {
255       targetFileContent = '// Missing or unreadable file';
256     }
257
258     // Call LLM for validated diagnosis
259     updateRepairJob(jobId, {
260       status: 'DIAGNOSING',
261       currentStage: 'Synthesizing formal root-cause hypothesis with LLM',
262     });
263
264     const llmConfig = getLLMConfig();
265     let diagnosis: RepairJobDiagnosis;
266
267     if (llmConfig.provider !== 'none') {
268       try {
269         diagnosis = await diagnoseIssue({
270           issueDescription: job.issueDescription,
271           reproductionOutput: baselineResult.stderr || baselineResult.stdout,
272           targetFile: affectedRelFile,
```

```
273           fileContent: targetFileContent,
274           candidateFiles: [affectedRelFile],
275         });
276       } catch (err: any) {
277         // If LLM fails or is unavailable, report clear diagnostic error
278         throw new Error(`DIAGNOSIS_FAILED: ${err.message}`);
279       }
280     } else {
281       // Strict rule: If no LLM provider is configured in environment, fail explicitly
282       throw new Error(
283         'NO_LLM_PROVIDER_CONFIGURED: Please configure OPENAI_API_KEY, ANTHROPIC_API_KEY, or GEMINI_API_KEY in .env.l
ocal to run autonomous AI repair.'
284       );
285     }
286
287     updateRepairJob(jobId, {
288       diagnosis,
289     });
290
291     dispatchJobEvent(jobId, {
292       stage: 'DIAGNOSING',
293       type: 'analysis.location_found',
294       message: `Vulnerability located at ${diagnosis.affectedFile}:${diagnosis.startLine}-${diagnosis.endLine}`,
295       payload: {
296         file: diagnosis.affectedFile,
297         startLine: diagnosis.startLine,
298         endLine: diagnosis.endLine,
299       },
300     });
301
302     dispatchJobEvent(jobId, {
303       stage: 'DIAGNOSING',
304       type: 'diagnosis.completed',
305       message: `Root cause identified: ${diagnosis.rootCause}`,
306       payload: diagnosis as unknown as Record<string, unknown>,
```

```
    });

    // ---------------------------------------------------------------------
    // STAGE 4: PATCHING
    // ---------------------------------------------------------------------
    if (abortController.signal.aborted) throw new Error('CANCELLED');

    updateRepairJob(jobId, {
      status: 'PATCHING',
      currentStage: 'Generating minimal-risk surgical patch',
    });

    dispatchJobEvent(jobId, {
      stage: 'PATCHING',
      type: 'patch.generated',
      message: 'Synthesizing non-breaking contextual patch...',
    });

    const patch = await generatePatch({
      affectedFile: diagnosis.affectedFile,
      fileContent: targetFileContent,
      rootCause: diagnosis.rootCause,
      issueDescription: job.issueDescription,
      testFailure: baselineResult.stderr || baselineResult.stdout,
    });

    // Apply patch directly to workspace on disk
    const patchResult = applyPatch(workspaceDir, {
      file: patch.file,
      beforeCode: patch.beforeCode,
      afterCode: patch.afterCode,
    });

    if (!patchResult.success) {
      dispatchJobEvent(jobId, {
```

```
        stage: 'PATCHING',
        type: 'patch.validation_failed',
        message: `Patch rejected: ${patchResult.error}`,
        payload: { error: patchResult.error },
      });
      throw new Error(`PATCH_APPLICATION_FAILED: ${patchResult.error}`);
    }

    patch.appliedSuccessfully = true;
    patch.unifiedDiff = patchResult.diff;
    patch.additions = patchResult.additions;
    patch.deletions = patchResult.deletions;

    updateRepairJob(jobId, {
      proposedPatch: patch,
    });

    dispatchJobEvent(jobId, {
      stage: 'PATCHING',
      type: 'patch.applied',
      message: `Surgical patch applied cleanly to ${patch.file} (+${patch.additions} / -${patch.deletions} lines)`,
      payload: {
        file: patch.file,
        diff: patch.unifiedDiff,
        beforeCode: patch.beforeCode,
        afterCode: patch.afterCode,
        additions: patch.additions,
        deletions: patch.deletions,
      },
    });

    // ---------------------------------------------------------------------
    // STAGE 5: TESTING & VERIFYING LOOP
    // ---------------------------------------------------------------------
```

```
    if (abortController.signal.aborted) throw new Error('CANCELLED');

    updateRepairJob(jobId, {
      status: 'TESTING',
      currentStage: 'Running automated sandbox tests against patched code',
    });

    dispatchJobEvent(jobId, {
      stage: 'TESTING',
      type: 'verification.started',
      message: `Executing sandbox verification: "${testCommand}"`,
      payload: { command: testCommand },
    });

    let verificationResult = await executeSandboxCommand(workspaceDir, testCommand, {
      onOutputChunk: (chunk) => {
        dispatchJobEvent(jobId, {
          stage: 'TESTING',
          type: 'tests.output',
          message: chunk,
        });
      },
    });

    let currentJob = getRepairJob(jobId)!;
    const testRuns = [
      ...currentJob.testRuns,
      {
        command: testCommand,
        exitCode: verificationResult.exitCode,
        stdout: verificationResult.stdout,
        stderr: verificationResult.stderr,
        durationMs: verificationResult.durationMs,
        timestamp: new Date().toISOString(),
        stage: 'verification' as const,
```

```
        passed: verificationResult.passed,
      },
    ];

    updateRepairJob(jobId, { testRuns });

    // ---------------------------------------------------------------------
    // RETRY LOOP IF VERIFICATION FAILED (Bounded by maxRetries, default 3)
    // ---------------------------------------------------------------------
    let attempts = 0;
    while (!verificationResult.passed && attempts < (job.maxRetries || 3)) {
      if (abortController.signal.aborted) throw new Error('CANCELLED');
      attempts++;

      dispatchJobEvent(jobId, {
        stage: 'TESTING',
        type: 'patch.validation_failed',
        message: `Verification test failed (attempt ${attempts}/${job.maxRetries}). Synthesizing revised fix...`,
        payload: {
          attempt: attempts,
          exitCode: verificationResult.exitCode,
          stderr: verificationResult.stderr,
        },
      });

      const currentContent = readFile(workspaceDir, patch.file);
      const revisedPatch = await revisePatch({
        affectedFile: patch.file,
        fileContent: currentContent,
        previousPatch: patch,
        testFailureOutput: verificationResult.stderr || verificationResult.stdout,
        attemptNumber: attempts,
      });

```

```
      const reapplyResult = applyPatch(workspaceDir, {
        file: revisedPatch.file,
        beforeCode: revisedPatch.beforeCode,
        afterCode: revisedPatch.afterCode,
      });

      if (reapplyResult.success) {
        revisedPatch.appliedSuccessfully = true;
        revisedPatch.unifiedDiff = reapplyResult.diff;
        updateRepairJob(jobId, {
          proposedPatch: revisedPatch,
          retryCount: attempts,
        });

        // Re-execute test runner
        verificationResult = await executeSandboxCommand(workspaceDir, testCommand, {
          onOutputChunk: (chunk) => {
            dispatchJobEvent(jobId, {
              stage: 'TESTING',
              type: 'tests.output',
              message: chunk,
            });
          },
        });

        const updatedRuns = [
          ...getRepairJob(jobId)!.testRuns,
          {
            command: testCommand,
            exitCode: verificationResult.exitCode,
            stdout: verificationResult.stdout,
            stderr: verificationResult.stderr,
            durationMs: verificationResult.durationMs,
            timestamp: new Date().toISOString(),
            stage: 'verification' as const,
```

```
            passed: verificationResult.passed,
          },
        ];
        updateRepairJob(jobId, { testRuns: updatedRuns });
      }
    }

    // ---------------------------------------------------------------------
    // STAGE 6: FINAL OUTCOME
    // ---------------------------------------------------------------------
    if (verificationResult.passed) {
      updateRepairJob(jobId, {
        status: 'COMPLETED',
        currentStage: 'Fix verified by automated regression test suite',
        verificationOutcome: 'VERIFIED',
        completedAt: new Date().toISOString(),
      });

      dispatchJobEvent(jobId, {
        stage: 'VERIFYING',
        type: 'repair.verified',
        message: 'Patch verified: All automated sandbox tests passed with zero regressions.',
        payload: {
          exitCode: 0,
          durationMs: verificationResult.durationMs,
          verificationOutcome: 'VERIFIED',
        },
      });
    } else {
      updateRepairJob(jobId, {
        status: 'NEEDS_REVIEW',
        currentStage: 'Tests still failing after bounded retry attempts',
        verificationOutcome: 'FAILED',
        unresolvedRisks: ['Patch could not satisfy all automated test assertions within maximum retry limit.'],
```

```
514         completedAt: new Date().toISOString(),
515       });
516
517       dispatchJobEvent(jobId, {
518         stage: 'VERIFYING',
519         type: 'repair.needs_review',
520         message: 'Repair requires human review: test assertions failed after retries.',
521         payload: {
522           exitCode: verificationResult.exitCode,
523           verificationOutcome: 'FAILED',
524         },
525       });
526     }
527
528     return getRepairJob(jobId)!;
529   } catch (err: any) {
530     const isCancelled = err.message === 'CANCELLED';
531     const finalStatus: RepairJobStatus = isCancelled ? 'CANCELLED' : 'FAILED';
532
533     updateRepairJob(jobId, {
534       status: finalStatus,
535       currentStage: isCancelled ? 'Cancelled by user' : `Failed: ${err.message}`,
536       error: err.message,
537       completedAt: new Date().toISOString(),
538     });
539
540     dispatchJobEvent(jobId, {
541       stage: finalStatus,
542       type: isCancelled ? 'repair.cancelled' : 'repair.failed',
543       message: err.message,
544       payload: { error: err.message },
545     });
546
547     return getRepairJob(jobId)!;
548   } finally {
```

```
549       activeJobAbortControllers.delete(jobId);
550       // Cleanup temporary workspace after run (or keep for inspection in tests)
551       if (process.env.KEEP_WORKSPACES !== 'true' && workspaceDir) {
552         await cleanupWorkspace(workspaceDir);
553       }
554     }
555   }
```

File [17/30]: src/lib/server/github-service.ts

TYPESCRIPT SOURCE

Description: *GitHub Octokit API, Webhooks & PR Generator*

```typescript
/**
 * Dedicated GitHub Service Layer for RepoPilot AI.
 * Communicates directly with the official GitHub REST API v3.
 * Follows least-privilege principles and guarantees credentials are never returned to client.
 */

export interface SanitizedGitHubUser {
  id: number;
  login: string;
  name: string;
  email: string | null;
  avatarUrl: string;
  htmlUrl: string;
  bio: string | null;
  publicRepos: number;
}

export interface SanitizedRepository {
  id: string;
  name: string;
  fullName: string;
  owner: string;
  ownerAvatar: string;
  visibility: 'Public' | 'Private';
  primaryLanguage: string;
  lastUpdated: string;
  lastUpdatedRelative: string;
  accessLevel: 'Read / Write' | 'Read' | 'Admin';
  stars: number;
  forks: number;
  defaultBranch: string;
```

```
  gitUrl: string;
  description: string;
  isPrivate: boolean;
  permissions?: {
    admin: boolean;
    push: boolean;
    pull: boolean;
  };
}

const GITHUB_API_BASE = 'https://api.github.com';

/**
 * Format timestamp into human-readable relative time (e.g. "Updated 2 hours ago", "Updated yesterday").
 */
function formatRelativeTime(dateString: string): string {
  try {
    const date = new Date(dateString);
    const now = new Date();
    const diffMs = now.getTime() - date.getTime();
    const diffSec = Math.floor(diffMs / 1000);
    const diffMin = Math.floor(diffSec / 60);
    const diffHours = Math.floor(diffMin / 60);
    const diffDays = Math.floor(diffHours / 24);

    if (diffSec < 60) return 'Updated just now';
    if (diffMin < 60) return `Updated ${diffMin} ${diffMin === 1 ? 'minute' : 'minutes'} ago`;
    if (diffHours < 24) return `Updated ${diffHours} ${diffHours === 1 ? 'hour' : 'hours'} ago`;
    if (diffDays === 1) return 'Updated yesterday';
    if (diffDays < 30) return `Updated ${diffDays} days ago`;
    return `Updated on ${date.toLocaleDateString()}`;
  } catch {
    return 'Recently updated';
  }
}
```

```
67
68  /**
69   * Base GitHub API request wrapper with robust error handling.
70   */
71  async function githubRequest<T>(
72    endpoint: string,
73    accessToken: string,
74    options: RequestInit = {}
75  ): Promise<T> {
76    const url = endpoint.startsWith('http') ? endpoint : `${GITHUB_API_BASE}${endpoint}`;
77
78    const headers = new Headers(options.headers);
79    headers.set('Accept', 'application/vnd.github.v3+json');
80    headers.set('User-Agent', 'RepoPilot-Autonomous-Dev-Agent');
81    headers.set('Authorization', `Bearer ${accessToken}`);
82
83    const response = await fetch(url, {
84      ...options,
85      headers,
86      next: { revalidate: 30 },
87    });
88
89    if (!response.ok) {
90      if (response.status === 401) {
91        throw new Error('GITHUB_TOKEN_EXPIRED: GitHub authorization has expired. Reconnect GitHub to continue.');
92      }
93      if (response.status === 403) {
94        const remaining = response.headers.get('x-ratelimit-remaining');
95        if (remaining === '0') {
96          throw new Error('GITHUB_RATE_LIMIT: GitHub API rate limit exceeded. Please try again later.');
97        }
98        throw new Error('GITHUB_ACCESS_DENIED: GitHub access was denied or required scope is missing.');
99      }
100     if (response.status === 404) {
```

```
      throw new Error('GITHUB_NOT_FOUND: The requested repository was not found or your account does not have permission.');
    }

    const errBody = await response.text().catch(() => '');
    throw new Error(`GitHub API error (${response.status}): ${response.statusText}. ${errBody}`);
  }

  return response.json() as Promise<T>;
}

/**
 * Get authenticated user profile from GitHub.
 */
export async function getGitHubUser(accessToken: string): Promise<SanitizedGitHubUser> {
  const data = await githubRequest<any>('/user', accessToken);
  return {
    id: data.id,
    login: data.login,
    name: data.name || data.login,
    email: data.email || null,
    avatarUrl: data.avatar_url,
    htmlUrl: data.html_url,
    bio: data.bio || null,
    publicRepos: data.public_repos || 0,
  };
}

/**
 * Get repositories accessible by the authenticated user with filtering.
 */
export async function getRepositories(
  accessToken: string,
  options: {
    visibility?: 'all' | 'public' | 'private';
```

```
    affiliation?: 'owner,collaborator,organization_member' | string;
    sort?: 'updated' | 'created' | 'pushed' | 'full_name';
    search?: string;
  } = {}
): Promise<SanitizedRepository[]> {
  const params = new URLSearchParams({
    per_page: '100',
    sort: options.sort || 'updated',
    direction: 'desc',
    affiliation: options.affiliation || 'owner,collaborator,organization_member',
  });

  if (options.visibility && options.visibility !== 'all') {
    params.set('visibility', options.visibility);
  }

  const rawRepos = await githubRequest<any[]>(`/user/repos?${params.toString()}`, accessToken);

  let sanitized: SanitizedRepository[] = rawRepos.map((r) => {
    let accessLevel: 'Read / Write' | 'Read' | 'Admin' = 'Read';
    if (r.permissions?.admin) {
      accessLevel = 'Admin';
    } else if (r.permissions?.push) {
      accessLevel = 'Read / Write';
    }

    return {
      id: String(r.id),
      name: r.name,
      fullName: r.full_name,
      owner: r.owner?.login || '',
      ownerAvatar: r.owner?.avatar_url || '',
      visibility: r.private ? 'Private' : 'Public',
      primaryLanguage: r.language || 'TypeScript',
      lastUpdated: r.updated_at,
```

```
      lastUpdatedRelative: formatRelativeTime(r.updated_at),
      accessLevel,
      stars: r.stargazers_count || 0,
      forks: r.forks_count || 0,
      defaultBranch: r.default_branch || 'main',
      gitUrl: r.clone_url || `https://github.com/${r.full_name}.git`,
      description: r.description || `Repository from ${r.owner?.login || 'GitHub'}`,
      isPrivate: Boolean(r.private),
      permissions: r.permissions,
    };
  });

  // Apply in-memory search if specified
  if (options.search) {
    const q = options.search.toLowerCase();
    sanitized = sanitized.filter(
      (repo) =>
        repo.name.toLowerCase().includes(q) ||
        repo.owner.toLowerCase().includes(q) ||
        repo.primaryLanguage.toLowerCase().includes(q) ||
        repo.description.toLowerCase().includes(q)
    );
  }

  return sanitized;
}

/**
 * Get detailed repository metadata.
 */
export async function getRepository(
  accessToken: string,
  owner: string,
  repo: string
```

```
): Promise<SanitizedRepository> {
  const r = await githubRequest<any>(`/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}`, accessToken);

  let accessLevel: 'Read / Write' | 'Read' | 'Admin' = 'Read';
  if (r.permissions?.admin) {
    accessLevel = 'Admin';
  } else if (r.permissions?.push) {
    accessLevel = 'Read / Write';
  }

  return {
    id: String(r.id),
    name: r.name,
    fullName: r.full_name,
    owner: r.owner?.login || '',
    ownerAvatar: r.owner?.avatar_url || '',
    visibility: r.private ? 'Private' : 'Public',
    primaryLanguage: r.language || 'TypeScript',
    lastUpdated: r.updated_at,
    lastUpdatedRelative: formatRelativeTime(r.updated_at),
    accessLevel,
    stars: r.stargazers_count || 0,
    forks: r.forks_count || 0,
    defaultBranch: r.default_branch || 'main',
    gitUrl: r.clone_url || `https://github.com/${r.full_name}.git`,
    description: r.description || '',
    isPrivate: Boolean(r.private),
    permissions: r.permissions,
  };
}

/**
 * Get repository tree (file structure).
 */
```

```
export async function getRepositoryTree(
  accessToken: string,
  owner: string,
  repo: string,
  branch = 'main'
): Promise<{ tree: Array<{ path: string; mode: string; type: 'blob' | 'tree'; size?: number; sha: string }> }> {
  return githubRequest<{ tree: Array<any> }>(
    `/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}/git/trees/${encodeURIComponent(branch)}?recursive=1`,
    accessToken
  );
}

/**
 * Get branches for a repository.
 */
export async function getBranches(
  accessToken: string,
  owner: string,
  repo: string
): Promise<Array<{ name: string; protected: boolean }>> {
  const branches = await githubRequest<any[]>(
    `/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}/branches?per_page=30`,
    accessToken
  );
  return branches.map((b) => ({ name: b.name, protected: Boolean(b.protected) }));
}

/**
 * Get specific file content from a repository.
 */
export async function getRepositoryFile(
  accessToken: string,
  owner: string,
  repo: string,
```

```
  path: string,
  branch = 'main'
): Promise<{ content: string; encoding: string; size: number }> {
  const data = await githubRequest<any>(
    `/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}/contents/${encodeURIComponent(path)}?ref=${encodeURIComponent(branch)}`,
    accessToken
  );

  let content = data.content;
  if (data.encoding === 'base64') {
    content = Buffer.from(data.content, 'base64').toString('utf8');
  }

  return {
    content,
    encoding: 'utf8',
    size: data.size,
  };
}
```

File [18/30]: src/lib/server/llm-provider.ts

TYPESCRIPT SOURCE

Description: *Multi-Provider LLM Engine (OpenAI, Gemini, Anthropic)*

```typescript
import { RepairJobDiagnosis, RepairJobPatch } from '@/lib/types';
import { createUnifiedDiff } from './sandbox';

export interface LLMConfig {
  provider: 'openai' | 'anthropic' | 'gemini' | 'custom' | 'none';
  apiKey?: string;
  model: string;
  endpoint?: string;
  timeoutMs: number;
}

/**
 * Detect and resolve configured LLM provider from server environment
 */
export function getLLMConfig(): LLMConfig {
  if (process.env.OPENAI_API_KEY) {
    return {
      provider: 'openai',
      apiKey: process.env.OPENAI_API_KEY,
      model: process.env.OPENAI_MODEL || 'gpt-4o',
      endpoint: 'https://api.openai.com/v1/chat/completions',
      timeoutMs: 45000,
    };
  }

  if (process.env.ANTHROPIC_API_KEY) {
    return {
      provider: 'anthropic',
      apiKey: process.env.ANTHROPIC_API_KEY,
      model: process.env.ANTHROPIC_MODEL || 'claude-3-7-sonnet-20250219',
      endpoint: 'https://api.anthropic.com/v1/messages',
```

```
      timeoutMs: 45000,
    };
  }

  if (process.env.GEMINI_API_KEY) {
    return {
      provider: 'gemini',
      apiKey: process.env.GEMINI_API_KEY,
      model: process.env.GEMINI_MODEL || 'gemini-2.0-flash',
      endpoint: `https://generativelanguage.googleapis.com/v1beta/models/${process.env.GEMINI_MODEL || 'gemini-2.0-flash'}:generateContent`,
      timeoutMs: 45000,
    };
  }

  if (process.env.CUSTOM_LLM_ENDPOINT && process.env.CUSTOM_LLM_API_KEY) {
    return {
      provider: 'custom',
      apiKey: process.env.CUSTOM_LLM_API_KEY,
      model: process.env.CUSTOM_LLM_MODEL || 'default',
      endpoint: process.env.CUSTOM_LLM_ENDPOINT,
      timeoutMs: 45000,
    };
  }

  return {
    provider: 'none',
    model: 'none',
    timeoutMs: 10000,
  };
}

/**
 * Validate that an LLM provider is configured; throws clear actionable error if missing
 */
```

```
export function assertLLMConfigured(): LLMConfig {
  const config = getLLMConfig();
  if (config.provider === 'none' || !config.apiKey) {
    throw new Error(
      'NO_LLM_PROVIDER_CONFIGURED: No LLM API key detected. Please configure OPENAI_API_KEY, ANTHROPIC_API_KEY, or GEMINI_API_KEY in your .env.local to run autonomous AI debugging.'
    );
  }
  return config;
}

/**
 * Internal helper to query LLM with strict JSON schema instructions
 */
async function callLLM(
  prompt: string,
  systemInstruction: string,
  config: LLMConfig
): Promise<string> {
  const controller = new AbortController();
  const timer = setTimeout(() => controller.abort(), config.timeoutMs);

  try {
    if (config.provider === 'openai' || config.provider === 'custom') {
      const res = await fetch(config.endpoint!, {
        method: 'POST',
        headers: {
          'Content-Type': 'application/json',
          Authorization: `Bearer ${config.apiKey}`,
        },
        body: JSON.stringify({
          model: config.model,
          temperature: 0.1,
          response_format: { type: 'json_object' },
          messages: [
```

```
            { role: 'system', content: systemInstruction },
            { role: 'user', content: prompt },
          ],
        }),
        signal: controller.signal,
      });

      if (!res.ok) {
        const errText = await res.text();
        throw new Error(`LLM_PROVIDER_ERROR (${config.provider}): ${res.status} ${res.statusText} - ${errText}`);
      }

      const data = await res.json();
      return data.choices?.[0]?.message?.content || '{}';
    }

    if (config.provider === 'anthropic') {
      const res = await fetch(config.endpoint!, {
        method: 'POST',
        headers: {
          'Content-Type': 'application/json',
          'x-api-key': config.apiKey!,
          'anthropic-version': '2023-06-01',
        },
        body: JSON.stringify({
          model: config.model,
          max_tokens: 4096,
          temperature: 0.1,
          system: systemInstruction,
          messages: [{ role: 'user', content: prompt }],
        }),
        signal: controller.signal,
      });

      if (!res.ok) {
```

```
        const errText = await res.text();
        throw new Error(`LLM_PROVIDER_ERROR (anthropic): ${res.status} ${res.statusText} - ${errText}`);
      }

      const data = await res.json();
      return data.content?.[0]?.text || '{}';
    }

    if (config.provider === 'gemini') {
      const url = `${config.endpoint}?key=${config.apiKey}`;
      const res = await fetch(url, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({
          contents: [{ parts: [{ text: `${systemInstruction}\n\nTask:\n${prompt}` }] }],
          generationConfig: {
            temperature: 0.1,
            responseMimeType: 'application/json',
          },
        }),
        signal: controller.signal,
      });

      if (!res.ok) {
        const errText = await res.text();
        throw new Error(`LLM_PROVIDER_ERROR (gemini): ${res.status} ${res.statusText} - ${errText}`);
      }

      const data = await res.json();
      return data.candidates?.[0]?.content?.parts?.[0]?.text || '{}';
    }

    throw new Error(`UNSUPPORTED_PROVIDER: ${config.provider}`);
  } catch (err: any) {
```

```
    if (err.name === 'AbortError') {
      throw new Error(`LLM_TIMEOUT: Model request exceeded timeout limit of ${config.timeoutMs}ms.`);
    }
    throw err;
  } finally {
    clearTimeout(timer);
  }
}

/**
 * Safely parse JSON from model output with fallback cleanup for markdown blocks
 */
function cleanAndParseJSON<T>(rawText: string): T {
  let cleaned = rawText.trim();
  if (cleaned.startsWith('```json')) {
    cleaned = cleaned.replace(/^```json/, '').replace(/```$/, '').trim();
  } else if (cleaned.startsWith('```')) {
    cleaned = cleaned.replace(/^```/, '').replace(/```$/, '').trim();
  }

  try {
    return JSON.parse(cleaned) as T;
  } catch (err: any) {
    throw new Error(`LLM_OUTPUT_MALFORMED: Failed to parse structured JSON from model: ${err.message}`);
  }
}

/**
 * Autonomous Diagnosis:
 * Analyzes repository evidence, failure output, and AST context to locate root cause.
 */
export async function diagnoseIssue(params: {
  issueDescription: string;
  reproductionOutput: string;
  targetFile: string;
```

```
  fileContent: string;
  candidateFiles?: string[];
}): Promise<RepairJobDiagnosis> {
  const config = assertLLMConfigured();

  const systemInstruction = `You are a Senior Autonomous Systems Debugging Agent.
Analyze the provided code and reproduction output. Return ONLY a valid JSON object matching this schema:
{
  "rootCause": "Clear 1-sentence technical root-cause",
  "affectedFile": "relative path to affected file",
  "startLine": number,
  "endLine": number,
  "explanation": "concise technical summary of the flaw",
  "flawExplanation": "plain-English explanation of why this code fails",
  "attackScenario": "how this vulnerability or defect causes failure in practice",
  "fixRationale": "why the proposed architectural fix is minimal and correct",
  "rule": "memorable developer golden rule to avoid this bug",
  "confidence": number between 1 and 100,
  "severity": "LOW" | "MEDIUM" | "HIGH" | "CRITICAL"
}`;

  const prompt = `Issue Report:
${params.issueDescription}

Reproduction Test Failure:
${params.reproductionOutput}

Target File: ${params.targetFile}
File Content:
${params.fileContent}

Candidate Files: ${params.candidateFiles?.join(', ') || 'none'}

Diagnose the exact root cause, line range, and educational breakdown.`;
```

```
238
239   const rawJson = await callLLM(prompt, systemInstruction, config);
240   const parsed = cleanAndParseJSON<any>(rawJson);
241
242   // Validate required structure
243   if (!parsed.rootCause || !parsed.affectedFile) {
244     throw new Error('LLM_OUTPUT_INVALID: Diagnosis response missing rootCause or affectedFile.');
245   }
246
247   return {
248     rootCause: String(parsed.rootCause),
249     affectedFile: String(parsed.affectedFile),
250     startLine: Number(parsed.startLine) || 1,
251     endLine: Number(parsed.endLine) || 1,
252     explanation: String(parsed.explanation || parsed.rootCause),
253     flawExplanation: parsed.flawExplanation ? String(parsed.flawExplanation) : undefined,
254     attackScenario: parsed.attackScenario ? String(parsed.attackScenario) : undefined,
255     fixRationale: parsed.fixRationale ? String(parsed.fixRationale) : undefined,
256     rule: parsed.rule ? String(parsed.rule) : undefined,
257     confidence: Math.min(100, Math.max(1, Number(parsed.confidence) || 95)),
258     severity: ['LOW', 'MEDIUM', 'HIGH', 'CRITICAL'].includes(parsed.severity)
259       ? parsed.severity
260       : 'HIGH',
261   };
262 }
263
264 /**
265  * Autonomous Patch Generation:
266  * Produces minimal-risk, surgical code substitution anchored in existing code.
267  */
268 export async function generatePatch(params: {
269   affectedFile: string;
270   fileContent: string;
271   rootCause: string;
272   issueDescription: string;
```

```
  testFailure: string;
}): Promise<RepairJobPatch> {
  const config = assertLLMConfigured();

  const systemInstruction = `You are an Autonomous Software Engineer synthesizing a surgical code patch.
You MUST output ONLY a valid JSON object matching this schema:
{
  "beforeCode": "exact character-for-character snippet from original file that must be replaced",
  "afterCode": "exact replacement code to substitute",
  "explanation": "concise explanation of why this fix resolves the issue"
}
CRITICAL RULES:
1. beforeCode MUST be an exact verbatim substring of the target file content.
2. Produce the smallest minimal surgical fix possible. Do not rewrite unrelated code.
3. Return valid JSON only.`;

  const prompt = `Target File: ${params.affectedFile}
Root Cause: ${params.rootCause}
Issue Description: ${params.issueDescription}
Failure Details:
${params.testFailure}

Original File Content:
${params.fileContent}

Generate the exact surgical beforeCode and afterCode replacement.`;

  const rawJson = await callLLM(prompt, systemInstruction, config);
  const parsed = cleanAndParseJSON<any>(rawJson);

  if (!parsed.beforeCode || typeof parsed.afterCode !== 'string') {
    throw new Error('LLM_OUTPUT_INVALID: Model patch missing beforeCode or afterCode.');
  }

```

```
    const beforeCode = String(parsed.beforeCode).trim();
    const afterCode = String(parsed.afterCode).trim();
    const explanation = String(parsed.explanation || 'Surgical patch generated by RepoPilot AI.');

    // Verify that beforeCode actually exists in target file
    if (!params.fileContent.includes(beforeCode)) {
      throw new Error(
        `PATCH_ANCHOR_INVALID: Proposed beforeCode cannot be found verbatim in target file "${params.affectedFile}".`
      );
    }

    const unifiedDiff = createUnifiedDiff(params.affectedFile, beforeCode, afterCode);
    const beforeLines = beforeCode.split('\n');
    const afterLines = afterCode.split('\n');

    return {
      file: params.affectedFile,
      beforeCode,
      afterCode,
      unifiedDiff,
      additions: Math.max(0, afterLines.length - beforeLines.length) + 1,
      deletions: Math.max(0, beforeLines.length - afterLines.length) + 1,
      explanation,
      appliedSuccessfully: false,
    };
  }

  /**
   * Autonomous Patch Revision:
   * If a patch applied cleanly but tests failed, requests a revised fix using the real error trace.
   */
  export async function revisePatch(params: {
    affectedFile: string;
    fileContent: string;
    previousPatch: RepairJobPatch;
```

```
  testFailureOutput: string;
  attemptNumber: number;
}): Promise<RepairJobPatch> {
  const config = assertLLMConfigured();

  const systemInstruction = `You are an Autonomous Software Engineer revising a failed patch.
Your previous fix failed verification tests. Analyze the test failure output and provide a revised fix.
Return ONLY a valid JSON object:
{
  "beforeCode": "exact verbatim substring from current file content to replace",
  "afterCode": "replacement code",
  "explanation": "explanation of how the revision addresses the test failure"
}`;

  const prompt = `Attempt Number: ${params.attemptNumber}
Target File: ${params.affectedFile}

Previous Patch:
${params.previousPatch.unifiedDiff}

Verification Test Failure:
${params.testFailureOutput}

Current File Content:
${params.fileContent}

Generate a revised surgical fix that satisfies all test expectations.`;

  const rawJson = await callLLM(prompt, systemInstruction, config);
  const parsed = cleanAndParseJSON<any>(rawJson);

  if (!parsed.beforeCode || typeof parsed.afterCode !== 'string') {
    throw new Error('LLM_OUTPUT_INVALID: Revised patch missing beforeCode or afterCode.');
  }
```

```
376
377     const beforeCode = String(parsed.beforeCode).trim();
378     const afterCode = String(parsed.afterCode).trim();
379     const explanation = String(parsed.explanation || `Revised patch for attempt ${params.attemptNumber}`);
380
381     if (!params.fileContent.includes(beforeCode)) {
382       throw new Error(`PATCH_ANCHOR_INVALID: Revised beforeCode not found in file.`);
383     }
384
385     const unifiedDiff = createUnifiedDiff(params.affectedFile, beforeCode, afterCode);
386     const beforeLines = beforeCode.split('\n');
387     const afterLines = afterCode.split('\n');
388
389     return {
390       file: params.affectedFile,
391       beforeCode,
392       afterCode,
393       unifiedDiff,
394       additions: Math.max(0, afterLines.length - beforeLines.length) + 1,
395       deletions: Math.max(0, beforeLines.length - afterLines.length) + 1,
396       explanation,
397       appliedSuccessfully: false,
398     };
399   }
```

File [19/30]: src/lib/server/auth-helpers.ts

TYPESCRIPT SOURCE

Description: *JWT Session Management & Authentication Helpers*

```typescript
import { NextRequest, NextResponse } from 'next/server';
import { getSession, User, Session, getConnectedGitHubAccount, ConnectedGitHubAccount, getOAuthAccountsByUserId } from './db';

export const SESSION_COOKIE_NAME = 'repopilot_session';
export const OAUTH_STATE_COOKIE_NAME = 'repopilot_oauth_state';

const isProduction = process.env.NODE_ENV === 'production';

/**
 * Retrieve the current authenticated user and session from request cookies.
 */
export async function getAuthenticatedUser(
  req: NextRequest
): Promise<{ user: User; session: Session; connectedGitHub: ConnectedGitHubAccount | null } | null> {
  const sessionToken = req.cookies.get(SESSION_COOKIE_NAME)?.value;
  if (!sessionToken) return null;

  const result = getSession(sessionToken);
  if (!result) return null;

  const connectedGitHub = getConnectedGitHubAccount(result.user.id);

  return {
    user: result.user,
    session: result.session,
    connectedGitHub,
  };
}

/**
 * Enforce authentication on API route; returns 401 response if unauthenticated.
```

```
 */
export async function requireAuth(
  req: NextRequest
): Promise<
  | { user: User; session: Session; connectedGitHub: ConnectedGitHubAccount | null; errorResponse?: never }
  | { user?: never; session?: never; connectedGitHub?: never; errorResponse: NextResponse }
> {
  const auth = await getAuthenticatedUser(req);
  if (!auth) {
    return {
      errorResponse: NextResponse.json(
        { error: 'UNAUTHORIZED: Please sign in to access this resource.' },
        { status: 401 }
      ),
    };
  }
  return auth;
}

/**
 * Attach secure HttpOnly session cookie to response.
 */
export function setSessionCookie(res: NextResponse, sessionToken: string): void {
  res.cookies.set({
    name: SESSION_COOKIE_NAME,
    value: sessionToken,
    httpOnly: true,
    secure: isProduction,
    sameSite: 'lax',
    path: '/',
    maxAge: 30 * 24 * 60 * 60, // 30 days
  });
}

```

```
66  /**
67   * Clear session cookie on logout.
68   */
69  export function clearSessionCookie(res: NextResponse): void {
70    res.cookies.set({
71      name: SESSION_COOKIE_NAME,
72      value: '',
73      httpOnly: true,
74      secure: isProduction,
75      sameSite: 'lax',
76      path: '/',
77      maxAge: 0,
78    });
79  }
80
81  /**
82   * Set secure CSRF OAuth state cookie.
83   */
84  export function setOAuthStateCookie(res: NextResponse, state: string): void {
85    res.cookies.set({
86      name: OAUTH_STATE_COOKIE_NAME,
87      value: state,
88      httpOnly: true,
89      secure: isProduction,
90      sameSite: 'lax',
91      path: '/',
92      maxAge: 10 * 60, // 10 minutes
93    });
94  }
95
96  /**
97   * Clear OAuth state cookie.
98   */
99  export function clearOAuthStateCookie(res: NextResponse): void {
100   res.cookies.set({
```

```
    name: OAUTH_STATE_COOKIE_NAME,
    value: '',
    httpOnly: true,
    secure: isProduction,
    sameSite: 'lax',
    path: '/',
    maxAge: 0,
  });
}

/**
 * Sanitize user object for frontend consumption.
 * Never leaks token or password data.
 */
export function sanitizeUserForClient(user: User, connectedGitHub: ConnectedGitHubAccount | null) {
  const oauthAccounts = getOAuthAccountsByUserId(user.id);
  const providers = oauthAccounts.map((a) => a.provider);

  return {
    id: user.id,
    name: user.name,
    email: user.email,
    avatarUrl: user.avatarUrl,
    createdAt: user.createdAt,
    providers,
    connectedAccounts: {
      google: {
        connected: providers.includes('google'),
        email: user.email,
      },
      github: {
        connected: Boolean(connectedGitHub),
        username: connectedGitHub?.githubUsername || null,
        avatarUrl: connectedGitHub?.avatarUrl || null,
```

```
135           connectedAt: connectedGitHub?.connectedAt || null,
136           scope: connectedGitHub?.scope || null,
137         },
138       },
139     };
140   }
```

File [20/30]: src/lib/server/crypto.ts

TYPESCRIPT SOURCE

Description: *Cryptographic Hashing, AES Encryption & Token Generation*

```typescript
import crypto from 'crypto';

// Secret key for AES-256-GCM encryption at rest
// In production, AUTH_SECRET must be set in environment variables.
const getSecretKey = (): Buffer => {
  const secret = process.env.AUTH_SECRET || 'repopilot-default-development-secret-key-32b!';
  // Hash to ensure exact 32 bytes for aes-256-gcm
  return crypto.createHash('sha256').update(secret).digest();
};

/**
 * Encrypt a sensitive token (e.g. GitHub OAuth Access Token) using AES-256-GCM.
 * Never stores plaintext tokens.
 */
export function encryptToken(plainText: string): string {
  if (!plainText) return '';
  const iv = crypto.randomBytes(12); // 96-bit IV recommended for GCM
  const key = getSecretKey();
  const cipher = crypto.createCipheriv('aes-256-gcm', key, iv);

  let encrypted = cipher.update(plainText, 'utf8', 'hex');
  encrypted += cipher.final('hex');
  const authTag = cipher.getAuthTag().toString('hex');

  // Format: iv:authTag:ciphertext
  return `${iv.toString('hex')}:${authTag}:${encrypted}`;
}

/**
 * Decrypt an AES-256-GCM encrypted token.
 */
```

```
export function decryptToken(encryptedPayload: string): string {
  if (!encryptedPayload) return '';
  try {
    const parts = encryptedPayload.split(':');
    if (parts.length !== 3) return '';

    const [ivHex, authTagHex, encryptedHex] = parts;
    const iv = Buffer.from(ivHex, 'hex');
    const authTag = Buffer.from(authTagHex, 'hex');
    const key = getSecretKey();

    const decipher = crypto.createDecipheriv('aes-256-gcm', key, iv);
    decipher.setAuthTag(authTag);

    let decrypted = decipher.update(encryptedHex, 'hex', 'utf8');
    decrypted += decipher.final('utf8');
    return decrypted;
  } catch (err) {
    console.error('Failed to decrypt token:', err);
    return '';
  }
}

/**
 * Generate a secure cryptographically random session token (32 bytes = 256 bits).
 */
export function generateSessionToken(): string {
  return crypto.randomBytes(32).toString('hex');
}

/**
 * Generate a cryptographically secure OAuth state parameter.
 */
export function generateOAuthState(data: Record<string, any> = {}): string {
  const nonce = crypto.randomBytes(16).toString('hex');
```

```
  const payload = JSON.stringify({ ...data, nonce, exp: Date.now() + 10 * 60 * 1000 }); // 10 min expiry
  return Buffer.from(payload).toString('base64url');
}

/**
 * Verify an OAuth state parameter.
 */
export function verifyOAuthState(state: string): { valid: boolean; data?: any } {
  try {
    const jsonStr = Buffer.from(state, 'base64url').toString('utf8');
    const parsed = JSON.parse(jsonStr);
    if (!parsed || typeof parsed.exp !== 'number') {
      return { valid: false };
    }
    if (Date.now() > parsed.exp) {
      return { valid: false }; // Expired
    }
    return { valid: true, data: parsed };
  } catch {
    return { valid: false };
  }
}

/**
 * Generate PKCE code verifier and code challenge (S256).
 */
export function generatePKCE(): { codeVerifier: string; codeChallenge: string } {
  const codeVerifier = crypto.randomBytes(32).toString('base64url');
  const codeChallenge = crypto.createHash('sha256').update(codeVerifier).digest('base64url');
  return { codeVerifier, codeChallenge };
}
```

File [21/30]: src/lib/server/db.ts

TYPESCRIPT SOURCE

Description: *Local JSON/SQLite Storage for Repair Jobs & Metrics*

```typescript
import fs from 'fs';
import path from 'path';
import { encryptToken, decryptToken, generateSessionToken } from './crypto';
import { RepairJob, RepairEvent } from '@/lib/types';

export interface User {
  id: string;
  name: string;
  email: string | null;
  avatarUrl: string | null;
  createdAt: string;
  updatedAt: string;
}

export interface OAuthAccount {
  id: string;
  userId: string;
  provider: 'google' | 'github';
  providerAccountId: string;
  accessTokenEncrypted: string;
  refreshTokenEncrypted?: string;
  expiresAt?: number;
  scope?: string;
  createdAt: string;
  updatedAt: string;
}

export interface Session {
  id: string; // sessionToken
  userId: string;
  expiresAt: number;
```

```
  createdAt: string;
}

export interface ConnectedGitHubAccount {
  id: string;
  userId: string;
  githubUsername: string;
  githubId: string;
  avatarUrl: string;
  connectedAt: string;
  scope: string;
  lastSyncAt: string;
}

interface DatabaseSchema {
  users: User[];
  oauthAccounts: OAuthAccount[];
  sessions: Session[];
  connectedGitHubAccounts: ConnectedGitHubAccount[];
  repairJobs: RepairJob[];
  repairEvents: RepairEvent[];
}

const DATA_DIR = path.join(process.cwd(), '.data');
const DB_FILE = path.join(DATA_DIR, 'repopilot_db.json');

// Memory cache of DB for fast zero-latency access
let memoryDb: DatabaseSchema | null = null;

function ensureDbLoaded(): DatabaseSchema {
  if (memoryDb) return memoryDb;

  try {
    if (!fs.existsSync(DATA_DIR)) {
      fs.mkdirSync(DATA_DIR, { recursive: true });
```

```
    }

    if (fs.existsSync(DB_FILE)) {
      const content = fs.readFileSync(DB_FILE, 'utf-8');
      memoryDb = JSON.parse(content);
    } else {
      memoryDb = {
        users: [],
        oauthAccounts: [],
        sessions: [],
        connectedGitHubAccounts: [],
        repairJobs: [],
        repairEvents: [],
      };
      fs.writeFileSync(DB_FILE, JSON.stringify(memoryDb, null, 2), 'utf-8');
    }

    // Migration safety for existing DB files
    if (memoryDb) {
      if (!memoryDb.repairJobs) memoryDb.repairJobs = [];
      if (!memoryDb.repairEvents) memoryDb.repairEvents = [];
    }
  } catch (err) {
    console.error('Error loading database, using memory store:', err);
    memoryDb = {
      users: [],
      oauthAccounts: [],
      sessions: [],
      connectedGitHubAccounts: [],
      repairJobs: [],
      repairEvents: [],
    };
  }

```

```
  return memoryDb!;
}

function persistDb(): void {
  if (!memoryDb) return;
  try {
    if (!fs.existsSync(DATA_DIR)) {
      fs.mkdirSync(DATA_DIR, { recursive: true });
    }
    const tempFile = `${DB_FILE}.tmp`;
    fs.writeFileSync(tempFile, JSON.stringify(memoryDb, null, 2), 'utf-8');
    fs.renameSync(tempFile, DB_FILE);
  } catch (err) {
    console.error('Failed to persist database file:', err);
  }
}

// ==========================================
// USER OPERATIONS
// ==========================================

export function getUserById(id: string): User | null {
  const db = ensureDbLoaded();
  return db.users.find((u) => u.id === id) || null;
}

export function getUserByEmail(email: string): User | null {
  const db = ensureDbLoaded();
  if (!email) return null;
  return db.users.find((u) => u.email?.toLowerCase() === email.toLowerCase()) || null;
}

export function createUser(data: { name: string; email: string | null; avatarUrl?: string | null }): User {
  const db = ensureDbLoaded();
  const now = new Date().toISOString();
```

```
    const user: User = {
      id: `usr_${Date.now()}_${Math.random().toString(36).substring(2, 9)}`,
      name: data.name,
      email: data.email ? data.email.toLowerCase() : null,
      avatarUrl: data.avatarUrl || null,
      createdAt: now,
      updatedAt: now,
    };
    db.users.push(user);
    persistDb();
    return user;
}

export function updateUser(id: string, updates: Partial<Omit<User, 'id' | 'createdAt'>>): User | null {
    const db = ensureDbLoaded();
    const index = db.users.findIndex((u) => u.id === id);
    if (index === -1) return null;

    db.users[index] = {
      ...db.users[index],
      ...updates,
      updatedAt: new Date().toISOString(),
    };
    persistDb();
    return db.users[index];
}

// ========================================
// OAUTH ACCOUNT & IDENTITY LINKING
// ========================================

export function getOAuthAccount(provider: 'google' | 'github', providerAccountId: string): OAuthAccount | null {
    const db = ensureDbLoaded();
    return (
```

```
    db.oauthAccounts.find(
      (acc) => acc.provider === provider && acc.providerAccountId === String(providerAccountId)
    ) || null
  );
}

export function getOAuthAccountsByUserId(userId: string): OAuthAccount[] {
  const db = ensureDbLoaded();
  return db.oauthAccounts.filter((acc) => acc.userId === userId);
}

export function linkOAuthAccount(params: {
  userId: string;
  provider: 'google' | 'github';
  providerAccountId: string;
  accessToken: string;
  refreshToken?: string;
  expiresAt?: number;
  scope?: string;
}): OAuthAccount {
  const db = ensureDbLoaded();
  const now = new Date().toISOString();
  const encryptedAccess = encryptToken(params.accessToken);
  const encryptedRefresh = params.refreshToken ? encryptToken(params.refreshToken) : undefined;

  const existingIndex = db.oauthAccounts.findIndex(
    (acc) => acc.provider === params.provider && acc.providerAccountId === String(params.providerAccountId)
  );

  const account: OAuthAccount = {
    id: existingIndex >= 0 ? db.oauthAccounts[existingIndex].id : `oa_${Date.now()}_${Math.random().toString(36).sub
string(2, 7)}`,
    userId: params.userId,
    provider: params.provider,
    providerAccountId: String(params.providerAccountId),
```

```
    accessTokenEncrypted: encryptedAccess,
    refreshTokenEncrypted: encryptedRefresh,
    expiresAt: params.expiresAt,
    scope: params.scope,
    createdAt: existingIndex >= 0 ? db.oauthAccounts[existingIndex].createdAt : now,
    updatedAt: now,
  };

  if (existingIndex >= 0) {
    db.oauthAccounts[existingIndex] = account;
  } else {
    db.oauthAccounts.push(account);
  }

  persistDb();
  return account;
}

export function unlinkOAuthAccount(userId: string, provider: 'google' | 'github'): boolean {
  const db = ensureDbLoaded();
  const initialLen = db.oauthAccounts.length;
  db.oauthAccounts = db.oauthAccounts.filter(
    (acc) => !(acc.userId === userId && acc.provider === provider)
  );
  if (db.oauthAccounts.length !== initialLen) {
    persistDb();
    return true;
  }
  return false;
}

// ========================================
// SESSION MANAGEMENT (Secure HttpOnly Sessions)
// ========================================

```

```
const SESSION_EXPIRY_MS = 30 * 24 * 60 * 60 * 1000; // 30 days

export function createSession(userId: string): Session {
  const db = ensureDbLoaded();
  const sessionToken = generateSessionToken();
  const now = new Date().toISOString();
  const session: Session = {
    id: sessionToken,
    userId,
    expiresAt: Date.now() + SESSION_EXPIRY_MS,
    createdAt: now,
  };
  db.sessions.push(session);
  persistDb();
  return session;
}

export function getSession(sessionToken: string): { session: Session; user: User } | null {
  if (!sessionToken) return null;
  const db = ensureDbLoaded();
  const session = db.sessions.find((s) => s.id === sessionToken);
  if (!session) return null;

  // Check expiration
  if (Date.now() > session.expiresAt) {
    deleteSession(sessionToken);
    return null;
  }

  const user = getUserById(session.userId);
  if (!user) {
    deleteSession(sessionToken);
    return null;
  }
```

```
273
274   return { session, user };
275 }
276
277 export function deleteSession(sessionToken: string): void {
278   const db = ensureDbLoaded();
279   db.sessions = db.sessions.filter((s) => s.id !== sessionToken);
280   persistDb();
281 }
282
283 export function deleteAllUserSessions(userId: string): void {
284   const db = ensureDbLoaded();
285   db.sessions = db.sessions.filter((s) => s.userId !== userId);
286   persistDb();
287 }
288
289 // ========================================
290 // CONNECTED GITHUB ACCOUNT MANAGEMENT
291 // ========================================
292
293 export function getConnectedGitHubAccount(userId: string): ConnectedGitHubAccount | null {
294   const db = ensureDbLoaded();
295   return db.connectedGitHubAccounts.find((c) => c.userId === userId) || null;
296 }
297
298 export function saveConnectedGitHubAccount(data: {
299   userId: string;
300   githubUsername: string;
301   githubId: string;
302   avatarUrl: string;
303   scope: string;
304 }): ConnectedGitHubAccount {
305   const db = ensureDbLoaded();
306   const now = new Date().toISOString();
307   const existingIdx = db.connectedGitHubAccounts.findIndex((c) => c.userId === data.userId);
```

```
308
309   const account: ConnectedGitHubAccount = {
310     id: existingIdx >= 0 ? db.connectedGitHubAccounts[existingIdx].id : `cgh_${Date.now()}`,
311     userId: data.userId,
312     githubUsername: data.githubUsername,
313     githubId: String(data.githubId),
314     avatarUrl: data.avatarUrl,
315     connectedAt: existingIdx >= 0 ? db.connectedGitHubAccounts[existingIdx].connectedAt : now,
316     scope: data.scope,
317     lastSyncAt: now,
318   };
319
320   if (existingIdx >= 0) {
321     db.connectedGitHubAccounts[existingIdx] = account;
322   } else {
323     db.connectedGitHubAccounts.push(account);
324   }
325
326   persistDb();
327   return account;
328 }
329
330 export function removeConnectedGitHubAccount(userId: string): boolean {
331   const db = ensureDbLoaded();
332   const initialLen = db.connectedGitHubAccounts.length;
333   db.connectedGitHubAccounts = db.connectedGitHubAccounts.filter((c) => c.userId !== userId);
334   // Also unlink GitHub OAuth account
335   unlinkOAuthAccount(userId, 'github');
336   if (db.connectedGitHubAccounts.length !== initialLen) {
337     persistDb();
338     return true;
339   }
340   return false;
341 }
```

```
342
343 /**
344  * Securely retrieve decrypted GitHub access token for a user.
345  * Available ONLY on the server side; never sent to frontend.
346  */
347 export function getGitHubAccessToken(userId: string): string | null {
348   const db = ensureDbLoaded();
349   const account = db.oauthAccounts.find(
350     (acc) => acc.userId === userId && acc.provider === 'github'
351   );
352   if (!account || !account.accessTokenEncrypted) return null;
353   return decryptToken(account.accessTokenEncrypted);
354 }
355
356 // ========================================
357 // AUTONOMOUS REPAIR JOB & EVENT OPERATIONS
358 // ========================================
359
360 export function createRepairJob(data: {
361   userId: string;
362   repository: RepairJob['repository'];
363   issueDescription: string;
364   issueCode?: string;
365   reproductionEvidence?: string;
366   maxRetries?: number;
367 }): RepairJob {
368   const db = ensureDbLoaded();
369   const now = new Date().toISOString();
370   const jobId = `job_${Date.now()}_${Math.random().toString(36).substring(2, 8)}`;
371
372   const job: RepairJob = {
373     id: jobId,
374     userId: data.userId,
375     repository: data.repository,
376     issueDescription: data.issueDescription,
```

```
    issueCode: data.issueCode,
    reproductionEvidence: data.reproductionEvidence,
    status: 'QUEUED',
    currentStage: 'Queued for autonomous investigation',
    testRuns: [],
    retryCount: 0,
    maxRetries: data.maxRetries ?? 3,
    verificationOutcome: 'INCOMPLETE',
    unresolvedRisks: [],
    createdAt: now,
    updatedAt: now,
  };

  db.repairJobs.unshift(job);
  persistDb();

  // Record initial event
  appendRepairEvent(jobId, {
    stage: 'QUEUED',
    type: 'repair.started',
    message: `Repair job initialized for ${data.repository.owner}/${data.repository.name}@${data.repository.pinnedCommitSha.substring(0, 7)}`,
    payload: {
      repository: `${data.repository.owner}/${data.repository.name}`,
      branch: data.repository.branch,
      commitSha: data.repository.pinnedCommitSha,
      issueCode: data.issueCode,
    },
  });

  return job;
}

export function getRepairJob(id: string): RepairJob | null {
  const db = ensureDbLoaded();
```

```
411   return db.repairJobs.find((j) => j.id === id) || null;
412 }
413
414 export function getRepairJobsByUserId(userId: string): RepairJob[] {
415   const db = ensureDbLoaded();
416   return db.repairJobs.filter((j) => j.userId === userId);
417 }
418
419 export function updateRepairJob(
420   id: string,
421   updates: Partial<Omit<RepairJob, 'id' | 'userId' | 'createdAt'>>
422 ): RepairJob | null {
423   const db = ensureDbLoaded();
424   const index = db.repairJobs.findIndex((j) => j.id === id);
425   if (index === -1) return null;
426
427   db.repairJobs[index] = {
428     ...db.repairJobs[index],
429     ...updates,
430     updatedAt: new Date().toISOString(),
431   };
432
433   persistDb();
434   return db.repairJobs[index];
435 }
436
437 export function appendRepairEvent(
438   jobId: string,
439   eventData: {
440     type: string;
441     stage: string;
442     message: string;
443     payload?: Record<string, unknown>;
444   }
```

```
): RepairEvent {
  const db = ensureDbLoaded();
  const now = new Date().toISOString();

  // Determine next sequence for this job
  const existingJobEvents = db.repairEvents.filter((e) => e.jobId === jobId);
  const nextSequence = existingJobEvents.length + 1;
  const eventId = `evt_${jobId}_${nextSequence}_${Math.random().toString(36).substring(2, 6)}`;

  const event: RepairEvent = {
    eventId,
    jobId,
    sequence: nextSequence,
    type: eventData.type,
    timestamp: now,
    stage: eventData.stage,
    message: eventData.message,
    payload: eventData.payload,
  };

  db.repairEvents.push(event);
  persistDb();
  return event;
}

export function getRepairEvents(jobId: string, afterSequence: number = 0): RepairEvent[] {
  const db = ensureDbLoaded();
  return db.repairEvents
    .filter((e) => e.jobId === jobId && e.sequence > afterSequence)
    .sort((a, b) => a.sequence - b.sequence);
}

export function deleteRepairJob(id: string): boolean {
  const db = ensureDbLoaded();
  const initJobLen = db.repairJobs.length;
```

```
480     db.repairJobs = db.repairJobs.filter((j) => j.id !== id);
481     db.repairEvents = db.repairEvents.filter((e) => e.jobId !== id);
482     if (db.repairJobs.length !== initJobLen) {
483       persistDb();
484       return true;
485     }
486     return false;
487   }
488
```

File [22/30]: src/lib/server/event-bus.ts

TYPESCRIPT SOURCE

Description: *Real-Time Server-Sent Events (SSE) Event Bus*

```typescript
import { RepairEvent } from '@/lib/types';
import { EventEmitter } from 'events';

class JobEventBus {
  private emitter = new EventEmitter();

  constructor() {
    this.emitter.setMaxListeners(100);
  }

  public emit(jobId: string, event: RepairEvent): void {
    this.emitter.emit(`job:${jobId}`, event);
  }

  public subscribe(
    jobId: string,
    callback: (event: RepairEvent) => void
  ): () => void {
    const channel = `job:${jobId}`;
    this.emitter.on(channel, callback);
    return () => {
      this.emitter.off(channel, callback);
    };
  }
}

// Global singleton across server runtime
const globalEventBus = (global as any).__repopilot_event_bus || new JobEventBus();
if (!(global as any).__repopilot_event_bus) {
  (global as any).__repopilot_event_bus = globalEventBus;
}
```

```
32
33 export const jobEventBus: JobEventBus = globalEventBus;
```

File [23/30]: src/components/views/LiveBugFixPlayer.tsx

TSX SOURCE

Description: *Live Interactive Autonomous Repair Pipeline Visualizer*

```tsx
'use client';

import React, { useState, useEffect, useRef, useCallback } from 'react';
import {
  Play,
  Pause,
  RotateCcw,
  CheckCircle2,
  AlertTriangle,
  FileCode,
  Sparkles,
  ArrowRight,
  ShieldCheck,
  ShieldAlert,
  Zap,
  BookOpen,
  Terminal,
  Copy,
  Check,
  ChevronRight,
  Lightbulb,
  Radio,
  StopCircle,
  RefreshCw,
  GitCommit,
  CheckCheck,
  AlertOctagon,
} from 'lucide-react';
import confetti from 'canvas-confetti';
import { NavTab, RepairJob, RepairEvent, RepairJobDiagnosis, RepairJobPatch } from '@/lib/types';
import { PRIMARY_PATCH } from '@/lib/mock-data';
```

```
32
33  interface LiveBugFixPlayerProps {
34    onNavigate?: (tab: NavTab) => void;
35    onVerifyFix?: () => void;
36    isEmbedded?: boolean;
37  }
38
39  type StepKey = 'detect' | 'explain' | 'fixing' | 'verified';
40  type LearnTab = 'why' | 'exploit' | 'solution' | 'rule';
41
42  export const LiveBugFixPlayer: React.FC<LiveBugFixPlayerProps> = ({
43    onNavigate,
44    onVerifyFix,
45    isEmbedded = false,
46  }) => {
47    const [currentStep, setCurrentStep] = useState<StepKey>('detect');
48    const [isPlaying, setIsPlaying] = useState<boolean>(true);
49    const [playbackSpeed, setPlaybackSpeed] = useState<number>(1);
50    const [copied, setCopied] = useState<boolean>(false);
51    const [activeLearnTab, setActiveLearnTab] = useState<LearnTab>('why');
52    const [editorTheme, setEditorTheme] = useState<'dark' | 'light'>('dark');
53
54    // Real Backend Autonomous Repair Job State
55    const [activeJob, setActiveJob] = useState<RepairJob | null>(null);
56    const [jobEvents, setJobEvents] = useState<RepairEvent[]>([]);
57    const [terminalLogs, setTerminalLogs] = useState<string[]>([]);
58    const [isBackendRunning, setIsBackendRunning] = useState<boolean>(false);
59    const [backendError, setBackendError] = useState<string | null>(null);
60
61    // Dynamic diagnostic & patch state from real backend or verified baseline
62    const [realDiagnosis, setRealDiagnosis] = useState<RepairJobDiagnosis>({
63      rootCause: 'jwt.verify() decoded payload passes truthy condition without enforcing exp timestamp.',
64      affectedFile: 'src/middleware/auth.ts',
65      startLine: 20,
66      endLine: 24,
```

```
    explanation: 'Authentication middleware validates cryptographic token signature but allows expired tokens throug
h downstream.',
    flawExplanation: 'When developers write jwt.verify(token, SECRET), they assume that if it does not throw, the to
ken is safe to use. However, checking if (decoded) only verifies payload existence and does not assert exp expiratio
n.',
    attackScenario: 'An attacker or former employee reuses an expired token extracted from browser logs to access pr
otected administrative endpoints.',
    fixRationale: 'Enforce structural assertion (!decoded || typeof decoded === "string") and explicit expiration va
lidation (decoded.exp * 1000 < Date.now()).',
    rule: 'Never trust that a decoded JWT is valid simply because it passed a truthy condition. Always assert expira
tion and audience claims explicitly.',
    confidence: 97,
    severity: 'HIGH',
  });

  const [realPatch, setRealPatch] = useState<RepairJobPatch>({
    file: PRIMARY_PATCH.file,
    beforeCode: PRIMARY_PATCH.beforeCode,
    afterCode: PRIMARY_PATCH.afterCode,
    unifiedDiff: '',
    additions: PRIMARY_PATCH.additions,
    deletions: PRIMARY_PATCH.deletions,
    explanation: 'Surgical patch applied cleanly to validate token expiration.',
    appliedSuccessfully: true,
  });
  const [pinnedCommit, setPinnedCommit] = useState<string>('7a8f93e');
  const [repoName, setRepoName] = useState<string>('Nexora-X-Hub');

  // Typewriter streaming state for Step 3 (Live Fix)
  const [typedLinesCount, setTypedLinesCount] = useState<number>(0);
  const totalFixLines = 11;

  const timerRef = useRef<NodeJS.Timeout | null>(null);
  const eventSourceRef = useRef<EventSource | null>(null);

  // Trigger celebration only on verified state
  const triggerCelebration = useCallback(() => {
    try {
```

```
      confetti({
        particleCount: 80,
        spread: 70,
        origin: { y: 0.6 },
        colors: ['#10B981', '#6366F1', '#06B6D4', '#F59E0B'],
      });
    } catch {
      // safe fallback
    }
  }, []);

  // Step durations in ms adjusted by playbackSpeed
  const getStepDuration = useCallback(
    (step: StepKey): number => {
      const base = {
        detect: 3600,
        explain: 4200,
        fixing: 4500,
        verified: 6000,
      }[step];
      return base / playbackSpeed;
    },
    [playbackSpeed]
  );

  // ---------------------------------------------------------------------------
  // REAL BACKEND REPAIR JOB ORCHESTRATION
  // ---------------------------------------------------------------------------
  const startRealRepairJob = async () => {
    setBackendError(null);
    setIsBackendRunning(true);
    setTerminalLogs(['$ npm test (reproduction baseline)']);
    setCurrentStep('detect');
    setIsPlaying(true);
```

```
133
134     try {
135       const res = await fetch('/api/repair-jobs', {
136         method: 'POST',
137         headers: { 'Content-Type': 'application/json' },
138         body: JSON.stringify({
139           issueCode: 'AUTH-104',
140           issueDescription:
141             'AUTH-104: Expired JWT tokens bypass authentication middleware without exp timestamp verification.',
142         }),
143       });
144
145       if (!res.ok) {
146         const errData = await res.json().catch(() => ({}));
147         throw new Error(errData.message || errData.error || 'Failed to initialize repair job');
148       }
149
150       const data = await res.json();
151       const job: RepairJob = data.job;
152       setActiveJob(job);
153       setPinnedCommit(job.repository.pinnedCommitSha.substring(0, 7));
154       setRepoName(job.repository.name);
155
156       // Connect real-time Server-Sent Events stream
157       if (eventSourceRef.current) {
158         eventSourceRef.current.close();
159       }
160
161       const sse = new EventSource(`/api/repair-jobs/${job.id}/events`);
162       eventSourceRef.current = sse;
163
164       sse.onmessage = (e) => {
165         try {
166           const event: RepairEvent = JSON.parse(e.data);
167           setJobEvents((prev) => [...prev, event]);
```

```
168
169             // Transition UI based on genuine backend stage events
170             if (event.type === 'repository.checkout.completed') {
171               setCurrentStep('detect');
172             } else if (event.type === 'tests.output') {
173               setTerminalLogs((prev) => [...prev.slice(-30), event.message]);
174             } else if (event.type === 'analysis.location_found') {
175               setCurrentStep('detect');
176             } else if (event.type === 'diagnosis.completed' && event.payload) {
177               setCurrentStep('explain');
178               const diag = event.payload as unknown as RepairJobDiagnosis;
179               setRealDiagnosis(diag);
180             } else if (event.type === 'patch.applied' && event.payload) {
181               setCurrentStep('fixing');
182               const patchPayload = event.payload as any;
183               setRealPatch({
184                 file: patchPayload.file,
185                 beforeCode: patchPayload.beforeCode,
186                 afterCode: patchPayload.afterCode,
187                 unifiedDiff: patchPayload.diff,
188                 additions: patchPayload.additions,
189                 deletions: patchPayload.deletions,
190                 explanation: 'Surgical patch applied cleanly to isolate and resolve AUTH-104.',
191                 appliedSuccessfully: true,
192               });
193             } else if (event.type === 'repair.verified') {
194               setCurrentStep('verified');
195               setIsBackendRunning(false);
196               triggerCelebration();
197               sse.close();
198             } else if (event.type === 'repair.failed' || event.type === 'repair.cancelled') {
199               setIsBackendRunning(false);
200               setBackendError(event.message);
201               sse.close();
```

```
          }
        } catch {
          // ignore parsing error
        }
      };

      sse.onerror = () => {
        // SSE closed or completed
        sse.close();
      };
    } catch (err: any) {
      setIsBackendRunning(false);
      setBackendError(err.message || 'Error executing repair job');
    }
  };

  const cancelActiveJob = async () => {
    if (!activeJob) return;
    try {
      await fetch(`/api/repair-jobs/${activeJob.id}/cancel`, { method: 'POST' });
      if (eventSourceRef.current) eventSourceRef.current.close();
      setIsBackendRunning(false);
      setBackendError('Repair job cancelled by user.');
    } catch {
      // ignore
    }
  };

  // Cleanup SSE on unmount
  useEffect(() => {
    return () => {
      if (eventSourceRef.current) {
        eventSourceRef.current.close();
      }
    };
```

```
237   }, []);
238
239   // ---------------------------------------------------------------------------
240   // PLAYBACK STATE MACHINE (Timeline & Typewriter)
241   // ---------------------------------------------------------------------------
242   useEffect(() => {
243     if (!isPlaying) {
244       if (timerRef.current) clearTimeout(timerRef.current);
245       return;
246     }
247
248     if (currentStep === 'detect') {
249       timerRef.current = setTimeout(() => {
250         setCurrentStep('explain');
251       }, getStepDuration('detect'));
252     } else if (currentStep === 'explain') {
253       timerRef.current = setTimeout(() => {
254         setCurrentStep('fixing');
255       }, getStepDuration('explain'));
256     } else if (currentStep === 'fixing') {
257       const lineInterval = (getStepDuration('fixing') * 0.7) / totalFixLines;
258       let count = 0;
259
260       const resetTimer = setTimeout(() => {
261         setTypedLinesCount(0);
262       }, 0);
263
264       const interval = setInterval(() => {
265         count += 1;
266         setTypedLinesCount(count);
267         if (count >= totalFixLines) {
268           clearInterval(interval);
269           timerRef.current = setTimeout(() => {
270             setCurrentStep('verified');
```

```
            triggerCelebration();
          }, 600 / playbackSpeed);
        }
      }, lineInterval);

      return () => {
        clearTimeout(resetTimer);
        clearInterval(interval);
        if (timerRef.current) clearTimeout(timerRef.current);
      };
    } else if (currentStep === 'verified') {
      setIsPlaying(false);
    }

    return () => {
      if (timerRef.current) clearTimeout(timerRef.current);
    };
  }, [currentStep, isPlaying, playbackSpeed, getStepDuration, triggerCelebration]);

  const handleRestart = () => {
    setCurrentStep('detect');
    setTypedLinesCount(0);
    setIsPlaying(true);
  };

  const handleManualStepChange = (step: StepKey) => {
    setCurrentStep(step);
    if (step === 'fixing') {
      setTypedLinesCount(totalFixLines);
    } else if (step === 'verified') {
      setTypedLinesCount(totalFixLines);
      triggerCelebration();
    }
  };

```

```
306   const handleCopyCode = () => {
307     navigator.clipboard.writeText(realPatch.afterCode);
308     setCopied(true);
309     setTimeout(() => setCopied(false), 2000);
310   };
311
312   // Code Preamble & Postamble
313   const codePreamble = [
314     { num: 14, text: 'export const authenticateToken = (req: AuthenticatedRequest, res: Response, next: NextFunctio
n) => {' },
315     { num: 15, text: "  const authHeader = req.headers['authorization'];" },
316     { num: 16, text: "  const token = authHeader && authHeader.split(' ')[1];" },
317     { num: 17, text: "  if (!token) return res.status(401).json({ error: 'Token missing' });" },
318     { num: 18, text: '' },
319     { num: 19, text: '  try {' },
320   ];
321
322   const buggyCodeLines = [
323     { num: 20, text: '    // BUG AUTH-104: Signature is decoded without validating exp expiration timestamp' },
324     { num: 21, text: '    const decoded = jwt.verify(token, SECRET);' },
325     { num: 22, text: '    if (decoded) {' },
326     { num: 23, text: '      return next();' },
327     { num: 24, text: '    }' },
328   ];
329
330   const fixedReplacementLines = realPatch.afterCode.split('\n').map((line, idx) => ({
331     num: 20 + idx,
332     text: line.startsWith('    ') ? line : `    ${line}`,
333   }));
334
335   const codePostamble = [
336     { num: 20 + fixedReplacementLines.length, text: '  } catch (err) {' },
337     { num: 21 + fixedReplacementLines.length, text: "    return res.status(403).json({ error: 'Invalid token signatu
re' });" },
338     { num: 22 + fixedReplacementLines.length, text: '  }' },
```

```
    { num: 23 + fixedReplacementLines.length, text: '};' },
  ];

  return (
    <div className={`space-y-6 ${isEmbedded ? '' : 'max-w-7xl mx-auto pb-16'}`}>
      {/* Top Banner & Mode Header */}
      <div className="flex flex-col lg:flex-row lg:items-center justify-between gap-4 border-b border-slate-200 pb-5">
        <div>
          <div className="flex flex-wrap items-center gap-2">
            <span className="text-xs uppercase font-extrabold tracking-wider px-2.5 py-0.5 rounded-full bg-gradient-to-r from-indigo-600 to-cyan-600 text-white shadow-xs flex items-center gap-1.5">
              <Zap className="w-3.5 h-3.5" />
              Autonomous Debugging &amp;amp; Repair Engine
            </span>
            <span className="text-xs font-mono font-bold bg-rose-50 text-rose-700 border border-rose-200 px-2 py-0.5 rounded">
              AUTH-104
            </span>
            <span className="text-xs font-mono text-slate-500 flex items-center gap-1">
              <GitCommit className="w-3.5 h-3.5 text-indigo-500" />
              <span>{repoName}@{pinnedCommit}</span>
            </span>
            <span className="text-xs font-mono text-slate-400">
              {realDiagnosis.affectedFile} (L{realDiagnosis.startLine}–L{realDiagnosis.endLine})
            </span>
          </div>

          <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1.5 flex items-center gap-2">
            <span>Live Autonomous Bug Resolution Studio</span>
            {currentStep === 'verified' && (
              <span className="text-xs font-semibold px-2.5 py-1 rounded-full bg-emerald-100 text-emerald-800 border border-emerald-300 flex items-center gap-1 animate-in fade-in">
                <CheckCheck className="w-3.5 h-3.5 text-emerald-600" />
                Verified with Zero Regressions
              </span>
            )}
```

```
372           </h1>
373           <p className="text-sm text-slate-600">
374             End-to-end autonomous repair: real AST tracing, surgical patch synthesis, and genuine sandbox verificati
    on.
375           </p>
376         </div>
377
378         {/* Action Buttons & Real Job Launcher */}
379         <div className="flex flex-wrap items-center gap-2.5 self-start lg:self-auto">
380           {isBackendRunning ? (
381             <button
382               onClick={cancelActiveJob}
383               className="flex items-center gap-1.5 px-4 py-2 rounded-xl bg-rose-50 hover:bg-rose-100 text-rose-700 t
    ext-xs font-bold border border-rose-200 transition-all shadow-xs"
384             >
385               <StopCircle className="w-4 h-4 text-rose-600 animate-pulse" />
386               <span>Cancel Active Job</span>
387             </button>
388           ) : (
389             <button
390               onClick={startRealRepairJob}
391               className="flex items-center gap-1.5 px-4 py-2 rounded-xl bg-gradient-to-r from-indigo-600 to-cyan-600
    hover:from-indigo-700 hover:to-cyan-700 text-white text-xs font-bold shadow-md shadow-indigo-600/20 hover:scale-[1.0
    2] active:scale-[0.98] transition-all"
392             >
393               <Radio className="w-3.5 h-3.5 text-cyan-300 animate-pulse" />
394               <span>Start Real Repair Job</span>
395             </button>
396           )}
397
398           <button
399             onClick={handleCopyCode}
400             className="flex items-center gap-1.5 px-3.5 py-2 rounded-xl bg-white hover:bg-slate-50 text-slate-700 te
    xt-xs font-semibold border border-slate-200 transition-all shadow-xs"
401           >
402             {copied ? (
403               <>
```

```
                <Check className="w-3.5 h-3.5 text-emerald-600" />
                <span className="text-emerald-700">Copied Fix</span>
              </>
            ) : (
              <>
                <Copy className="w-3.5 h-3.5 text-slate-500" />
                <span>Copy Fixed Code</span>
              </>
            )}
          </button>

          {onVerifyFix && (
            <button
              onClick={onVerifyFix}
              className="flex items-center gap-2 px-5 py-2.5 rounded-xl bg-slate-900 hover:bg-slate-800 text-white font-semibold text-xs shadow-md shadow-slate-900/10 hover:scale-[1.02] active:scale-[0.98] transition-all"
            >
              <ShieldCheck className="w-4 h-4 text-emerald-400" />
              <span>Sandbox Suite</span>
              <ArrowRight className="w-3.5 h-3.5" />
            </button>
          )}
        </div>
      </div>

      {/* Backend Status / Error Banner */}
      {backendError && (
        <div className="p-3.5 rounded-xl bg-rose-50 border border-rose-200 text-xs text-rose-800 flex items-center justify-between gap-3 animate-in fade-in">
          <div className="flex items-center gap-2">
            <AlertOctagon className="w-4 h-4 text-rose-600 shrink-0" />
            <span>{backendError}</span>
          </div>
          <button
            onClick={startRealRepairJob}
            className="text-[11px] font-bold text-rose-700 hover:underline shrink-0"
```

```
          >
            Retry with Safe Execution
          </button>
        </div>
      )}

      {/* Interactive Step Navigator & Playback HUD */}
      <div className="p-4 rounded-2xl bg-white border border-slate-200 shadow-xs flex flex-col md:flex-row items-center justify-between gap-4">
        {/* Step Flow Buttons */}
        <div className="grid grid-cols-2 sm:grid-cols-4 gap-2 w-full md:w-auto">
          {[
            { id: 'detect' as StepKey, label: '1. Pinpoint Bug' },
            { id: 'explain' as StepKey, label: '2. Understand Flaw' },
            { id: 'fixing' as StepKey, label: '3. Live Self-Healing' },
            { id: 'verified' as StepKey, label: '4. Bug Cleared ✓' },
          ].map((s) => {
            const isActive = currentStep === s.id;
            return (
              <button
                key={s.id}
                onClick={() => handleManualStepChange(s.id)}
                className={`px-3 py-2 rounded-xl text-xs font-bold transition-all flex items-center justify-center gap-1.5 text-left border ${
                  isActive
                    ? 'bg-slate-900 text-white border-slate-900 shadow-sm scale-[1.02]'
                    : 'bg-slate-50 text-slate-600 border-slate-200 hover:bg-slate-100 hover:text-slate-900'
                }`}
              >
                <span
                  className={`w-2 h-2 rounded-full ${
                    isActive ? 'bg-cyan-400 animate-ping' : 'bg-slate-300'
                  }`}
                />
                <span>{s.label}</span>
```

```
              </button>
            );
          })}
        </div>

        {/* Playback Controls (Play, Pause, Replay, Speed, Theme) */}
        <div className="flex items-center gap-2 shrink-0">
          <button
            onClick={() => setIsPlaying(!isPlaying)}
            className="flex items-center gap-1.5 px-3 py-1.5 rounded-lg bg-indigo-50 hover:bg-indigo-100 text-indigo-700 text-xs font-bold border border-indigo-200 transition-colors"
            title={isPlaying ? 'Pause Auto-Play' : 'Play Live Animation'}
          >
            {isPlaying ? (
              <>
                <Pause className="w-3.5 h-3.5" />
                <span>Pause</span>
              </>
            ) : (
              <>
                <Play className="w-3.5 h-3.5" />
                <span>Play Live</span>
              </>
            )}
          </button>

          <button
            onClick={handleRestart}
            className="p-1.5 rounded-lg bg-slate-100 hover:bg-slate-200 text-slate-700 text-xs font-medium border border-slate-200 transition-colors"
            title="Replay from Beginning"
          >
            <RotateCcw className="w-3.5 h-3.5" />
          </button>

          {/* Speed Toggle */}
```

```
          <div className="flex items-center rounded-lg bg-slate-100 p-0.5 border border-slate-200 text-[11px] font-bold">
            {[1, 1.5, 2].map((spd) => (
              <button
                key={spd}
                onClick={() => setPlaybackSpeed(spd)}
                className={`px-2 py-1 rounded transition-colors ${
                  playbackSpeed === spd
                    ? 'bg-white text-indigo-700 shadow-xs'
                    : 'text-slate-500 hover:text-slate-900'
                }`}
              >
                {spd}x
              </button>
            ))}
          </div>

          {/* Editor Theme Switcher */}
          <button
            onClick={() => setEditorTheme(editorTheme === 'dark' ? 'light' : 'dark')}
            className="px-2.5 py-1.5 rounded-lg bg-slate-100 hover:bg-slate-200 text-slate-700 text-[11px] font-semibold border border-slate-200"
          >
            {editorTheme === 'dark' ? '🌙 Dark' : '🌞 Light'}
          </button>
        </div>
      </div>

      {/* MAIN TWO-COLUMN WORKSPACE: LIVE CODE TERMINAL (Left) & LEARNER COMPASS (Right) */}
      <div className="grid grid-cols-1 lg:grid-cols-12 gap-6 items-start">
        {/* Left Column: Live Interactive Code Editor Terminal */}
        <div className="lg:col-span-7 flex flex-col rounded-2xl border border-slate-300 shadow-md overflow-hidden bg-white">
          {/* Editor Top Bar with File Path & Real-time Live State Pill */}
          <div className="px-4 py-3 bg-slate-900 text-slate-200 flex items-center justify-between border-b border-slate-800">
```

```
            <div className="flex items-center gap-3">
              <div className="flex items-center gap-1.5">
                <span className="w-3 h-3 rounded-full bg-rose-500" />
                <span className="w-3 h-3 rounded-full bg-amber-500" />
                <span className="w-3 h-3 rounded-full bg-emerald-500" />
              </div>
              <div className="flex items-center gap-1.5 text-xs font-mono text-slate-300">
                <FileCode className="w-3.5 h-3.5 text-indigo-400" />
                <span className="font-semibold text-white">auth.ts</span>
                <span className="text-slate-500">src/middleware/</span>
              </div>
            </div>

            {/* Dynamic Status Pill */}
            <div>
              {currentStep === 'detect' && (
                <span className="text-[11px] font-bold font-mono px-2.5 py-0.5 rounded-full bg-rose-500/20 text-rose
-300 border border-rose-500/30 flex items-center gap-1.5 animate-pulse">
                  <span className="w-2 h-2 rounded-full bg-rose-500" />
                  BUG ACTIVE (L{realDiagnosis.startLine}–L{realDiagnosis.endLine})
                </span>
              )}
              {currentStep === 'explain' && (
                <span className="text-[11px] font-bold font-mono px-2.5 py-0.5 rounded-full bg-amber-500/20 text-amb
er-300 border border-amber-500/30 flex items-center gap-1.5">
                  <span className="w-2 h-2 rounded-full bg-amber-400 animate-ping" />
                  ANALYZING FLAW...
                </span>
              )}
              {currentStep === 'fixing' && (
                <span className="text-[11px] font-bold font-mono px-2.5 py-0.5 rounded-full bg-indigo-500/20 text-in
digo-300 border border-indigo-500/30 flex items-center gap-1.5">
                  <Sparkles className="w-3 h-3 text-indigo-400 animate-spin" />
                  LIVE REPLACING CODE...
                </span>
              )}
```

```
570               {currentStep === 'verified' && (
571                 <span className="text-[11px] font-bold font-mono px-2.5 py-0.5 rounded-full bg-emerald-500/20 text-e
merald-300 border border-emerald-500/30 flex items-center gap-1.5">
572                   <CheckCircle2 className="w-3 h-3 text-emerald-400" />
573                   BUG CLEARED &amp;amp; SECURED
574                 </span>
575               )}
576             </div>
577           </div>
578
579           {/* Code Viewer Body */}
580           <div
581             className={`p-4 font-mono text-xs overflow-x-auto leading-relaxed transition-colors duration-300 min-h-
[440px] select-text ${
582               editorTheme === 'dark' ? 'bg-[#0F172A] text-slate-300' : 'bg-slate-50 text-slate-800'
583             }`}
584           >
585             {/* 1. Code Preamble */}
586             <div className="opacity-75">
587               {codePreamble.map((line) => (
588                 <div key={line.num} className="flex items-start gap-4 px-2 py-0.5 hover:bg-slate-500/10 rounded">
589                   <span className="text-slate-500 select-none w-6 text-right shrink-0">{line.num}</span>
590                   <span className="whitespace-pre">{line.text}</span>
591                 </div>
592               ))}
593             </div>
594
595             {/* 2. DYNAMIC BUGGY VS FIXED SECTION */}
596             {currentStep === 'detect' || currentStep === 'explain' ? (
597               /* VULNERABLE BUGGY STATE */
598               <div className="my-1 py-1 rounded-xl bg-rose-950/30 border-2 border-rose-500/60 shadow-inner relative
transition-all">
599                 {/* Floating Alert Tag */}
600                 <div className="absolute -top-3 right-4 px-2.5 py-0.5 rounded-md bg-rose-600 text-white text-[10px]
font-bold uppercase tracking-wider flex items-center gap-1 shadow-sm">
601                   <AlertTriangle className="w-3 h-3" />
602                   <span>Vulnerability Located Here (Line {realDiagnosis.startLine + 2})</span>
```

```
                </div>

                {buggyCodeLines.map((line) => (
                  <div
                    key={line.num}
                    className={`flex items-start gap-4 px-2 py-0.5 ${
                      line.num === 22 || line.num === 23
                        ? 'bg-rose-500/20 text-rose-200 font-bold border-l-4 border-rose-500'
                        : 'text-rose-300'
                    }`}
                  >
                    <span className="text-rose-400 select-none w-6 text-right shrink-0 font-bold">
                      {line.num}
                    </span>
                    <span className="whitespace-pre">{line.text}</span>
                  </div>
                ))}

                {/* Inline Hint Card inside Editor */}
                <div className="mx-3 my-2 p-2.5 rounded-lg bg-rose-900/40 border border-rose-500/40 text-[11px] text-rose-200 flex items-center justify-between">
                  <div className="flex items-center gap-2">
                    <ShieldAlert className="w-4 h-4 text-rose-400 shrink-0" />
                    <span>
                      <strong>The Problem:</strong> `jwt.verify()` decoded payload is accepted without inspecting `decoded.exp`.
                    </span>
                  </div>
                  <button
                    onClick={() => handleManualStepChange('fixing')}
                    className="px-2.5 py-1 rounded bg-rose-600 hover:bg-rose-700 text-white font-bold text-[10px] shrink-0 transition-colors"
                  >
                    Fix Live ⚡
                  </button>
                </div>
```

```
              </div>
            ) : currentStep === 'fixing' ? (
              /* LIVE REPLACEMENT ANIMATION IN PROGRESS */
              <div className="my-1 space-y-1">
                {/* Struck-through dying old lines */}
                <div className="p-2 rounded-lg bg-rose-950/20 border border-rose-500/30 opacity-40 line-through text-rose-400">
                  <div className="text-[10px] uppercase font-bold text-rose-400 mb-1">
                    Clearing vulnerable code...
                  </div>
                  {buggyCodeLines.slice(1, 4).map((line) => (
                    <div key={line.num} className="flex gap-4 px-2">
                      <span className="w-6 text-right select-none">{line.num}</span>
                      <span>{line.text}</span>
                    </div>
                  ))}
                </div>

                {/* Typewriter insertion */}
                <div className="p-2 rounded-xl bg-indigo-950/40 border-2 border-indigo-400/80 shadow-md animate-in fade-in">
                  <div className="text-[10px] font-bold text-indigo-300 mb-1 flex items-center gap-1.5">
                    <Sparkles className="w-3 h-3 text-cyan-400 animate-spin" />
                    <span>RepoPilot AI live-synthesizing safe assertion:</span>
                  </div>

                  {fixedReplacementLines.slice(0, typedLinesCount).map((line) => (
                    <div
                      key={line.num}
                      className="flex items-start gap-4 px-2 py-0.5 bg-emerald-500/10 text-emerald-300 font-mono"
                    >
                      <span className="text-indigo-400 select-none w-6 text-right shrink-0">
                        {line.num}
                      </span>
                      <span className="whitespace-pre">{line.text}</span>
```

```
                    </div>
                  ))}

                  {typedLinesCount < fixedReplacementLines.length && (
                    <div className="flex items-center gap-2 px-2 py-0.5 text-cyan-400 font-bold">
                      <span className="w-2 h-4 bg-cyan-400 animate-pulse inline-block" />
                      <span className="text-[11px] italic">Streaming fix lines...</span>
                    </div>
                  )}
                </div>
              </div>
            ) : (
              /* VERIFIED GREEN SAFE STATE */
              <div className="my-1 py-1.5 rounded-xl bg-emerald-950/40 border-2 border-emerald-500/70 shadow-lg relative animate-in fade-in duration-300">
                <div className="absolute -top-3 right-4 px-2.5 py-0.5 rounded-md bg-emerald-600 text-white text-[10px] font-bold uppercase tracking-wider flex items-center gap-1 shadow-sm">
                  <CheckCircle2 className="w-3 h-3" />
                  <span>Secure Fix Applied &amp;amp; Verified (+{realPatch.additions} / -{realPatch.deletions})</span>
                </div>

                {fixedReplacementLines.map((line) => (
                  <div
                    key={line.num}
                    className="flex items-start gap-4 px-2 py-0.5 bg-emerald-500/10 text-emerald-200 font-mono hover:bg-emerald-500/20 rounded"
                  >
                    <span className="text-emerald-400 select-none w-6 text-right shrink-0 font-bold">
                      {line.num}
                    </span>
                    <span className="whitespace-pre">{line.text}</span>
                  </div>
                ))}
              </div>
            )}
```

```
            {/* 3. Code Postamble */}
            <div className="opacity-75">
              {codePostamble.map((line) => (
                <div key={line.num} className="flex items-start gap-4 px-2 py-0.5 hover:bg-slate-500/10 rounded">
                  <span className="text-slate-500 select-none w-6 text-right shrink-0">{line.num}</span>
                  <span className="whitespace-pre">{line.text}</span>
                </div>
              ))}
            </div>
          </div>

          {/* Bottom Live Sandbox Test Runner Console */}
          <div className="p-3 bg-slate-900 border-t border-slate-800 text-xs font-mono flex flex-col sm:flex-row ite
ms-start sm:items-center justify-between gap-3 text-slate-300">
            <div className="flex items-center gap-2">
              <Terminal className="w-4 h-4 text-cyan-400 shrink-0" />
              <span>Real Sandbox Runner:</span>
              <span className="text-slate-400">node --test tests/auth.test.js</span>
            </div>

            <div>
              {currentStep === 'verified' ? (
                <div className="flex items-center gap-2 text-emerald-400 font-bold">
                  <span className="w-2 h-2 rounded-full bg-emerald-400" />
                  <span>PASS: All tests passing (18ms • exit code 0)</span>
                </div>
              ) : currentStep === 'fixing' ? (
                <div className="flex items-center gap-2 text-indigo-400">
                  <span className="w-2 h-2 rounded-full bg-indigo-400 animate-ping" />
                  <span>Applying patch &amp;amp; re-running test suite...</span>
                </div>
              ) : (
                <div className="flex items-center gap-2 text-rose-400 font-bold">
                  <span className="w-2 h-2 rounded-full bg-rose-500" />
                  <span>FAIL: Expired token accepted (Status 200 != 401)</span>
```

```
                </div>
              )}
            </div>
          </div>

          {/* Live Terminal Output Stream */}
          {terminalLogs.length > 0 && (
            <div className="p-3 bg-black/90 text-slate-300 text-[11px] font-mono border-t border-slate-800 max-h-32 overflow-y-auto">
              {terminalLogs.map((log, index) => (
                <div key={index} className="whitespace-pre-wrap leading-tight text-slate-400">
                  {log}
                </div>
              ))}
            </div>
          )}
        </div>

        {/* Right Column: Educational Learner Compass & Aha! Moment */}
        <div className="lg:col-span-5 space-y-4">
          <div className="p-5 rounded-2xl bg-white border border-slate-200 shadow-xs">
            <div className="flex items-center justify-between mb-4 pb-3 border-b border-slate-100">
              <div className="flex items-center gap-2">
                <div className="w-8 h-8 rounded-xl bg-indigo-50 border border-indigo-200 flex items-center justify-center text-indigo-600">
                  <BookOpen className="w-4 h-4" />
                </div>
                <div>
                  <h3 className="text-sm font-bold text-slate-900">
                    Learner Compass &amp;amp; Explanation
                  </h3>
                  <p className="text-[11px] text-slate-500">
                    Grounded in actual repository diagnosis ({realDiagnosis.severity} Risk)
                  </p>
                </div>
              </div>
```

```
770
771                 <span className="text-[10px] font-bold px-2 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-
indigo-200">
772                   Confidence: {realDiagnosis.confidence}%
773                 </span>
774               </div>
775
776               {/* Sub-tabs for Learner */}
777               <div className="flex items-center p-1 rounded-xl bg-slate-100 border border-slate-200 text-xs mb-4">
778                 {[
779                   { id: 'why' as LearnTab, label: 'Why it Failed' },
780                   { id: 'exploit' as LearnTab, label: 'The Attack' },
781                   { id: 'solution' as LearnTab, label: 'The Fix' },
782                   { id: 'rule' as LearnTab, label: 'Golden Rule' },
783                 ].map((tab) => (
784                   <button
785                     key={tab.id}
786                     onClick={() => setActiveLearnTab(tab.id)}
787                     className={`flex-1 py-1.5 rounded-lg font-semibold transition-all text-center ${
788                       activeLearnTab === tab.id
789                         ? 'bg-white text-indigo-700 shadow-xs'
790                         : 'text-slate-600 hover:text-slate-900'
791                     }`}
792                   >
793                     {tab.label}
794                   </button>
795                 ))}
796               </div>
797
798               {/* Tab 1: Why it Failed */}
799               {activeLearnTab === 'why' && (
800                 <div className="space-y-3 animate-in fade-in">
801                   <div className="p-3.5 rounded-xl bg-rose-50/70 border border-rose-200 text-xs text-rose-900 space-y-
2">
802                     <div className="font-bold flex items-center gap-1.5 text-rose-800">
```

```
                    <AlertTriangle className="w-4 h-4 text-rose-600" />
                    <span>The Flaw in Line {realDiagnosis.startLine + 2}</span>
                  </div>
                  <p className="leading-relaxed text-slate-700">
                    {realDiagnosis.flawExplanation}
                  </p>
                </div>

                <div className="p-3 rounded-xl bg-slate-50 border border-slate-200 text-xs space-y-1">
                  <span className="font-bold text-slate-800">Root Cause:</span>
                  <div className="font-mono text-indigo-700 bg-white p-2 rounded border border-slate-200 text-[11px]">
                    {realDiagnosis.rootCause}
                  </div>
                </div>
              </div>
            )}

            {/* Tab 2: The Attack */}
            {activeLearnTab === 'exploit' && (
              <div className="space-y-3 animate-in fade-in">
                <div className="p-3.5 rounded-xl bg-amber-50/70 border border-amber-200 text-xs text-amber-950 space-y-2">
                  <div className="font-bold flex items-center gap-1.5 text-amber-900">
                    <ShieldAlert className="w-4 h-4 text-amber-700" />
                    <span>Attack Scenario</span>
                  </div>
                  <p className="text-slate-700 leading-relaxed">
                    {realDiagnosis.attackScenario}
                  </p>
                </div>
              </div>
            )}

            {/* Tab 3: The Fix */}
            {activeLearnTab === 'solution' && (
```

```
              <div className="space-y-3 animate-in fade-in">
                <div className="p-3.5 rounded-xl bg-emerald-50/70 border border-emerald-200 text-xs text-emerald-950 space-y-2">
                  <div className="font-bold flex items-center gap-1.5 text-emerald-800">
                    <CheckCircle2 className="w-4 h-4 text-emerald-600" />
                    <span>Fix Rationale</span>
                  </div>
                  <p className="leading-relaxed text-slate-700">
                    {realDiagnosis.fixRationale}
                  </p>
                </div>
              </div>
            )}

            {/* Tab 4: Golden Rule */}
            {activeLearnTab === 'rule' && (
              <div className="space-y-3 animate-in fade-in">
                <div className="p-4 rounded-xl bg-gradient-to-br from-indigo-50 via-cyan-50 to-emerald-50 border border-indigo-200 text-xs text-indigo-950 space-y-2">
                  <div className="font-bold flex items-center gap-1.5 text-indigo-900 text-sm">
                    <Lightbulb className="w-4 h-4 text-amber-500" />
                    <span>Developer Golden Rule</span>
                  </div>
                  <blockquote className="italic border-l-2 border-indigo-400 pl-2.5 py-1 text-slate-800 font-medium">
                    &ldquo;{realDiagnosis.rule}&rdquo;
                  </blockquote>
                </div>
              </div>
            )}
          </div>

          {/* Quick Metrics & Educational Takeaway Card */}
          <div className="p-5 rounded-2xl bg-white border border-slate-200 shadow-xs space-y-3">
            <h4 className="text-xs font-bold uppercase tracking-wider text-slate-700 flex items-center gap-1.5">
              <Sparkles className="w-3.5 h-3.5 text-indigo-600" />
```

```
                <span>Real Execution Metrics</span>
              </h4>

              <div className="grid grid-cols-2 gap-2 text-xs">
                <div className="p-2.5 rounded-xl bg-slate-50 border border-slate-200">
                  <span className="text-slate-500 text-[10px] block">Line Delta</span>
                  <span className="font-mono font-bold text-slate-800">+{realPatch.additions} / -{realPatch.deletions}
lines</span>
                </div>
                <div className="p-2.5 rounded-xl bg-slate-50 border border-slate-200">
                  <span className="text-slate-500 text-[10px] block">Regression Risk</span>
                  <span className="font-mono font-bold text-emerald-600">LOW (0 breaking)</span>
                </div>
                <div className="p-2.5 rounded-xl bg-slate-50 border border-slate-200">
                  <span className="text-slate-500 text-[10px] block">Job Lifecycle</span>
                  <span className="font-mono font-bold text-indigo-600">{activeJob?.status || 'VERIFIED'}</span>
                </div>
                <div className="p-2.5 rounded-xl bg-slate-50 border border-slate-200">
                  <span className="text-slate-500 text-[10px] block">Events Dispatched</span>
                  <span className="font-mono font-bold text-slate-800">{jobEvents.length || 8} events</span>
                </div>
              </div>

              {/* Bottom Navigation CTA */}
              {onNavigate && (
                <div className="pt-2 flex items-center justify-between">
                  <button
                    onClick={() => onNavigate('fix')}
                    className="text-xs font-semibold text-slate-600 hover:text-indigo-600 flex items-center gap-1"
                  >
                    View Full Patch Diff
                  </button>
                  <button
                    onClick={() => onNavigate('verification')}
                    className="text-xs font-bold text-indigo-600 hover:text-indigo-700 flex items-center gap-1"
```

```
                >
                  <span>Verification Suite</span>
                  <ChevronRight className="w-3.5 h-3.5" />
                </button>
              </div>
            )}
          </div>
        </div>
      </div>
    </div>
  );
};
```

File [24/30]: src/components/views/FixView.tsx

TSX SOURCE

Description: *Unified Issue Investigation & Fix Workspace*

```tsx
'use client';

import React, { useState } from 'react';
import {
  CheckCircle2,
  ShieldCheck,
  ArrowRight,
  Copy,
  Check,
  Activity,
  Cpu,
  Zap,
} from 'lucide-react';
import { PRIMARY_PATCH, PRIMARY_REGRESSION_TEST } from '@/lib/mock-data';
import { NavTab } from '@/lib/types';

interface FixViewProps {
  onVerifyFix: () => void;
  onNavigate: (tab: NavTab) => void;
}

export const FixView: React.FC<FixViewProps> = ({
  onVerifyFix,
  onNavigate,
}) => {
  const [copiedTest, setCopiedTest] = useState(false);
  const [copiedPatch, setCopiedPatch] = useState(false);

  const agentSteps = [
    'Loaded repository context',
    'Located authentication middleware',
```

```
    'Traced token validation',
    'Identified missing expiration handling',
    'Generated minimal patch',
    'Generated regression test',
  ];

  const handleCopyTest = () => {
    navigator.clipboard.writeText(PRIMARY_REGRESSION_TEST.code);
    setCopiedTest(true);
    setTimeout(() => setCopiedTest(false), 2000);
  };

  const handleCopyPatch = () => {
    navigator.clipboard.writeText(PRIMARY_PATCH.afterCode);
    setCopiedPatch(true);
    setTimeout(() => setCopiedPatch(false), 2000);
  };

  return (
    <div className="space-y-8 max-w-7xl mx-auto pb-16">
      {/* Header */}
      <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-200 pb-5">
        <div>
          <div className="flex items-center gap-2">
            <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-emerald-50 text-emerald-700 border border-emerald-200">
              Fix & Regression Suite
            </span>
            <span className="text-xs text-slate-500 font-mono">AUTH-104 Patch Ready</span>
          </div>
          <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1">
            Automated Patch & Regression Verification
          </h1>
          <p className="text-sm text-slate-600">
            Surgical non-breaking diff synthesized alongside automated regression test suites.
```

```
          </p>
        </div>

        <div className="flex flex-col sm:flex-row items-center gap-2.5 self-start md:self-auto">
          <button
            onClick={() => onNavigate('live-fix')}
            className="flex items-center gap-2 px-5 py-3 rounded-xl bg-indigo-50 hover:bg-indigo-100 text-indigo-700 font-bold text-sm border border-indigo-200 shadow-xs transition-all"
          >
            <Zap className="w-4 h-4 text-indigo-600" />
            <span>Watch Live Fix Player</span>
          </button>

          <button
            onClick={onVerifyFix}
            className="flex items-center gap-2 px-6 py-3 rounded-xl bg-gradient-to-r from-cyan-600 via-indigo-600 to-emerald-600 hover:from-cyan-700 hover:to-emerald-700 text-white font-semibold text-sm shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all"
          >
            <ShieldCheck className="w-4 h-4" />
            <span>Verify Fix in Sandbox</span>
            <ArrowRight className="w-4 h-4" />
          </button>
        </div>
      </div>

      {/* AI Agent Progress Log */}
      <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs">
        <div className="flex items-center justify-between mb-4">
          <div className="flex items-center gap-2.5">
            <div className="w-7 h-7 rounded-lg bg-indigo-50 border border-indigo-200 flex items-center justify-center">
              <Cpu className="w-4 h-4 text-indigo-600" />
            </div>
            <div>
              <h3 className="text-sm font-bold text-slate-900 tracking-wide">
```

```
                RepoPilot Agent
              </h3>
              <p className="text-[11px] text-slate-500">
                Deterministic autonomous synthesis pipeline
              </p>
            </div>
          </div>

          <span className="text-xs font-semibold px-2.5 py-0.5 rounded-full bg-emerald-50 text-emerald-700 border bo
rder-emerald-200 flex items-center gap-1.5">
            <span className="w-1.5 h-1.5 rounded-full bg-emerald-500" />
            Completed in 480ms
          </span>
        </div>

        <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-3">
          {agentSteps.map((step, idx) => (
            <div
              key={idx}
              className="p-3 rounded-xl bg-slate-50 border border-slate-200 flex items-center gap-2.5"
            >
              <div className="w-5 h-5 rounded-full bg-emerald-100 text-emerald-700 flex items-center justify-center
shrink-0 border border-emerald-200">
                <CheckCircle2 className="w-3.5 h-3.5" />
              </div>
              <span className="text-xs font-medium text-slate-800">{step}</span>
            </div>
          ))}
        </div>
      </div>

      {/* PATCH READY METRICS */}
      <div className="p-6 rounded-2xl bg-gradient-to-r from-emerald-50/80 via-white to-indigo-50/80 border border-em
erald-200 shadow-xs">
        <div className="flex items-center justify-between pb-4 border-b border-slate-200 mb-5">
          <div className="flex items-center gap-2">
```

```
            <span className="text-xs uppercase font-extrabold tracking-wider px-2.5 py-0.5 rounded bg-emerald-600 te
xt-white shadow-xs">
              PATCH READY
            </span>
            <span className="text-xs text-slate-500 font-mono">
              Ready for verification testing
            </span>
          </div>

          <button
            onClick={handleCopyPatch}
            className="flex items-center gap-1.5 px-3 py-1 rounded-lg bg-white hover:bg-slate-100 text-slate-700 tex
t-xs font-medium border border-slate-200 transition-colors shadow-xs"
          >
            {copiedPatch ? (
              <>
                <Check className="w-3.5 h-3.5 text-emerald-600" />
                <span className="text-emerald-700 font-semibold">Copied</span>
              </>
            ) : (
              <>
                <Copy className="w-3.5 h-3.5" />
                <span>Copy Patch</span>
              </>
            )}
          </button>
        </div>

        <div className="grid grid-cols-2 sm:grid-cols-4 gap-4">
          <div className="p-4 rounded-xl bg-white border border-slate-200 shadow-xs">
            <div className="text-[11px] text-slate-500 uppercase tracking-wider mb-1 font-semibold">
              File
            </div>
            <div className="text-sm font-mono font-bold text-slate-900 truncate" title={PRIMARY_PATCH.file}>
              {PRIMARY_PATCH.file}
            </div>
```

```
            <div className="text-[10px] text-slate-500 mt-1">Single module impact</div>
          </div>

          <div className="p-4 rounded-xl bg-white border border-slate-200 shadow-xs">
            <div className="text-[11px] text-slate-500 uppercase tracking-wider mb-1 font-semibold">
              Lines Changed
            </div>
            <div className="text-2xl font-bold text-emerald-600 font-mono">
              8
            </div>
            <div className="text-[10px] text-slate-500 mt-1">+11 added / -3 removed</div>
          </div>

          <div className="p-4 rounded-xl bg-white border border-slate-200 shadow-xs">
            <div className="text-[11px] text-slate-500 uppercase tracking-wider mb-1 font-semibold">
              Risk Assessment
            </div>
            <div className="flex items-center gap-1.5 mt-0.5">
              <span className="w-2 h-2 rounded-full bg-emerald-500" />
              <span className="text-xl font-bold text-emerald-700 uppercase">
                {PRIMARY_PATCH.risk}
              </span>
            </div>
            <div className="text-[10px] text-slate-500 mt-1">Zero breaking API changes</div>
          </div>

          <div className="p-4 rounded-xl bg-white border border-slate-200 shadow-xs">
            <div className="text-[11px] text-slate-500 uppercase tracking-wider mb-1 font-semibold">
              Confidence
            </div>
            <div className="text-2xl font-bold text-indigo-600 font-mono">
              {PRIMARY_PATCH.confidence}%
            </div>
            <div className="text-[10px] text-slate-500 mt-1">High semantic alignment</div>
```

```
          </div>
        </div>
      </div>

      {/* REGRESSION TEST DISPLAY */}
      <div className="rounded-2xl border border-slate-200 bg-white overflow-hidden shadow-xs">
        <div className="p-4 bg-slate-50 border-b border-slate-200 flex flex-wrap items-center justify-between gap-4">
          <div className="flex items-center gap-3">
            <div className="flex items-center gap-1.5">
              <span className="w-3 h-3 rounded-full bg-rose-400" />
              <span className="w-3 h-3 rounded-full bg-amber-400" />
              <span className="w-3 h-3 rounded-full bg-emerald-400" />
            </div>

            <div className="flex items-center gap-2">
              <Activity className="w-4 h-4 text-indigo-600" />
              <span className="font-mono text-xs font-bold text-slate-900">
                {PRIMARY_REGRESSION_TEST.filename}
              </span>
              <span className="text-[11px] font-mono text-indigo-700 bg-indigo-50 border border-indigo-200 px-2 py-0.5 rounded font-semibold">
                {PRIMARY_REGRESSION_TEST.framework}
              </span>
            </div>
          </div>

          <div className="flex items-center gap-2">
            <button
              onClick={handleCopyTest}
              className="flex items-center gap-1.5 px-3 py-1 rounded-lg bg-white hover:bg-slate-100 text-slate-700 text-xs font-medium border border-slate-200 transition-colors shadow-xs"
            >
              {copiedTest ? (
                <>
                  <Check className="w-3.5 h-3.5 text-emerald-600" />
```

```
                  <span className="text-emerald-700 font-semibold">Copied</span>
                </>
              ) : (
                <>
                  <Copy className="w-3.5 h-3.5" />
                  <span>Copy Test</span>
                </>
              )}
            </button>
          </div>
        </div>

        <div className="p-5 font-mono text-xs text-slate-800 leading-relaxed overflow-x-auto bg-[#fafafa]">
          <pre className="space-y-1">
            {PRIMARY_REGRESSION_TEST.code.split('\n').map((line, idx) => (
              <div key={idx} className="flex items-start gap-4">
                <span className="text-slate-400 select-none w-6 text-right shrink-0">
                  {idx + 1}
                </span>
                <span className="whitespace-pre">{line}</span>
              </div>
            ))}
          </pre>
        </div>
      </div>

      {/* Verify Fix Action Card */}
      <div className="p-6 md:p-8 rounded-2xl bg-gradient-to-r from-indigo-50/80 via-white to-sky-50/80 border border-indigo-200 flex flex-col sm:flex-row items-start sm:items-center justify-between gap-6 shadow-xs">
        <div className="space-y-1">
          <div className="flex items-center gap-2">
            <ShieldCheck className="w-5 h-5 text-indigo-600" />
            <h3 className="text-lg font-bold text-slate-900">
              Ready for Automated Sandbox Verification
            </h3>
```

```
          </div>
          <p className="text-xs text-slate-600 max-w-xl">
            RepoPilot will run the full test suite, execute the newly generated regression test, and verify that AUTH-104 is completely resolved without regressions.
          </p>
        </div>

        <button
          onClick={onVerifyFix}
          className="flex items-center gap-2 px-8 py-4 rounded-xl bg-gradient-to-r from-cyan-600 via-indigo-600 to-emerald-600 hover:from-cyan-700 hover:to-emerald-700 text-white font-bold text-sm shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all whitespace-nowrap"
        >
          <ShieldCheck className="w-4 h-4" />
          <span>Verify Fix</span>
          <ArrowRight className="w-4 h-4" />
        </button>
      </div>
    </div>
  );
};
```

File [25/30]: src/components/views/InvestigationView.tsx

TSX SOURCE

Description: *AST & Dependency Graph Issue Investigation View*

```tsx
'use client';

import React, { useState } from 'react';
import {
  CheckCircle2,
  AlertCircle,
  FileCode,
  ArrowRight,
  ShieldAlert,
  Sparkles,
  Copy,
  Check,
  Zap,
} from 'lucide-react';
import { PRIMARY_ISSUE, PRIMARY_PATCH } from '@/lib/mock-data';
import { NavTab } from '@/lib/types';

interface InvestigationViewProps {
  onGenerateFix: () => void;
  onNavigate: (tab: NavTab) => void;
}

export const InvestigationView: React.FC<InvestigationViewProps> = ({
  onGenerateFix,
  onNavigate,
}) => {
  const [diffViewMode, setDiffViewMode] = useState<'split' | 'unified'>('split');
  const [copied, setCopied] = useState(false);

  const reasoningSteps = [
    {
```

```
      id: 'step-1',
      text: 'Repository context loaded',
      detail: 'Indexed 184 files and parsed AST symbol references.',
      status: 'completed',
    },
    {
      id: 'step-2',
      text: 'Relevant files identified',
      detail: 'Identified src/middleware/auth.ts and user route chains.',
      status: 'completed',
    },
    {
      id: 'step-3',
      text: 'Authentication flow traced',
      detail: 'Traced Bearer token extraction -> jwt.verify() -> next() middleware execution.',
      status: 'completed',
    },
    {
      id: 'step-4',
      text: 'Root cause identified',
      detail: 'Missing explicit expiration condition allows expired tokens through.',
      status: 'completed',
    },
    {
      id: 'step-5',
      text: 'Fix strategy generated',
      detail: 'Enforce payload verification and decoded.exp timestamp validation against Date.now().',
      status: 'completed',
    },
    {
      id: 'step-6',
      text: 'Regression test generated',
      detail: 'Synthesized automated Supertest spec with expired JWT token payload.',
      status: 'completed',
    },
```

```
  ];

  const handleCopyCode = () => {
    navigator.clipboard.writeText(PRIMARY_PATCH.afterCode);
    setCopied(true);
    setTimeout(() => setCopied(false), 2000);
  };

  return (
    <div className="space-y-8 max-w-7xl mx-auto pb-16">
      {/* Header */}
      <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-200 pb-5">
        <div>
          <div className="flex items-center gap-2">
            <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-indigo-200">
              AI Investigation
            </span>
            <span className="text-xs font-mono font-semibold text-rose-700 bg-rose-50 px-2 py-0.5 rounded border border-rose-200">
              {PRIMARY_ISSUE.code}
            </span>
            <span className="text-xs text-slate-500 font-mono">Confidence: 96%</span>
          </div>
          <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1">
            {PRIMARY_ISSUE.title}
          </h1>
          <p className="text-sm text-slate-600">
            Automated execution-path tracing and root-cause vulnerability diagnosis.
          </p>
        </div>

        <div className="flex flex-col sm:flex-row items-center gap-2.5 self-start md:self-auto">
          <button
            onClick={() => onNavigate('live-fix')}
```

```
            className="flex items-center gap-2 px-5 py-3 rounded-xl bg-gradient-to-r from-indigo-600 via-indigo-500 to-cyan-600 hover:from-indigo-700 hover:to-cyan-700 text-white font-semibold text-sm shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all"
          >
            <Zap className="w-4 h-4 text-cyan-300" />
            <span>Watch Live Fix &amp;amp; Learn</span>
          </button>

          <button
            onClick={onGenerateFix}
            className="flex items-center gap-2 px-5 py-3 rounded-xl bg-slate-900 hover:bg-slate-800 text-white font-semibold text-sm shadow-md shadow-slate-900/10 hover:scale-[1.02] active:scale-[0.98] transition-all"
          >
            <Sparkles className="w-4 h-4 text-emerald-400" />
            <span>Generate Fix</span>
            <ArrowRight className="w-4 h-4" />
          </button>
        </div>
      </div>

      {/* AI Reasoning Timeline */}
      <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs">
        <div className="flex items-center justify-between mb-5">
          <div className="flex items-center gap-2">
            <div className="w-2.5 h-2.5 rounded-full bg-emerald-500 animate-pulse"></div>
            <h2 className="text-sm font-bold uppercase tracking-wider text-slate-800">
              AI Reasoning Timeline
            </h2>
          </div>
          <span className="text-xs text-emerald-700 font-semibold bg-emerald-50 px-2.5 py-0.5 rounded-full border border-emerald-200">
            6/6 Analysis Steps Completed
          </span>
        </div>

        <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-3">
```

```
          {reasoningSteps.map((step) => (
            <div
              key={step.id}
              className="p-3.5 rounded-xl bg-slate-50/80 border border-slate-200 hover:border-indigo-300 transition-all flex items-start gap-3 shadow-xs"
            >
              <div className="w-6 h-6 rounded-full bg-emerald-100 text-emerald-700 flex items-center justify-center shrink-0 mt-0.5 border border-emerald-200">
                <CheckCircle2 className="w-3.5 h-3.5" />
              </div>
              <div className="space-y-0.5">
                <div className="text-xs font-bold text-slate-900 flex items-center gap-1.5">
                  <span className="text-emerald-600">✓</span>
                  <span>{step.text}</span>
                </div>
                <div className="text-[11px] text-slate-500 leading-snug">
                  {step.detail}
                </div>
              </div>
            </div>
          ))}
        </div>
      </div>

      {/* Root Cause & Impact Cards */}
      <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
        {/* ROOT CAUSE */}
        <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs relative overflow-hidden">
          <div className="flex items-center justify-between mb-3">
            <span className="text-xs font-bold uppercase tracking-wider text-rose-700 flex items-center gap-1.5">
              <ShieldAlert className="w-4 h-4 text-rose-600" />
              Root Cause
            </span>
            <span className="text-xs text-slate-400 font-mono">auth.ts:38-42</span>
          </div>

```

```
166           <p className="text-sm text-slate-800 leading-relaxed font-medium">
167             "{PRIMARY_ISSUE.rootCause}"
168           </p>
169
170           <div className="mt-4 pt-4 border-t border-slate-100 flex items-center gap-2 text-xs text-slate-500">
171             <span>Affected File:</span>
172             <span className="font-mono text-indigo-700 bg-indigo-50 px-2 py-0.5 rounded border border-indigo-200 fon
    t-semibold">
173               {PRIMARY_ISSUE.affectedFile}
174             </span>
175           </div>
176         </div>
177
178         {/* IMPACT */}
179         <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs relative overflow-hidden">
180           <div className="flex items-center justify-between mb-3">
181             <span className="text-xs font-bold uppercase tracking-wider text-amber-800 flex items-center gap-1.5">
182               <AlertCircle className="w-4 h-4 text-amber-600" />
183               Security Impact
184             </span>
185             <span className="text-xs font-semibold px-2 py-0.5 rounded bg-rose-50 text-rose-700 border border-rose-2
    00">
186               HIGH RISK
187             </span>
188           </div>
189
190           <p className="text-sm text-slate-800 leading-relaxed font-medium">
191             "{PRIMARY_ISSUE.impact}"
192           </p>
193
194           <div className="mt-4 pt-4 border-t border-slate-100 flex items-center gap-2 text-xs text-slate-500">
195             <span>Vulnerability Class:</span>
196             <span className="font-mono text-amber-800 bg-amber-50 px-2 py-0.5 rounded border border-amber-200 font-s
    emibold">
197               CWE-287: Improper Authentication
198             </span>
```

```
          </div>
        </div>
      </div>

      {/* Live Bug Fix & Learner Mode Callout */}
      <div className="p-4 rounded-2xl bg-gradient-to-r from-indigo-900 via-indigo-800 to-slate-900 text-white flex flex-col sm:flex-row items-start sm:items-center justify-between gap-4 shadow-md">
        <div className="flex items-center gap-3">
          <div className="w-10 h-10 rounded-xl bg-white/10 flex items-center justify-center border border-white/20 shrink-0">
            <Zap className="w-5 h-5 text-cyan-300 animate-pulse" />
          </div>
          <div>
            <h4 className="text-sm font-bold tracking-tight">Interactive Live Bug Healing &amp;amp; Learning Studio</h4>
            <p className="text-xs text-indigo-200">Watch the AI pinpoint the exact line, dissolve the bug in real-time, and explain how to prevent it.</p>
          </div>
        </div>
        <button
          onClick={() => onNavigate('live-fix')}
          className="flex items-center gap-2 px-5 py-2.5 rounded-xl bg-white text-indigo-900 hover:bg-indigo-50 font-bold text-xs shadow-md hover:scale-[1.02] active:scale-[0.98] transition-all shrink-0"
        >
          <Zap className="w-3.5 h-3.5 text-indigo-600" />
          <span>Launch Live Fix Player</span>
          <ArrowRight className="w-3.5 h-3.5" />
        </button>
      </div>

      {/* CODE DIFF (Visually Impressive Split or Unified) */}
      <div className="rounded-2xl border border-slate-200 bg-white overflow-hidden shadow-xs">
        {/* Diff Header */}
        <div className="p-4 bg-slate-50 border-b border-slate-200 flex flex-wrap items-center justify-between gap-4">
          <div className="flex items-center gap-3">
            <div className="flex items-center gap-1.5">
              <span className="w-3 h-3 rounded-full bg-rose-400" />
```

```
              <span className="w-3 h-3 rounded-full bg-amber-400" />
              <span className="w-3 h-3 rounded-full bg-emerald-400" />
            </div>

            <div className="flex items-center gap-2">
              <FileCode className="w-4 h-4 text-indigo-600" />
              <span className="font-mono text-xs font-bold text-slate-900">
                {PRIMARY_PATCH.file}
              </span>
              <span className="text-[11px] font-mono text-slate-600 px-2 py-0.5 rounded bg-slate-200/70">
                +11 / -3 lines
              </span>
            </div>
          </div>

          <div className="flex items-center gap-2">
            {/* View toggle */}
            <div className="flex items-center p-0.5 rounded-lg bg-slate-100 border border-slate-200 text-xs">
              <button
                onClick={() => setDiffViewMode('split')}
                className={`px-2.5 py-1 rounded font-medium transition-colors ${
                  diffViewMode === 'split'
                    ? 'bg-white text-indigo-700 shadow-xs font-bold'
                    : 'text-slate-600 hover:text-slate-900'
                }`}
              >
                Split View
              </button>
              <button
                onClick={() => setDiffViewMode('unified')}
                className={`px-2.5 py-1 rounded font-medium transition-colors ${
                  diffViewMode === 'unified'
                    ? 'bg-white text-indigo-700 shadow-xs font-bold'
                    : 'text-slate-600 hover:text-slate-900'
                }`}
```

```
              >
                Unified Diff
              </button>
            </div>

            {/* Copy button */}
            <button
              onClick={handleCopyCode}
              className="flex items-center gap-1.5 px-3 py-1 rounded-lg bg-white hover:bg-slate-100 text-slate-700 t
ext-xs font-medium border border-slate-200 transition-colors shadow-xs"
            >
              {copied ? (
                <>
                  <Check className="w-3.5 h-3.5 text-emerald-600" />
                  <span className="text-emerald-700 font-semibold">Copied</span>
                </>
              ) : (
                <>
                  <Copy className="w-3.5 h-3.5" />
                  <span>Copy Fix</span>
                </>
              )}
            </button>
          </div>
        </div>

        {/* Diff Body */}
        {diffViewMode === 'split' ? (
          <div className="grid grid-cols-1 md:grid-cols-2 divide-y md:divide-y-0 md:divide-x divide-slate-200 font-m
ono text-xs">
            {/* BEFORE (Red Removal) */}
            <div className="p-4 bg-rose-50/30">
              <div className="flex items-center justify-between pb-3 mb-3 border-b border-rose-200">
                <span className="text-xs font-bold uppercase tracking-wider text-rose-700 flex items-center gap-1.
5">
                  <span className="w-2 h-2 rounded-full bg-rose-500" />
```

```
                  BEFORE (VULNERABLE)
                </span>
                <span className="text-[11px] text-slate-500">Lines 38–44</span>
              </div>

              <div className="space-y-1 text-slate-700">
                <div className="text-slate-600 px-2 py-0.5">
                  <span className="text-slate-400 select-none mr-3">37</span>
                  <span className="text-indigo-600 font-semibold">const</span> decoded = jwt.verify(token, SECRET);
                </div>
                <div className="bg-rose-100 text-rose-950 px-2 py-1 rounded border-l-2 border-rose-500 flex items-st
art gap-2">
                  <span className="text-rose-600 select-none font-bold">-</span>
                  <span className="text-slate-500 select-none">38</span>
                  <pre className="whitespace-pre font-mono">if (decoded) {'{'}</pre>
                </div>
                <div className="bg-rose-100 text-rose-950 px-2 py-1 rounded border-l-2 border-rose-500 flex items-st
art gap-2">
                  <span className="text-rose-600 select-none font-bold">-</span>
                  <span className="text-slate-500 select-none">39</span>
                  <pre className="whitespace-pre font-mono">    return next();</pre>
                </div>
                <div className="bg-rose-100 text-rose-950 px-2 py-1 rounded border-l-2 border-rose-500 flex items-st
art gap-2">
                  <span className="text-rose-600 select-none font-bold">-</span>
                  <span className="text-slate-500 select-none">40</span>
                  <pre className="whitespace-pre font-mono">{'}'}</pre>
                </div>
              </div>
            </div>

            {/* AFTER (Green Addition) */}
            <div className="p-4 bg-emerald-50/30">
              <div className="flex items-center justify-between pb-3 mb-3 border-b border-emerald-200">
                <span className="text-xs font-bold uppercase tracking-wider text-emerald-700 flex items-center gap-
1.5">
```

```
                  <span className="w-2 h-2 rounded-full bg-emerald-500" />
                  AFTER (VERIFIED FIX)
                </span>
                <span className="text-[11px] text-slate-500">Lines 38–52</span>
              </div>

              <div className="space-y-1 text-slate-700">
                <div className="text-slate-600 px-2 py-0.5">
                  <span className="text-slate-400 select-none mr-3">37</span>
                  <span className="text-indigo-600 font-semibold">const</span> decoded = jwt.verify(token, SECRET);
                </div>

                <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
                  <span className="text-emerald-700 select-none font-bold">+</span>
                  <span className="text-slate-500 select-none">38</span>
                  <pre className="whitespace-pre font-mono">if (!decoded || typeof decoded === "string") {'{'}</pre>
                </div>
                <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
                  <span className="text-emerald-700 select-none font-bold">+</span>
                  <span className="text-slate-500 select-none">39</span>
                  <pre className="whitespace-pre font-mono">    return res.status(401).json({'{'}</pre>
                </div>
                <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
                  <span className="text-emerald-700 select-none font-bold">+</span>
                  <span className="text-slate-500 select-none">40</span>
                  <pre className="whitespace-pre font-mono">        error: "Invalid token"</pre>
                </div>
                <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
                  <span className="text-emerald-700 select-none font-bold">+</span>
                  <span className="text-slate-500 select-none">41</span>
                  <pre className="whitespace-pre font-mono">    {'}'});</pre>
                </div>
```

```
363                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
364                   <span className="text-emerald-700 select-none font-bold">+</span>
365                   <span className="text-slate-500 select-none">42</span>
366                   <pre className="whitespace-pre font-mono">{'}'}</pre>
367                 </div>
368
369                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
370                   <span className="text-emerald-700 select-none font-bold">+</span>
371                   <span className="text-slate-500 select-none">43</span>
372                   <pre className="whitespace-pre font-mono">if (decoded.exp && decoded.exp * 1000 &lt; Date.now())
{'{'}</pre>
373                 </div>
374                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
375                   <span className="text-emerald-700 select-none font-bold">+</span>
376                   <span className="text-slate-500 select-none">44</span>
377                   <pre className="whitespace-pre font-mono">    return res.status(401).json({'{'}</pre>
378                 </div>
379                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
380                   <span className="text-emerald-700 select-none font-bold">+</span>
381                   <span className="text-slate-500 select-none">45</span>
382                   <pre className="whitespace-pre font-mono">        error: "Token expired"</pre>
383                 </div>
384                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
385                   <span className="text-emerald-700 select-none font-bold">+</span>
386                   <span className="text-slate-500 select-none">46</span>
387                   <pre className="whitespace-pre font-mono">    {'}'});</pre>
388                 </div>
389                 <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500
flex items-start gap-2">
390                   <span className="text-emerald-700 select-none font-bold">+</span>
391                   <span className="text-slate-500 select-none">47</span>
392                   <pre className="whitespace-pre font-mono">{'}'}</pre>
393                 </div>
394
```

```
                <div className="bg-emerald-100/80 text-emerald-950 px-2 py-0.5 rounded border-l-2 border-emerald-500 flex items-start gap-2">
                  <span className="text-emerald-700 select-none font-bold">+</span>
                  <span className="text-slate-500 select-none">48</span>
                  <pre className="whitespace-pre font-mono">return next();</pre>
                </div>
              </div>
            </div>
          </div>
        ) : (
          /* Unified Diff View */
          <div className="p-4 font-mono text-xs space-y-0.5 bg-[#fafafa]">
            {PRIMARY_PATCH.diffLines.map((line, idx) => {
              if (line.type === 'deletion') {
                return (
                  <div
                    key={idx}
                    className="flex items-center gap-4 px-2 py-0.5 rounded bg-rose-100 text-rose-950 border-l-2 border-rose-500"
                  >
                    <span className="text-slate-400 select-none w-6 text-right">
                      {line.oldLineNumber}
                    </span>
                    <span className="text-rose-600 select-none font-bold">-</span>
                    <span className="whitespace-pre">{line.content.replace(/^-\s*/, '')}</span>
                  </div>
                );
              }

              if (line.type === 'addition') {
                return (
                  <div
                    key={idx}
                    className="flex items-center gap-4 px-2 py-0.5 rounded bg-emerald-100 text-emerald-950 border-l-2 border-emerald-500"
                  >
```

```
                    <span className="text-slate-400 select-none w-6 text-right">
                      {line.newLineNumber}
                    </span>
                    <span className="text-emerald-700 select-none font-bold">+</span>
                    <span className="whitespace-pre">{line.content.replace(/^\+\s*/, '')}</span>
                  </div>
                );
              }

              return (
                <div key={idx} className="flex items-center gap-4 px-2 py-0.5 text-slate-600">
                  <span className="text-slate-400 select-none w-6 text-right">
                    {line.newLineNumber || line.oldLineNumber}
                  </span>
                  <span className="text-slate-400 select-none font-bold"> </span>
                  <span className="whitespace-pre">{line.content}</span>
                </div>
              );
            })}
          </div>
        )}
      </div>

      {/* Footer action trigger */}
      <div className="flex items-center justify-between p-6 rounded-2xl bg-indigo-50 border border-indigo-200 shadow-xs">
        <div>
          <h4 className="text-base font-bold text-slate-900">Ready to proceed to Fix Generation?</h4>
          <p className="text-xs text-slate-600">
            RepoPilot will construct the minimal patch and synthesize automated regression specs.
          </p>
        </div>

        <button
          onClick={onGenerateFix}
```

```
462           className="flex items-center gap-2 px-6 py-3 rounded-xl bg-indigo-600 hover:bg-indigo-700 text-white font-semibold text-sm shadow-md shadow-indigo-600/20 transition-all"
463         >
464           <Sparkles className="w-4 h-4" />
465           <span>Generate Fix & Tests</span>
466           <ArrowRight className="w-4 h-4" />
467         </button>
468       </div>
469     </div>
470   );
471 };
```

File [26/30]: src/components/views/VerificationView.tsx

TSX SOURCE

Description: *Autonomous Test Suite Verification View*

```tsx
'use client';

import React, { useState, useEffect } from 'react';
import {
  Terminal,
  CheckCircle2,
  ShieldCheck,
  GitPullRequest,
  RotateCcw,
  Sparkles,
  ArrowRight,
  Check,
} from 'lucide-react';
import confetti from 'canvas-confetti';
import { NavTab } from '@/lib/types';

interface VerificationViewProps {
  onCreatePR: () => void;
  onNavigate: (tab: NavTab) => void;
  isAutoRunning?: boolean;
}

export const VerificationView: React.FC<VerificationViewProps> = ({
  onCreatePR,
}) => {
  const [isRunning, setIsRunning] = useState(false);
  const [isCompleted, setIsCompleted] = useState(false);
  const [visibleLogCount, setVisibleLogCount] = useState(0);

  const verificationLogs = [
    '$ npm test -- --runInBand',
```

```
    'Initializing verification...',
    '✓ Loading patched repository',
    '✓ Running authentication tests',
    '✓ Running regression tests',
    '✓ Running API tests',
    '✓ Checking affected modules',
    'All 24 test suites executed cleanly in 1,240ms.',
  ];

  const runVerification = () => {
    setIsRunning(true);
    setIsCompleted(false);
    setVisibleLogCount(0);

    let count = 0;
    const interval = setInterval(() => {
      count++;
      setVisibleLogCount(count);
      if (count >= verificationLogs.length) {
        clearInterval(interval);
        setIsRunning(false);
        setIsCompleted(true);

        // Confetti burst for verified triumph!
        try {
          confetti({
            particleCount: 80,
            spread: 70,
            origin: { y: 0.6 },
            colors: ['#10B981', '#6366F1', '#06B6D4', '#F59E0B'],
          });
        } catch (e) {
          // ignore if canvas-confetti is not loaded in SSR
        }
      }
```

```
    }, 240);
  };

  useEffect(() => {
    runVerification();
  }, []);

  return (
    <div className="space-y-8 max-w-7xl mx-auto pb-16">
      {/* Header */}
      <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-200 pb-5">
        <div>
          <div className="flex items-center gap-2">
            <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-cyan-50 text-cyan-700 border border-cyan-200">
              Closed-Loop Execution
            </span>
            <span className="text-xs text-slate-500 font-mono">Sandbox Sandbox-v2.1</span>
          </div>
          <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1">
            Verification Center
          </h1>
          <p className="text-sm text-slate-600">
            Realtime automated regression runner, security check validator, and sandbox outcome verification.
          </p>
        </div>

        <div className="flex items-center gap-3 self-start md:self-auto">
          <button
            onClick={runVerification}
            disabled={isRunning}
            className="flex items-center gap-2 px-4 py-2.5 rounded-xl bg-white hover:bg-slate-50 border border-slate-300 text-slate-700 text-xs font-semibold disabled:opacity-50 transition-colors shadow-xs"
          >
            <RotateCcw className={`w-3.5 h-3.5 ${isRunning ? 'animate-spin' : ''}`} />
```

```
            <span>Re-run Verification</span>
          </button>

          {isCompleted && (
            <button
              onClick={onCreatePR}
              className="flex items-center gap-2 px-6 py-2.5 rounded-xl bg-gradient-to-r from-emerald-600 to-indigo-600 hover:from-emerald-700 hover:to-indigo-700 text-white font-semibold text-xs shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all"
            >
              <GitPullRequest className="w-4 h-4" />
              <span>Create Pull Request</span>
              <ArrowRight className="w-3.5 h-3.5" />
            </button>
          )}
        </div>
      </div>

      {/* Terminal-like execution view */}
      <div className="rounded-2xl border border-slate-800 bg-[#0d1322] overflow-hidden shadow-md">
        <div className="px-4 py-3 bg-slate-900 border-b border-slate-800 flex items-center justify-between">
          <div className="flex items-center gap-2">
            <span className="w-3 h-3 rounded-full bg-rose-500/80" />
            <span className="w-3 h-3 rounded-full bg-amber-500/80" />
            <span className="w-3 h-3 rounded-full bg-emerald-500/80" />
            <span className="text-xs font-mono text-slate-300 ml-2 flex items-center gap-1.5 font-semibold">
              <Terminal className="w-3.5 h-3.5 text-indigo-400" />
              terminal — sandbox-runner
            </span>
          </div>

          <div className="flex items-center gap-2 text-xs font-mono">
            {isRunning ? (
              <span className="text-cyan-400 flex items-center gap-1.5">
                <span className="w-2 h-2 rounded-full bg-cyan-400 animate-ping" />
```

```
                Executing tests...
              </span>
            ) : isCompleted ? (
              <span className="text-emerald-400 flex items-center gap-1.5 font-semibold">
                <Check className="w-3.5 h-3.5" />
                Exit code: 0 (All passed)
              </span>
            ) : null}
          </div>
        </div>

        <div className="p-6 font-mono text-xs text-slate-200 space-y-2 min-h-[220px] bg-[#070c18]">
          {verificationLogs.slice(0, visibleLogCount).map((log, index) => {
            const isCommand = log.startsWith('$');
            const isSuccess = log.includes('✓') || log.includes('passed');

            return (
              <div
                key={index}
                className={`flex items-start gap-2 ${
                  isCommand
                    ? 'text-cyan-400 font-bold text-sm'
                    : isSuccess
                    ? 'text-emerald-400 font-semibold'
                    : 'text-slate-300'
                }`}
              >
                <span>{log}</span>
              </div>
            );
          })}

          {isRunning && (
            <div className="inline-block w-2 h-4 bg-cyan-400 animate-pulse ml-1" />
          )}
```

```
        </div>
      </div>

      {/* Outcome Cards Grid */}
      <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
        {/* TEST RESULTS */}
        <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs">
          <div className="text-xs font-bold uppercase tracking-wider text-slate-500 mb-3">
            TEST RESULTS
          </div>
          <div className="flex items-baseline gap-3">
            <span className="text-4xl font-extrabold text-emerald-600 font-mono">
              24
            </span>
            <span className="text-sm font-semibold text-emerald-700">
              passed
            </span>
            <span className="text-slate-300">|</span>
            <span className="text-2xl font-bold text-slate-400 font-mono">
              0
            </span>
            <span className="text-xs text-slate-500">
              failed
            </span>
          </div>

          <div className="mt-4 pt-4 border-t border-slate-100 text-xs text-slate-600 flex items-center gap-2">
            <CheckCircle2 className="w-4 h-4 text-emerald-600 shrink-0" />
            <span>100% of repository test suites green</span>
          </div>
        </div>

        {/* SECURITY CHECK */}
        <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs">
```

```
202           <div className="flex items-center justify-between mb-3">
203             <div className="text-xs font-bold uppercase tracking-wider text-slate-500">
204               SECURITY CHECK
205             </div>
206             <span className="text-xs font-extrabold px-2.5 py-0.5 rounded bg-emerald-50 text-emerald-700 border bord
    er-emerald-200">
207               PASS
208             </span>
209           </div>
210
211           <div className="text-sm font-bold text-slate-900 mb-1">
212             JWT expiration handling
213           </div>
214
215           <p className="text-xs text-slate-600 leading-relaxed">
216             Expired cryptographic signatures and elapsed timestamps immediately rejected with HTTP 401.
217           </p>
218
219           <div className="mt-4 pt-4 border-t border-slate-100 text-xs text-emerald-700 font-semibold flex items-cent
    er gap-2">
220             <ShieldCheck className="w-4 h-4 text-emerald-600 shrink-0" />
221             <span>CWE-287 Bypass Neutralized</span>
222           </div>
223         </div>
224
225         {/* REGRESSION TEST */}
226         <div className="p-6 rounded-2xl bg-white border border-slate-200 shadow-xs">
227           <div className="flex items-center justify-between mb-3">
228             <div className="text-xs font-bold uppercase tracking-wider text-slate-500">
229               REGRESSION TEST
230             </div>
231             <span className="text-xs font-extrabold px-2.5 py-0.5 rounded bg-emerald-50 text-emerald-700 border bord
    er-emerald-200">
232               PASS
233             </span>
234           </div>
```

```
235
236           <div className="text-sm font-bold text-slate-900 mb-1">
237             Expired token rejected
238           </div>
239
240           <p className="text-xs text-slate-600 leading-relaxed">
241             Targeted Supertest suite confirmed that simulated expired bearer token fails with 401 status.
242           </p>
243
244           <div className="mt-4 pt-4 border-t border-slate-100 text-xs text-indigo-700 font-semibold flex items-cente
r gap-2">
245             <Sparkles className="w-4 h-4 text-indigo-600 shrink-0" />
246             <span>tests/auth.test.ts (+1 verified test)</span>
247           </div>
248         </div>
249       </div>
250
251       {/* LARGE SUCCESS CARD (FIX VERIFIED) */}
252       <div className="p-8 md:p-10 rounded-3xl bg-gradient-to-br from-emerald-50/80 via-white to-sky-50/80 border-2 b
order-emerald-300 shadow-sm relative overflow-hidden">
253         {/* Glow orb */}
254         <div className="absolute top-0 right-0 w-80 h-80 bg-emerald-200/40 rounded-full blur-3xl pointer-events-non
e" />
255
256         <div className="flex flex-col lg:flex-row lg:items-center justify-between gap-8 relative z-10">
257           <div className="space-y-4">
258             <div className="inline-flex items-center gap-2 px-3.5 py-1 rounded-full bg-emerald-100 border border-eme
rald-200 text-emerald-800 text-xs font-extrabold tracking-wider uppercase">
259               <CheckCircle2 className="w-4 h-4 text-emerald-600" />
260               <span>Fix Verified & Validated</span>
261             </div>
262
263             <h2 className="text-4xl font-extrabold text-slate-900 tracking-tight">
264               FIX VERIFIED
265             </h2>
266
267             <p className="text-base text-slate-600 max-w-2xl leading-relaxed">
```

```
              The patch has been automatically compiled, applied to a virtual clone of <span className="font-semibol
d text-slate-900">commerce-api</span>, and validated across all 24 tests with zero regressions.
            </p>

            {/* Metrics Pills */}
            <div className="flex flex-wrap items-center gap-4 pt-2">
              <div className="px-4 py-2 rounded-xl bg-white border border-slate-200 shadow-xs">
                <div className="text-[11px] text-slate-500 font-medium">Confidence</div>
                <div className="text-xl font-black text-indigo-700 font-mono">97%</div>
              </div>

              <div className="px-4 py-2 rounded-xl bg-white border border-slate-200 shadow-xs">
                <div className="text-[11px] text-slate-500 font-medium">Test Outcomes</div>
                <div className="text-xl font-black text-emerald-600 font-mono">24 / 24 Passed</div>
              </div>

              <div className="px-4 py-2 rounded-xl bg-white border border-slate-200 shadow-xs">
                <div className="text-[11px] text-slate-500 font-medium">Files Affected</div>
                <div className="text-xl font-black text-slate-900 font-mono">1 File Changed</div>
              </div>

              <div className="px-4 py-2 rounded-xl bg-white border border-slate-200 shadow-xs">
                <div className="text-[11px] text-slate-500 font-medium">Risk Assessment</div>
                <div className="text-xl font-black text-emerald-600">LOW</div>
              </div>
            </div>
          </div>

          <div className="flex flex-col gap-3 shrink-0">
            <button
              onClick={onCreatePR}
              className="flex items-center justify-center gap-3 px-8 py-4 rounded-xl bg-gradient-to-r from-emerald-6
00 via-indigo-600 to-cyan-600 hover:from-emerald-700 hover:to-cyan-700 text-white font-extrabold text-base shadow-md
shadow-indigo-600/25 hover:scale-[1.03] active:scale-[0.98] transition-all whitespace-nowrap"
            >
              <GitPullRequest className="w-5 h-5" />
```

```
              <span>Create Pull Request</span>
              <ArrowRight className="w-5 h-5" />
            </button>
            <p className="text-[11px] text-center text-slate-500">
              Generates ready-to-merge pull request with tests & docs
            </p>
          </div>
        </div>
      </div>
    </div>
  );
};
```

File [27/30]: src/components/views/HistoryView.tsx

TSX SOURCE

Description: *Historical Repair Audit Log & Performance Analytics*

```tsx
'use client';

import React from 'react';
import {
  CheckCircle2,
  AlertTriangle,
  ShieldCheck,
  Clock,
  ArrowRight,
  GitBranch,
} from 'lucide-react';
import { INITIAL_HISTORY } from '@/lib/mock-data';
import { NavTab } from '@/lib/types';

interface HistoryViewProps {
  onNavigate: (tab: NavTab) => void;
}

export const HistoryView: React.FC<HistoryViewProps> = ({ onNavigate }) => {
  return (
    <div className="space-y-6 max-w-7xl mx-auto pb-16">
      {/* Header */}
      <div className="border-b border-slate-200 pb-5">
        <div className="flex items-center gap-2">
          <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-indigo-200">
            Audit Trail
          </span>
          <span className="text-xs text-slate-500 font-mono">5 events logged</span>
        </div>
        <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1">
          Activity & Verification History
```

```
        </h1>
        <p className="text-sm text-slate-600">
          Historical timeline of autonomous diagnostic runs, patches synthesized, and verified test outcomes.
        </p>
      </div>

      {/* History Items List */}
      <div className="rounded-2xl border border-slate-200 bg-white divide-y divide-slate-100 overflow-hidden shadow-xs">
        {INITIAL_HISTORY.map((item) => {
          const isVerified = item.action === 'Fixed + Verified';
          const isAnalyzed = item.action === 'Analyzed';

          return (
            <div
              key={item.id}
              className="p-5 hover:bg-slate-50 transition-colors flex flex-col sm:flex-row sm:items-center justify-between gap-4"
            >
              <div className="flex items-start gap-4">
                <div className="flex items-center gap-1.5 text-slate-400 font-mono text-xs w-14 shrink-0 pt-0.5">
                  <Clock className="w-3.5 h-3.5 text-slate-400" />
                  <span>{item.timestamp}</span>
                </div>

                <div className="space-y-1">
                  <div className="flex flex-wrap items-center gap-2">
                    <span className="text-xs font-mono font-bold text-indigo-700 bg-indigo-50 border border-indigo-200 px-2 py-0.5 rounded">
                      {item.issueCode}
                    </span>
                    <h3 className="text-sm font-bold text-slate-900">
                      {item.title}
                    </h3>
                  </div>
```

```
                  <div className="flex flex-wrap items-center gap-3 text-xs text-slate-500">
                    <span className="flex items-center gap-1">
                      <GitBranch className="w-3 h-3 text-slate-400" />
                      <span>{item.repository}</span>
                    </span>
                    <span className="text-slate-300">•</span>
                    <span className="font-mono text-slate-600">
                      {item.file}
                    </span>
                  </div>
                </div>
              </div>

              <div className="flex items-center gap-3 self-end sm:self-center">
                {isVerified ? (
                  <span className="inline-flex items-center gap-1.5 px-3 py-1 rounded-full text-xs font-bold bg-emerald-50 text-emerald-700 border border-emerald-200">
                    <CheckCircle2 className="w-3.5 h-3.5" />
                    <span>Fixed + Verified ({item.confidence}%)</span>
                  </span>
                ) : isAnalyzed ? (
                  <span className="inline-flex items-center gap-1.5 px-3 py-1 rounded-full text-xs font-bold bg-amber-50 text-amber-800 border border-amber-200">
                    <AlertTriangle className="w-3.5 h-3.5" />
                    <span>Analyzed</span>
                  </span>
                ) : (
                  <span className="inline-flex items-center gap-1.5 px-3 py-1 rounded-full text-xs font-bold bg-cyan-50 text-cyan-700 border border-cyan-200">
                    <ShieldCheck className="w-3.5 h-3.5" />
                    <span>Completed</span>
                  </span>
                )}

                {item.issueCode === 'AUTH-104' && (
                  <button
```

```
                    onClick={() => onNavigate('verification')}
                    className="p-1.5 rounded-lg bg-slate-100 hover:bg-slate-200 text-slate-600 hover:text-slate-900 transition-colors"
                    title="View Verification Details"
                  >
                    <ArrowRight className="w-4 h-4" />
                  </button>
                )}
              </div>
            </div>
          );
        })}
      </div>
    </div>
  );
};
```

File [28/30]: src/components/views/RepositoryView.tsx

TSX SOURCE

Description: *Connected GitHub Repositories Explorer*

```tsx
'use client';

import React, { useState } from 'react';
import {
  Folder,
  FolderOpen,
  FileCode,
  FileText,
  AlertTriangle,
  CheckCircle2,
  ChevronRight,
  ChevronDown,
  Sparkles,
  ArrowRight,
  Layers,
  FileJson,
  FolderGit2,
  Zap,
} from 'lucide-react';
import { ACTIVE_REPO, REPO_FILE_TREE, PRIMARY_ISSUE } from '@/lib/mock-data';
import { FileNode, NavTab, RepositoryData } from '@/lib/types';

interface RepositoryViewProps {
  onNavigate: (tab: NavTab) => void;
  onInvestigateIssue: () => void;
  currentRepo?: RepositoryData;
  onOpenRepoModal?: () => void;
}

export const RepositoryView: React.FC<RepositoryViewProps> = ({
  onNavigate,
```

```
  onInvestigateIssue,
  currentRepo = ACTIVE_REPO,
  onOpenRepoModal,
}) => {
  const [selectedFile, setSelectedFile] = useState<FileNode>(() => {
    // Default select auth.ts
    const src = REPO_FILE_TREE.find((n) => n.name === 'src');
    const middleware = src?.children?.find((n) => n.name === 'middleware');
    const authFile = middleware?.children?.find((n) => n.name === 'auth.ts');
    return authFile || REPO_FILE_TREE[0];
  });

  const [expandedFolders, setExpandedFolders] = useState<Record<string, boolean>>({
    src: true,
    'src/middleware': true,
    'src/controllers': false,
    'src/services': false,
    'src/routes': false,
    'src/utils': false,
    tests: false,
  });

  const toggleFolder = (path: string) => {
    setExpandedFolders((prev) => ({
      ...prev,
      [path]: !prev[path],
    }));
  };

  const renderTree = (nodes: FileNode[], depth = 0) => {
    return nodes.map((node) => {
      const isFolder = node.type === 'folder';
      const isExpanded = expandedFolders[node.path];
      const isSelected = selectedFile?.path === node.path;

```

```
      if (isFolder) {
        return (
          <div key={node.path} className="select-none">
            <div
              onClick={() => toggleFolder(node.path)}
              className="flex items-center gap-1.5 px-2 py-1.5 rounded-md hover:bg-slate-100 cursor-pointer text-xs font-medium text-slate-700 transition-colors"
              style={{ paddingLeft: `${depth * 14 + 8}px` }}
            >
              {isExpanded ? (
                <ChevronDown className="w-3.5 h-3.5 text-slate-400" />
              ) : (
                <ChevronRight className="w-3.5 h-3.5 text-slate-400" />
              )}
              {isExpanded ? (
                <FolderOpen className="w-3.5 h-3.5 text-indigo-600" />
              ) : (
                <Folder className="w-3.5 h-3.5 text-indigo-500" />
              )}
              <span className="font-mono">{node.name}/</span>
            </div>

            {isExpanded && node.children && (
              <div>{renderTree(node.children, depth + 1)}</div>
            )}
          </div>
        );
      }

      return (
        <div
          key={node.path}
          onClick={() => setSelectedFile(node)}
          className={`flex items-center justify-between px-2 py-1.5 rounded-md cursor-pointer text-xs font-mono transition-all ${
```

```
            isSelected
              ? 'bg-indigo-50 text-indigo-900 font-semibold border border-indigo-200'
              : 'text-slate-600 hover:text-slate-900 hover:bg-slate-100/80'
          }`}
          style={{ paddingLeft: `${depth * 14 + 14}px` }}
        >
          <div className="flex items-center gap-2 truncate">
            {node.language === 'json' ? (
              <FileJson className="w-3.5 h-3.5 text-amber-600 shrink-0" />
            ) : node.language === 'markdown' ? (
              <FileText className="w-3.5 h-3.5 text-cyan-600 shrink-0" />
            ) : (
              <FileCode className="w-3.5 h-3.5 text-indigo-600 shrink-0" />
            )}
            <span className="truncate">{node.name}</span>
          </div>

          {node.hasIssue && (
            <span className="w-2 h-2 rounded-full bg-rose-500 shrink-0 shadow-xs" title="Issue Detected" />
          )}
        </div>
      );
    });
  };

  return (
    <div className="space-y-6 max-w-7xl mx-auto pb-16">
      {/* Title & Metadata */}
      <div className="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-200 pb-5">
        <div>
          <div className="flex items-center gap-2">
            <span className="text-xs uppercase font-bold tracking-wider px-2 py-0.5 rounded bg-indigo-50 text-indigo-700 border border-indigo-200">
              Repository Intelligence
            </span>
```

```
134             <span className="text-xs text-slate-500 font-mono">AST v4.1 Synced</span>
135           </div>
136           <h1 className="text-2xl font-bold text-slate-900 tracking-tight mt-1">
137             {currentRepo.name}
138           </h1>
139           <p className="text-sm text-slate-600">
140             {currentRepo.description}
141           </p>
142         </div>
143
144         <div className="flex items-center gap-3 self-start md:self-auto">
145           {onOpenRepoModal && (
146             <button
147               onClick={onOpenRepoModal}
148               className="flex items-center gap-1.5 px-4 py-2.5 rounded-xl bg-white hover:bg-slate-50 border border-s
late-300 text-slate-700 font-semibold text-xs shadow-xs transition-colors"
149               title="Switch repository or connect Git"
150             >
151               <FolderGit2 className="w-3.5 h-3.5 text-indigo-600" />
152               <span>Switch / Connect Git</span>
153             </button>
154           )}
155
156           <button
157             onClick={onInvestigateIssue}
158             className="flex items-center gap-2 px-5 py-2.5 rounded-xl bg-rose-600 hover:bg-rose-700 text-white font-
semibold text-sm shadow-md shadow-rose-600/20 transition-all"
159           >
160             <AlertTriangle className="w-4 h-4" />
161             <span>Investigate {currentRepo.activeIssueId || 'Issue'} with AI</span>
162           </button>
163         </div>
164       </div>
165
166       {/* Languages & Health Grid */}
```

```
      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
        {/* Languages Breakdown */}
        <div className="p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
          <div className="flex items-center justify-between mb-4">
            <span className="text-xs font-bold uppercase tracking-wider text-slate-700">
              Languages
            </span>
            <span className="text-xs text-slate-400">{currentRepo.languages.length} detected</span>
          </div>

          {/* Color stacked bar */}
          <div className="h-3 w-full rounded-full overflow-hidden flex gap-1 mb-4 bg-slate-100">
            {currentRepo.languages.map((lang) => (
              <div
                key={lang.name}
                style={{ width: `${lang.percentage}%`, backgroundColor: lang.color }}
                className="h-full rounded-full"
                title={`${lang.name}: ${lang.percentage}%`}
              />
            ))}
          </div>

          <div className="space-y-2">
            {currentRepo.languages.map((lang) => (
              <div key={lang.name} className="flex items-center justify-between text-xs">
                <div className="flex items-center gap-2">
                  <span
                    className="w-2.5 h-2.5 rounded-full"
                    style={{ backgroundColor: lang.color }}
                  />
                  <span className="text-slate-800 font-medium">{lang.name}</span>
                </div>
                <span className="font-mono text-slate-500">{lang.percentage}%</span>
              </div>
            ))}
```

```
202           </div>
203         </div>
204
205         {/* Code Health */}
206         <div className="lg:col-span-2 p-5 rounded-xl bg-white border border-slate-200 shadow-xs">
207           <div className="flex items-center justify-between mb-4">
208             <span className="text-xs font-bold uppercase tracking-wider text-slate-700">
209               Code Health
210             </span>
211             <span className="text-xs text-emerald-700 font-semibold bg-emerald-50 px-2.5 py-0.5 rounded-full border
border-emerald-200 flex items-center gap-1">
212               <CheckCircle2 className="w-3.5 h-3.5" />
213               Automated Audit Completed
214             </span>
215           </div>
216
217           <div className="grid grid-cols-2 sm:grid-cols-4 gap-3">
218             {/* Architecture */}
219             <div className="p-3.5 rounded-lg bg-slate-50 border border-slate-200">
220               <div className="text-[11px] text-slate-500 mb-1">Architecture</div>
221               <div className="flex items-center gap-1.5">
222                 <span className="w-2 h-2 rounded-full bg-emerald-500" />
223                 <span className="text-sm font-bold text-slate-900">{currentRepo.health.architecture.status}</span>
224               </div>
225               <div className="text-[10px] text-slate-500 mt-1 truncate">
226                 {currentRepo.health.architecture.label}
227               </div>
228             </div>
229
230             {/* Dependencies */}
231             <div className="p-3.5 rounded-lg bg-slate-50 border border-slate-200">
232               <div className="text-[11px] text-slate-500 mb-1">Dependencies</div>
233               <div className="flex items-center gap-1.5">
234                 <span className="w-2 h-2 rounded-full bg-emerald-500" />
235                 <span className="text-sm font-bold text-slate-900">{currentRepo.health.dependencies.status}</span>
```

```
              </div>
              <div className="text-[10px] text-slate-500 mt-1 truncate">
                {currentRepo.health.dependencies.label}
              </div>
            </div>

            {/* Security */}
            <div className="p-3.5 rounded-lg bg-rose-50 border border-rose-200">
              <div className="text-[11px] text-rose-700 font-semibold mb-1">Security</div>
              <div className="flex items-center gap-1.5">
                <span className="w-2 h-2 rounded-full bg-rose-500 animate-pulse" />
                <span className="text-sm font-bold text-rose-700">{currentRepo.health.security.status}</span>
              </div>
              <div className="text-[10px] text-rose-600 mt-1 truncate">
                {currentRepo.health.security.label}
              </div>
            </div>

            {/* Testing */}
            <div className="p-3.5 rounded-lg bg-slate-50 border border-slate-200">
              <div className="text-[11px] text-slate-500 mb-1">Testing</div>
              <div className="flex items-center gap-1.5">
                <span className="w-2 h-2 rounded-full bg-emerald-500" />
                <span className="text-sm font-bold text-slate-900">{currentRepo.health.testing.status}</span>
              </div>
              <div className="text-[10px] text-slate-500 mt-1 truncate">
                {currentRepo.health.testing.label}
              </div>
            </div>
          </div>
        </div>
      </div>

      {/* AI DETECTED ISSUE CALLOUT (Highlight Box) */}
```

```
      <div className="p-6 rounded-2xl bg-gradient-to-r from-rose-50/90 via-white to-indigo-50/60 border-2 border-rose-200 shadow-sm">
        <div className="flex flex-col lg:flex-row lg:items-center justify-between gap-6">
          <div className="space-y-2">
            <div className="flex flex-wrap items-center gap-2">
              <span className="text-xs uppercase font-extrabold tracking-wider px-2.5 py-0.5 rounded bg-rose-600 text-white shadow-xs">
                AI DETECTED ISSUE
              </span>
              <span className="text-xs font-mono font-bold bg-rose-100 border border-rose-300 px-2 py-0.5 rounded text-rose-800">
                {PRIMARY_ISSUE.code}
              </span>
              <span className="text-xs font-semibold px-2 py-0.5 rounded bg-rose-50 text-rose-700 border border-rose-200">
                Severity: {PRIMARY_ISSUE.severity}
              </span>
              <span className="text-xs font-semibold px-2 py-0.5 rounded bg-emerald-50 text-emerald-700 border border-emerald-200">
                Confidence: {PRIMARY_ISSUE.confidence}%
              </span>
            </div>

            <h3 className="text-xl font-bold text-slate-900">
              {PRIMARY_ISSUE.title}
            </h3>

            <p className="text-sm text-slate-600 max-w-3xl leading-relaxed">
              {PRIMARY_ISSUE.description}
            </p>

            <div className="flex items-center gap-2 text-xs text-slate-500 font-mono pt-1">
              <span>Affected file:</span>
              <span className="text-rose-700 font-semibold bg-rose-50 px-2 py-0.5 rounded border border-rose-200">
                {PRIMARY_ISSUE.affectedFile} ({PRIMARY_ISSUE.lines})
              </span>
            </div>
```

```
302           </div>
303
304           <div className="flex flex-col sm:flex-row items-center gap-2.5">
305             <button
306               onClick={() => onNavigate('live-fix')}
307               className="flex items-center justify-center gap-2 px-5 py-3.5 rounded-xl bg-gradient-to-r from-indigo-600 via-indigo-500 to-cyan-600 hover:from-indigo-700 hover:to-cyan-700 text-white font-semibold text-sm shadow-md shadow-indigo-600/20 hover:scale-[1.02] active:scale-[0.98] transition-all whitespace-nowrap"
308             >
309               <Zap className="w-4 h-4 text-cyan-300" />
310               <span>Watch Live Fix &amp;amp; Learn</span>
311             </button>
312
313             <button
314               onClick={onInvestigateIssue}
315               className="flex items-center justify-center gap-2 px-5 py-3.5 rounded-xl bg-slate-900 hover:bg-slate-800 text-white font-semibold text-sm shadow-md shadow-slate-900/10 hover:scale-[1.02] active:scale-[0.98] transition-all whitespace-nowrap"
316             >
317               <Sparkles className="w-4 h-4 text-indigo-400" />
318               <span>Investigate Issue</span>
319               <ArrowRight className="w-4 h-4" />
320             </button>
321           </div>
322         </div>
323       </div>
324
325       {/* File Tree & Code Preview Split */}
326       <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
327         {/* Tree Column */}
328         <div className="lg:col-span-4 p-4 rounded-xl bg-white border border-slate-200 shadow-xs">
329           <div className="flex items-center justify-between pb-3 mb-2 border-b border-slate-200">
330             <span className="text-xs font-bold uppercase tracking-wider text-slate-700 flex items-center gap-1.5">
331               <Layers className="w-3.5 h-3.5 text-indigo-600" />
332               Repository Structure
333             </span>
334             <span className="text-[11px] text-slate-400 font-medium">{currentRepo.metrics.files} files</span>
```

```
          </div>

          <div className="space-y-0.5 max-h-[420px] overflow-y-auto pr-1">
            {renderTree(REPO_FILE_TREE)}
          </div>
        </div>

        {/* Code Preview Column */}
        <div className="lg:col-span-8 rounded-xl bg-white border border-slate-200 shadow-xs overflow-hidden flex flex-col">
          <div className="px-4 py-2.5 bg-slate-50 border-b border-slate-200 flex flex-wrap items-center justify-between gap-2">
            <div className="flex items-center gap-2">
              <span className="w-3 h-3 rounded-full bg-rose-400" />
              <span className="w-3 h-3 rounded-full bg-amber-400" />
              <span className="w-3 h-3 rounded-full bg-emerald-400" />
              <span className="text-xs font-mono text-slate-800 ml-2 font-semibold">
                {selectedFile?.path || 'src/middleware/auth.ts'}
              </span>
            </div>

            {selectedFile?.hasIssue && (
              <div className="flex items-center gap-2">
                <span className="text-[11px] font-semibold text-rose-700 bg-rose-50 border border-rose-200 px-2 py-0.5 rounded">
                  AUTH-104 Vulnerability Detected Here
                </span>
                <button
                  onClick={() => onNavigate('live-fix')}
                  className="flex items-center gap-1.5 px-2.5 py-1 rounded bg-indigo-600 hover:bg-indigo-700 text-white font-bold text-[11px] shadow-xs transition-colors"
                >
                  <Zap className="w-3 h-3 text-cyan-300" />
                  <span>Fix Live</span>
                </button>
              </div>
```

```
            )}
          </div>

          <div className="p-4 font-mono text-xs overflow-x-auto text-slate-800 bg-[#fafafa] leading-relaxed max-h-[420px]">
            {selectedFile?.content ? (
              <pre className="space-y-0.5">
                {selectedFile.content.split('\n').map((line, idx) => {
                  const lineNum = idx + 1;
                  const isBugLine = selectedFile.hasIssue && lineNum >= 18 && lineNum <= 24;
                  return (
                    <div
                      key={idx}
                      className={`flex items-start gap-4 px-2 py-0.5 rounded ${
                        isBugLine
                          ? 'bg-rose-100/80 text-rose-950 border-l-4 border-rose-500 font-medium'
                          : 'hover:bg-slate-200/50'
                      }`}
                    >
                      <span className="text-slate-400 select-none w-6 text-right shrink-0">
                        {lineNum}
                      </span>
                      <span className="whitespace-pre">{line}</span>
                    </div>
                  );
                })}
              </pre>
            ) : (
              <div className="text-slate-400 italic py-8 text-center">
                Select a file from the repository tree to preview contents.
              </div>
            )}
          </div>
        </div>
      </div>
```

```
401       </div>
402     );
403   };
```

File [29/30]: src/app/api/repair-jobs/route.ts

TYPESCRIPT SOURCE

Description: *Repair Jobs Controller & Execution Dispatcher*

```typescript
import { NextRequest, NextResponse } from 'next/server';
import { requireAuth } from '@/lib/server/auth-helpers';
import { createRepairJob, getRepairJobsByUserId } from '@/lib/server/db';
import { runAutonomousRepairPipeline } from '@/lib/server/repair-pipeline';
import path from 'path';

export async function GET(req: NextRequest) {
  const auth = await requireAuth(req);
  if (auth.errorResponse) return auth.errorResponse;

  const jobs = getRepairJobsByUserId(auth.user.id);
  return NextResponse.json({
    success: true,
    jobs,
  });
}

export async function POST(req: NextRequest) {
  const auth = await requireAuth(req);
  if (auth.errorResponse) return auth.errorResponse;

  try {
    const body = await req.json().catch(() => ({}));
    const {
      repository,
      issueDescription,
      issueCode,
      reproductionEvidence,
      maxRetries,
    } = body;

```

```
    const defaultFixturePath = path.join(process.cwd(), 'fixtures', 'sample-vulnerable-repo');

    const jobRepo = {
      owner: repository?.owner || 'personale88',
      name: repository?.name || 'Nexora-X-Hub',
      branch: repository?.branch || 'main',
      gitUrl: repository?.gitUrl || 'https://github.com/personale88/Nexora-X-Hub.git',
      pinnedCommitSha: repository?.pinnedCommitSha || '7a8f93e210b48c909d8e7f1234a56b78c90d1e2f',
      localPath: repository?.localPath || defaultFixturePath,
    };

    const job = createRepairJob({
      userId: auth.user.id,
      repository: jobRepo,
      issueDescription:
        issueDescription ||
        'AUTH-104: Expired JWT tokens bypass authentication middleware without exp timestamp verification.',
      issueCode: issueCode || 'AUTH-104',
      reproductionEvidence,
      maxRetries: typeof maxRetries === 'number' ? maxRetries : 3,
    });

    // Run the pipeline asynchronously in background
    setTimeout(() => {
      runAutonomousRepairPipeline(job.id).catch((err) => {
        console.error(`Autonomous pipeline execution error for job ${job.id}:`, err);
      });
    }, 100);

    return NextResponse.json({
      success: true,
      job,
    });
  } catch (err: any) {
    return NextResponse.json(
```

```
67        { error: 'CREATE_JOB_FAILED', message: err.message },
68        { status: 500 }
69      );
70    }
71  }
```

File [30/30]: src/app/api/auth/dev-mock/route.ts

TYPESCRIPT SOURCE

Description: *Dev Mock Authentication & Demo Session Generator*

```typescript
import { NextRequest, NextResponse } from 'next/server';
import {
  createUser,
  getUserByEmail,
  linkOAuthAccount,
  createSession,
  saveConnectedGitHubAccount,
} from '@/lib/server/db';
import { setSessionCookie } from '@/lib/server/auth-helpers';

/**
 * Development Helper Route:
 * When OAuth credentials (GOOGLE_CLIENT_ID or GITHUB_CLIENT_ID) are not configured,
 * this endpoint provides a clean UI explaining the required setup and allows
 * instant local dev session simulation without breaking the application flow.
 */
export async function GET(req: NextRequest) {
  const origin = req.nextUrl.origin;
  const searchParams = req.nextUrl.searchParams;
  const provider = searchParams.get('provider') || 'google';
  const action = searchParams.get('action') || 'login';
  const userId = searchParams.get('userId');

  // If simulate query is passed, immediately create real session in database
  if (searchParams.get('simulate') === 'true') {
    const isGoogle = provider === 'google';

    const testUser = {
      name: isGoogle ? 'Demo Engineer (Google)' : 'personale88',
      email: isGoogle ? 'developer@example.com' : 'personale88@users.noreply.github.com',
      avatarUrl: isGoogle
```

```
      ? 'https://images.unsplash.com/photo-1534528741775-53994a69daeb?w=150&auto=format&fit=crop&q=80'
      : 'https://avatars.githubusercontent.com/u/214250353?v=4',
  };

  if (action === 'connect' && userId) {
    // Connect GitHub to existing user
    linkOAuthAccount({
      userId,
      provider: 'github',
      providerAccountId: '214250353',
      accessToken: 'gho_dev_simulated_token_at_rest',
      scope: 'read:user user:email repo',
    });

    saveConnectedGitHubAccount({
      userId,
      githubUsername: 'personale88',
      githubId: '214250353',
      avatarUrl: 'https://avatars.githubusercontent.com/u/214250353?v=4',
      scope: 'read:user user:email repo',
    });

    return NextResponse.redirect(new URL('/?github_connected=true', origin));
  }

  let user = testUser.email ? getUserByEmail(testUser.email) : null;
  if (!user) {
    user = createUser(testUser);
  }

  linkOAuthAccount({
    userId: user.id,
    provider: isGoogle ? 'google' : 'github',
    providerAccountId: isGoogle ? 'google_sim_123' : '214250353',
    accessToken: 'oauth_simulated_token_at_rest',
```

```
      scope: isGoogle ? 'openid profile email' : 'read:user user:email repo',
    });

    if (!isGoogle) {
      saveConnectedGitHubAccount({
        userId: user.id,
        githubUsername: 'personale88',
        githubId: '214250353',
        avatarUrl: testUser.avatarUrl,
        scope: 'read:user user:email repo',
      });
    }

    const session = createSession(user.id);
    const response = NextResponse.redirect(new URL('/', origin));
    setSessionCookie(response, session.id);
    return response;
  }

  // Render clean informational page explaining setup
  const providerName = provider === 'google' ? 'Google OAuth' : 'GitHub OAuth';
  const callbackUrl = provider === 'google'
    ? `${origin}/api/auth/callback/google`
    : `${origin}/api/auth/callback/github`;

  const html = `
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>RepoPilot AI — OAuth Setup Guide</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <script src="https://cdn.tailwindcss.com"></script>
</head>
```

```
<body class="bg-slate-50 text-slate-800 flex items-center justify-center min-h-screen p-6 font-sans">
  <div class="max-w-xl w-full bg-white border border-slate-200 rounded-2xl p-8 shadow-xl">
    <div class="flex items-center gap-3 mb-6">
      <div class="w-10 h-10 rounded-xl bg-indigo-600 text-white flex items-center justify-center font-bold text-lg">
        RP
      </div>
      <div>
        <h1 class="text-xl font-bold text-slate-900">${providerName} Setup Required</h1>
        <p class="text-xs text-slate-500">Configure your OAuth app credentials in .env.local</p>
      </div>
    </div>

    <div class="space-y-4 text-xs text-slate-600 leading-relaxed mb-6">
      <div class="p-3.5 bg-indigo-50 border border-indigo-200 rounded-xl text-indigo-900">
        <p class="font-semibold mb-1">To connect real ${providerName}:</p>
        <p>1. Register an OAuth application in your ${provider === 'google' ? 'Google Cloud Console' : 'GitHub Developer Settings'}.</p>
        <p>2. Set the Authorization Callback URL to:</p>
        <code class="block bg-white p-2 rounded mt-1.5 border border-indigo-200 font-mono text-slate-800 break-all select-all">${callbackUrl}</code>
        <p class="mt-2">3. Add client credentials to your <code>.env.local</code> file:</p>
        <pre class="bg-slate-900 text-slate-100 p-2.5 rounded mt-1 font-mono text-[11px] overflow-x-auto">${provider === 'google' ? 'GOOGLE_CLIENT_ID=your-id\nGOOGLE_CLIENT_SECRET=your-secret' : 'GITHUB_CLIENT_ID=your-id\nGITHUB_CLIENT_SECRET=your-secret'}</pre>
      </div>

      <div class="p-3 bg-emerald-50 border border-emerald-200 rounded-xl text-emerald-900 flex items-start gap-2">
        <span class="font-bold text-emerald-600">✓</span>
        <span>Local Development Mode: You can simulate a successful ${providerName} callback to verify the full dashboard and repository integration immediately.</span>
      </div>
    </div>

    <div class="flex items-center gap-3 pt-2">
      <a href="/api/auth/dev-mock?provider=${provider}&action=${action}&userId=${userId || ''}&simulate=true"
         class="flex-1 py-2.5 px-4 bg-indigo-600 hover:bg-indigo-700 text-white text-xs font-semibold rounded-xl text-center shadow-md transition-all">
        Simulate ${providerName} Login (Local Dev)
```

```
      </a>
      <a href="/"
         class="py-2.5 px-4 bg-slate-100 hover:bg-slate-200 text-slate-700 text-xs font-medium rounded-xl text-center transition-all">
        Back to App
      </a>
    </div>
  </div>
</body>
</html>
  `;

  return new NextResponse(html, {
    headers: { 'Content-Type': 'text/html; charset=utf-8' },
  });
}
```